

Fusion Compiler™ Graphical User Interface User Guide

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Preface

This preface includes the following sections:

- [About This User Guide](#)
- [Customer Support](#)

About This User Guide

The Synopsys Fusion Compiler tool provides a complete netlist-to-GDSII design solution, which combines proprietary design planning, physical synthesis, clock tree synthesis, and routing for logical and physical design implementations throughout the design flow.

This guide describes the Fusion Compiler implementation and integration flow. For more information about the Fusion Compiler tool, see the following companion volumes:

- *Library Manager User Guide*
- *Fusion Compiler Implementation User Guide*
- *Fusion Compiler Design Planning User Guide*
- *Fusion Compiler Data Model User Guide*
- *Fusion Compiler Timing Analysis User Guide*

Audience

This user guide is for design engineers who use the Fusion Compiler tool to implement designs.

To use the Fusion Compiler tool, you need to be skilled in physical design and synthesis and be familiar with the following:

- Physical design principles
- The Linux or UNIX operating system
- The tool command language (Tcl)

Related Publications

For additional information about the Fusion Compiler tool, see the documentation on the Synopsys SolvNet[®] online support site at the following address:

<https://solvnet.synopsys.com/DocsOnWeb>

You might also want to see the documentation for the following related Synopsys products:

- Design Compiler[®]
- IC Validator
- PrimeTime[®] Suite

Release Notes

Information about new features, enhancements, changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the *Fusion Compiler Release Notes* on the SolvNet site.

To see the *Fusion Compiler Release Notes*,

1. Go to the SolvNet Download Center located at the following address:

<https://solvnet.synopsys.com/DownloadCenter>

2. Select Fusion Compiler, and then select a release in the list that appears.

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>w- rite_file design_list</code> .
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
[]	Denotes optional arguments in syntax, such as <code>wri- te_file [-format fmt]</code>

Convention	Description
...	Indicates that arguments can be repeated as many times as needed, such as <i>pin1 pin2 ... pinN</i> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy.

Customer Support

Customer support is available through SolvNet online customer support and through contacting the Synopsys Technical Support Center.

Accessing SolvNet

The SolvNet site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNet site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNet site, go to the following address:

<https://solvnet.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNet site, click HELP in the top-right menu bar.

Contacting the Synopsys Technical Support Center

If you have problems, questions, or suggestions, you can contact the Synopsys Technical Support Center in the following ways:

- Open a support case to your local support center online by signing in to the SolvNet site at <https://solvnet.synopsys.com>, clicking Support, and then clicking “Open A Support Case.”
- Send an e-mail message to your local support center.
 - E-mail support_center@synopsys.com from within North America.
 - Find other local support center e-mail addresses at <https://www.synopsys.com/support/global-support-centers.html>
- Telephone your local support center.
 - Call (800) 245-8005 from within North America.
 - Find other local support center telephone numbers at <https://www.synopsys.com/support/global-support-centers.html>

1

Getting Started with the GUI

The Fusion Compiler GUI provides a flexible working environment with multiple windows for performing different tasks. The GUI windows operate independently in your X windows environment, but they all share the block and global selection data in the tool.

The following topics describe how to work with the GUI:

- [Starting the GUI](#)
- [Closing the GUI](#)
- [Exiting the Tool From the GUI](#)
- [Determining Whether the GUI is Running in a Script](#)
- [Using GUI Windows](#)
- [Working With the Console](#)
- [Setting Application Preferences](#)
- [Using Command Search](#)
- [Viewing and Editing Application Options](#)
- [Working in the Block Window](#)
- [Changing the View of the Design](#)
- [Editing the Physical Layout](#)
- [Viewing Metal Fill Data](#)

Starting the GUI

The Fusion Compiler GUI provides both menu commands and a command-line interface. It is an advanced analysis and physical editing tool. The Fusion Compiler tool can perform certain tasks, such as accurately displaying a block and providing visual analysis tools, only from the GUI. The look and feel of the Fusion Compiler GUI is consistent with the look and feel of other Synopsys GUIs.

The Fusion Compiler GUI uses the `DISPLAY` environment variable to determine the display location. Before using the GUI, ensure that this variable is set to the name of your UNIX or Linux system display.

To start the Fusion Compiler tool in the GUI,

1. Make sure the path to the bin directory is included in your `PATH` variable.
2. Enter the `fc_shell` command with the `-gui` option in a UNIX or Linux shell.

```
% fc_shell-mode-gui
```

where *mode* is either *fe* for front-end mode or *be* for back-end mode.

To start the GUI for the Library Manager tool, use the `icc2_lm_shell` command with the `-gui` option as follows.

```
% icc2_lm_shell -gui
```

You can include other options on the command line when you start the Fusion Compiler GUI. For example, you can use

- `-help` to display a list of all available options
- `-f script_file_name` to execute a script
- `-x command` to execute an `fc_shell` command

When you start the GUI, the tool

1. Runs the commands in the setup files

The Fusion Compiler tool loads the `.synopsys_fc.setup` and `.synopsys_fc_gui/setup.tcl` setup files; the Library Manager tool loads the `.synopsys_icc2_lm.setup` and `.synopsys_icc2_lm_gui/setup.tcl` setup files. The tool first runs the setup files in your home directory followed by the files in the local directory where you ran the `fc_shell` or `icc2_lm_shell` command.

2. Runs the command specified by the `-x` option or the script specified by the `-f` option
3. Opens a main window that displays the splash screen

By default, the console appears attached (docked) to the main window above the status bar.

The Fusion Compiler tool performs the same startup tasks when you start the tool in the command-line interface and open a block.

If you start either tool without the GUI, you can open the GUI by entering the `gui_start` command at the command prompt.

```
gui_start
```

To execute a script before opening the GUI, use the `-file` option. For example,

```
gui_start -file my_gui_setup.tcl
```

If you open a block at the `fc_shell` prompt and then open the GUI, the Fusion Compiler tool automatically opens a block window and displays the block in the layout view. If you have multiple blocks open when you open the GUI, the Fusion Compiler tool opens a separate block window for each block. The active window (the window that has the mouse focus) is the window with the current block.

The Fusion Compiler GUI provides menu commands and dialog boxes for most graphic features, such as viewing, selecting, and highlighting objects in layout views. Menu commands are grouped by function on the menus in each GUI window.

Closing the GUI

You can open or close the GUI at any time during the session. For example, if you need to conserve system resources, you can close the GUI and continue the session in `fc_shell` or `icc2_lm_shell`.

To close the GUI, do one of the following:

- Choose File > Close GUI in the block window.
- Enter the `gui_stop` command.

```
gui_stop
```

When you are ready to use the GUI again, enter the `gui_start` command on the command line.

Exiting the Tool From the GUI

You can end the session and exit the Fusion Compiler or Library Manager tool at any time. To exit the tool, enter the `exit` or `quit` command or choose File > Exit. Enter the `quit!` command to exit and ignore any prompts issued by the tool.

When the GUI is open, the tool behavior depends on whether there are unsaved changes to any open block.

- If there are unsaved changes in any open block, the Exit Fusion Compiler dialog box or Exit Library Manager dialog box appears, which allows you to save the changes to the modified blocks.
- If there are no unsaved block changes, a message box appears instead of the Exit Fusion Compiler dialog box or Exit Library Manager dialog box. If you see this message box, click OK to exit.

Note:

The tool does not save view settings when you exit. If you have changed view settings for layout or schematic views and want to save them, do so before exiting the tool, as explained in [Setting and Saving View Properties](#).

Running the GUI in Batch Mode Without an X Server

The tool supports a mode that allows you to run GUI-related commands without running an X server. When you run the `fc_shell` command with the `-batch` option, the GUI can run in a “headless” mode that does not require an X server or display window when you specify the `-gui` option or the `start_gui` command. In this mode, you supply Tcl commands to the tool in a script file specified with the `-file` option or by specifying the commands on the command line with the `-x` option.

For example, the following script loads a design, creates a screenshot of the layout, and exits the tool:

```
# capture_layout.tcl
open_block ./orca:orca.design
start_gui
gui_write_window_image -file ./layout.png -size {1000 1000} \
    -window [gui_get_current_window -types Layout -mru] -format png
quit!
```

To run the script in headless mode, specify the `-batch` option followed by the `-f` or `-x` option as shown in the following examples:

```
% fc_shell -batch -f capture_layout.tcl
% fc_shell -batch -x "source capture_layout.tcl"
% fc_shell -batch \
    -x 'open_block ./orca:orca.design; start_gui; \
        gui_write_window_image -file ./layout.png -format png; \
        quit!'
```

Instead of using the `-batch` option, you can use the following commands and options to run the tool in headless mode:

- Specify the `-offscreen 1` option with the `gui_start` command.

```
# capture_layout.tcl using gui_start -offscreen 1
open_block ./orca:orca.design
start_gui -offscreen 1
gui_write_window_image -file ./layout.png -size {1000 1000} \
    -window [gui_get_current_window -types Layout -mru] -format png
quit!
```

- Set the `gui.batch_x_display` application option to `offscreen`.

```
# capture_layout.tcl using gui.batch_x_display offscreen
open_block ./orca:orca.design
set_app_options -list {gui.batch_x_display offscreen}
start_gui
gui_write_window_image -file ./layout.png -size {1000 1000} \
    -window [gui_get_current_window -types Layout -mru] -format png
quit!
```

In each example, the tool runs the commands without displaying a GUI window and does not require a running X server.

Determining Whether the GUI is Running in a Script

If you are planning to execute commands in the GUI from a script, use the `in_gui_session` application variable to determine if the GUI is active or not. The variable is `true` when the GUI is active and `false` when inactive as shown in the following example.

```
fc_shell> gui_start
fc_shell> get_app_var in_gui_session
true
fc_shell> gui_stop
To restart the GUI, type 'gui_start'.
fc_shell> get_app_var in_gui_session
false
```

Using GUI Windows

Each top-level GUI window displays design information in view windows and provides menus and dialog boxes (with keyboard shortcuts), toolbars and panels, interactive mouse button tools, a command console (with a command-line interface), and a status bar.

The Fusion Compiler GUI provides the following types of windows:

- Block window
- Clock tree analysis window
- Message browser window
- Relative placement (RP) window
- Timing analysis window

Each top-level GUI window is independent of other GUI windows. The title bar displays the product name (Fusion Compiler), the window name (BlockWindow.1 for the window that appears when you start the GUI), the current task, and the name of the current library

and block. Design objects you select in one window are automatically selected in the other windows.

You can open and close GUI windows, and you can open multiple instances of each window to work with different block settings. Each window instance has a unique name that includes an incremented number.

To open a new instance of a GUI window, choose Window > *window-type*.

To close a GUI window, choose Window > Close Window in the window you want to close.

Note:

If you close all the open GUI windows, you end the GUI session but you remain in the `fc_shell` and your blocks remain open. You can reopen the GUI by using the `gui_start` command.

You can move, resize, minimize, or maximize GUI windows by using the window management tools on your UNIX or Linux desktop.

The following topics describe the GUI window components:

- [Menu Bar](#)
- [Toolbars](#)
- [Status Bar](#)
- [View Windows](#)
- [Panels](#)

Menu Bar

The menu bar at the top of a GUI window contains menus with common tasks and commands. You can display a brief message in the status bar about the action that a menu command performs by holding the pointer over the command. For menu commands that can also be used by clicking a toolbar button or pressing a keyboard shortcut, the menus show representations of those alternatives.

Note:

If a window is not wide enough to display all the menu names on the menu bar, the window displays all the menu names that fit, from left to right, followed by an overflow button (»). To access the other menus, click the overflow button.

Toolbars

Each top-level window provides toolbars with buttons you can use to quickly perform frequently used operations or tasks. To determine the function of a toolbar button, hold the pointer over the button. A tooltip displays the name of the button, and the status bar displays a brief description of its use. You cannot disable these messages.

Toolbars are always attached to a window edge. If a window edge is not long enough to display all the toolbars attached to it, the GUI displays the full toolbars that fit and shortened versions of the other toolbars. A shortened toolbar displays an overflow button (). To access the other buttons on a shortened toolbar, click the overflow button.

You can move a toolbar to another location by dragging the double bars on one of its sides. You can also disable a toolbar, hiding it from view.

To display or hide a toolbar or panel, choose View > Toolbars > *toolbar_name*. A check mark beside the name of a toolbar on the Toolbars menu indicates that the toolbar is visible.

Status Bar

By default, each GUI window displays a status bar at the bottom of the window. The status bar displays the following information:

- Status information such as the number and type of selected objects
- Information about the action performed by the toolbar button, menu command, or workspace tab under the pointer

If you hold the pointer over a menu command, a toolbar button, or a tab in the workspace, the status bar displays a brief message about the action that the command, button, or tab performs.

- The relative coordinates of the pointer position over an active layout view (block windows only)

To quickly display the list of selected objects in the Selection List dialog box, click the  button at the right end of the status bar.

To display or hide the status bar, choose View > Status Bar. A check mark next to the Status Bar command on the View menu indicates that the status bar is visible.

View Windows

The Fusion Compiler tool displays design information in view windows, which are child windows that open within the workspace area of a GUI window. View windows display specific graphic or textual design information.

Most menu commands and toolbar buttons in a GUI window operate in the active view, which is the view window that has the mouse focus. Some view windows contain two or more panes, with a view in each pane. You can drag the split bars between two panes to adjust their relative sizes within the window.

You can resize a view window or move it to another location inside the workspace area of its parent window.

To change the size of a view window,

1. Ensure that the window is restored (it does not take up the entire workspace area).
If the view window is maximized, click the Restore Down button at the far right of the menu bar to restore it.
2. Hold the pointer over an edge or corner of the window until the pointer changes shape.
3. Drag the edge or corner to resize the window.

To move a view window,

1. Move the pointer over the title bar.
2. Drag the window to a different location.

You can organize view windows by tiling them within the workspace area. To tile the view windows in the workspace area, choose Window > Tile Views.

You can also minimize individual view windows, or maximize a view window to fill the workspace area.

In most views, you can use navigation keys (the arrow keys, Page Up, Page Down, Home, and End) to scroll through and navigate the view.

Panels

Panels are enhanced toolbars that open within the workspace area of a top-level window to provide tools or views for performing particular tasks. Most panels are associated with a particular view window and operate on the active view (the view window that has the mouse focus). An exception is the console, which contains a command line and its own views.

Some panels contain tabs that you can click to access different tools or views. The first time that you open a panel during a session, it displays the tools or view for its default tab.

Move a panel to another location on or off the parent window by dragging the double bars on one of its sides. You can also dock a panel (attach it to a window edge) or let it float where you leave it.

Disable a panel to hide it. (A panel that is associated with a particular view is automatically hidden when you close the view window).

To display or hide a toolbar or panel, choose View > Toolbars > *panel_name*. A check mark beside the name of a panel on the Toolbars menu indicates that the panel is visible.

Working With the Console

The console provides a command-line interface and two views, a log view that displays the session transcript (the default) and a history view that displays the command list. To open the console, choose View > Toolbars > Console from the menu.

You can use the console to

- Enter `fc_shell` commands on the console command line
- View either the session transcript (log view) or the command history list (history view) by clicking the Log or History tab above the command line
- Display an error message man page in the man page viewer by clicking the message number in the console log view
- Copy and edit or reuse commands
- Search for, select, and save commands or messages in the log view or the history view

You can enter any `fc_shell` command on the console command line. For example, if you enter `get_selection`, the console log view displays a list of the names of all selected objects. When you enter a command on the console command line, make sure the console has the mouse focus.

To enter a command on the console command line,

1. Type the command.
2. Click the `ifc_shell>` prompt button or press Enter.

You can open (or close) one console in each top-level window. When the console is open, you can dock it to the bottom or top of its parent window, or move it over or away from the window.

Viewing the Session Log

Within the Console toolbar, the console log view in the Log tab displays a transcript of session information. The tool records the commands you entered and the tool output and messages resulting from your commands. You can use this information to check on tool status after performing functions, to troubleshoot problems you encounter, and to look up information about past functions. You can reenter commands you have already used by copying them from the log to the `fc_shell` command line.

To view the session transcript, click the Log tab in the console. The console changes to the log view, which is displayed by default whenever the console is opened.

The log view displays the commands you entered at the command prompt. Warnings and error messages are noted with “Warning” or “Error” as the first word. If you need to see information not currently displayed in the display area of the log view, use the scroll bars to scroll through the session information.

You can view commands in the Options menu by clicking the Options button () on the right side of the console above the command line. Use the Options menu commands to find text in the transcript, copy text in the transcript, and save text from the transcript in a text file.

To find text in the transcript,

1. Click the Options button and choose Find.

The Find Text in Console dialog box appears.

2. Type the text you want to find in the Find box.

You must type the text exactly as it appears in the transcript, including uppercase and lowercase letters and blank spaces.

3. Search forward in the transcript by clicking Next, or backward by clicking Prev.

The console highlights the next or previous instance of the text. Repeat this step as needed to find additional instances of the text.

To copy text from the transcript to the command line,

1. Select the command or portion of the command that you want to copy.
2. Move your mouse to the command line and click the middle mouse button.

If you already have text on the command line, middle-click where you want to paste the selected information.

To copy text from the transcript that you want to paste somewhere else, such as in a file,

1. Select the text.

To select all the text in the transcript, choose **Select All** from the **Options** menu.

2. Choose **Copy** from the **Options** menu.

To save text from the transcript in a text file,

1. Do one of the following:

- To save the transcript, choose **Save Contents As** from the **Options** menu.
- To save selected text, select the text and choose **Save Selection As** from the **Options** menu.
- To save error and warning messages, choose **Save Errors/Warnings As** from the **Options** menu.

2. In the dialog box that appears, navigate to the directory where you want to save the file.

3. Enter the file name in the “File name” box.

4. Click **Save**.

Viewing the History View

The console history view lists shell, menu, and stroke commands you have used in the current session. You can do the following with this list:

- Review the commands you have used in this session
- Copy commands in the list
- Save the list to use at a later time

To review the command history for the current session,

- Click the **History** tab in the command console.

The console displays the list of commands, each with a number showing the order in which you used it.

Use the scroll bars to view information not currently in the display area of the history view. Alternatively, you can enter the `history` command on the command line. This displays the list of commands in the log view.

To reuse a command listed in the history view, you can do any of the following:

- Double-click the command in the history view.

- Select the command in the view and click either Execute (to immediately rerun the command) or Edit (to paste the command on the command line where you can edit it).
- Enter an exclamation point character followed by the number of the command on the `fc_shell` command line. For example, to reuse the fifth command, use the following command:

```
fc_shell> !5
```

You can copy commands in the list that you want to paste somewhere else, such as in a file.

To copy commands in the history list,

1. Select the text.

To select all the text in the transcript, choose Select All from the Options menu.

2. Choose Copy from the Options menu.

To save the command history list (or a portion of the list) in a text file,

1. (Optional) To save a portion of the list, select the commands you want to save.
2. Do one of the following:
 - If you are saving the entire list, click the Save Contents As button or choose Save Contents As from the Options menu.

The Save Contents As dialog box appears.

- If you are saving just a portion of the list, choose Save Selection As from the Options menu.

The Save Selection As dialog box appears.

3. If necessary, navigate to the directory where you want to save the file.
4. Enter the file name in the “File name” box.
5. Click Save.

You can set options in the Global Settings category of the Application Preferences dialog box to control which types of GUI commands are included in the history list. To open the Application Preferences dialog box, choose View > Preferences.

Setting Application Preferences

Application preferences are used to set the font, icon size, default colors and color scheme, and other display settings for the GUI. To change the application preferences, choose View > Preferences to open the Application Preferences dialog box.

Changing Icon Size

To increase the size of the toolbar icons from the default size of 16 point, choose View > Preferences and select the Global Settings entry in the Categories box as shown in [Figure 1](#). In the Icon sizes section, select a larger value in the Large icon size box. [Figure 2](#) shows a toolbar displayed with 16, 24, and 32 point icon sizes.

Figure 1 Application Preferences for Icon Size

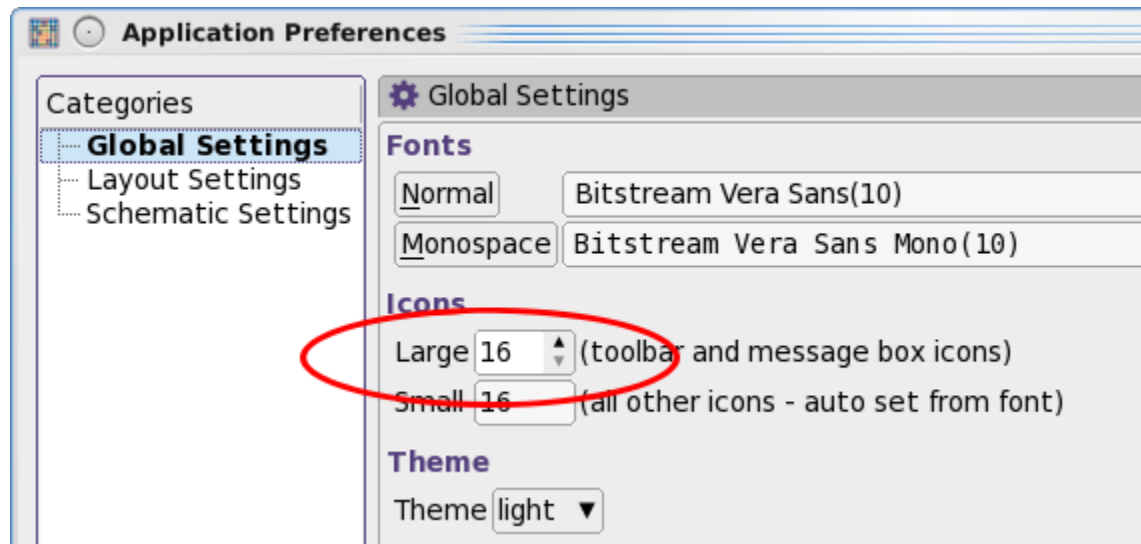
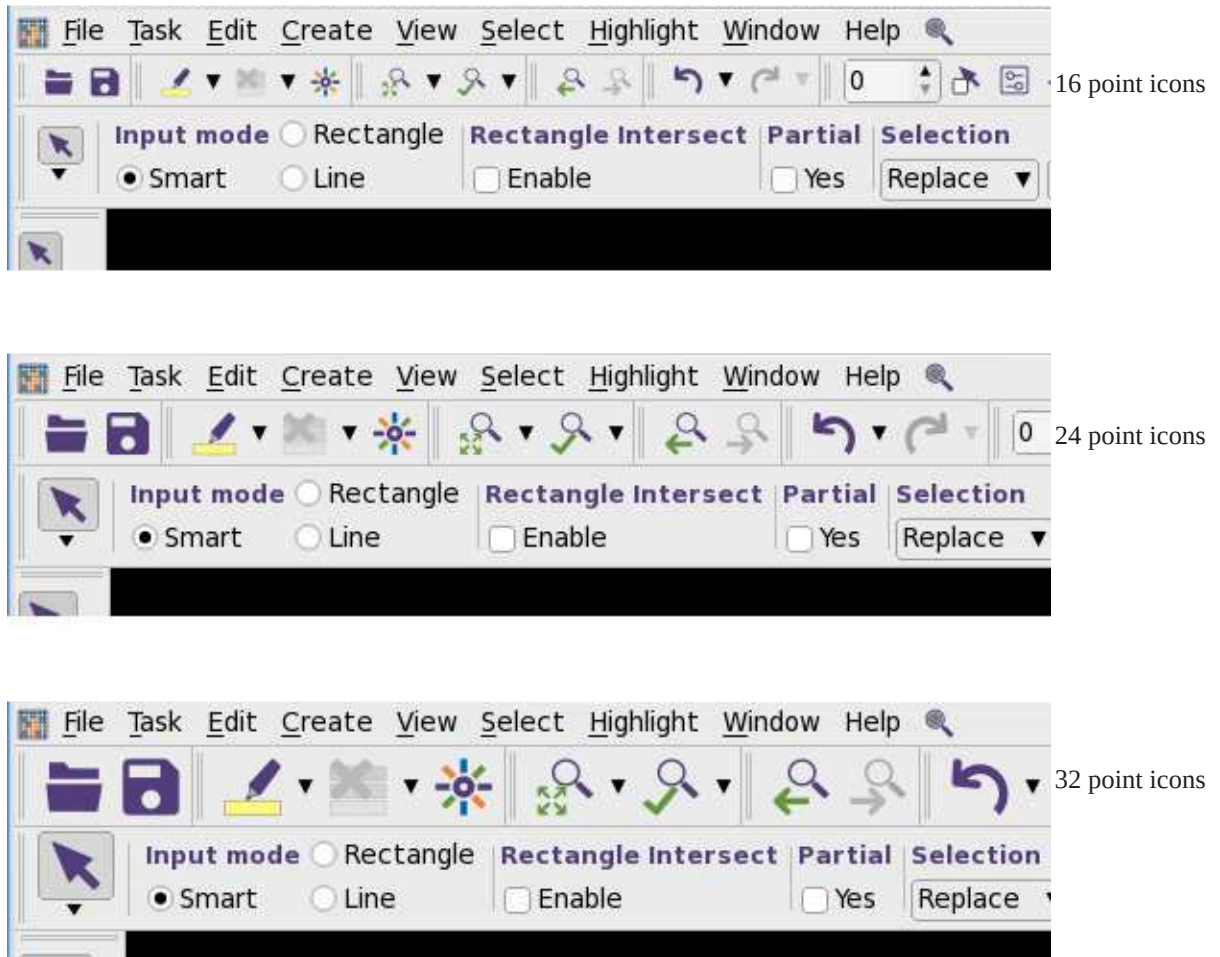


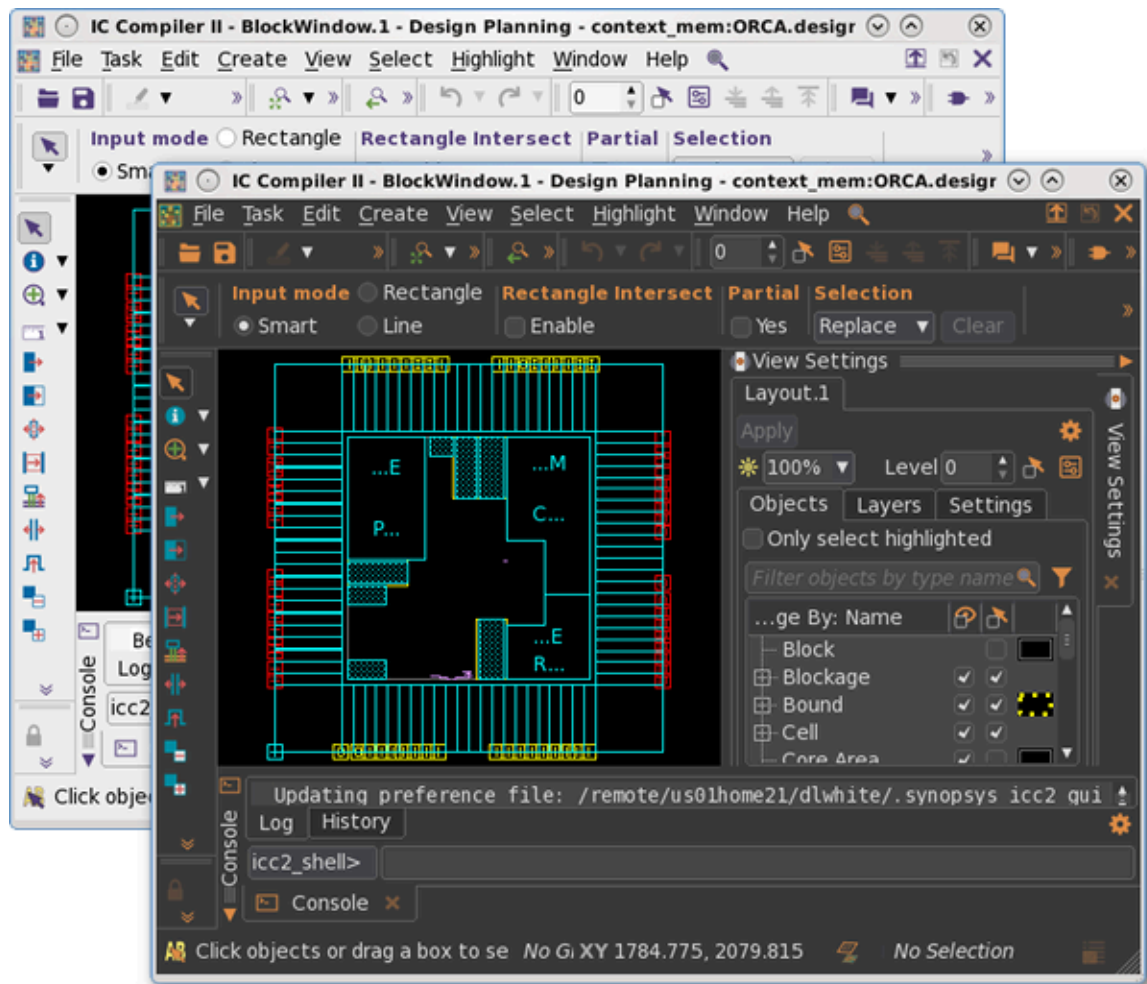
Figure 2 Toolbars Displayed Using 16, 24, and 32 Point Icons



Selecting a Light or Dark Display Theme

To change the background from the default light gray theme to a dark gray theme, choose View > Preferences and select dark in the Theme box. [Figure 3](#) shows both light and dark themes.

Figure 3 Light and Dark Themes



Using Command Search

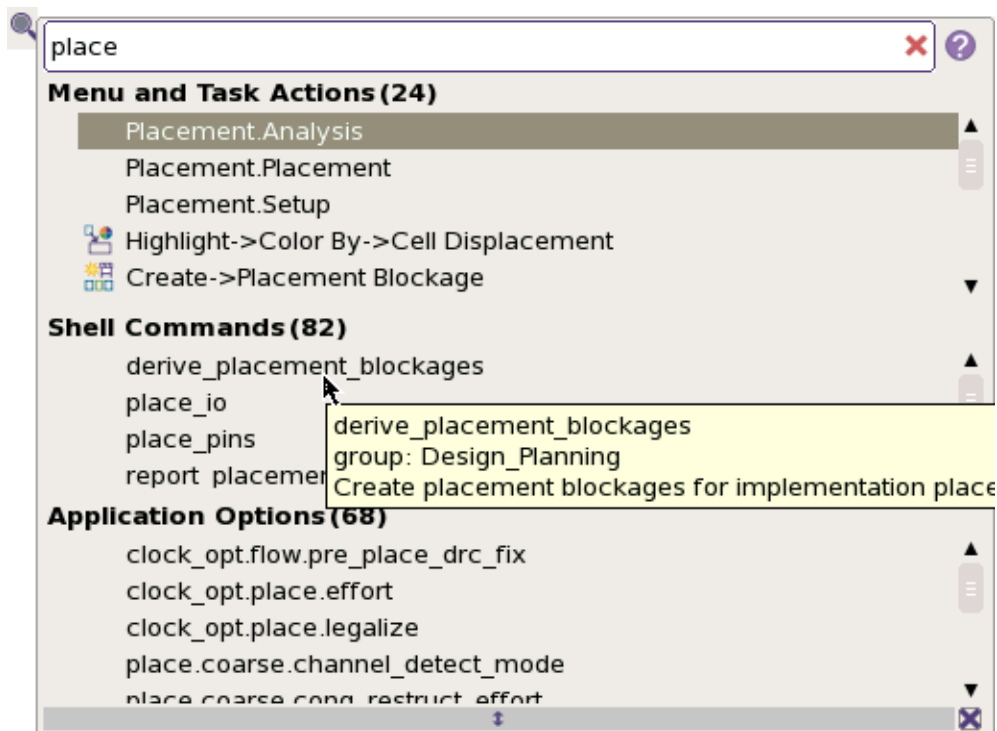
The Fusion Compiler GUI provides a command search feature to quickly locate menu and task actions, shell commands, and application options.

To search for a command,

1. Click the magnifying glass icon (🔍) on the menu bar.
2. Click one or more options in the Search for: section at the top of the form.
3. Click in the search box and type a few letters of the command name or topic, or type multiple terms separated by space characters, to return a list of matching results.

As you type, the tool returns a list of menu and task actions, shell commands, and application options related to the search term, based on the options you selected in step 2. For shell commands, the tool searches a list of command names and command options, and sorts the search results based on the quality of the match. Command name matches are placed higher in the search results than matches for command options. Move the pointer over the item in the search results to view an infotip that contains information about the result as shown in [Figure 4](#).

Figure 4 Command Search Results



The command search supports full or partial matching of one or more search terms in the search box. For example, to search for results related to block shaping, you can enter "shape", "block", "shape", "shape_blocks", and so on. If the tool returns a large number of results, a scroll bar is displayed on the right to navigate the list. Alternatively, edit your search term to narrow the search results. Finally, click the item from the list that matches your search.

Click an item in the Menu and Task Actions list to run that item. If a dialog box or panel is available for the action, the tool opens it. The tool dynamically generates a dialog box if you click an item from the Shell Commands section. For more information, see [Using the Command Dialog](#).

If you click an item in the Application Options list, the tool opens the Application Options dialog box and displays the application option you selected. Set the value of the

application option as needed and click OK to save the changes, or Cancel to leave the option unchanged. Alternatively, you can open the Application Options dialog box by choosing File > Application Options from the menu bar.

As you use the command search, the tool remembers the search terms and actions you select for the current tool session. The tool displays the history of previous selections in the Recently found section, and the search terms for this session in the Recently searched section. Click an item in the Recently found section to perform the action again, or click the item in the Recently searched for section to reproduce the search.

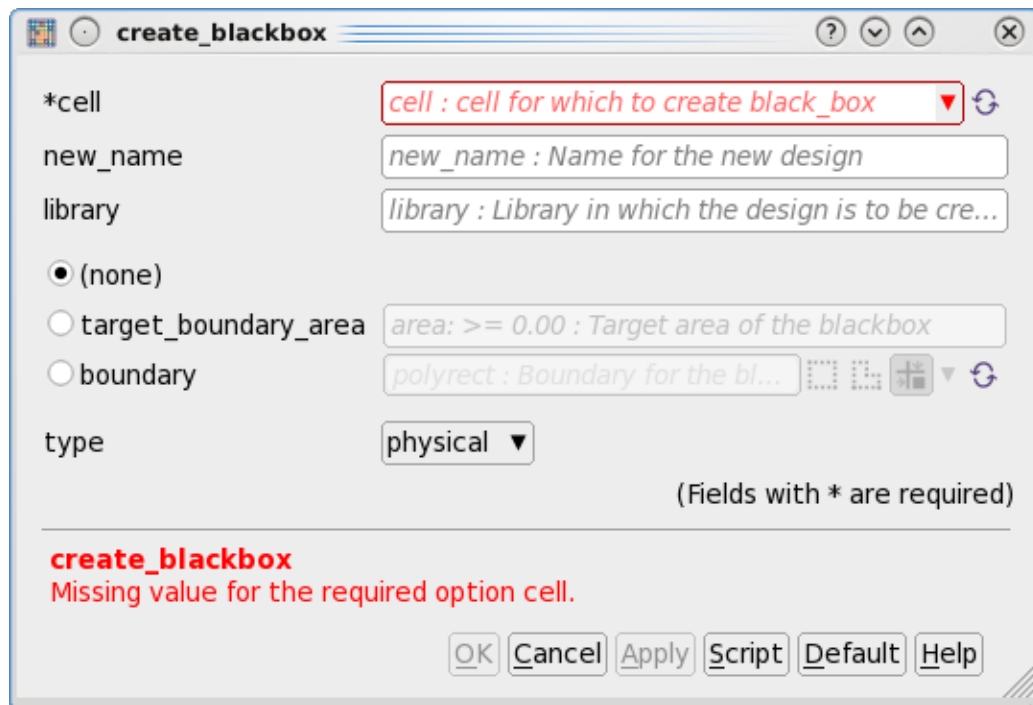
The command search tool supports these additional features:

- Click the book icon to the right of the search box to open a Web browser with a link to search SolvNet for the search term.
- Click the (X) to the right of the search box to clear any text in the search box (only if the search box contains text).
- Click the Tab key to move the focus between the search bar and three sections in the command search tool.
- Hover your pointer between the Menu and Task Actions section and Shell Commands section, or between the Shell Commands and Application Options sections, to reveal a splitter; click and drag the splitter to resize the section.
- Click and drag the bar at the bottom of the command search tool to resize the dialog box.
- Click outside the command search tool, or press the Esc key, to dismiss the tool.

Using the Command Dialog

In the Fusion Compiler tool, the graphical user interface generates dialog boxes on-the-fly for Tcl commands in the tool. When you click an item in the Shell Commands list in the command search, the tool generates a dialog box for the command as shown in [Figure 5](#).

Figure 5 *create_blackbox Dialog Box*



The dialog box lists options and data entry fields in the top section, and displays status information and a command preview in the bottom section. The bottom row contains buttons to perform tasks related to this dialog box.

In the top section, required fields in the dialog box are marked with an asterisk (*). As you type in a text box, the tool validates your input. If no error is detected in your input, the tool displays your text in blue and updates the command preview. If your input contains an error, the tool displays your text in red and displays an error message in the command preview area. Click the Default button to restore the default values in the dialog box. For more information about the purpose of a particular text box, hover the pointer over the box to view the infotip.

You can also enter Tcl expressions and variables in command dialog boxes. For example, you can enter the following in the cell text box in [Figure 5](#):

- The actual cell name, such as "u0_0"
- The "\$cell" variable, to specify a variable that you defined which contains the cell name
- A Tcl expression, such as "[get_cells u0_0]"

When possible, the dialog box provides tools to select bounding boxes, layers, and enumerated values to help you specify the command. To specify a boundary, do one of the following:

- To define a rectangular bound, select the  button, and either type the x- and y-coordinates for the upper-left and lower-right corners of the rectangle in the Coordinates box or drag the pointer in the layout view to form the rectangle where you need it.
- To define a rectilinear bound, select the  button, and either type the x- and y-coordinates for each corner or direction change in the Coordinates box or drag the pointer in the layout view to form the sides of the polygon where you need it.
- To enter coordinates manually, click the  button and type the x- and y-coordinates for the upper-left and lower-right corners of the rectangle in the Coordinates box.
- To enable or disable grid snapping, select or deselect the  button.
- To select options from a list, click the  button and choose an option on the menu that appears.

If the command option requires that you specify one or more layers, click the  button and choose the layers. Alternatively, click the  button and type the layer names.

If the command option supports multiple, enumerated values, click the  button and choose an option on the menu that appears. For all other form entries, type the value in the text box.

In the bottom section, the tool provides buttons to perform different actions.

- Click the OK button to run the command and close the dialog box. If the dialog box contains an error, you must correct the error before clicking OK.
- Click Cancel to stop the command.
- Click Apply to run the command and keep the dialog box open.
- Click Script to copy the command to the Script Editor window without running the command.
- Click the Default button to restore the default values in the dialog box.
- Click Help to open the man page for the command in the Man Page Viewer.

Using Topic Search

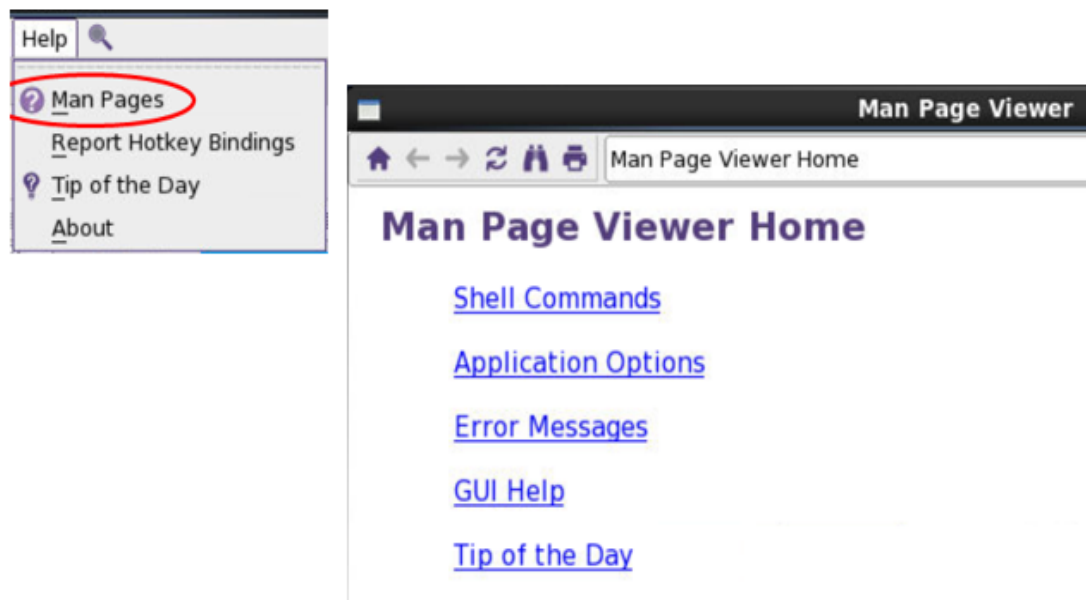
The tool provides an interface for searching help topics and other documentation available in the installation directory. Searchable documents include man pages, tips-of-the-day, and help topics for GUI features that are accessed through the menu or by clicking a button. Matches are returned in the search results if the search term is contained in the command name, application option name, document title, or document keyword list. Note that the search tool does not consider online documentation available through SolvNet.

To open the search tool,

1. Choose Help > Man Pages from the menu to open the man page viewer.

The first time you open the search tool, the tool displays links for viewing shell commands, application options, error messages, GUI help, and tip-of-the-day topics as shown in the following figure.

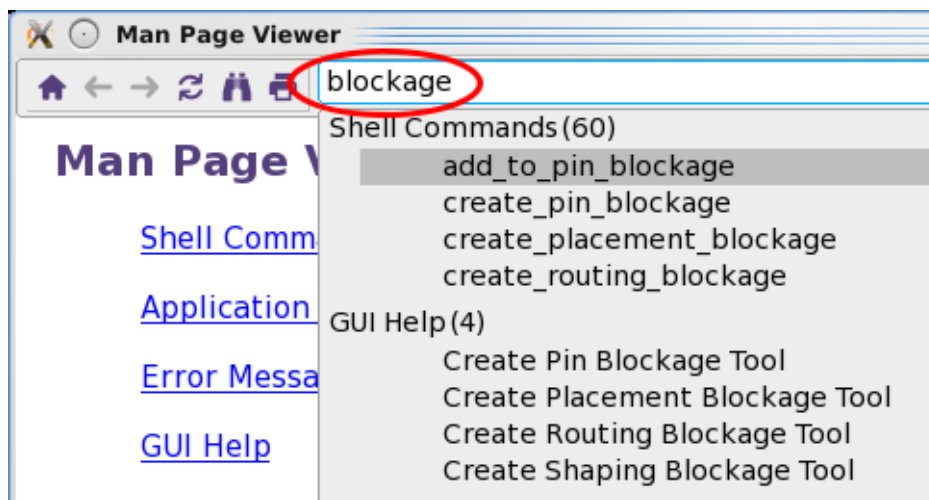
Figure 6 *Man Page Viewer*



2. (Optional) Click on a link to view a table of contents for that category.

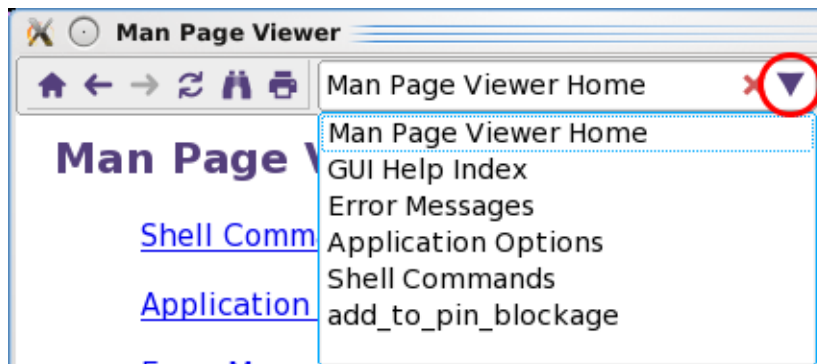
3. Enter a search term in the title bar to display commands, application options, and help topics that match the term.

Figure 7 Search Term



4. Click the down arrow at the right of the search box to view the search history.

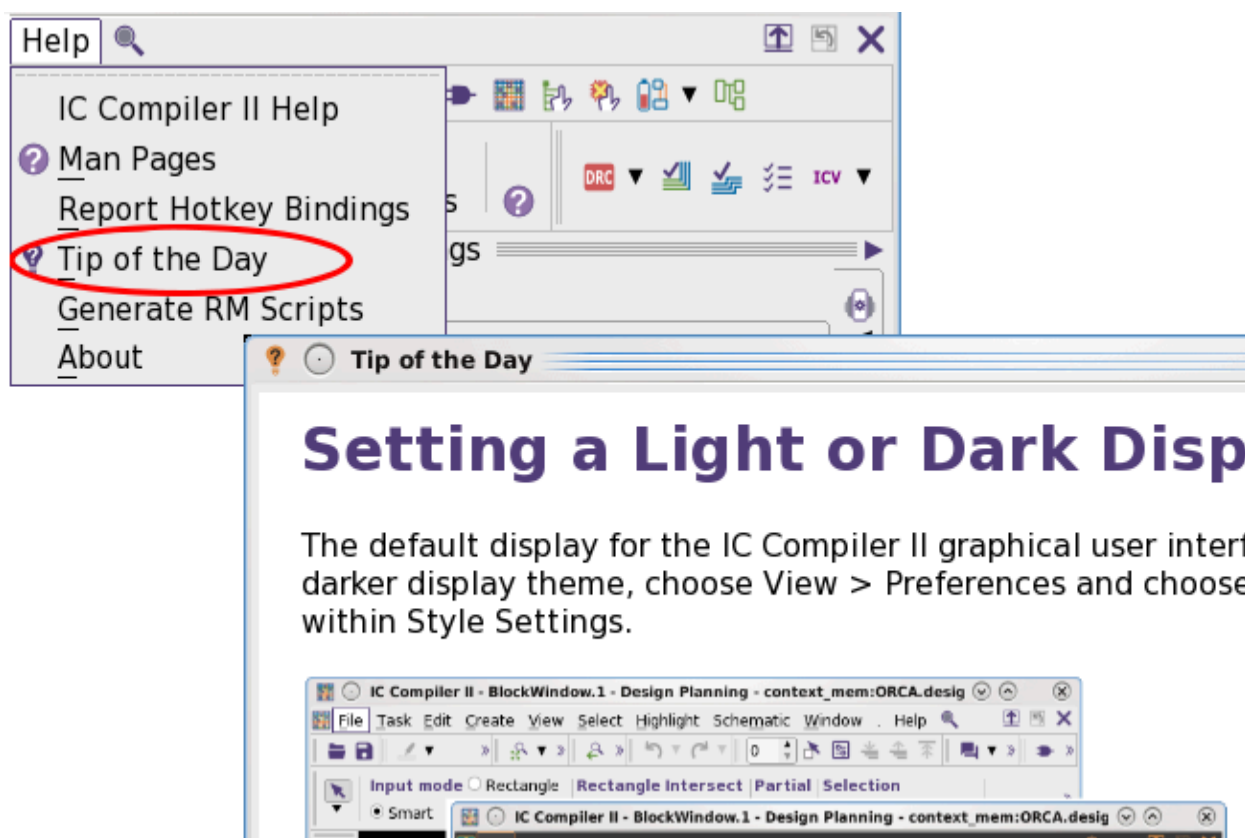
Figure 8 Search History



Viewing the Tip of the Day

The tool provides a set of helpful hints that you can use to improve your workflow. To view the tip of the day, choose Help > Tip of the Day from the menu. The tool displays a random tip of the day. Use the Prev and Next buttons to view additional tips.

Figure 9 Tip of the Day



Creating Reference Methodology Scripts

The Synopsys reference methodology for the Fusion Compiler tool is a set of scripts that capture the latest recommended methodology for using the tool for various vendors and process nodes. These scripts are a starting point from which you can adapt and build based on your own design requirements. The scripts contain embedded comments and alternate settings to help you customize the flow as needed.

To generate RM scripts for the Fusion Compiler tool with the GUI,

1. Choose Help > Generate RM Scripts from the menu.
2. Select the Technology node for your design.
3. Enter the Foundry name for your design.
4. Select the version type for the scripts: Full RM or JumpStart. JumpStart is the starting point for new users with a block that is ready for the `place_opt` command; the scripts run the recommended default place-and-route flow for flat designs.

5. Enter an output directory for the scripts. By default, the compressed tar file is written to the current directory.
6. Click the box next to “I agree to the above Terms and Conditions.”
7. Click OK.

These steps write a compressed tar file to the directory you specified in step 5. After writing out the file, use the `tar xzf filename` command to extract the files. You can also create the Reference Methodology scripts with the `generate_rm` command.

Note:

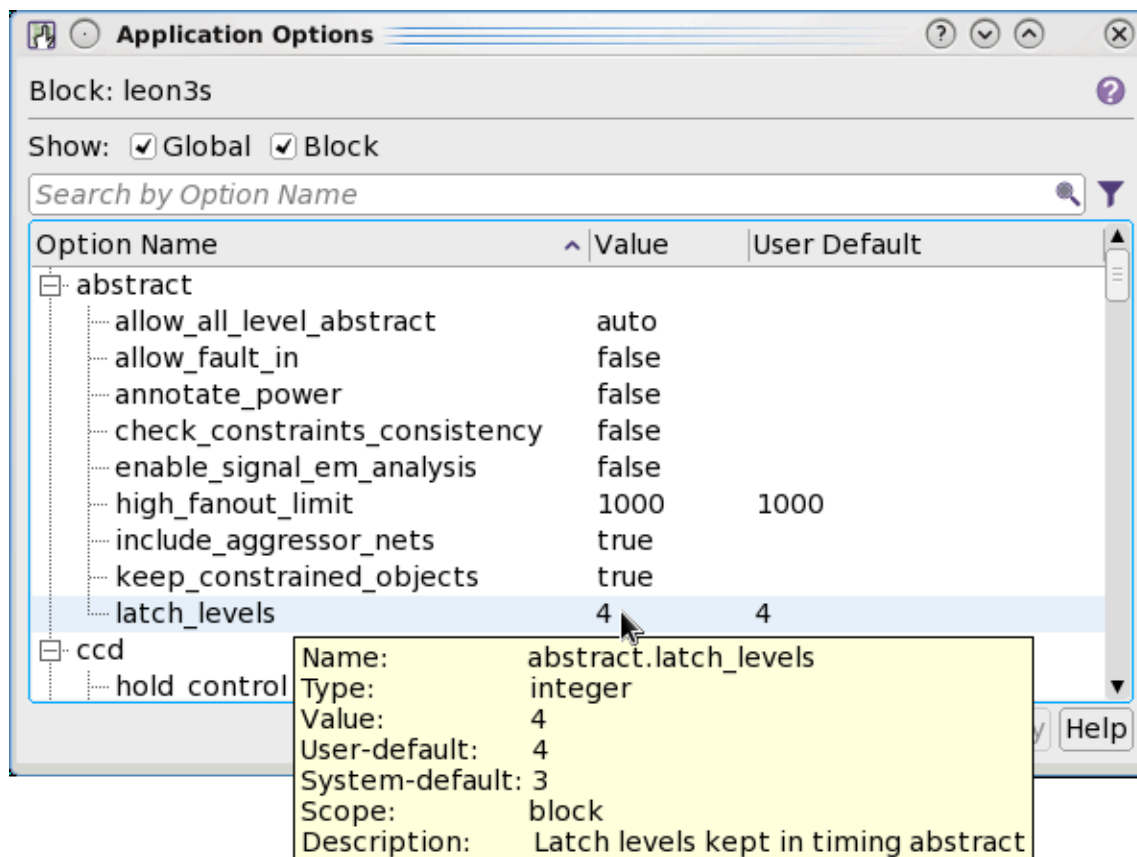
Some technology nodes require additional configuration files from the foundry. Contact your Synopsys representative for help in obtaining the files.

Viewing and Editing Application Options

The Fusion Compiler tool uses application options to control the tool behavior. To manage application options, the Fusion Compiler GUI provides a tool to display current application option settings, change settings, restore settings to default values, and get help on specific application options.

To open the application options tool, choose File > Application Options, or click an application option in the command search results.

Figure 10 Application Options Dialog Box



Application options use the following naming convention when specifying them in a Tcl command:

category[.subcategory].option_name

where *category* is the text that generally refers to the engine affected by the application option. Some application option categories have optional subcategories to further refine the area affected by the application option.

Application options have either a global scope or a block scope.

- Block-scoped application options apply only to the block on which they are set. They are saved in the design library and are persistent across tool sessions.
- Global-scoped application options apply to all blocks, but only within the current session. They are not saved in the design library; you must specify them in each `fc_shell` session. You might want to consider adding the global settings to your `.synopsys_fc.setup` file.

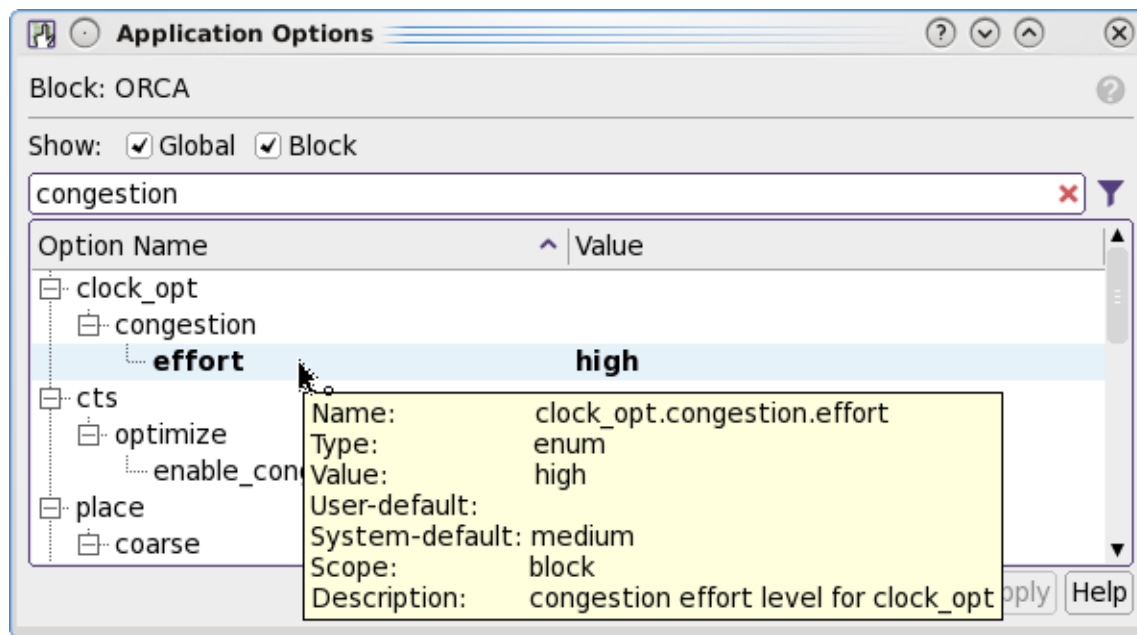
Viewing Application Options

After opening the Application Options dialog box, the tool displays the current application option names and settings. Select the scope of the application options to display by clicking the Global option, Block option, or both options. To filter the list of application options, enter a search term in the search box. As you type, the tool filters the application options and displays only the option names that contain the search term. To filter by the Value column, enter "Value:" followed by the search term. For example, enter "Value:true" to display only application options with a setting of "true" in the Value column.

To further limit the application options displayed in the list, click the filter button to the right of the search box. Deselect the "Show if value is default" option to hide all default application option settings. Deselect the "Show read-only" option to hide all read-only application options.

Application options which have a nondefault setting are displayed in bold font. Move the pointer over the text to display an infotip. The infotip shows the option name, option type, current setting, user default setting, system default setting, scope, and a brief description.

Figure 11 Nondefault Options



To display the man page for an application option, click the option to select it, then click  near the top right of the dialog box. The tool opens the man page for the selected application option.

Editing Application Options

To change the value of an application option,

1. Click the text for the option value.

For enumerated and Boolean options, the tool displays a list of possible settings. For integer type, float type, and text options, the tool displays an edit box.

2. Select the option from the list, or enter a new value in the text box.
3. Click OK to save the setting and close the dialog box, or click Apply to save the setting.

If you applied a nondefault setting, the tool displays the application option and setting in bold font. To revert a nondefault setting to its default value,

1. Click the option value.
2. Click  to the right of the edit box to set the value to its default.
3. Click OK or Apply to apply the setting.

To set an application option value that is based on a Tcl variable, use the `set_app_options` and `list` commands as follows:

```
fc_shell> set scriptdir "myscriptdir"
myscriptdir

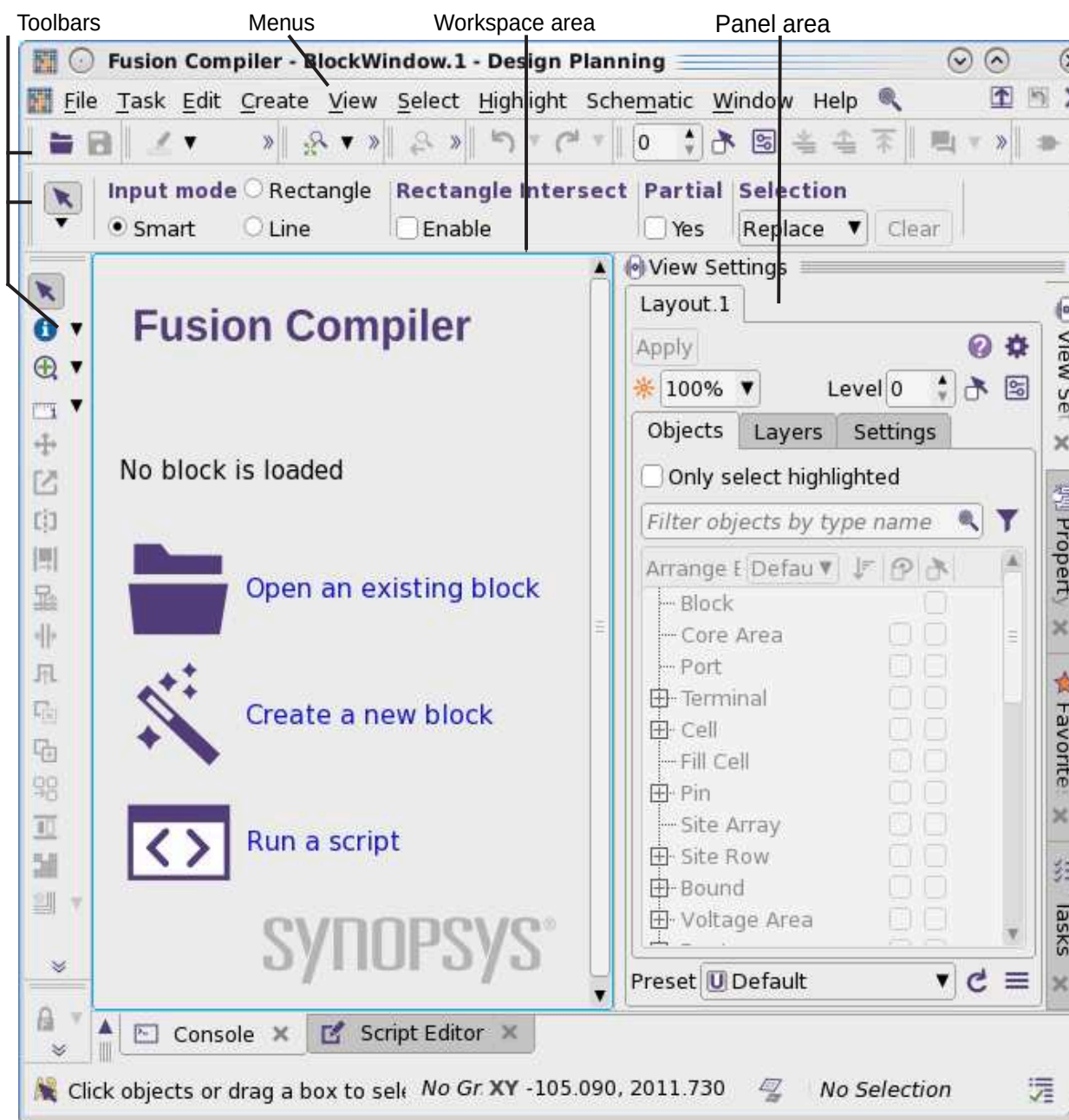
fc_shell> set_app_options \
    -list [list signoff.fix_drc.run_dir $scriptdir]
signoff.fix_drc.run_dir myscriptdir

fc_shell> report_app_options signoff.fix_drc.run_dir
*****
Report : app_option
Design : ORCA
*****
Name                               Type    Value
-----
signoff.fix_drc.run_dir string  myscriptdir
-----
1
```

Working in the Block Window

When you start the Fusion Compiler tool and open the GUI, a block window appears. The following figure identifies the major features of the block window.

Figure 12 Block Window Features



The block window provides the working environment for the GUI. For details about viewing and editing blocks, see the following topics:

- [Changing the View of the Design](#)
- [Editing the Physical Layout](#)

Changing the View of the Design

Design data can be densely packed with overlapping objects on different layers. You can control the visibility (display or hide) and selection (enable or disable) of individual object types or layers by setting options on the View Settings panel. You can also customize object or layer properties (color, fill pattern, and so forth) and set other layout and object display options. If you open multiple views, you can set different options for each view.

You can visualize the design as explained in the following topics:

- [Setting and Saving View Properties](#)
- [Displaying Grid Lines in a Layout View](#)
- [Drawing Timing Rulers in a Layout View](#)
- [Drawing Physical Rulers in a Layout View](#)
- [Using Strokes to Activate Commands](#)

Setting and Saving View Properties

The View Settings panel provides options you can use to set display properties in the active view. You can also save the current settings in your preferences file, or load settings from the preferences file. If you open multiple views, you can set different options for each view.

To display the View Settings panel, do one of the following

- Choose View > Toolbars > View Settings
- Press the F8 key

If the View Settings panel is hidden behind another panel, select the View Settings tab () in the panel area.

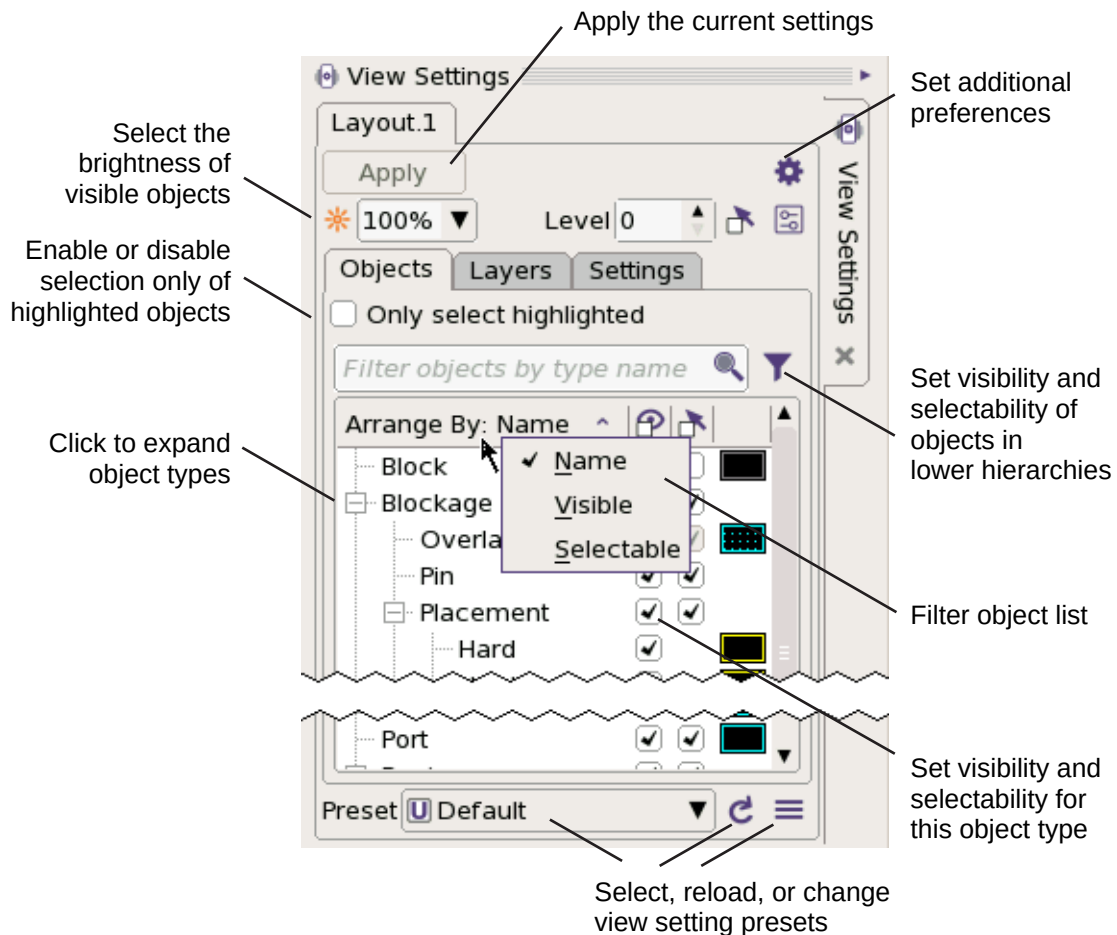
You can use the View Settings panel to

- Control object or layer visibility and selection
- Change object or layer style properties such as colors or fill patterns

- Set display options for the view or for objects in the view

Figure 13 shows the View Settings panel for a layout view, and explains what you can do with it.

Figure 13 View Settings Panel for Objects in Layout Views



The default display characteristics for design objects are set for optimal viewing. These default characteristics work well for most blocks. However, you can change the display characteristics for individual object types if you need to customize your views for a particular block or working environment.

By displaying or hiding particular object types or layers, you can visually inspect just the data that you are interested in viewing while ignoring unrelated data. By enabling or disabling selection for particular object types or layers, you can control which types of objects are selected when you click or drag the pointer in a view.

You can customize how objects appear in the view by changing their style properties. Object styles set the appearance of objects in the active view. You can set the color, fill pattern, outline style, outline width, or exaggeration value for individual object types or layers.

Note:

The exact properties that you can change depend on an object's graphic representation in the view.

You can set options individually, or you can apply a predefined set of view settings called a *preset*. You can apply a system preset provided by the tool or a custom preset that you create. For more information about presets, see [Applying Predefined View Settings](#).

To change layout view display properties in the active view,

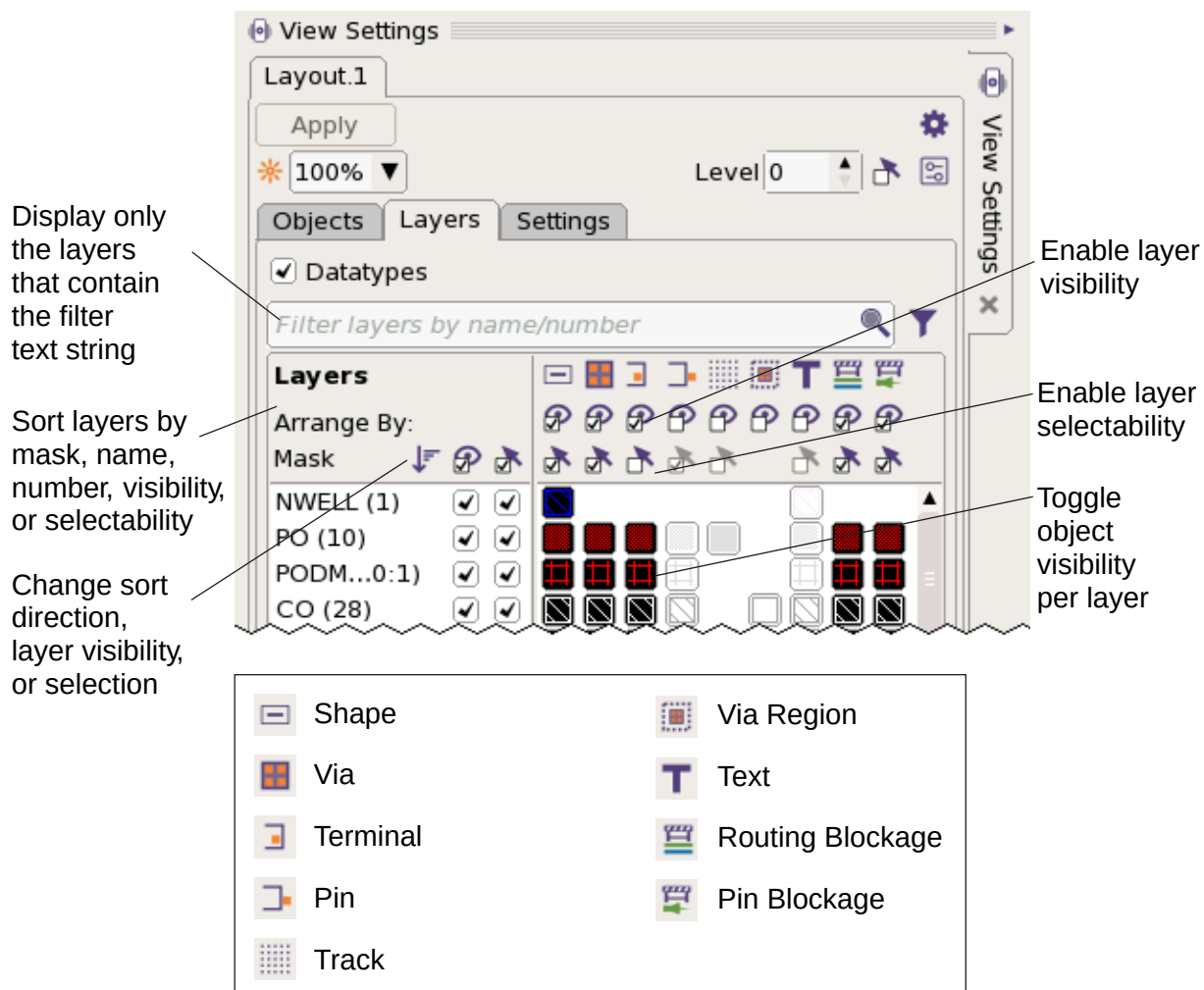
1. Set options as needed on the View Settings panel.
2. Click Apply.

By default, to change the settings, you must click Apply. If you prefer, you can set the panel to automatically apply your changes as soon as you make them.

To enable or disable the automatic apply mechanism, click the Options button () and choose "Auto apply." To reverse unapplied changes, click the Options button and choose "Cancel changes."

Figure 14 shows the View Settings panel as it appears when you click the Layers tab to set options for layers in a layout view.

Figure 14 View Settings Panel for Layers in Layout Views



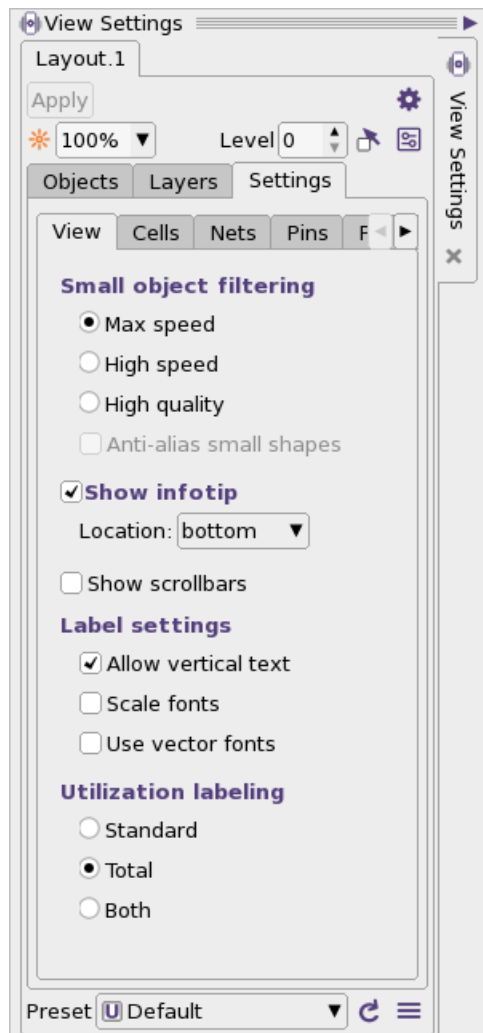
To filter the list of layers, enter a text string in the layer text filter box. As you type, the layer settings panel updates to show only layers that contain the text you entered. Only the list of layers is updated; the design shown in the block window is not affected by layer filtering. Click the (X) to the right side of the text box to clear the filter.

To sort the list of layers, click Arrange By and choose a sort option to reorder the list of layers. The tool updates the Layer panel and sorts the layers by mask, name, layer name, layer number, visibility, or selectability, based on the option you choose.

By default, the layers are sorted in ascending order. Click the (^) to the right of the Arrange By field.

Figure 15 shows the View Settings panel as it appears when you click the Settings tab in a layout view. The Settings panel has tabs to set display options for the view, cells, nets, pins, paths, and mask colors.

Figure 15 View Settings Panel for Settings in Layout Views



Saving and Restoring View Settings

The tool does not save view settings when you exit the session. If you change view settings during a session and want to use the same settings in a future session, you can save them in your preferences file. You can also restore previously saved view settings by loading them from your preferences file.

To save the current settings for the active view, go to the View Settings panel, click the options button () and choose Preferences > Save To Preferences on the View Settings panel. To restore the most recently saved view settings, click the Options button and choose Preferences > Set From Preferences on the View Settings panel.

Applying Predefined View Settings

You can configure a view by applying a predefined set of view properties called a *preset*. You can apply a system preset provided by the tool, and you can create and apply your own custom presets.

By default, when you open a block window, the tool configures the view by using the view settings saved in your preferences file. To apply view settings from a preset, select the preset in the Preset list at the bottom of the View Settings panel.

For example, to apply the system preset that configures a layout view for design planning, select “floorplan” in the Preset list.



When you change settings manually by setting options on the View Settings panel, an asterisk sign (*) appears in front of the preset name in the Preset list.



If you need to reapply the view settings from the preset, click the  button. To reconfigure the view with the default view settings, select Default in the Preset list.

When you create a custom preset, you specify a preset name and the view settings that you want to include in the preset. You can create a new preset or redefine an existing custom preset. You cannot redefine a system preset.

To create a custom preset,

1. Click the  button and choose “Save preset.”
The Write Layout Setting Preset dialog box appears.
2. Select an existing preset, or type a new preset name in the “Select or type a preset name” box.
3. Set options to control which settings you want to include.

For a layout view, by default, the object type filters and object display properties from the Objects tab, the layer type filters from the Layers tab, and the layout display settings from the Settings tab are included. You can set options to include the layer

display properties from the Layers tab or the object style properties from the Objects and Layers tabs.

4. Click OK.

The tool saves the preset and adds the preset name to the Preset list on the View Settings panel.

You can specify a preset as the default configuration when you open a new block window or view. You can remove a custom preset but not a system preset.

Displaying Grid Lines in a Layout View

By default, the grid lines are not displayed in the layout view. You can display the lithography grid or the user grid. In addition, when the grid lines are displayed, you can cycle between the default grid spacing and ten times the default grid spacing.

- To display or hide the lithography grid, choose View > Grid > Show Litho Grid.
- To display or hide the user grid, choose View > Grid > Show User Grid.
- To display or hide the FinFET grid, choose View > Grid > Show Finfet Grid.
- To switch between the default grid spacing and ten times the default grid spacing, choose View > Grid > “Cycle grid spacing.”

Drawing Timing Rulers in a Layout View

Timing rulers are used to measure timing delays between points in the layout. The timing is based on parasitic parameters set by the `read_parasitic_tech` and `set_parasitic_parameters` commands, or on net estimation rules set by the `set_net_estimation_rule` command. A timing ruler can be composed of one or more horizontal or vertical segments. By default, the timing delays from the beginning of the ruler are labeled at the ends of each segment and at every tenth tick mark within a segment.

Note:

Timing ruler drawing operations cannot be reversed by choosing Edit > Undo.

To draw timing rulers in a layout view,

1. Click the Timing Ruler button () on the Mouse Tools toolbar or choose View > Mouse Tool > Timing Ruler.

When you enable the Timing Ruler tool in a layout view, the Timing Ruler Tool options appear on the Mouse Tool Options toolbar and the pointer changes shape. By default, cross hair rulers appear attached to the pointer in the layout view.

2. (Optional) Set options as needed on the Mouse Tool Options toolbar.
 - Select Attach to attach a point on the ruler to an object
 - Click the Use option in the Net Estimation Rule section and select a rule name from the box to calculate timing. To calculate the timing based on metal layer and buffer size, leave this option deselected.
 - Select the horizontal and vertical layer
 - Change the utilization
 - Change the buffer type and spacing
 - Specify a nondefault routing rule (NDR)
 - Change the grid snapping option for the ruler segments
3. Click in the layout view where you want to begin the ruler. If the Attach option is selected, objects are highlighted as you move the pointer over them; the ruler attaches to highlighted objects.
4. Click where you want to end the ruler segment.
5. (Optional) To draw another ruler segment, repeat steps 2 through 4.
6. To end the ruler, press the Enter key.
7. (Optional) To draw another ruler, repeat steps 2 through 6.

Before you click points to define the ruler, you can select or deselect the Attach option to limit the objects to which you attach the ruler. You can remove individual rulers or all rulers in the layout view, but you cannot remove or modify individual ruler segments.

For information on removing timing rulers, see [Drawing Physical Rulers in a Layout View](#).

Drawing Physical Rulers in a Layout View

You measure distances in a layout view by drawing physical rulers. A physical ruler can be composed of one or more horizontal, vertical, or diagonal segments. By default, the distances from the beginning of the ruler are labeled at the ends of each segment and at every tenth tick mark within a segment.

Note:

Physical ruler drawing operations cannot be reversed by choosing Edit > Undo.

To draw rulers in a layout view,

1. Click the  button on the Mouse Tools toolbar or choose View > Mouse Tool > Ruler.

When you enable the Ruler tool in a layout view, the Ruler Tool options appear on the Mouse Tool Options toolbar and the pointer changes shape. By default, cross hair rulers appear attached to the pointer in the layout view.

2. (Optional) Set options as needed on the Mouse Tool Options toolbar.

- Select the ruler mode

By default, the Ruler tool draws a single-segment ruler ("Two point" mode). To draw a multisegment ruler, select the "Multi point" mode.

- Select a direction option
- Display or hide the cross hair rulers attached to the pointer when you start a new ruler, the labels that show the distance from the ruler origin to the endpoint of each ruler segment, and the labels that show the segment length at the midpoint of each ruler segment
- Control whether the distances displayed by the ruler labels are absolute measurements from the design origin (Coordinate), relative measurements from the ruler origin (Ruler Distance), or relative measurements from each segment origin (Segment Distance)
- Change the grid snapping option for the ruler segments
- Record the ruler segment lengths or distances in the session transcript (console log view)

3. (Optional) Select attach to attach the startpoint to an object.
4. Click in the layout view where you want to begin the ruler. If the Attach option is selected, objects are highlighted as you move the pointer over them; the ruler attaches to highlighted objects.
5. Click where you want to end the ruler segment.
6. (Optional) To draw another ruler segment, repeat steps 2 through 4.
7. To end the ruler, press the Enter key.
8. (Optional) To draw another ruler, repeat steps 2 through 6.

Before you click points to define the ruler, you can select or deselect the Attach option to limit the objects to which you attach the ruler. You can remove individual rulers or all rulers in the layout view, but you cannot remove or modify individual ruler segments.

To remove a ruler from the layout view,

1. Click the  button on the Mouse Tools toolbar or choose View > Mouse Tool > Ruler.
2. Move the pointer over the ruler you need to remove.
3. Right-click and choose Remove Nearest Ruler.

To remove all rulers from the layout view,

- Click the  button on the Mouse Tools toolbar or choose View > Mouse Tool > Remove All Rulers.

Drawing Circular Rulers

You can create circular rulers in the tool by using the `gui_add_annotation` command. For more information, see [Creating Annotations](#).

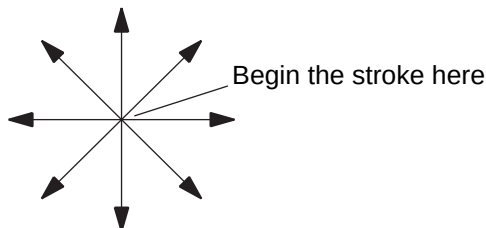
Using Strokes to Activate Commands

You can stroke the pointer to activate preset commands in the active view. By default, the strokes are set up to activate commands that do the following:

- Increase or decrease the magnification of the block
- Pan the block
- Display the entire block

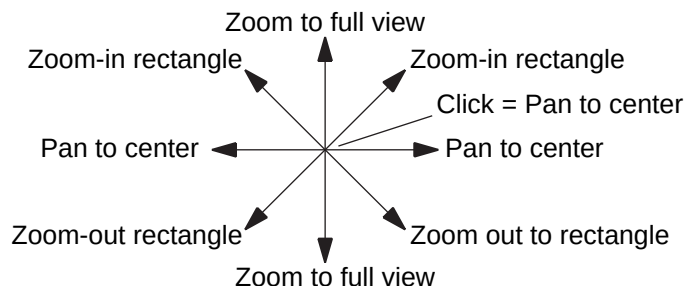
You can also define your own strokes. You can define up to 72 basic strokes and many more multisegmented strokes.

A basic stroke is a single click of the middle mouse button with the pointer in the layout view. A point-to-point stroke (dragging the pointer from point to point in one of the following directions while holding down the middle mouse button).

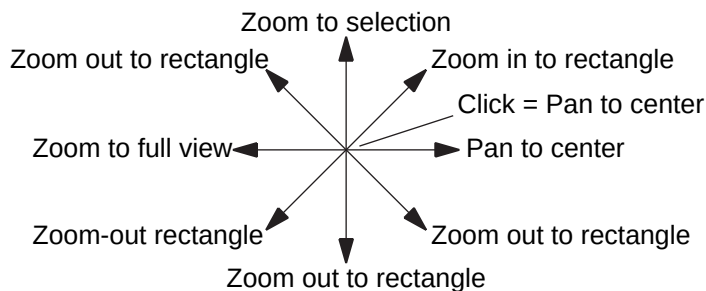


You can activate other commands for each of these strokes by also pressing a modifier key (Shift, Ctrl, or Alt), or a combination of these keys as shown in the following figures:

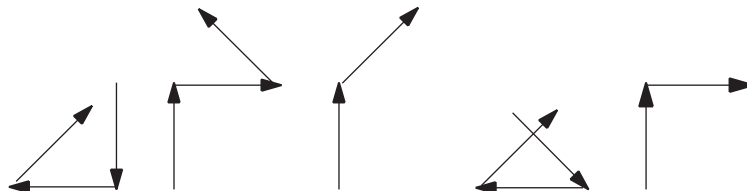
Basic strokes made without pressing Shift, Ctrl, or Alt



Basic strokes made while pressing Ctrl



In addition to the basic and modified stroke commands, you can perform multisegmented strokes. Each segment can go in any one of the eight directions used for the basic stroke commands. The following drawing shows some simple multisegmented strokes you can define:



To bind a command to a basic or multisegmented stroke, use the `set_gui_stroke_binding` command. By default, the GUI is set up to use basic strokes instead of multisegmented strokes. You can change this preference by using the `set_gui_stroke_preferences` command. For details about using these commands, see their man pages.

Editing the Physical Layout

The block window provides interactive tools and commands for editing the layout of blocks in a library workspace.

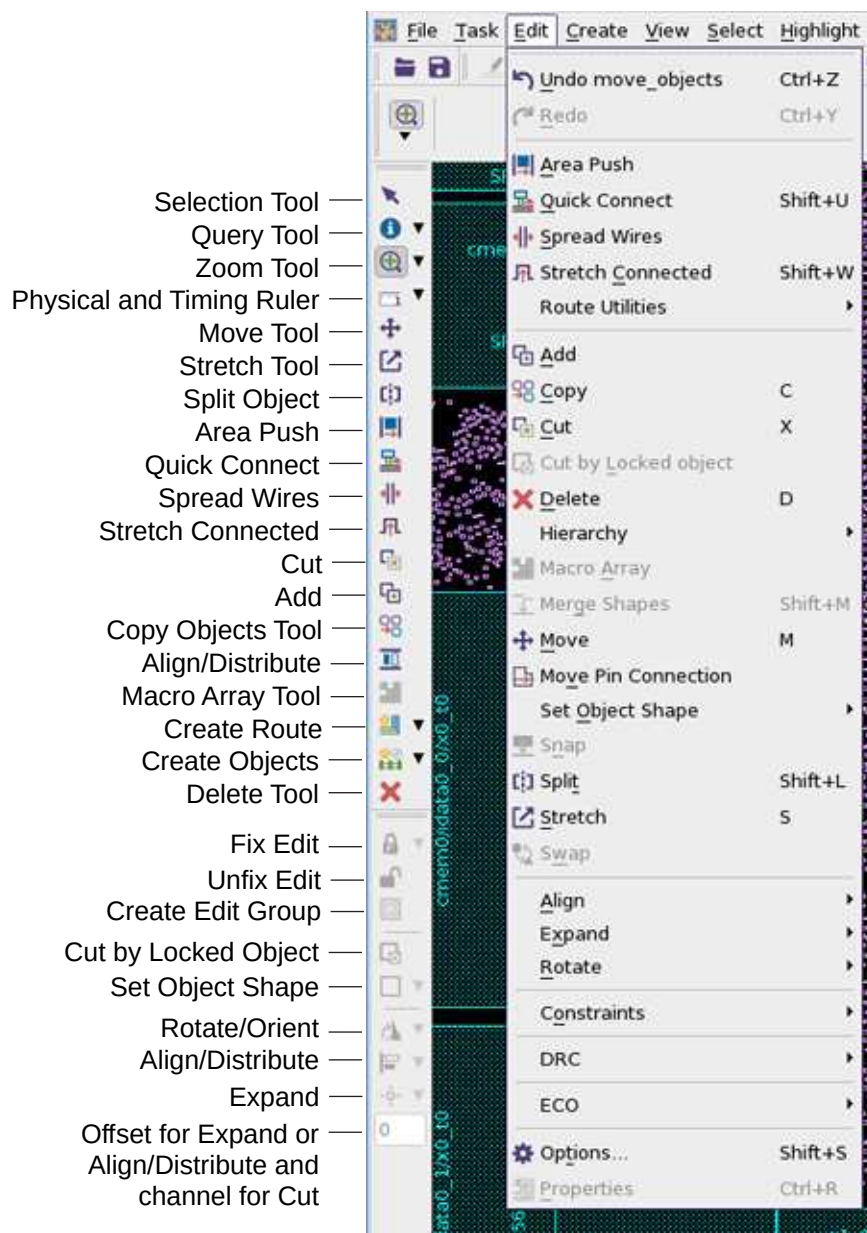
You can use the editing tools and commands to

- Move, resize, copy, and remove objects
- Rotate and flip objects, and change cell orientations
- Reshape objects
- Align, distribute, and spread objects
- Expand objects
- Create routing blockages, net shapes, port terminals, text objects, and vias

Typically, you select the objects that you want to edit. You can select the objects interactively with the Selection tool or by typing their names on the Select By Name toolbar. The interactive editing tools also allow you to select objects when the tool is active.

You can access editing tools and commands by clicking buttons on the Edit or Mouse Tools toolbar or by choosing commands on the Edit or Create menu. [Figure 16](#) shows these toolbars and the Edit menu. The toolbars group related editing functions together in lists that open when you click the down arrow button.

Figure 16 GUI Editing Commands



For information about editing physical design objects and constraints, see the following topics:

- [Editing Objects Interactively](#)
- [Working in the Editing Environment](#)

Editing Objects Interactively

The layout editing mouse tools allow you to interactively create, edit, or remove physical objects in a layout view. The block window provides a consistent editing environment for all the tools. You can

- Set tool options on the Mouse Tool Options toolbar
- Use keyboard shortcuts for frequent or repetitive operations
- Preview objects for selection and cycle through overlapping objects

If you need to magnify or traverse the design or measure distances when an editing tool is active, you can temporarily activate the Zoom In tool, Zoom Out tool, Pan tool, or Ruler tool without deactivating the editing tool.

When you activate a mouse tool in a layout view, the options for using the tool appear on the Mouse Tool Options toolbar.

You can perform frequent or repetitive actions by using keyboard shortcuts. [Table 1](#) shows the standard key bindings for the interactive editing tools.

Table 1 *Interactive Editing Tool Conventions*

Action	Shortcut
Add points to perform the operation	Click or Drag
Add point and apply changes	Return
Remove last point	Backspace
Cancel pending input, if any, or exit tool	Escape
Cycle forward through overlapping objects	F1
Cycle backward through overlapping objects	Shift+F1
Reverse (undo) most recent operation	Ctrl+X
Reapply (redo) most recently applied operation	Ctrl+Y

Many of the editing tools operate on selected objects. You can select the objects before or after you activate a tool. If any objects are selected when you activate an editing tool, you can Ctrl-click objects to add them to the current selection and Shift-click objects to remove them from the current selection. Selected objects appear in the selection color, which is white by default.

Note:

If you have trouble selecting or previewing small objects, try magnifying the view. In a large design, some objects or object outlines are too small to click at low magnification.

All the operations that you can perform with the interactive editing tools are logged and can be replayed with the `gui_mouse_tool` command. Similarly, options that you set on the Mouse Tool Options toolbar are logged and can be replayed with the `gui_set_mouse_tool_option` command. For information about these commands and the

options for a particular tool, click the  button to see the Help page for that tool.

For information about the interactive editing operations that you can perform, see the following topics:

- [Moving, Modifying, Creating, and Removing Objects](#)
- [Expanding and Rotating Objects](#)

Moving, Modifying, Creating, and Removing Objects

You can move, resize, copy, split, reshape, create, and remove objects.

[Table 2](#) shows the editing tools you can use to modify the physical design.

Table 2 *Editing Tools for Design Objects*

To perform this operation	Use this tool
Move objects	Click  or choose Edit > Move, and then select the objects and click or drag to the new location.
Snap objects to the grid	Choose Edit > Snap.
Align objects relative to a side or center of an anchor object or location	Click  or choose Edit > Align > Align/Distribute Tool, click the Align tab on the Mouse Tool Options toolbar, select the objects you want to align, specify the anchor object or location, and click the toolbar button for the alignment operation you want to perform. You can set options as needed on the Mouse Tool Options toolbar.

To perform this operation	Use this tool
Distribute objects relative to a side of an anchor object or location	Click  or choose Edit > Align > Align/Distribute Tool, click the Distribute tab on the Mouse Tool Options toolbar, select the objects you want to distribute, specify the anchor object or location, and click the toolbar button for the distribution operation you want to perform. You can set options as needed on the Mouse Tool Options toolbar.
Resize objects	Click  or choose Edit > Stretch, and then select an object and drag an edge or corner to resize.
Copy objects	Click  or choose Edit > Copy, and then select and drag the objects.
Split objects	Click  or choose Edit > Split, and then drag a line or rectangle across the objects where the split is to occur. You can set options as needed on the Mouse Tool Options toolbar.
Reshape objects by cutting an area with an optional gap	Click  or choose Edit > Cut, optionally set the gap size, and then drag a rectangle across the object where you want to cut a piece. You can set options as needed on the Mouse Tool Options toolbar.
Reshape objects by adding an area with an optional gap	Click  or choose Edit > Add, optionally set the gap size, and then drag a rectangle across the object where you want to add a piece. You can set options as needed on the Mouse Tool Options toolbar.
Create a bound	Click  or choose Create > Bound, and then click and drag a rectangle for the bound. You can set options as needed on the Mouse Tool Options toolbar.
Create a routing blockage	Click  or choose Create > Routing Blockage, and then click and drag a rectangle for the routing blockage. You can set options as needed on the Mouse Tool Options toolbar.
Create a net shape	Click  or choose Create > Shape, and then click and drag a rectangle for the net shape. You can set options as needed on the Mouse Tool Options toolbar.
Create a terminal	Click  or choose Create > Terminal, specify a pin or port, and then click and drag a rectangle for the terminal. You can set options as needed on the Mouse Tool Options toolbar.

To perform this operation	Use this tool
Create a text object	Click  or choose Create > Text, specify the text, and then click the location for the text. You can set options as needed on the Mouse Tool Options toolbar.
Create a via or via array	Click  or choose Create > Via, and then click the location for the via or via array. You can set options as needed on the Mouse Tool Options toolbar.
Remove objects	Click  or choose Edit > Delete, and then click or drag a rectangle to delete objects.

Expanding and Rotating Objects

[Table 3](#) explains how you can expand, rotate, and flip design objects and orient cells.

Table 3 *Editing Commands for Design Objects*

To perform this operation	Do this
Expand selected objects to objects on one or all sides	Click an expansion tool ( ,  ,  ,  , or ) or choose Edit > Expand > <i>expansion_direction</i> . Alternatively, you can use the <code>expand_objects</code> command.
Rotate selected objects 90 or 180 degrees	Click a rotation tool ( ,  , or ) or choose Edit > Rotate > <i>rotation</i> . Alternatively, you can use the <code>rotate_objects</code> command.
Flip selected objects on an x- or y-axis	Click a flip tool ( or ) or choose Edit > Rotate > <i>axis</i> . Alternatively, you can use the <code>flip_objects</code> command.

Working in the Editing Environment

The editing environment provides an incremental undo and redo facility that allows you to reverse or reapply the most recent editing operation or a series of operations. You can also constrain (fix) objects during editing operations, control grid snapping, set editing options, and view and edit object properties. [Table 4](#) explains the global operations that you can use with any editing tool or command.

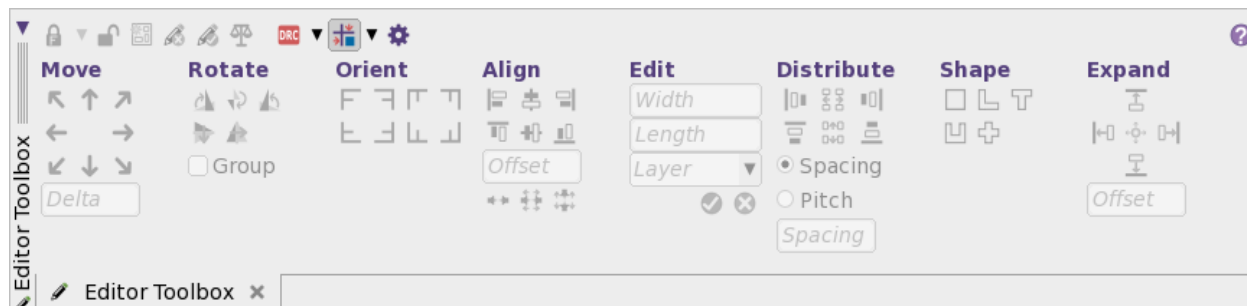
Table 4 *Global Editing Operations*

To perform this operation	Use this tool
Reverse the previous edit change	Click the  button or choose Edit > Undo. Alternatively, you can use the <code>undo</code> command.
Reapply the subsequent edit change	Click the  button or choose Edit > Redo. Alternatively, you can use the <code>redo</code> command.
Constrain (fix) selected objects to prevent changes from interactive edits	Click the  button or choose Edit > Constraints > Fix Edit.
Remove (unfix) the editing constraint on selected objects to allow changes from interactive edits	Click the  button or choose Edit > Constraints > Unfix Edit.
Enable or disable grid snapping	Click the  button or choose Edit > Options and select or de-select “Snapping enabled” on the Snapping tab.
View or edit object attribute values for selected objects	Click the  button or choose Edit > Properties.

Using the Editor Toolbox

The Editor Toolbox provides easy access to common editing tasks such as move, rotate, align, distribute, spread, and so on. Choose Edit > Editor Toolbox from the menu to start the Editor Toolbox. The Editor Toolbox is shown in the following figure.

Figure 17 *Editor Toolbox*



Editing Pins With the Editor Toolbox

You can use the Editor Toolbox to move, distribute, align, and perform other operations on block pins as follows:

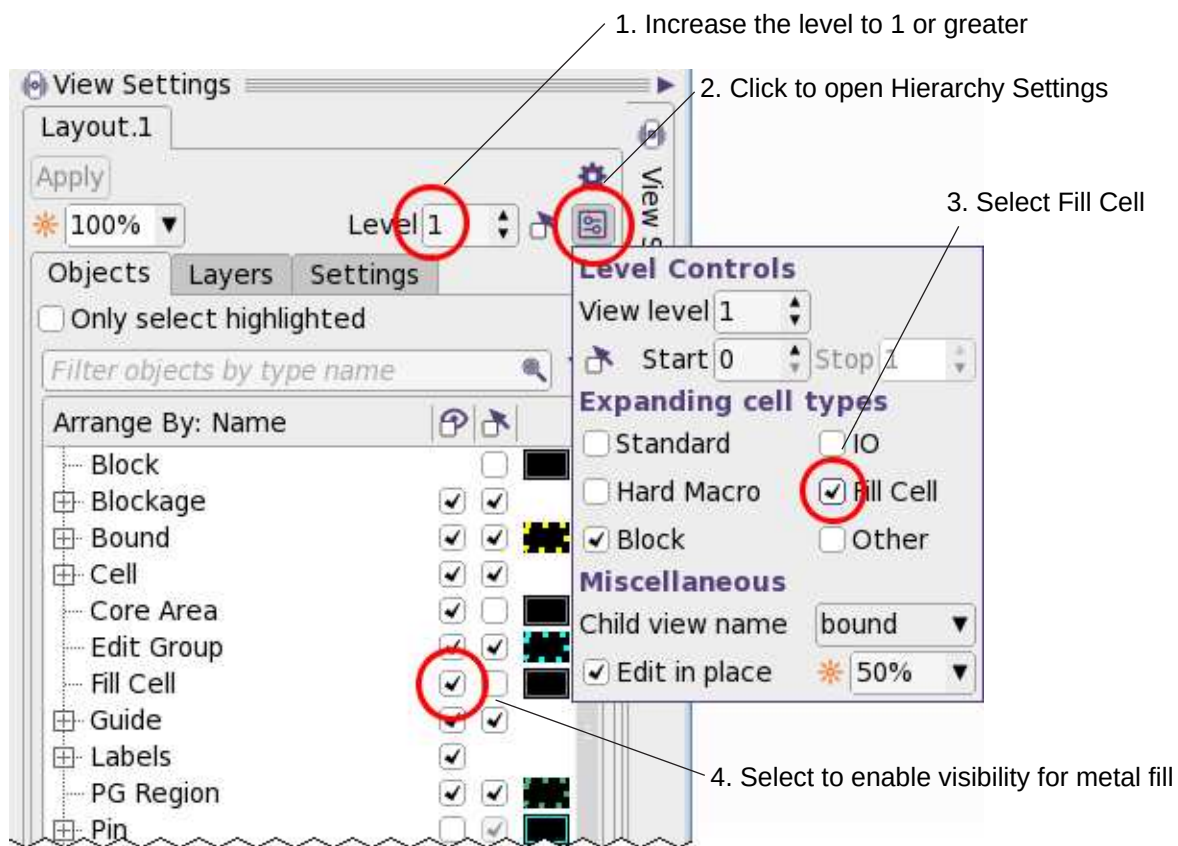
Table 5 *Pin Editing Operations*

To perform this operation	Use this tool
Enable pin editing on a block	Click the  button. Alternatively, you can use the <code>set_editability -blocks [get_selection] -value true</code> command.
Disable pin editing on a block	Click the  button. Alternatively, you can use the <code>set_editability -blocks [get_selection] -value false</code> command.
Legalize the selected block pins, or the block pins connected to the selected nets or cells	Click the  button. Alternatively, you can use the <code>place_pins -legalize</code> command.
Align the selected pins	Click the  ,  , or  , button. The second and third buttons change the layer width and layer height respectively to match the layer properties. Alternatively, you can use the <code>align_pins</code> , <code>align_pins -change_layer_width</code> , or <code>align_pins -change_layer_height</code> command.
Change the width of the selected pin	Enter a new width value in the Width box in the Edit section and click the checkmark button to commit the change.
Change the length of the selected pin	Enter a new value in the Length box in the Edit section and click the checkmark button to commit the change.
Change the layer of the selected pin	Select a new layer value in the Layer box in the Edit section and click the checkmark button to commit the change.

Viewing Metal Fill Data

The Fusion Compiler tool supports editing design and metal fill data in the same context. To enable metal fill editing, open the View Settings panel and change the metal fill settings as shown in [Figure 18](#). Metal fill data appears as one or more fill instances within the block; the data is stored with the block in the library. To edit the metal fill data, enable the All Levels Active option in the View Settings panel.

Figure 18 Metal Fill Display Settings



2

Using Toolbars and the Selection List

The Fusion Compiler GUI provides tools for querying information, editing and recording scripts, and performing other tasks.

The following topics describe design planning tasks supported by the GUI:

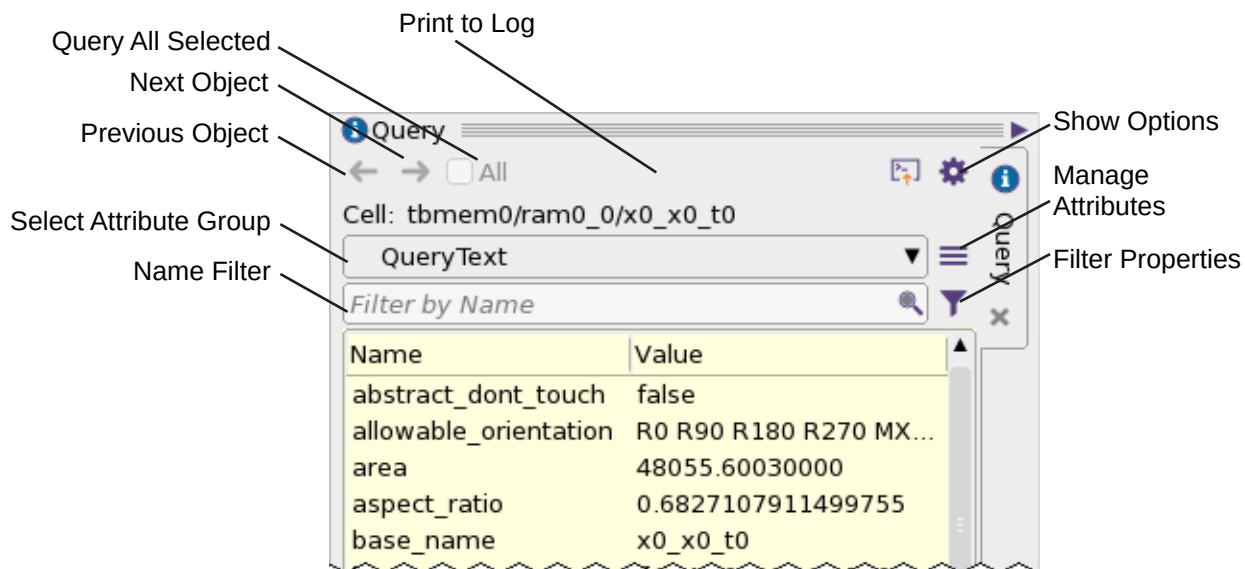
- [Using the Query Tool](#)
- [Using the Recent Menu Button](#)
- [Using the Script Editor](#)
- [Using the Select By Name Toolbar](#)
- [Using the Overview Toolbar](#)
- [Using the Selection List and Work List](#)
- [Using User Tables](#)

Using the Query Tool

The Query panel displays information about objects in the active layout or schematic view. The panel opens automatically when you enable the Query tool as the active mouse tool. When you click an object, the attribute information for the object is displayed in the Query panel.

To activate the Query tool, choose View > Mouse Tool > Query Tool or click the Query tool button on the Mouse Tools toolbar. The pointer becomes the Query Tool pointer and the tool opens the Query panel on the right of the layout window.

Figure 19 Query Panel



To display information for an object, click the object. The tool populates the query panel with information about the object.

To display information for an object, click the object. The tool populates the query panel with information about the object.

To display the Query tool and populate with information for selected objects, choose Select > Query Selection.

To further control the display of information in the Query panel, click the Show Options button and change the option settings:

- Select “Auto Print Text to Log” to write out the contents of the Query panel to the console each time you query a new object
- Select “Auto Show when Query Tool is Chosen” to automatically open the Query panel when you click an object when the Query Tool is active
- Click “Customize Text” to open the Customize Query Toolbar dialog box; see [Using the Customize Query Toolbar](#) for more information
- Select “Warn in selection exceeds size limit” and set a limit to issue a warning if the selection exceeds the specified limit

- Select “Show attribute group” to display the attribute group selection box in the Query panel
- Click “Show attribute filter” to display a text filter box in the Query panel

Select an attribute group in the attribute group box to display only the attributes that belong to that group. To get additional information about which attributes are available for a specific group, click the Activate Attribute Group Manager button to the right of the attribute group box. See [Using the Attribute Group Manager](#) for more information about attribute groups.

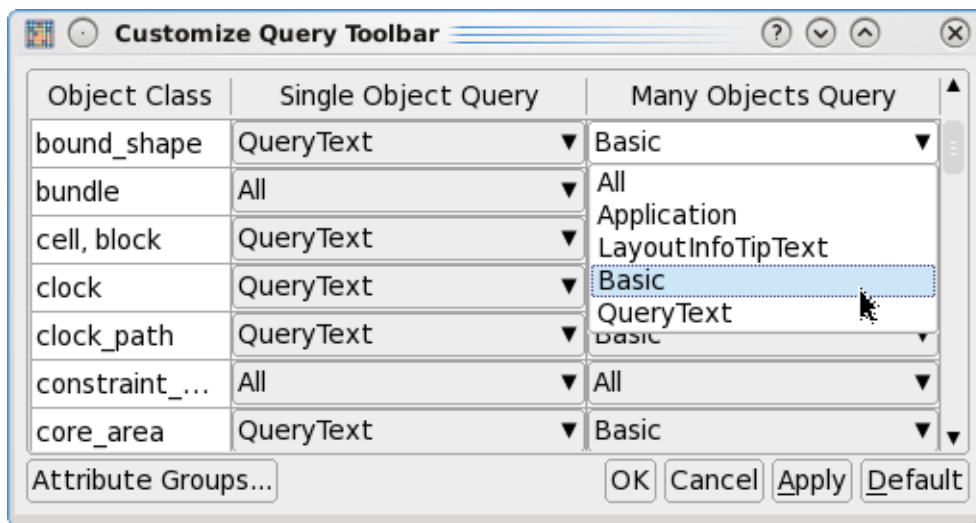
Enter a simple text string in the filter box to display only attributes where the filter text matches a portion of the attribute name. To clear the string, click the (X) at the right of the filter text box. Click the Filter properties button at the right of the filter box and select Show Read Only Properties to display only attributes with read-only values.

Click the Name or Value column header to change the sort order for the column.

Using the Customize Query Toolbar

To open the Customize Query Toolbar, click Customize Text in the Show Options panel in the Query Panel. The Customize Query Toolbar is shown in [Figure 20](#).

Figure 20 Customize Query Toolbar Dialog



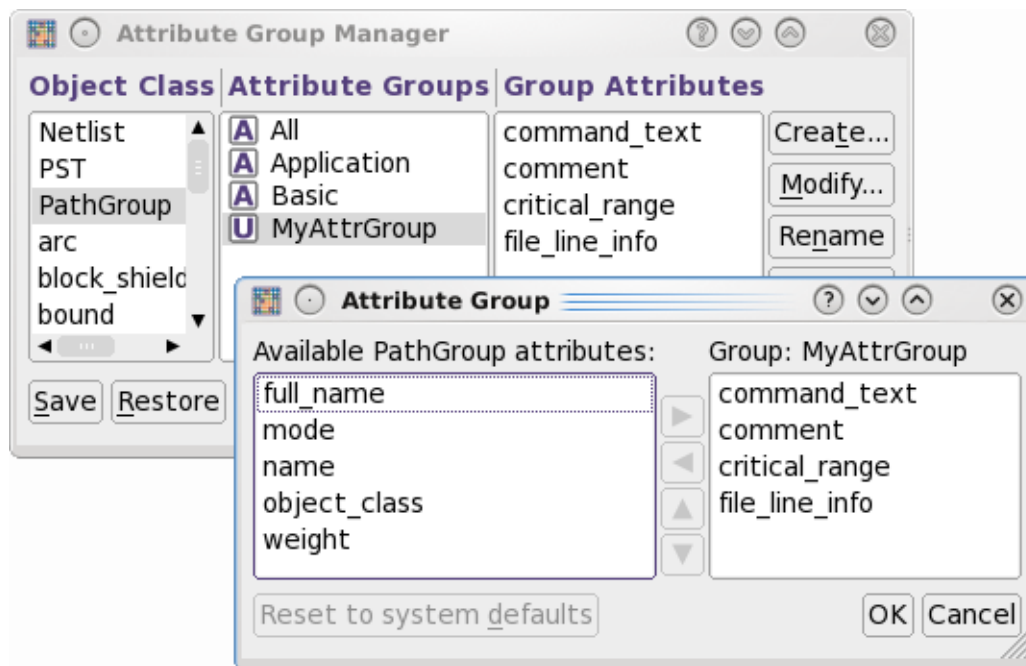
The Customize Query Toolbar displays a table of Object Classes and the attribute group in which they appear. To change the setting, click the text in the Single Object Query or Many Objects Query column and select a new value. Select All to enable the value to appear in all groups. After configuring the list, click OK or Apply to apply the changes.

By default, objects can be members of groups such as Application, Basic, QueryText, and so on. When the group name is selected in the Query Panel, the information about the object type appears in the Name and Value column of Query Panel. You can create additional groups by clicking the Attribute Groups button and using the Attribute Group Manager dialog box. See [Using the Attribute Group Manager](#) for more information.

Using the Attribute Group Manager

Use the Attribute Group Manager to define new attribute groups and add or remove objects from an attribute group.

Figure 21 Attribute Group Manager and Attribute Group Editor



To use the Attribute Group Manager,

- Click the Create button to create a new attribute group; enter the attribute group name in the dialog box that appears and click OK to save the new attribute group
- Click an Object Class, click an Attribute Group name, and click the Modify button to change the attributes that are displayed when you query the object in the Query panel; see [Using the Attribute Group Editor](#) for more information
- Click an Attribute Group name and click the Rename button, or double-click the Attribute Group name, to rename a user-defined attribute group name; Attribute Group names defined by the system cannot be changed

- Click an Attribute Group name and click the Copy button create a new attribute group and copy the group attributes to the new attribute group
- Click the Delete button to remove the selected user-defined attribute group from the system
- Click the Save button to write out the current attribute group settings to the GUI user preferences file
- Click the Restore button to read in the attribute group settings from the GUI user preferences file
- Click the Reset to system defaults button to reset the content of the application-defined groups to their defaults; user-defined groups are not affected

Using the Attribute Group Editor

Use the Attribute Group Editor to add or remove attributes from the list of attributes associated with a user-defined attribute group. The Attribute Group Editor is opened by clicking the Modify button in the Attribute Group Manager.

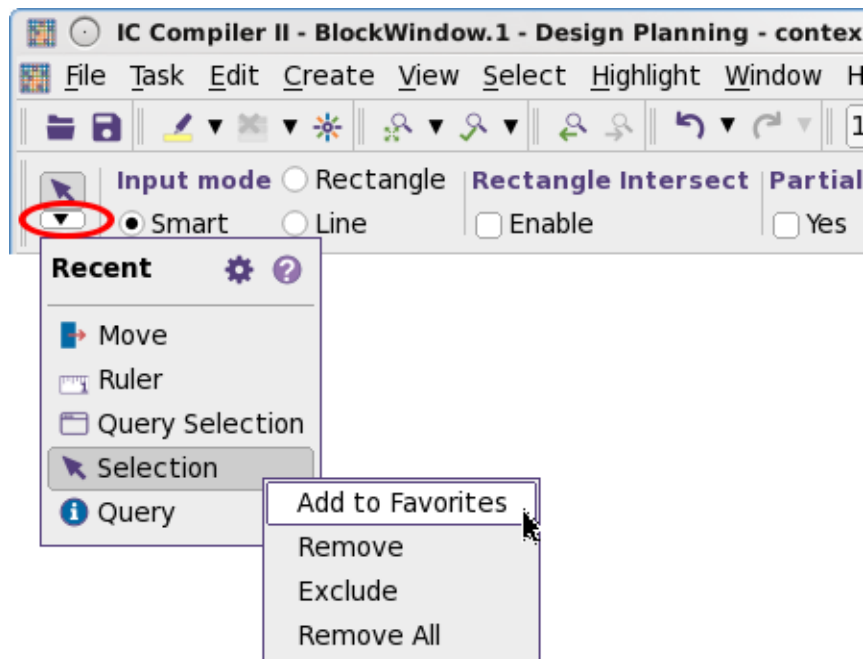
Do the following to modify the list of attributes for the selected attribute group:

- Select an attribute in the left column and click the right arrow to move the entry to the right and make it available for display with this attribute group
- Select an attribute in the right column and click the left arrow to move the entry to the left and make it unavailable for this attribute group
- Select one or more entries in the right column and click the up arrow to move the entries up in the list; use the down arrow to move the entries down
- Press Control-A to select all entries in the column
- Click OK or Cancel to save the results or exit the dialog box without saving

Using the Recent Menu Button

The recent menu button displays icons and text that represent the most recent actions you performed in the GUI. The recent menu button is displayed by clicking the down arrow below the current mouse tool icon near the top left of the layout window as shown in [Figure 22](#).

Figure 22 Recent Menu Button



As you perform tasks in the GUI, the tool inserts an icon and text at the top of the list of recent actions. By default, the tool shows the last five commands; you can change this number by clicking the Show Options button (⚙️) and selecting a new value for Maximum Commands.

The tool provides shortcut keys to access the last nine most recently used commands. Press Alt-1 to recall the most recent command, Alt-2 to recall the second most recent, and so on.

You can modify the list of most recently used commands by clicking the right mouse button and selecting one of the following options from the context menu:

- Choose Add to Favorites to add this command to the list of commands in the Favorites Toolbar.
- Click Remove to delete this command from the most recently used commands list.
- Click Exclude to delete this command and prevent it from being added to this list in the future; to remove the command from the exclude list, click the Show Options button and click Reset Excluded.
- Click Remove All to remove all commands from the most recently used command list.

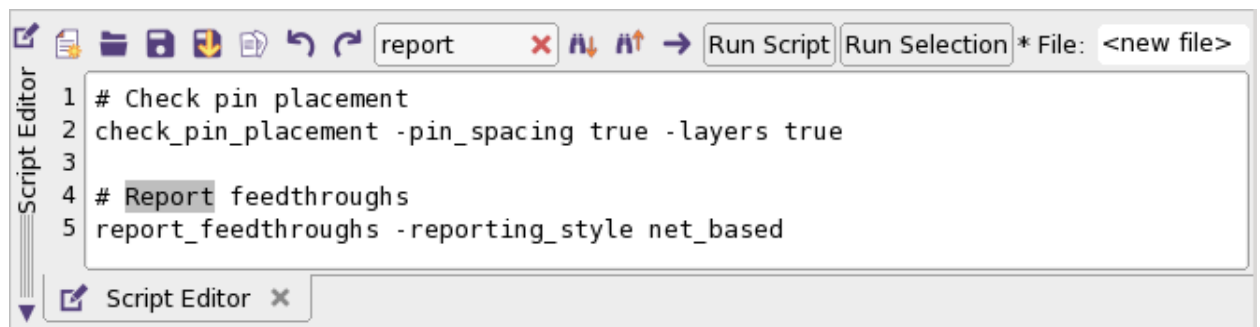
Click the Show Options button to change other settings for the Recent toolbar:

- Use Maximum Commands to select a new value for the maximum number of commands to be saved in the list of recent actions.
- Click Reset Exclude to remove all commands from the exclude list; this button is only available if you have excluded one or more commands.

Using the Script Editor

The Script Editor is an easy-to-use tool for loading, editing, running, and saving Tcl scripts in the Fusion Compiler tool. Activate the Script Editor by choosing View > Toolbars > Script Editor or by moving your mouse to the status bar at the bottom of the layout view, clicking the right mouse key, and selecting Script Editor from the context menu.

Figure 23 Script Editor



Move the pointer over the icons to view their function in the infotip. You can perform the following operations in the Script Editor:

- Click the New button to clear the contents of the script window.
- Click the Open button and enter a file name in the Open dialog box to read in an existing script.
- Click the Save button to save the script.
- Click the Save As button to save the script by using a new file name.
- Click the Undo button to revert a change to the script.
- Click the Redo button to reverse the last undo action.
- Enter text in the search box and click the Find next button to search forward in the file; click the Find previous button to search backward in the script.
- Enter an integer value in the search box and click the Go to line button to scroll the editor to the specified line number.

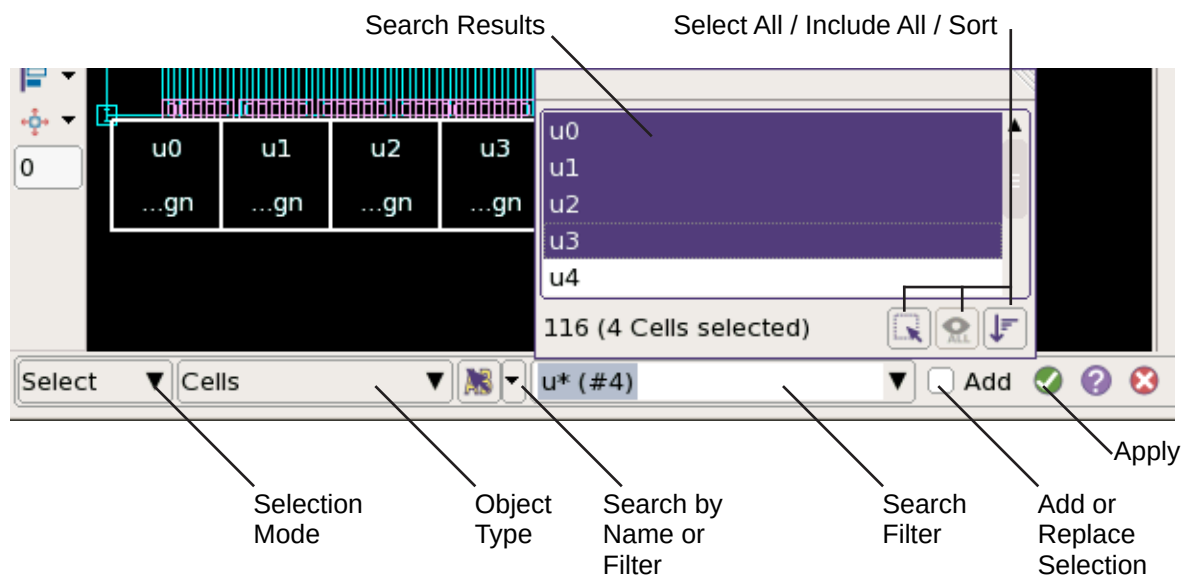
- Click Run Script to save the contents of the script editor to a temporary file and run the file as Tcl commands.
- Use the mouse to select words or lines in the script editor and click Run Selection to save the selection to a file and execute it as Tcl commands.
- Edit to contents in the script editor by typing, cutting and pasting, or deleting text.

Using the Select By Name Toolbar

The Select By Name Toolbar selects or highlights objects by object name or by a search filter expression. To activate the tool, choose Select > By Name Toolbar from the menu.

Alternatively, click the Select By Name button () at the lower-left corner of the layout window.

Figure 24 Select By Name Toolbar



The toolbar supports keyboard shortcuts to efficiently search for and select objects. A typical workflow using the Select By Name toolbar is as follows:

1. Press Ctrl+/ to display the Select By Name toolbar.
2. (Optional) Change the selection mode to Highlight to highlight the selected objects.
3. (Optional) Choose an object type from the Object Type list.
4. Press F3 to search by name or F4 to search by filter.

5. Enter a partial search term that matches the beginning of the object name for which you are searching; if no matches are found, the tool applies a light red color to the box.
6. Press Tab to expand the search term to the matching object name, or display all matching results if the search term matches more than one object.
7. Press Insert to toggle between Add to Selection and Replace Selection.
8. Press Enter or click Apply to select the objects.

Entering Search Terms for Multiple Objects

The search box provides a flexible interface for finding multiple objects. To search for multiple object names, separate the search terms with a space. If the object name contains a space, use curly braces ({}) to separate the terms. Press the Tab key to search for the term preceding the text cursor. To search for the other term, move the cursor and press the Tab key again. Click the (X) to the right of the search box to clear the search term.

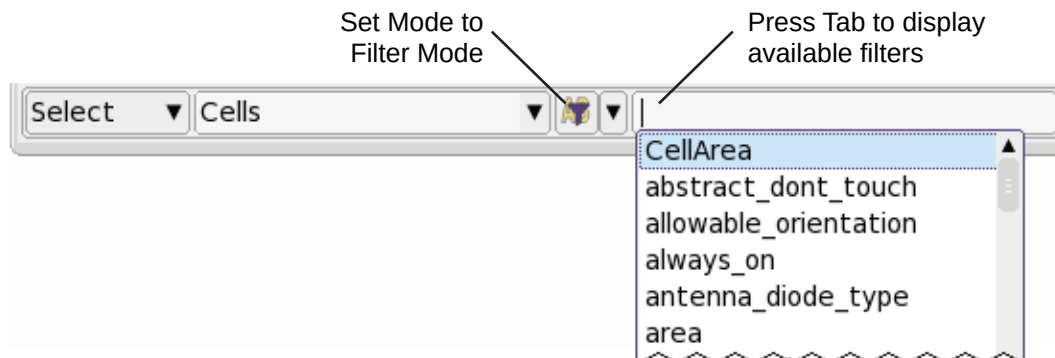
If the search term matches more than one object, the tool displays a scrollable search results list. Continue typing to create a longer search term and narrow the search results. Press the Up arrow key or Down arrow key to scroll up or down the list. Press Enter to select a single entry in the list, or press the Control key and click additional results to select them. Click the Select All button, or press Ctrl+A to select all results in the list. Click the Sort button to reverse the sort order of the list. Press Esc to exit the list.

After selecting multiple objects in the list, the search box shows an abbreviated search term, such as "(#4)", where "4" is the number of objects you chose. To select the objects in the layout, press the Enter key. To add additional objects to the selection, click the Add option or press the Insert key, then enter a new search term and repeat the previous steps.

Searching by Filter

You can use the Select By Name tool to search for objects by using a filter expression as shown in the following figure.

Figure 25 Select by Filter



By typing the first few letters of the search expression and pressing the Tab key, you can quickly build the filter expression. A typical workflow to build a filter search expression is as follows:

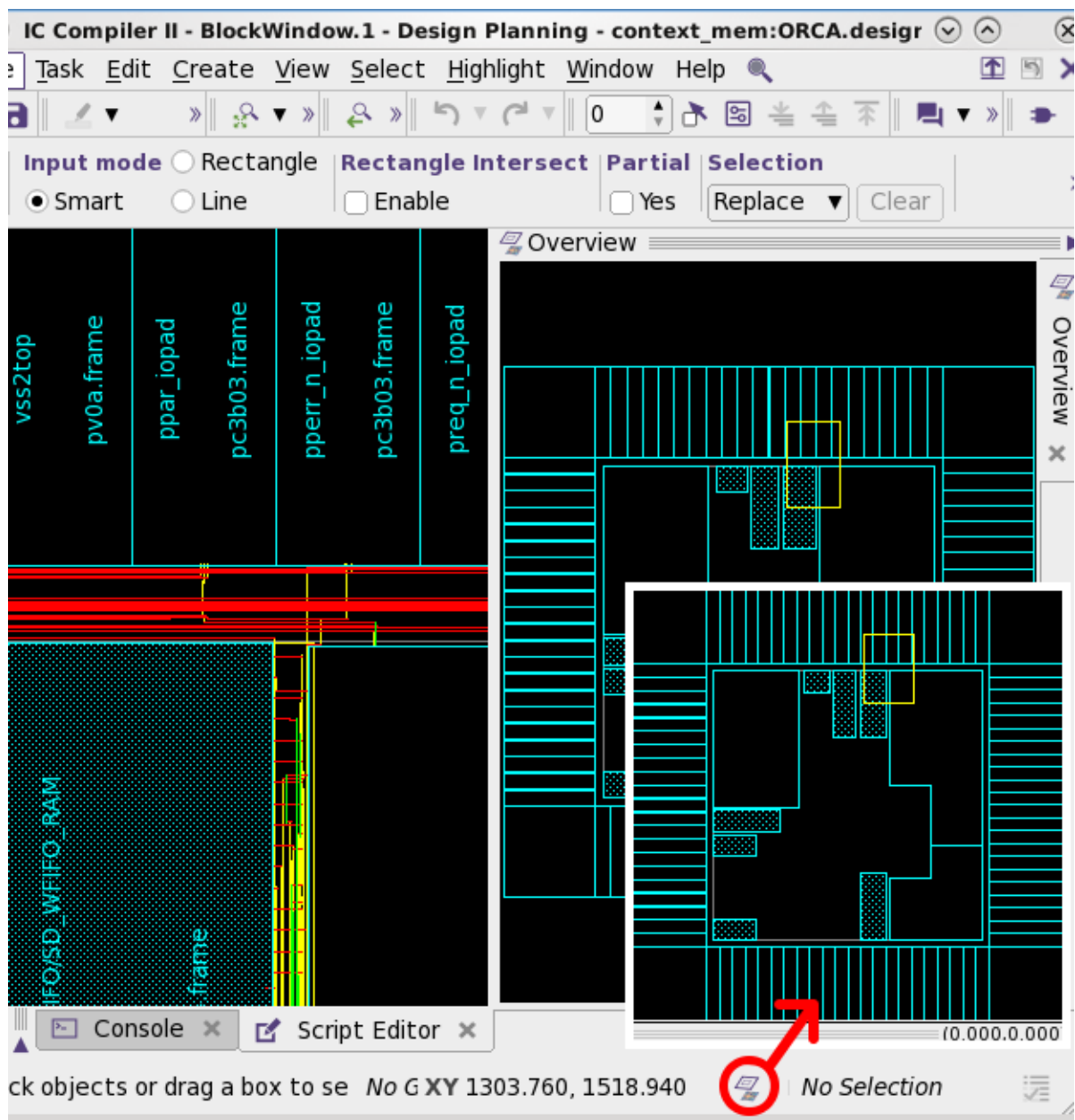
1. Type the first few letters of the search expression, for example, "a".
2. Press the Tab key to show the list of attributes which match the search expression.
3. Continue typing to narrow the search, or click the attribute in the list.
4. Press the Tab key again to display the legal operators for the attribute.
5. Continue typing to narrow the search, or click the attribute in the list.
6. Press the Tab key again; the tool displays the legal values for an attribute with enumerated values, or inserts empty quotation marks, or inserts a value of 0.0 into the search box.
7. Continue using the Tab key to repeat these steps to append additional information to the expression, or press Enter to commit the expression and perform the search.

Using the Overview Toolbar

The Overview toolbar displays an outline of the entire layout view with a yellow highlight that shows the area currently displayed in the layout view. To activate the Overview toolbar and display the Overview panel, choose View > Toolbars > Overview from the menu.

To display the overview in a small pop-up window, click the Overview button () at the bottom of the layout window.

Figure 26 Overview Toolbar



To change the region of the design that is displayed in the layout window, click and drag the yellow highlight region in the overview panel or window. After you release the mouse, the layout view updates to display the area within the highlight.

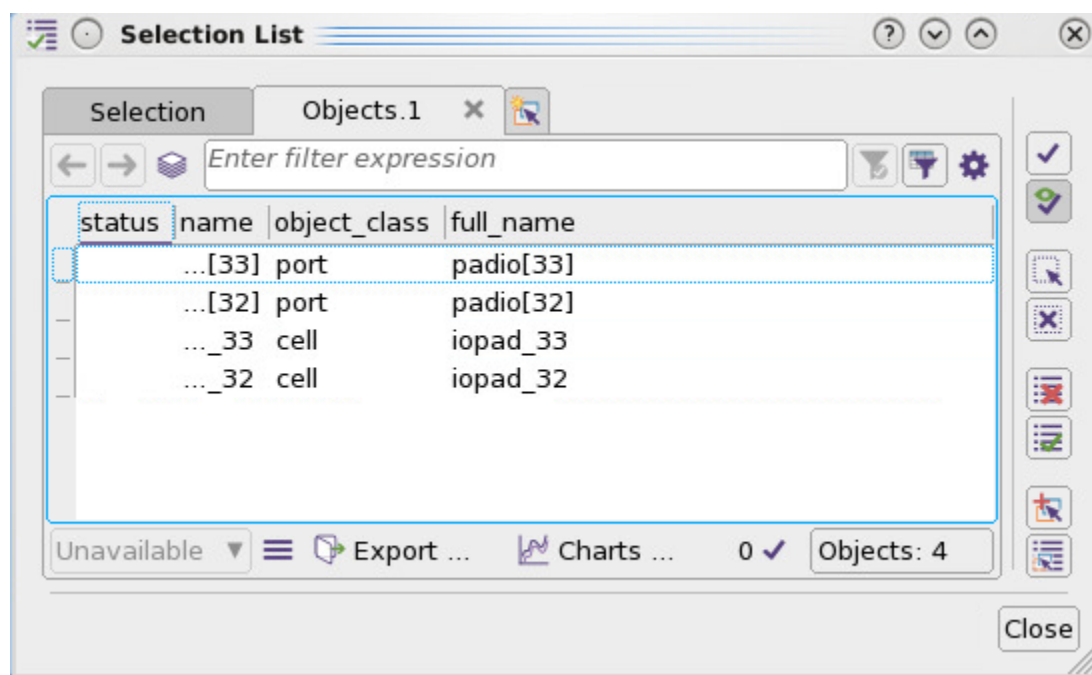
To change the zoom factor in the layout view, move the pointer to the boundary of the yellow highlight region. The pointer changes to allow you to resize the highlight. Click and drag the edge to zoom in or zoom out.

Using the Selection List and Work List

The selection list displays a list of the currently selected objects in the design. If you select objects of the same type, the selection list displays the Selection List group of attributes for the objects. If the objects are not of the same type, the selection list displays the name, object_class, and full_name attributes for the objects.

The following figure shows the Selection List dialog box with a work list tab selected.

Figure 27 Selection List Dialog Box



The following actions are available in the selection list:

- Sort the list of objects by clicking a column header; click the header a second time to reverse the sorting order.
- Enter a filter expression to reduce the number of items in the list. For example, enter the following expression to display cells with a name that begins with "C" and an orientation of R270.

```
name=~C && orientation==R270
```

- Click the forward and backward arrows next to the filter box to scroll forward and backward through previous filter expressions. Alternatively, click the filter menu button () to view the entire filter history; click an entry in the list to filter the list by using the expression.
- Click the Toggle between filter expression button () to show or hide filters for individual columns.
- Click the options button () to change whether the filter expression is interpreted as a wildcard expression or a regular expression and whether expressions are ANDed or ORed.
- Select an entry in the attribute group selection box to display a new set of columns for the selected objects. By default, the selection list entry is displayed. Use the Attribute Group Manager to view or modify the list of attributes displayed for a particular attribute group.
- Click the Modify Attribute Group button () to open the modify, manage, or save attribute groups.
- Click the Charts button ( Charts ...) to analyze design data in chart plots. See [Charts](#) for more information.
- Click the Export button ( Export ...) to export the list, or the selected list data, to the User Tables import form or a comma-separated values file.

Work List

The work list is a persistent list of objects based on the current selection and is created by clicking the Create Work List tab button; this action copies the currently selected objects to the work list. Clicking an item in a work list automatically selects the object in the current layout, schematic, or other view. The basic functions of the work list are similar to the selection list, however, the work list does not change as you select or deselect objects in the layout, schematic, or other view. You can create and manage multiple work lists in the same tool session, and each work list is available in a tab in the Selection List dialog box.

The following additional actions are available in a work list tab:

- Select one or more rows and click the Toggle Selected Items Fixed button () to mark or unmark the selected items with a check mark.
- Select one or more rows and click the Show/Hide Fixed Objects in List button () to show or hide the rows with the check mark annotation.

- Click the Add Global Selection button () to add the currently selected items to the current work list.
- Click the Select Items to match Global Selection button () to select items in the current work list that are also selected in the layout, schematic, or other view.

Charts

To analyze design data in a chart plot, click the Charts button ( Charts ...) in the Selection List dialog box or the work list tab and choose from the following chart plots:

- Distribution plot Organizes the values of a column into bins and displays the number of values in each bin.
- Box plot
Displays the distribution of values in one or more columns based on the five-number summary: minimum, first quartile, median, third quartile, and maximum. Outliers (outside the 1.5 quartile range) are shown as individual points.
- Scatter plot
Displays values from two columns in a two-dimensional plot.
- Matrix plot
Displays an array of scatter plots, allowing you to select two or more columns to produce a matrix of plots that summarize the relationship between the columns.

Each chart plot dialog box has a context menu that allows you to perform common operations on the plot. In addition, the dialog box includes

- Plot area
- Settings panel
- Data table (not shown by default)
- Status bar

You can specify the columns to be used in the plot, column groups, plot-specific controls, and controls for the selection behavior.

Using User Tables

User Tables provide a way to import, define, interact, and save tables of data. You can save a user table to a file that exists apart from the design database and reload the file in

a later tool session. User tables can be generated from a selection list or from a comma- or tab-separated file.

User tables support many common spreadsheet operations, such as editing, sorting, filtering, summing, viewing distributions of column values, and so on. In addition, you can change the current selection by clicking rows in the user table.

Creating a User Table

To create a user table from an existing selection set in the design,

1. Open the Selection List dialog box by clicking the Selection List button () or choose Select > Selection List from the menu.
2. Click the Export button to create the user table based on the current values in the selection list.

The tool copies the values to the user table import form.

3. (Optional) Change the table name in the Table Name box.
4. Click OK to import the data and build the user table.

To create a user table by importing a comma- or tab-separated text file,

1. Choose View > User Tables from the menu
2. Click the Open button () near the top of the user table form.
3. Complete the form by selecting the separator (comma or tab). Optionally, enter a table name.
4. Enter or choose a file name.
5. In the Configure Table Columns form, set the data type for each column as needed.
6. Click OK to create the user table.

Saving and Reloading a User Table

To save a user table,

1. Choose Save in the Action box near the top of the form.
2. Enter the file name in which to save the database. User tables typically have a .utable extension.
3. Click OK

The tool writes out the user table data to the specified file.

To load a saved user table,

1. Click the Open button () near the top of the user table form.
2. Choose Open UserTable file from the Select box.
3. Enter the user table file name to be loaded.
4. Click OK.

Analyzing Data in the User Table

After creating the user table, the following operations can be performed:

- Sort or reverse the sort of the table by clicking a column header.
- Filter the table by clicking the filter button () and entering a filter expression.
- Rearrange the order of columns in the table by clicking and dragging the column header. Alternatively, right-click a column header and choose Configure Columns from the context menu to open the Configure Columns dialog box.
- Select a user table to display by clicking the Tables box near the top of the user table form and selecting a new table name.
- Change the currently selected objects in the layout or schematic view by right clicking a cell in the table, choosing Auto Selection Mode, and clicking a cell to select or deselect.

Creating a Report Summary

The summary report in a user table provides basic statistics on data in the table columns. To generate a summary report for a user table, choose Report(summary) from the Action box near the top of the user table form. The tool displays statistics (min, max, average, and so on) for columns of numerical data in the table. A link to a histogram of numeric and categorical values is shown in the first row for each column.

Click any link in the first row to display the distribution report for that column. In the distribution report, click a column header to sort the table by the values in the column. Choose Report(summary) in the Action box to return to the summary report.

3

Using GUI Assistant Tools

The Fusion Compiler GUI provides the Task Assistant, Net Assistant, Pin Assistant, and Design Assistant to help you manage different objects in your design. For example, the Task Assistant consolidates commonly used commands into different flows to support design planning, timing analysis, placement, and other flows. The GUI Assistant Tools can simplify the design process.

The following topics describe how to work with the GUI:

- [Using the Task Assistant](#)
- [Using the Design Assistant](#)
- [Setting Constraints With the Relative Location Constraints Tool](#)
- [Using the Netlist Tracer](#)
- [Using the Track Assistant](#)

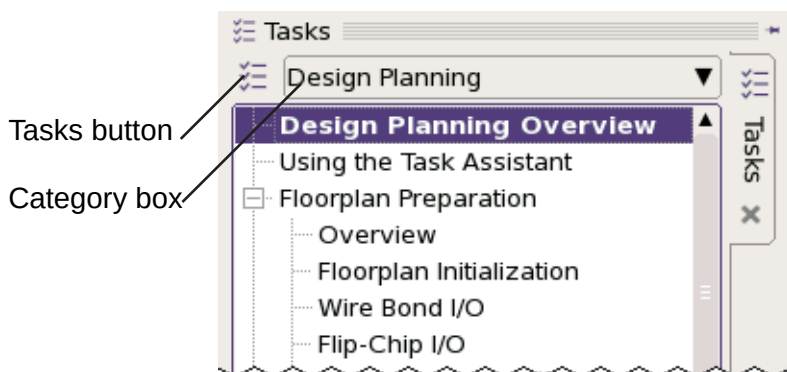
Using the Task Assistant

The Task Assistant is an easy-to-use tool for navigating the design planning and other flows and executing individual tasks to create your design. The following sections describe the major features of the Task Assistant.

Starting the Tasks Toolbar and Task Assistant

To open the Tasks Toolbar, choose View > Toolbars > Tasks from the menu. The tool displays the tasks panel to the right of the layout view as shown in [Figure 28](#).

Figure 28 Tasks Panel

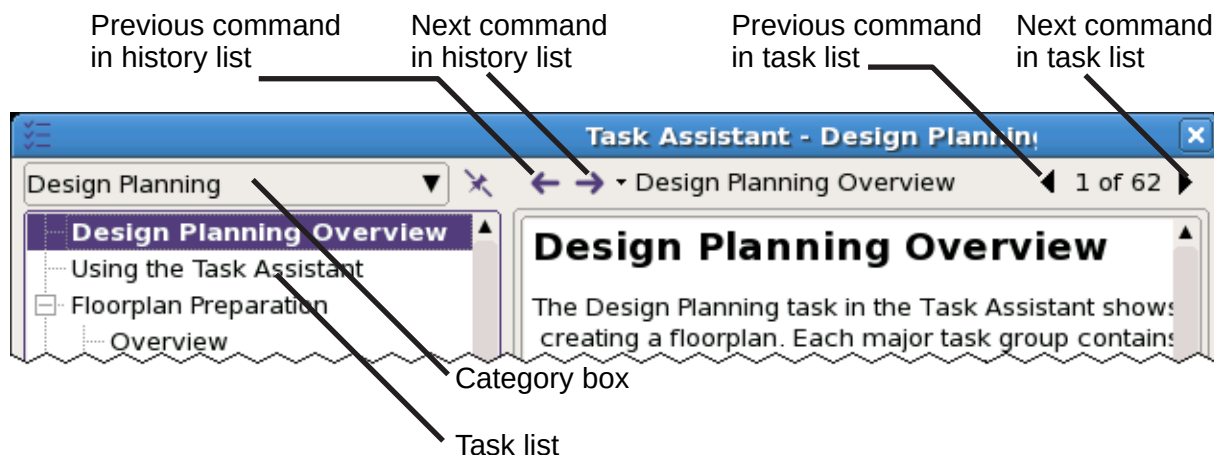


To display the list of tasks for a new category, click the down arrow to the right of the category box. To display the Task Assistant dialog, click the tasks button () to the left of the category box. Alternatively, choose Task > Task Assistant from the menu.

Navigating the Tasks in the Task Assistant Dialog

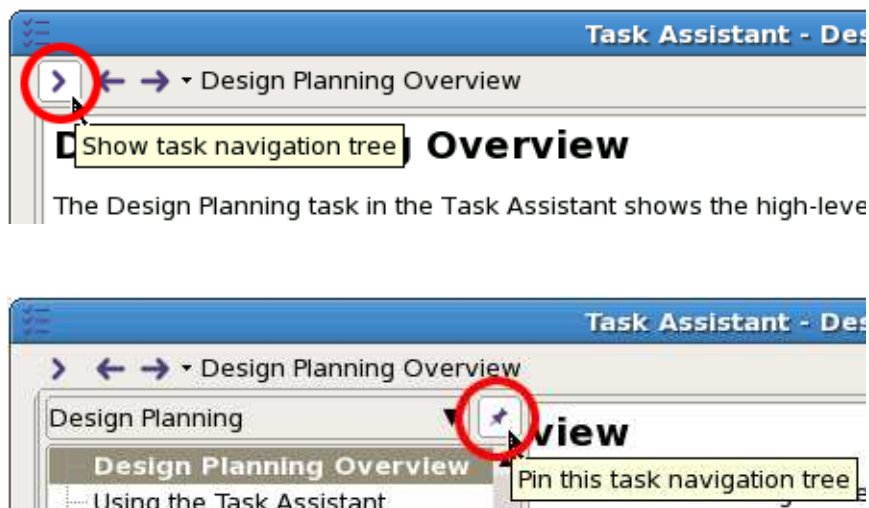
The Task Assistant dialog provides buttons to navigate to the next or previous task in the task list, or to go backward or forward in the history list. On the left of the Task Assistant window, the tool provides a navigation pane with a table of contents. In the navigation pane, select the category at the top left to load the list of tasks for the category. Click any task in the navigation pane to load the corresponding form into the Task Assistant. Click the [-] icon in the navigation pane to collapse a task subcategory, click the [+] icon to expand the subcategory. [Figure 29](#) shows some navigation features of the Task Assistant.

Figure 29 Task Assistant Navigation



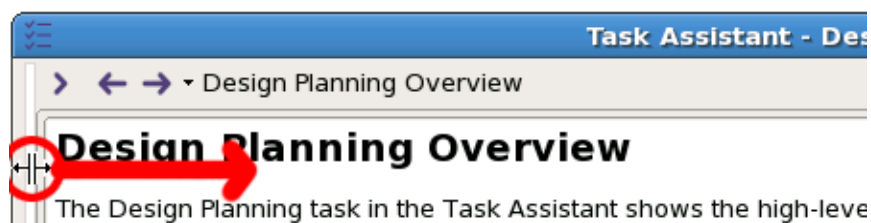
If the list of tasks is hidden as shown in [Figure 30](#), click the arrow to open it and click the pin to display the list of tasks.

Figure 30 Pinning the Navigation Pane



Alternatively, move your pointer to the left of the Task Assistant until you see two vertical bars. Click and drag the bar to the right to display the navigation bar.

Figure 31 Showing the Navigation Bar



Previewing, Capturing, and Issuing Commands

After you complete the form data, check the preview window near the bottom of the form and verify the command to be run as shown in the following figure. The window updates automatically as you enter data and select options in the form. Based on the type of data allowed for a given entry, the text box might restrict you from entering incorrect information.

Figure 32 Tcl Command Preview Window

The screenshot shows a GUI window titled "Tcl Command Preview Window". It contains several input fields and a command preview area. At the top, there are radio buttons for "Side size control: Ratio" (selected) and "Length". Below this are input fields for "Core utilization: 0.7" and "Aspect ratio: 1". A checkbox labeled "Pad limited" is also present. Under the heading "Sides", there are two rows of input fields. The first row has "a: 3", "b: 4", and "c: 5". The second row has "d: 6", "e:", and "f:". Red circles are drawn around the input fields for "a: 3", "b: 4", and "c: 5". At the bottom, a text box displays the command: **initialize_floorplan** -shape L -side_ratio {3 4 5 6}. The curly braces in the command are also circled in red.

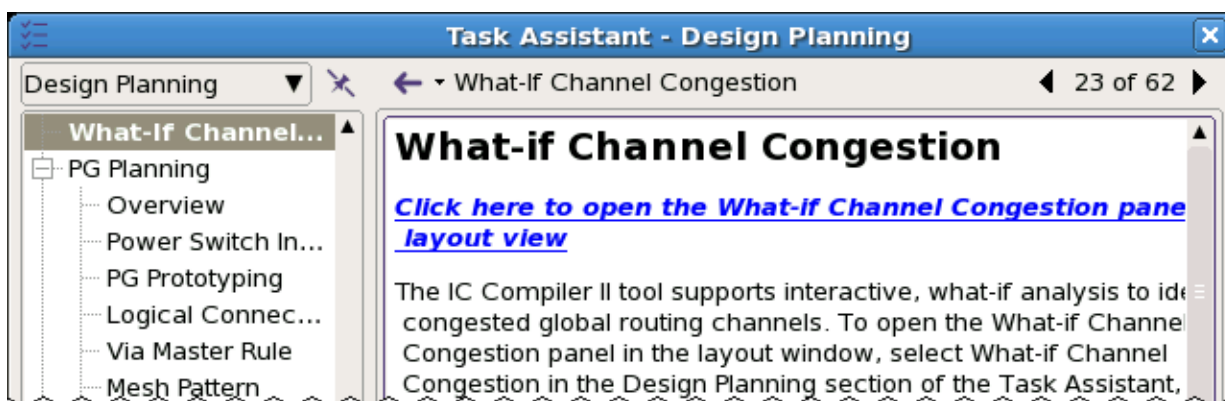
You can change the command by changing a selection or text box entry. Click the Script button to copy the command to the Script Editor in the layout window in the Fusion Compiler GUI. You can use the Script Editor tools to edit and save the command as needed.

After previewing the command, click Apply to run the command. The tool issues a warning message if a problem is detected in the command syntax. If a problem occurs, correct the form and try again.

Using Overview Pages

Some tasks in the task list are operated from panels in the GUI. For example, the Hierarchy Exploration and What-if Congestion Analysis tasks are performed from the View > Assistants > Hierarchy Exploration and View > Map > What-if Channel Congestion tools in the GUI. For these tasks, the Task Assistant provides overview pages to help you understand the tool. [Figure 33](#) shows the overview page for What-if Channel Congestion.

Figure 33 What-if Channel Congestion Overview Page



Click the link at the top of the page, or use the menu commands as described in the page, to open the tool in the GUI. Use the instructions in the overview page to guide you in performing the task.

Editing Data in Tables

For tasks related to setting or reviewing constraint information, the Task Assistant provides editable tables for viewing or modifying the constraint data. The following shows the list of blocks displayed for the Block Shaping > Shaping Constraints task.

Figure 34 Instance Search

← ▼ Block Shaping->Shaping Constraints

Enter search regular expression

7 block items

Instance Name	Reference Name	Outer Hard Keepout	Outer Hard Keepout
I_ORCA_TOP/I_BLENDER_2	BLENDER_6	0.82 0.82 0.82 0.82	0 0
I_ORCA_TOP/I_BLENDER_1	BLENDER_0	0.82 0.82 0.82 0.82	0 0
I_ORCA_TOP/I_BLENDER_7	BLENDER_1	0.82 0.82 0.82 0.82	0 0
I_ORCA_TOP/I_BLENDER_6	BLENDER_2	0.82 0.82 0.82 0.82	0 0
I_ORCA_TOP/I_BLENDER_5	BLENDER_3	0.82 0.82 0.82 0.82	0 0
I_ORCA_TOP/I_BLENDER_4	BLENDER_4	0.82 0.82 0.82 0.82	0 0
I_ORCA_TOP/I_BLENDER_3	BLENDER_5	0.82 0.82 0.82 0.82	0 0

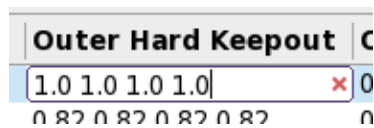
To edit a constraint,

1. Double-click a cell in the table
2. Use the Backspace key to delete the current entry
3. Enter a new constraint for the table cell

The tool displays the text in red if it is incomplete or incorrect. Valid values are displayed in black.

4. Press Return to confirm the new values

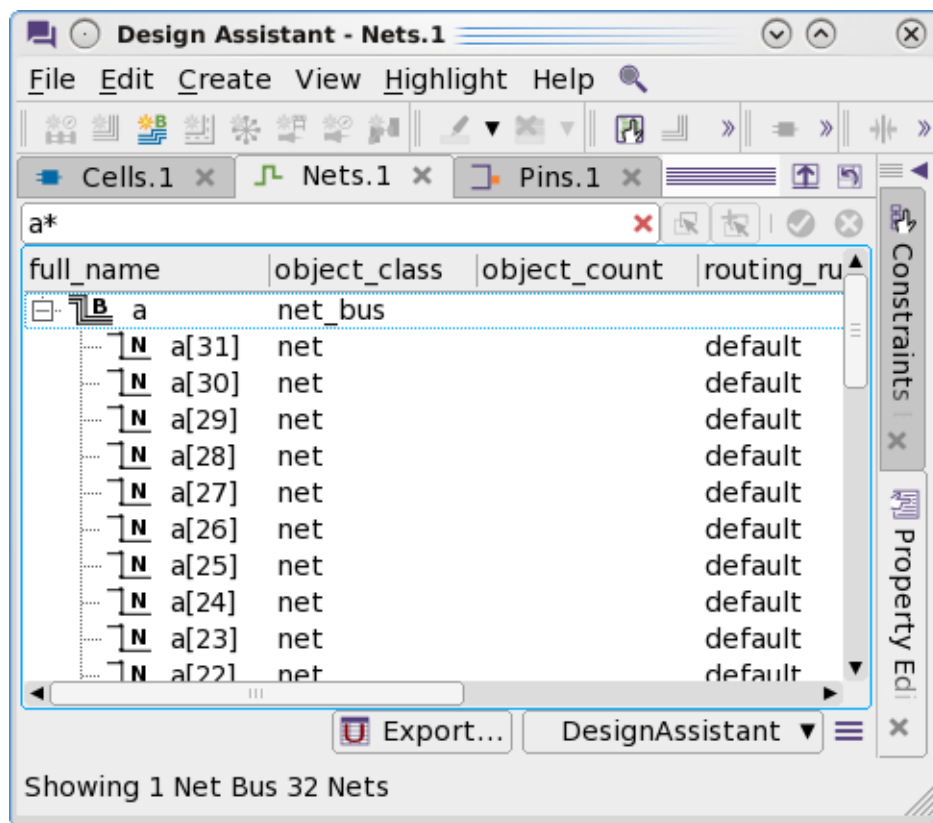
Figure 35 Editing Table Cell Data



Using the Design Assistant

The Design Assistant helps you to locate and select objects in the layout view. To start the Design Assistant, choose View > Assistants > Design Assistant from the menu, or click the Design Assistant button. The following sections describe how to use the Design Assistant.

Figure 36 Design Assistant



Listing and Selecting Objects

The Design Assistant supports a search box for listing cell, net, and pin objects. To populate the list of objects, click the Cells, Nets, or Pins tab, enter a search term in the box, and press Enter. You can specify a full string, a substring, a wildcard expression, a Tcl command, or a Tcl variable. For example, specify any of the following entries in the search box to match the cells u0_0, u0_1, u0_2, u0_3 (and possibly other cells).

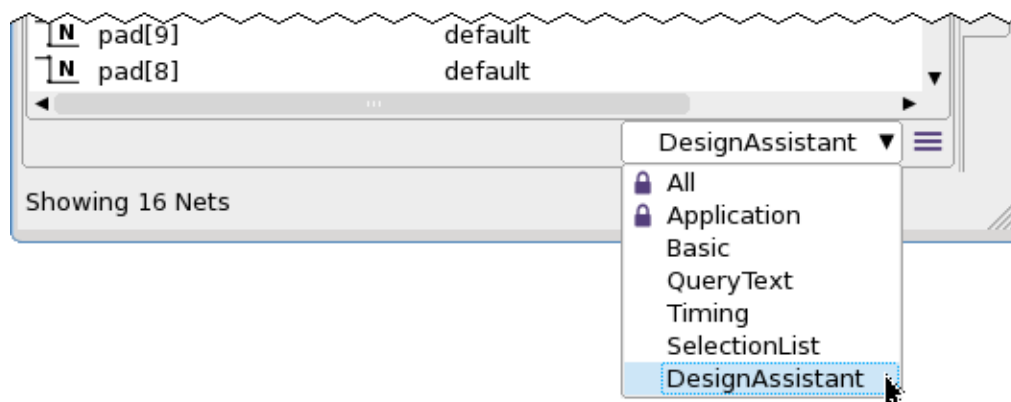
- Strings only: 'u', 'u0' or 'u0_'
- Wildcards: 'u0_*', 'u0_?'
- A Tcl command: `get_cells "*" -filter name=~"u0*"`
- A Tcl variable: `$my_cells` (after performing `set my_cells [get_cells "*" -filter name=~"u0*"]` in the tool)
- A selection in the GUI: `get_selection` (after selecting u0_0, u0_1, u0_2, and u0_3 in the layout view or Hierarchy Browser); you can also click the Get current selection button to the right of the search box to populate the list with the selected objects

After the list is populated, click the cell instance in the list to select the cell in the layout. Click and drag the pointer to select more than one cell, or press the Shift or Control key while clicking the cell to add the cell to the selection. To clear the selection, click the (X) at the right of the search box, or delete the characters in the search box and press Enter.

Configuring Columns in the Design Assistant

To change the columns displayed in the Design Assistant, click the down arrow in the attribute group selector box near the bottom of the dialog box and click an entry in the list that appears. The list contains available attribute groups for this object type as shown in the following figure.

Figure 37 Design Assistant Attribute Group Selector

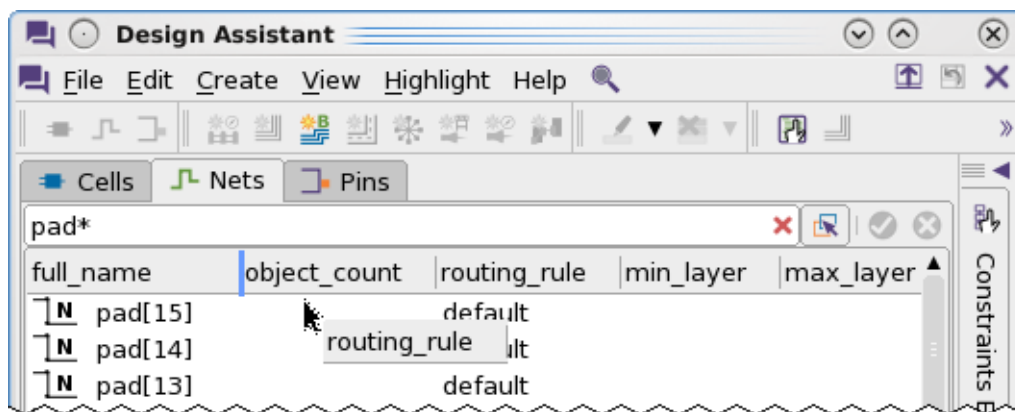


The entries in the table can be configured by using the Attribute Group Manager. Click the Modify Attribute Group button (≡) at the bottom right of the dialog box and click Manage All to open the Attribute Group Manager. For more information on the Attribute Group Manager, see [Using the Attribute Group Manager](#).

You can also perform the following actions on column headers in the table:

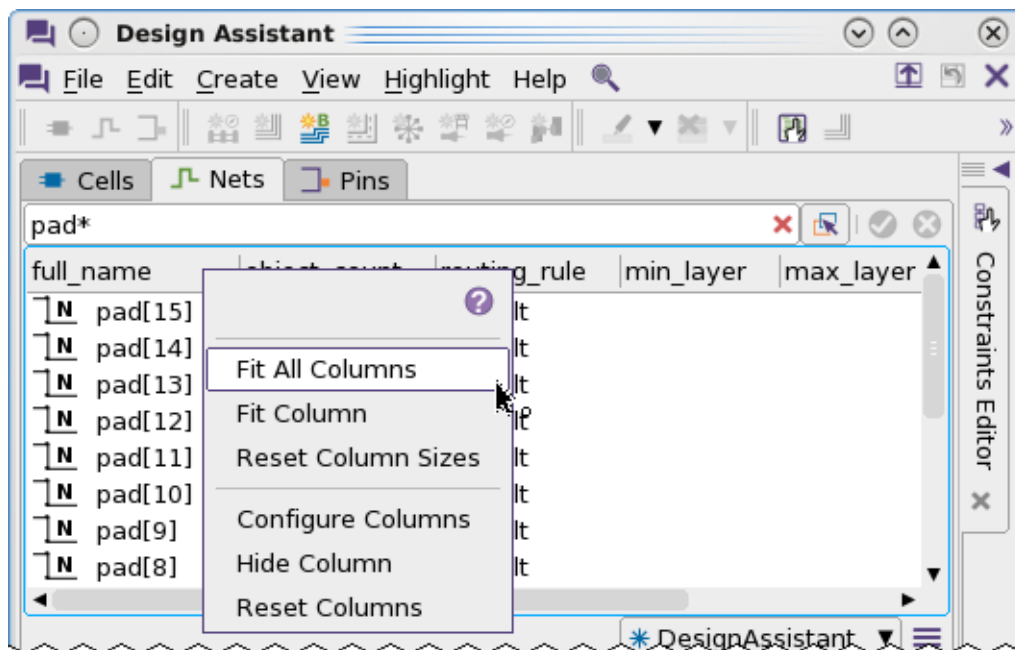
- Sort the data in the list by clicking a column header.
- Reverse the sort order by clicking the column header again.
- Change the column order by clicking a column header and dragging it to its new location as shown in the following figure.

Figure 38 Rearranging Columns in the Design Assistant



- Sort columns by right-clicking the column header and selecting Sort Increasing or Sort Decreasing.
- Fit column widths to the Design Assistant dialog box by right-clicking the column header and selecting Fit All Columns or Fit Column as shown in the following figure.

Figure 39 Fit Column Widths

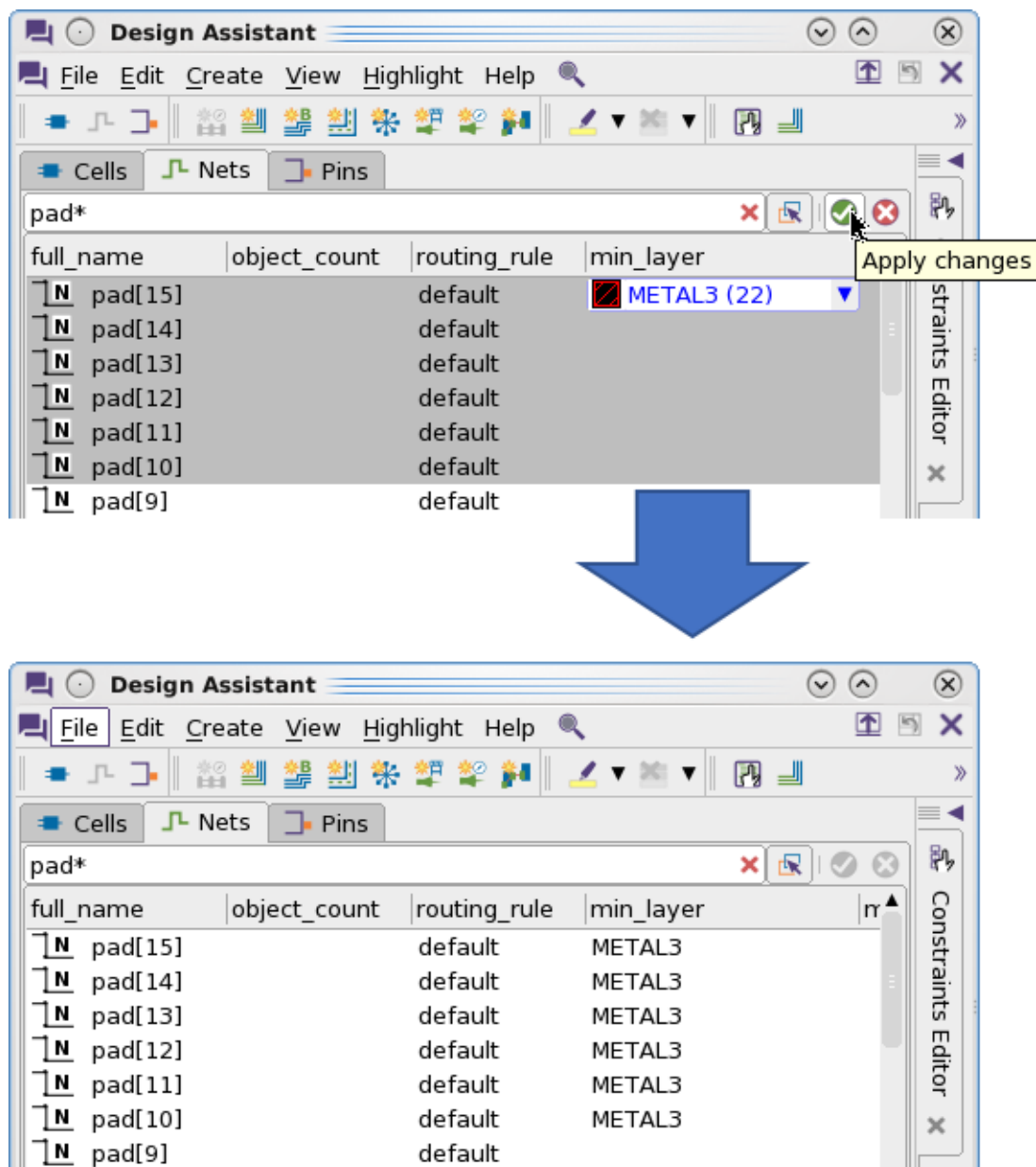


Changing Property Settings for Multiple Objects

You can apply changes to property settings for multiple objects at one time. To change the settings for a property for multiple selected objects,

1. Select multiple objects in the Design Assistant by clicking the rows in the table while holding the Shift key. Alternatively, click and hold while dragging the mouse to select multiple rows.
2. Change the property for a single object by clicking its row and changing the value.
3. Click the Apply button to apply the changes to all selected objects.

Figure 40 Design Assistant Apply Changes to Multiple Objects



Performing Related Tasks

Each tab supports a context menu which displays different tasks you can perform on the list of objects. Move the pointer over the entry in the list and click the right mouse button to view the context menu. The context menu displays tasks related to the selected object. For example, if you open the context menu in the Cells tab, the context menu displays

actions to Get Nets, Get Hierarchical Nets, Get Pins, and Create Bounds for the selected object. Click the item in the context menu to perform the action. After you select the action from the context menu, the tool opens the appropriate tab and displays the information.

Note that you can perform the same related actions by choosing it from the menu bar. For example, to display nets for the selected item in the Cells tab, choose Cells > Get Nets from the Design Assistant menu bar. The items in the menu bar change as you select different tabs in the Design Assistant.

You can also use the context menu to highlight an object, zoom in to view the object, write a Tcl command to a script file to select the object, and perform other tasks. Click the item in the context menu to perform the action.

Viewing Net and Shaping Constraints

Use the Constraints Editor panel to display and modify net and shaping constraints. Net constraints include nondefault routing rules, topology objects, and routing corridors associated with objects selected in the Nets tab. Shaping constraints include target location, parent object, boundary, utilization, and so on set by using the `create_shaping_constraint` command.

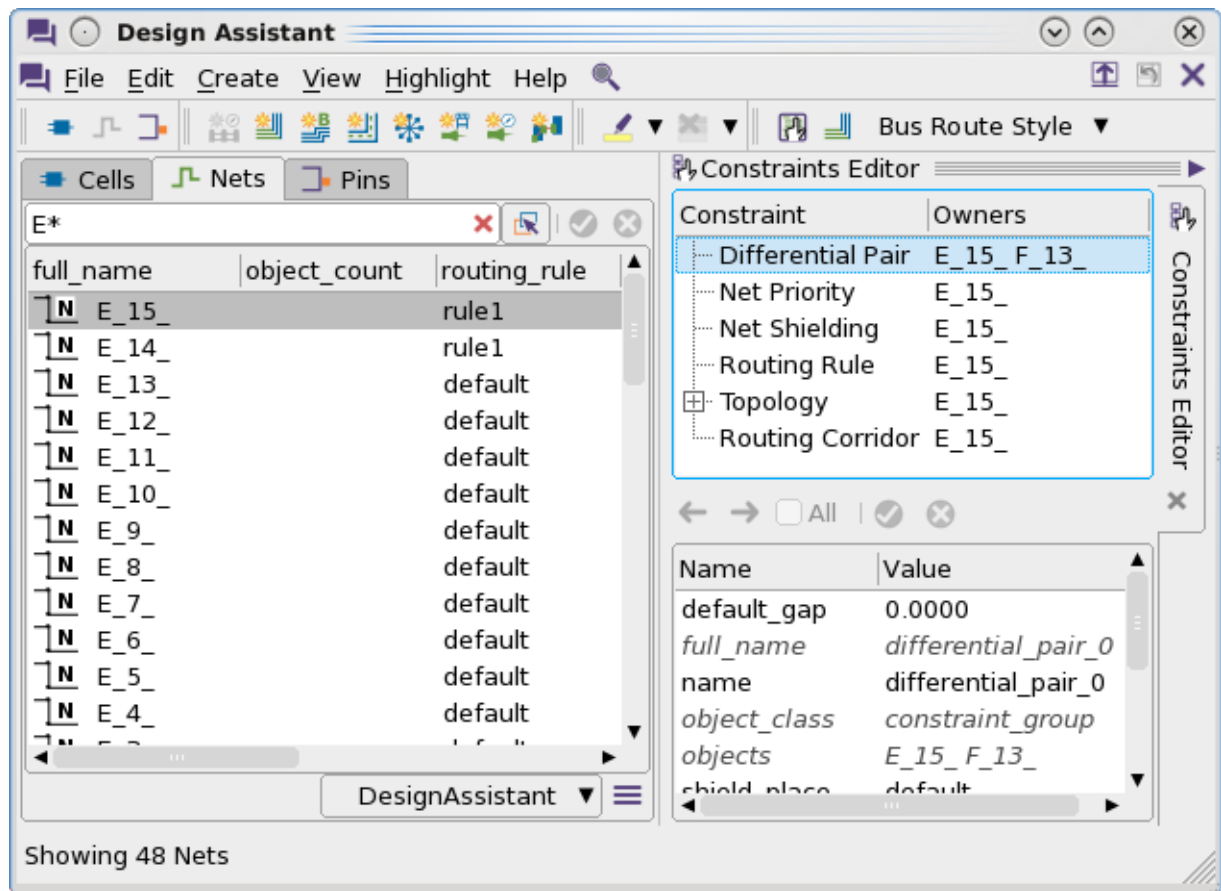
To use the panel to review and modify shaping constraints,

1. Choose View > Toolbars > Constraints Editor from the Design Assistant menu or click the Constraints Editor tab.
2. Click a net name in the table.
3. Click the constraint type in the constraint list to view the properties for that constraint.
4. (Optional) Modify the properties for the constraint in the properties editor table.
5. (Optional) Right-click over the constraint type in the constraint table and choose Select, Select Owners, or Delete to select or delete the constraint type.

To use the panel to review and modify net constraints,

1. Choose View > Shapeable Objects from the Design Assistant menu.
2. Choose View > Toolbars > Constraints Editor from the Design Assistant menu or click the Constraints Editor tab.
3. Click an object name in the table.
4. Click the [+] symbol in the constraint list to view the properties for that constraint.
5. (Optional) Modify the properties for the constraint in the properties editor table.
6. (Optional) Click the right mouse button over the constraint type in the constraint table and choose Select, Select Owners, or Delete to select or delete the constraint type.

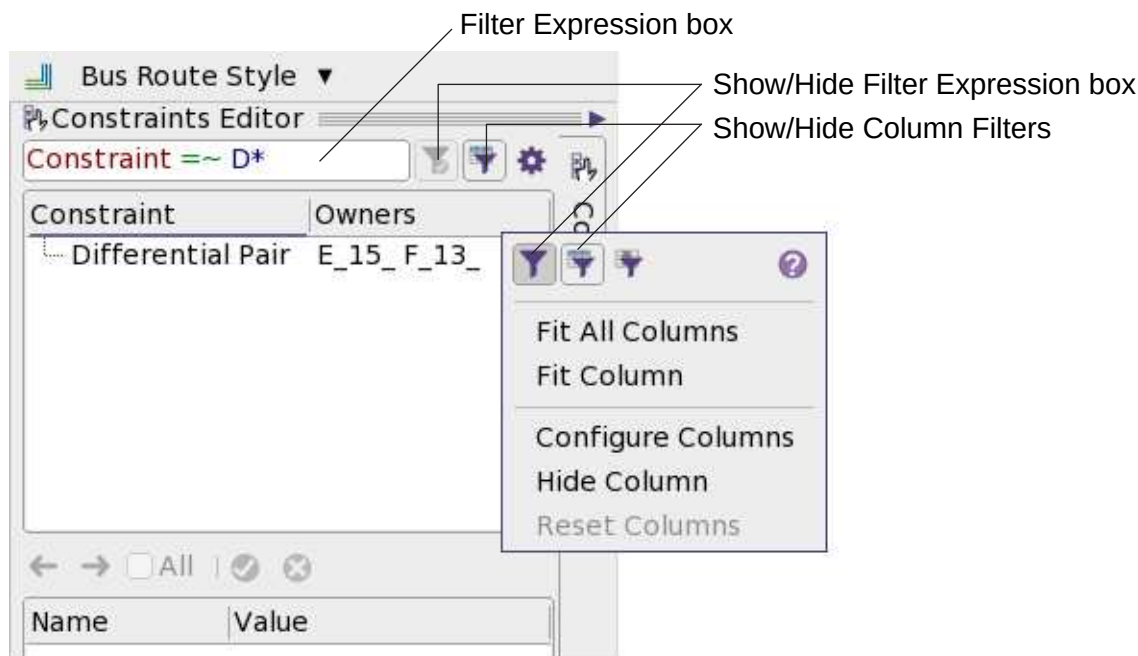
Figure 41 Design Assistant Net Constraints



The Constraints Editor supports expression-based filtering and column filtering. Click on a constraint and press the forward slash (/) key to display the filter expression box. Alternatively, click the right mouse key over a column header and click the Show/Hide Filter Control button to display the filter expression box. Enter an expression in the box and press the Enter key to filter the results.

To perform column filtering, click the Show/Hide column filter control button. The tool displays a funnel icon to the right of each column heading. Click the icon to display the column filter menu as shown in the following figure.

Figure 42 Design Assistant Column Filtering



Highlighting Objects

To highlight objects in the layout, move the pointer over the entry in the list, click the right mouse button to view the context menu, and click Highlight to highlight the object. Alternatively, select the object in the list and choose Highlight > Highlight from the Design Assistant menu bar. To clear a highlight, select the object and choose Highlight > Clear Selected from the menu bar. Choose Highlight > Clear All to clear all highlights.

Reading and Writing Object Selection Scripts

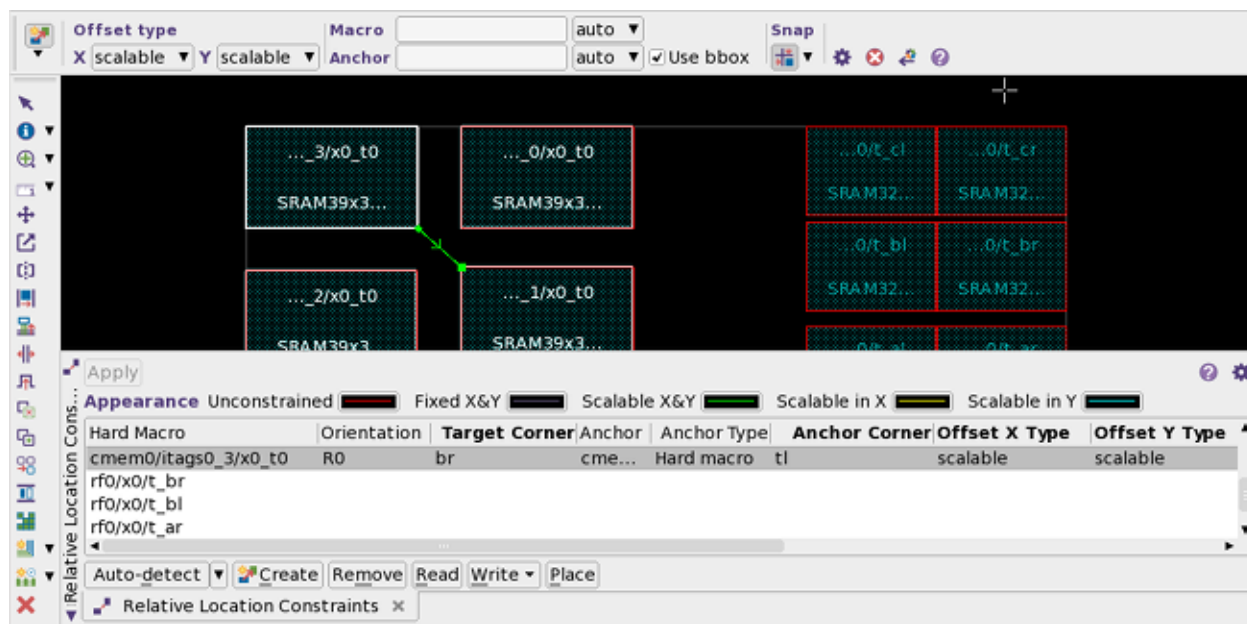
To save a Tcl command that returns the list of currently selected objects in the current tab, choose File > Write Script from the menu bar or click the right mouse button and select Write Script from the context menu. Note that you must have at least one object selected to perform this action. Navigate to the directory in which to save the file, enter the file name and click Save to write the file. The Design Assistant writes a script that contains a `get_pins`, `get_nets`, or `get_cells` command followed by the names of currently selected objects.

To read a Tcl script from the Design Assistant, select File > Read Script or click the right mouse button and select Read Script from the context menu.

Setting Constraints With the Relative Location Constraints Tool

The Relative Location Constraints tool helps you set, review, and remove placement constraints for hard macros. To start the tool, choose View > Assistants > Relative Location Constraints from the menu. The Relative Location Constraints panel appears at the bottom of the layout as shown in [Figure 43](#). The list box displays the hard macro instances in the design together with any relative location constraints that are specified. Sort the table by clicking a column header; click again to sort in the reverse order.

Figure 43 Relative Location Constraints Panel



Manually Creating Relative Location Constraints

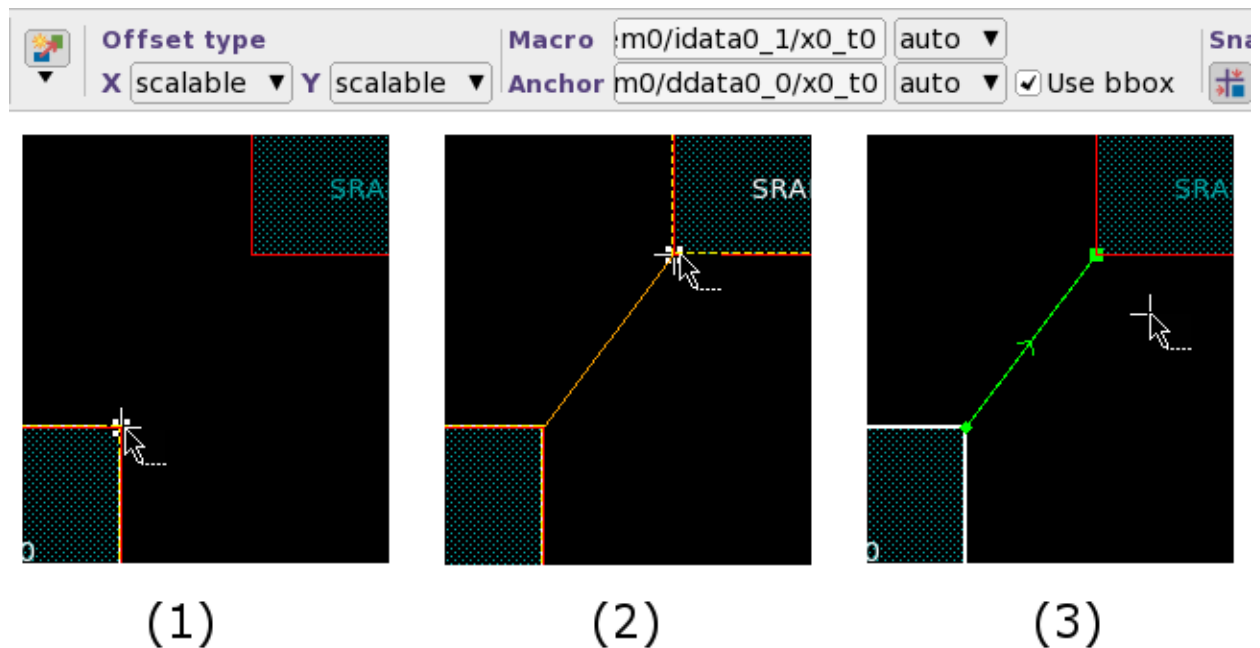
To create a relative location constraint,

1. Click the Create button to begin defining the constraint.
2. (Optional) Change the Offset type and other settings in the Mouse tool options toolbar as needed.
3. Click the corner of the target macro to be constrained as shown in [Figure 44](#) (1).
4. Click the corner of the anchor object to define the anchor corner as shown in [Figure 44](#) (2). If you have trouble, make sure the anchor object type is selectable in the View Settings panel.

The tool displays a constraint flyline between the corners of the target macro and anchor object as shown in Figure 44 (3); the constraints table in the Relative Location Constraints panel is updated to display the new constraint.

5. (Optional) Repeat steps 2 through 4 to define additional macro relative location constraints.
6. Press Esc to exit the Create Macro Relative Location Constraint tool.

Figure 44 Manually Creating a Relative Macro Constraint



Automatically Creating Relative Location Constraints

The tool supports automatic creation of relative location constraints. To create the constraints for a selected set of macros,

1. Select one or more macros using one of the following methods:
 - Click on a row in the table.
 - Click on a row in the table and drag to select more than one macro.
 - Click in the table and press Ctrl+A to select all macros.
 - Click on one or more macros in the layout.

2. Click the Auto-detect button.

The tool creates relative location constraints for all selected macros.

Placing Macros Using Relative Location Constraints

To place the macros based on relative location constraints, click the Place button and click OK in the dialog box that appears. The tool runs the `create_macro_relative_location_placement` command to update the macro placement based on the relative location constraints.

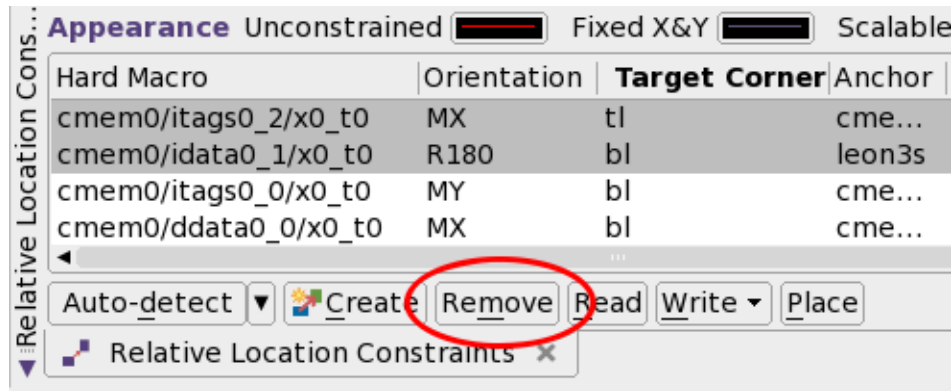
Removing Relative Location Constraints

To remove one or more relative location constraints,

1. Select one or more macros.
2. Click the Remove button to delete the constraints.

Figure 45 shows a constraint table with two macros selected. In this example, clicking the Remove button removes the constraints for the selected macros.

Figure 45 Removing Relative Location Constraints



Writing Out Relative Location Constraints

To write out relative location constraints for all macros with defined constraints, click the Write button, choose Macro relative location, and enter a file name in which to save the constraints. The tool writes out a `set_macro_relative_location` command for each macro with a defined macro relative constraint. The following example shows two defined constraints:

```
# macro_relative_constraints.tcl
set_macro_relative_location \
```

```
-target_object [get_cells u1/REG_FILE_D_RAM] \  
-target_orientation R0 -target_corner tr \  
-anchor_object [get_blocks ORCA] -anchor_corner tr -offset { \  
-0.1594 -0.2488} -offset_type scalable \  
-used_length {0.0000 0.0000} -scale_edge {0 0}  
  
set_macro_relative_location \  
-target_object [get_cells u2/SD_FIFO_RAM] \  
-target_orientation MXR90 -target_corner br \  
-anchor_object [get_blocks ORCA] -anchor_corner br -offset { \  
-0.1715 0.2576} -offset_type scalable \  
-used_length {0.0000 0.0000} -scale_edge {0 0}  
  
set_attribute -objects [get_blocks test:ORCA.design] \  
-name edge_number -value 4 -quiet
```

Writing Out Port Relative Location Constraints

To write out port relative location constraints for all I/O ports, click the Write button, choose Port relative location, and enter a file name in which to save the constraints. The tool writes out a `set_individual_pin_constraints` command for each I/O port with a defined relative constraint. An example file for a design with two defined constraints is as follows:

```
# Port relative location constraints  
# Generated on Feb 3 12:51:21  
...  
set_individual_pin_constraints -ports [get_ports pad[15]] \  
-sides 2 -offset [mrl_get_offset 2 0.5]  
set_individual_pin_constraints -ports [get_ports pad[14]] \  
-sides 2 -offset [mrl_get_offset 2 0.5]  
...
```

Reading In Relative Location Constraints

To read in relative location constraints for macros or I/O ports, click the Read button and enter the script file name that contains the macro relative location constraints or I/O port relative location constraints. The tool reads the file and applies the constraints.

Changing the Color of the Constraint Flylines

The tool displays constraint flylines between macros and objects when a macro relative constraint is defined. To change the color of the constraint flyline,

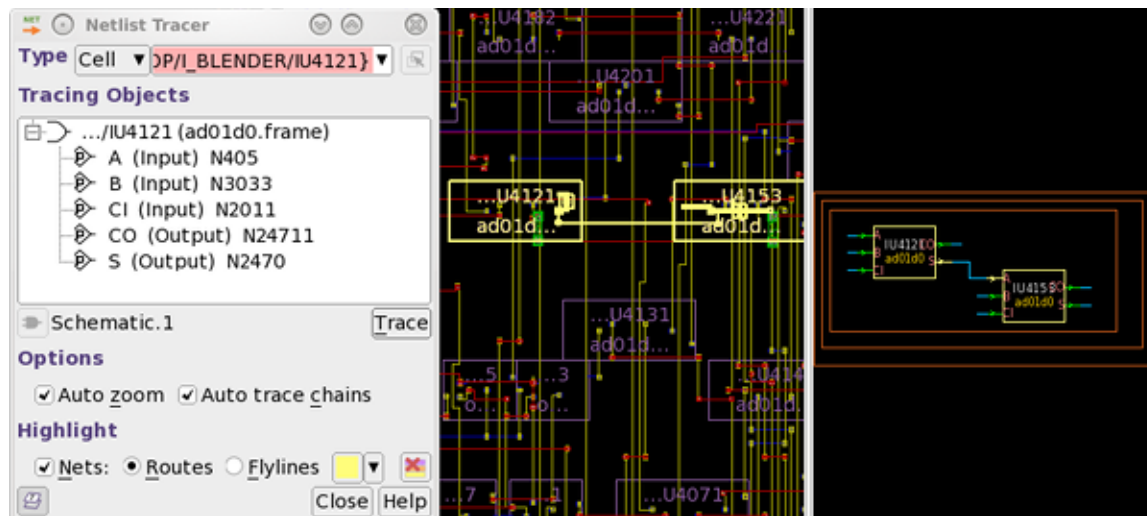
1. Click a color bar in the Appearance section near the top of the panel.
2. Select a new color from the Select Style dialog box and click OK.
3. Click Apply to apply your changes.

The tool redisplayes the constraint flylines using the new colors.

Using the Netlist Tracer

The Fusion Compiler tool supports the Netlist Tracer for tracing the path of a net through the design. Start the Netlist Tracer by choosing View > Assistants > Netlist Tracer from the menu. Use the Netlist Tracer to trace a net by traversing its path through the design. Begin by selecting a cell, net, pin, or port from which to trace. You can traverse the path of nets that connect to the object, highlight the path, and zoom in on related cells, nets, and pins. As you trace the path, the Netlist Tracer displays flylines between the current object and any connected object.

Figure 46 Netlist Tracer



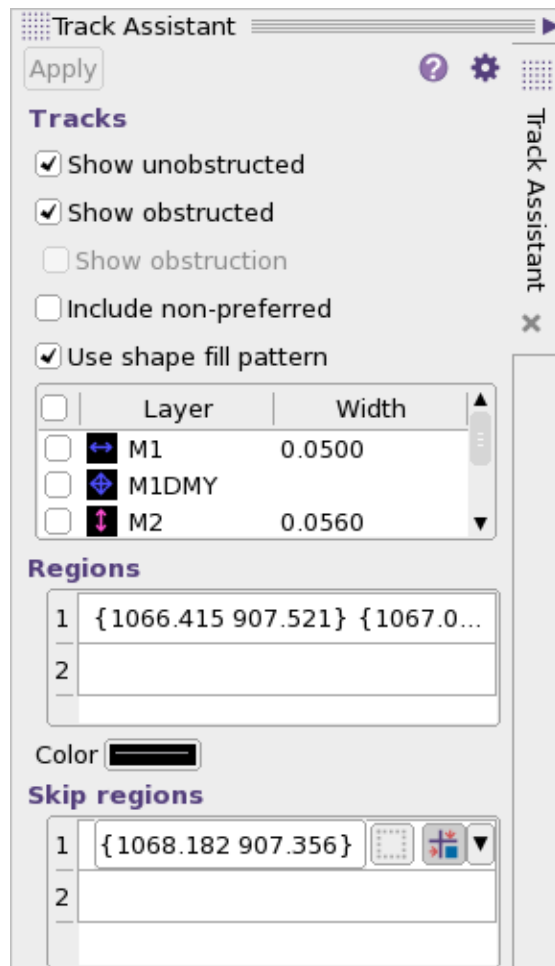
Using the Track Assistant

Use the Track Assistant panel to examine routing tracks in the current design. The Track Assistant shows the bounding boxes for tracks on routing layers that you select and

displays a track number to indicate the track position relative to the selected regions. You can display tracks by layer, width, preferred and nonpreferred direction, and obstruction.

To start the Track Assistant, choose View > Assistants > Track Assistant from the menu, or click the Track Assistant button (). The following sections describe how to use the Track Assistant. The following figure shows the Track Assistant panel.

Figure 47 Track Assistant



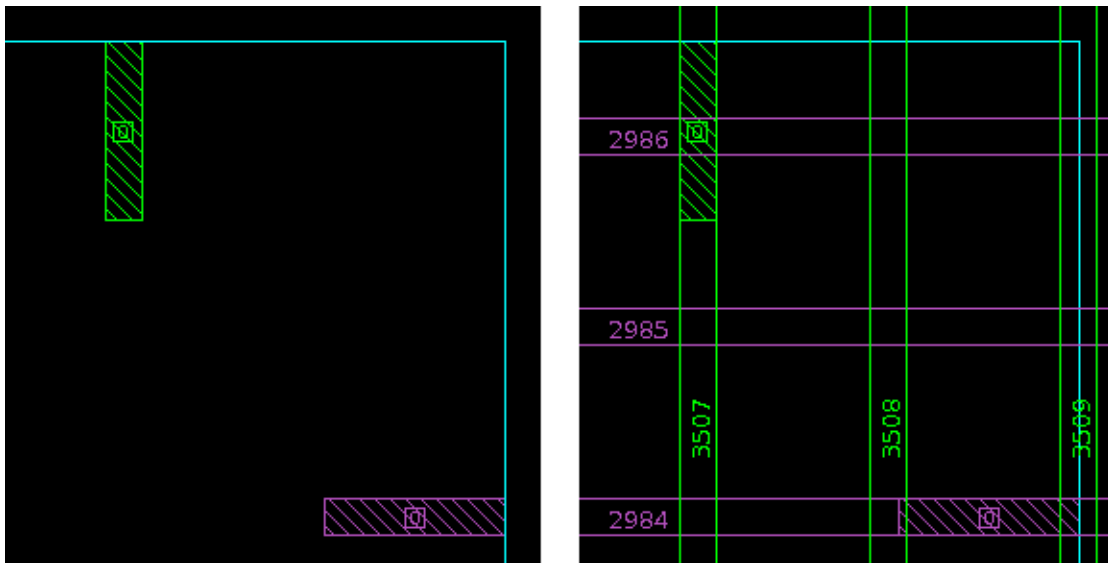
Displaying Tracks

To display tracks by using the Track Assistant,

1. Start the Track Assistant by choosing View > Assistants > Track Assistant from the menu.
2. Zoom to the region of the layout to examine.
3. Select the layers for which to display tracks.

The tool displays the routing track using the track width for the selected layers as shown in the following figure. By default, the tool displays both unobstructed and obstructed tracks.

Figure 48 Before and After Track Display



Displaying Track Information in Specific Regions

Use options available in the Track Assistant to limit the track display to specific regions of the design. To display track information only for certain regions in the design,

1. Start the Track Assistant and select layers as described in the previous section.
2. Click in an empty row in the Regions section and enter the bounding box for the region to display.

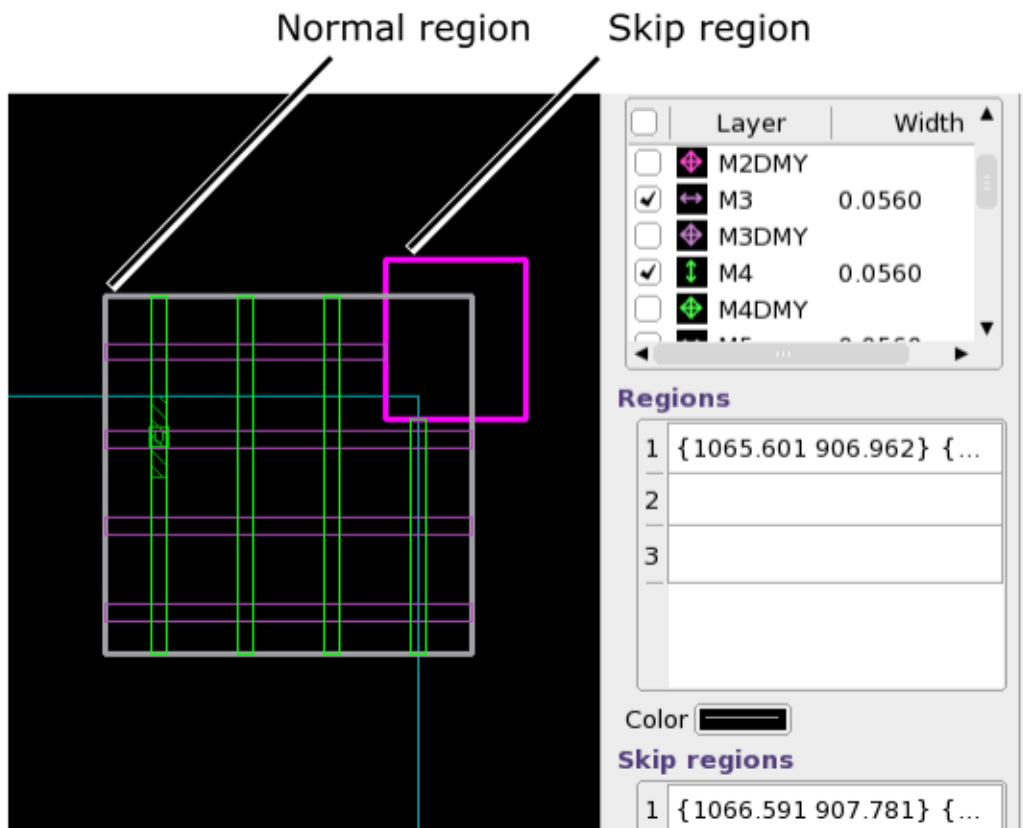
Alternatively, click the Specify Rectangle button () and draw the region on the layout. To clear the region, move your mouse pointer over the bounding box coordinates in the panel and click the Clear region button ().

3. (Optional) Repeat the previous step to draw additional regions as needed.
4. (Optional) Create a region in which to hide the track display by clicking in an empty row in the Skip regions section and entering the bounding box for the region.

Alternatively, click the Specify Rectangle button () and draw the skip region on the layout.

The following figure shows the layout and Track Assistant panel displaying only a specific region together with a skip region. Note that tracks are not displayed outside the defined region.

Figure 49 Track Assistant With Normal and Skip Regions

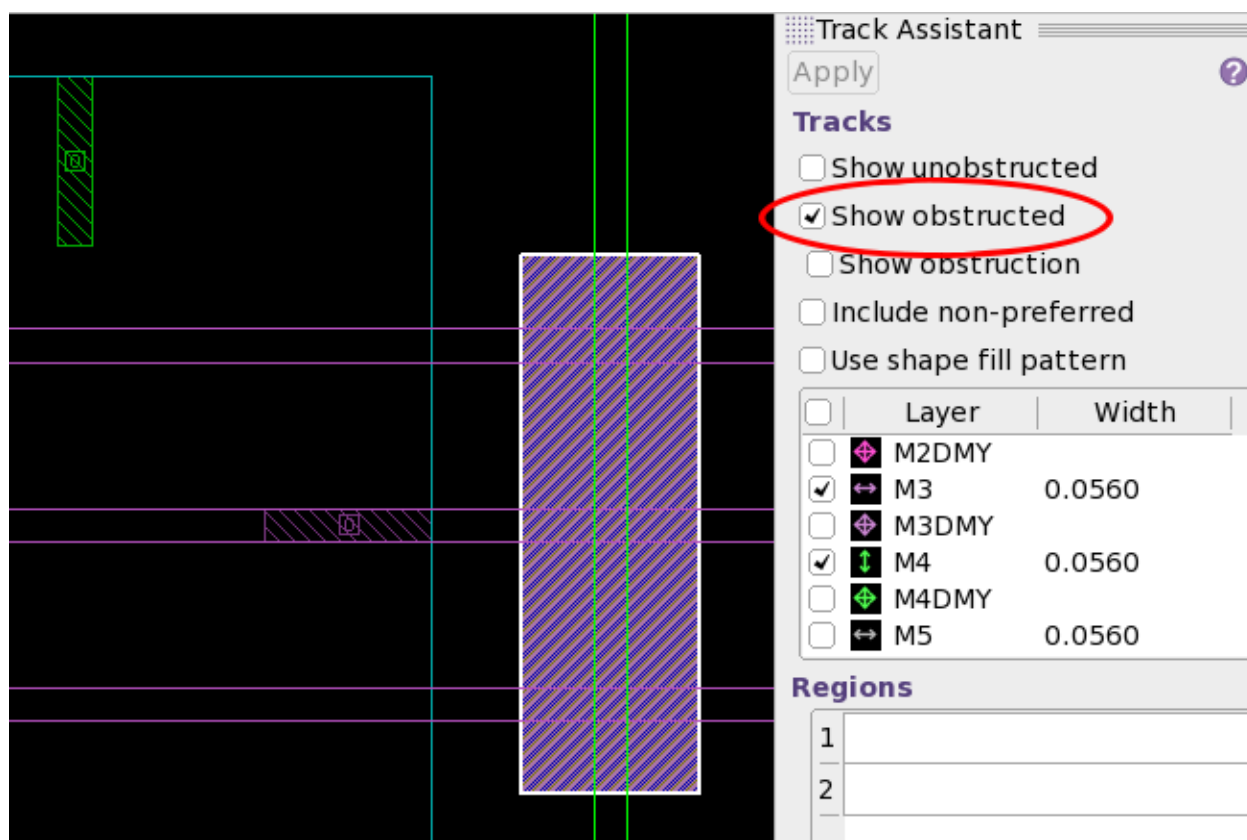


Displaying Obstructed or Unobstructed Tracks

When you start the Track Assistant, the tool displays both obstructed and unobstructed tracks. To display only unobstructed tracks, select the Show unobstructed option and deselect the Show obstructed option. To display only obstructed tracks, deselect the Show unobstructed option and select the Show obstructed option. To highlight the obstruction, select the Show obstruction option.

The following figure shows the panel settings and layout when displaying obstructed tracks.

Figure 50 Obstructed Tracks



Setting Track Assistant Options

The Track Assistant panel supports the following additional options:

- Display tracks in both the preferred and nonpreferred routing directions by choosing the Include non-preferred option.
- Display the routing tracks with the same pattern as the metal layers by choosing the Use shape fill pattern option.
- Set the color for the normal or skip regions by clicking the color bar below the region list and choosing a new color from the Select Style dialog box.

4

Using Design Planning Tools

The Fusion Compiler GUI provides tools to support various tasks in Design Planning. The What-if Channel Congestion tool supports interactive, what-if analysis to identify congested global routing channels. The Hierarchy Exploration tool supports several tasks related to module boundaries.

The following topics describe design planning tasks supported by the GUI:

- [Hierarchy Browser](#)
- [Name Based Hierarchy Browser](#)
- [What-if Channel Congestion](#)
- [Hierarchy Exploration](#)
- [Topological Pin Constraints](#)

Hierarchy Browser

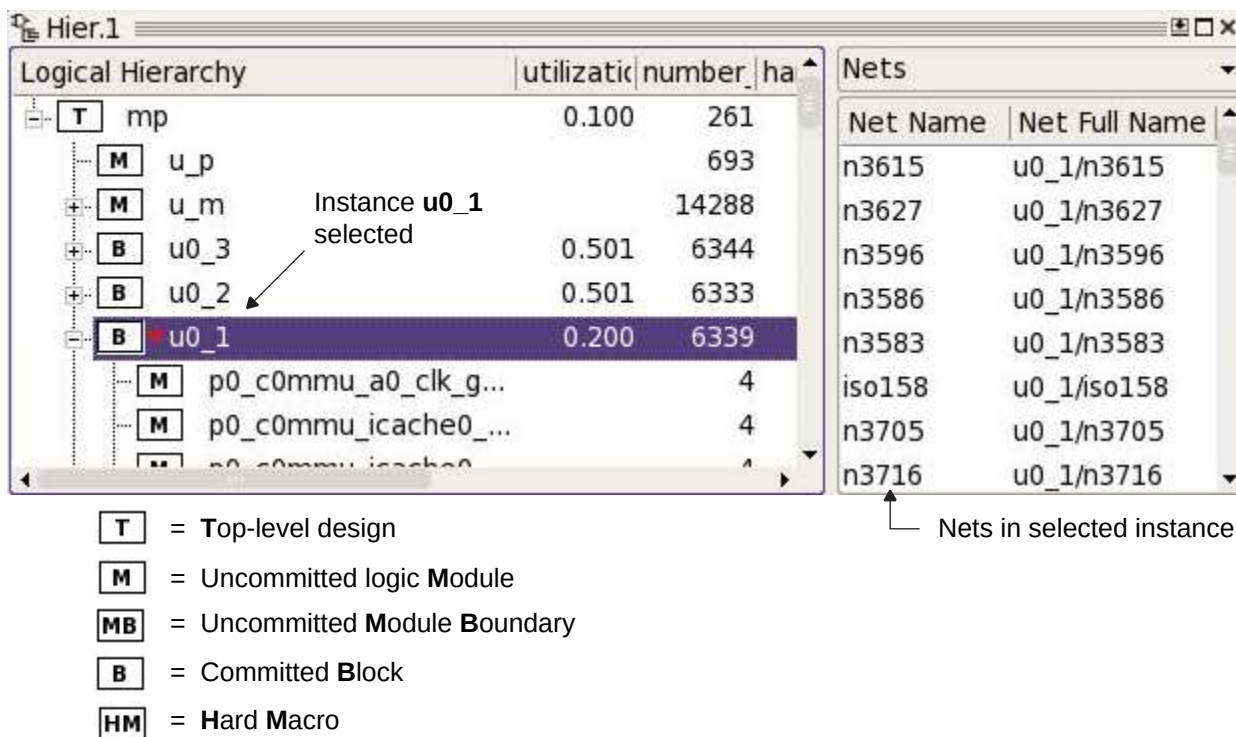
The Fusion Compiler tool provides tools to explore your design hierarchy and determine which logical hierarchy cells to commit to physical hierarchy blocks. The hierarchy browser in the GUI displays a view of the design hierarchy and provides other information about each hierarchy cell.

To explore the design hierarchy with the Hierarchy Browser,

1. Choose View > Hierarchy Browser in the GUI.
2. Expand or collapse the hierarchy by clicking the [+] symbols next to the hierarchy cells.
3. Rearrange the display by clicking the column headers.
4. Click the block name to perform additional operations on the block.

An example hierarchy as displayed in the Hierarchy Browser is shown in [Figure 51](#).

Figure 51 Hierarchy Browser



You can perform the following actions with the context menu; right-click the mouse over a logical hierarchy to open the menu.

- Select Create Module Boundary to create a module boundary for the specified level of hierarchy.
- Select Create Bound to create a module boundary and start the Create Bound tool in the layout window. Draw the bound in the layout window to complete the task.
- Select Color Cells by Hierarchy and select Color Child Hierarchies, Color Boundaries, Color Selected Cells, Remove All Hierarchy Colors, or Remove Hierarchy Color on Selected Cells to modify the color for a hierarchy.
- Select Color Cells by Hierarchy > Set Selected Color and choose a color from the menu to change to a specific color.
- Select Clear Selection to deselect all selected objects in the hierarchy.
- Select the Select Related Objects menu item and choose an item from the list to select the connections, macros, or terminals associated with the object.
- Select Properties to open a dialog and display the properties for the logical hierarchy.

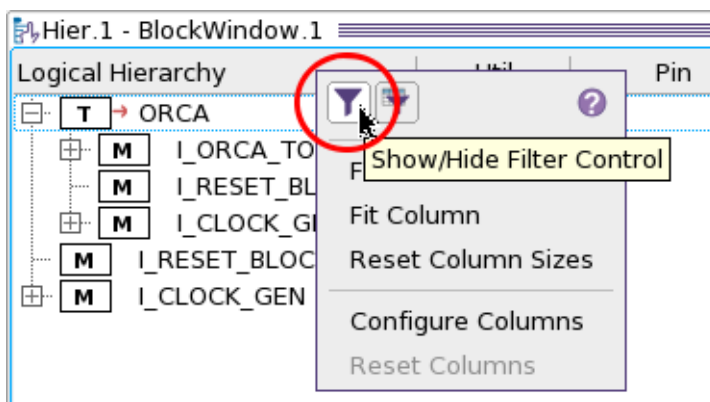
- Select Expand Tree to display the sub hierarchies for the selected logical hierarchy.
- Select View Options > Show Child list to open a panel and display the hierarchy only for the selected object.

Filtering Information in the Hierarchy Browser

The hierarchy browser supports column filtering and a filter box to filter the information in the table. To display the filter expression box, do one of the following:

- Press the forward slash key (/), or
- Click the right mouse button over a column header, then click the Show/Hide Filter Control button as shown in the following figure.

Figure 52 Enable Filtering



Enter a filter expression in the filter expression box to display only the entries that match the expression. In the following example, the expression "Pin > 16" shows only table entries where the number of pins is greater than 16.

Figure 53 Filter Expression

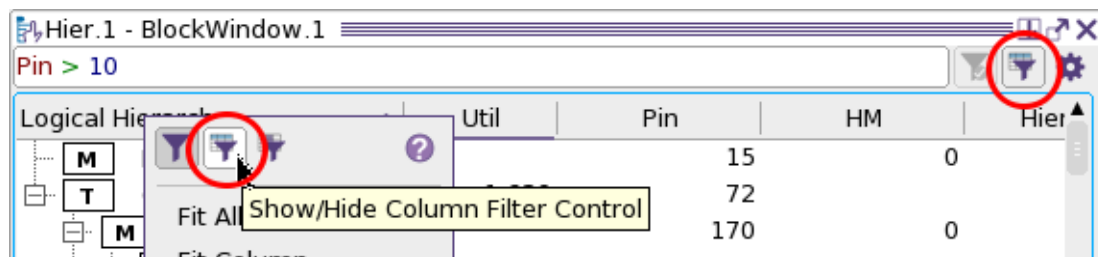
Hier.1 - BlockWindow.1

Pin > 16

Logical Hierarchy	Util	Pin
T ORCA	1.629	72
M I_ORCA_TOP		170
B I_BLENDER_1	0.849	141
B I_BLENDER_2	0.852	163
B I_BLENDER_3	0.854	353
B I_BLENDER_4	0.835	180

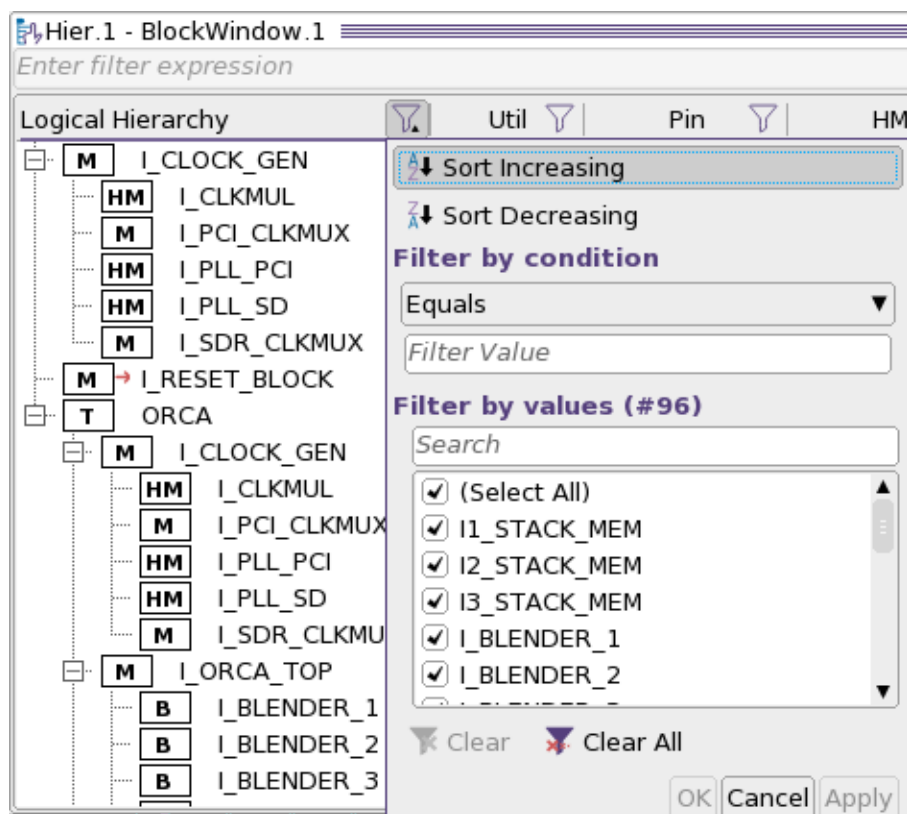
To enable column filters, click the Show/Hide Column filter control button to the right of the filter box. Alternatively, click the right mouse button over a column header and click the button in the context menu as shown in the following figure.

Figure 54 Enable Column Filters



To view the Column Filter menu, click the funnel icon. Use the options in the menu to sort the column in increasing or decreasing order, create a filter condition, or filter the column by selecting specific values in the column. The filter menu is shown in the following figure.

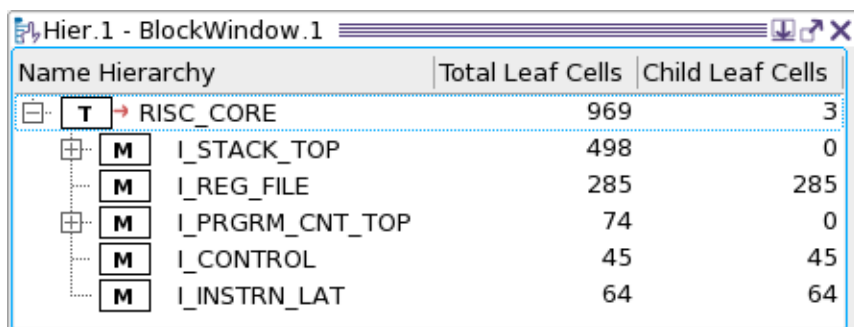
Figure 55 Column Filter Menu



Name Based Hierarchy Browser

For flat designs, the tool provides the Name Based Hierarchy Browser to display the implicit hierarchy in the design and allows you to perform actions on the hierarchies. To start the Name Based Hierarchy Browser, choose View > Name Based Hierarchy Browser. The tool displays the browser as shown in [Figure 56](#). If your design contains hierarchy, the selection is disabled and you cannot use this feature.

Figure 56 Name Based Hierarchy Browser



The screenshot shows a window titled 'Hier.1 - BlockWindow.1'. It contains a table with three columns: 'Name Hierarchy', 'Total Leaf Cells', and 'Child Leaf Cells'. The table lists a hierarchy of components. The first row is 'RISC_CORE' with 969 total leaf cells and 3 child leaf cells. It is expanded to show five sub-components: 'I_STACK_TOP' (498 total, 0 child), 'I_REG_FILE' (285 total, 285 child), 'I_PRGRM_CNT_TOP' (74 total, 0 child), 'I_CONTROL' (45 total, 45 child), and 'I_INSTRN_LAT' (64 total, 64 child). Each sub-component is preceded by a small icon consisting of a square with an 'M' inside.

Name Hierarchy	Total Leaf Cells	Child Leaf Cells
T → RISC_CORE	969	3
M I_STACK_TOP	498	0
M I_REG_FILE	285	285
M I_PRGRM_CNT_TOP	74	0
M I_CONTROL	45	45
M I_INSTRN_LAT	64	64

The tool extracts the hierarchy based on the setting of the `gui.name_based_hier_delimiter` application option. By default, the value is "/". You can convert your design to a flat design with the `ungroup_cells -all -flatten` command.

Similar to the Hierarchy Browser, you can perform different actions by using the context menu; right-click the mouse over a hierarchy to display the context menu:

- Select Set Colors On Top Hierarchies to assign a color to each of the top-level hierarchies
- Select Clear All Hierarchy Colors to remove all colors from the hierarchies
- Select Set Selected Hierarchy Color and choose a color to assign a color to the selected hierarchy
- Select Clear Selected Hierarchy Color to remove the color from the currently selected Hierarchy
- Select Show Child List to split the dialog box and display a list of cells in the selected hierarchy
- Select Expand All Hierarchy Trees and select the number of levels to show the subhierarchies for all top-level hierarchies

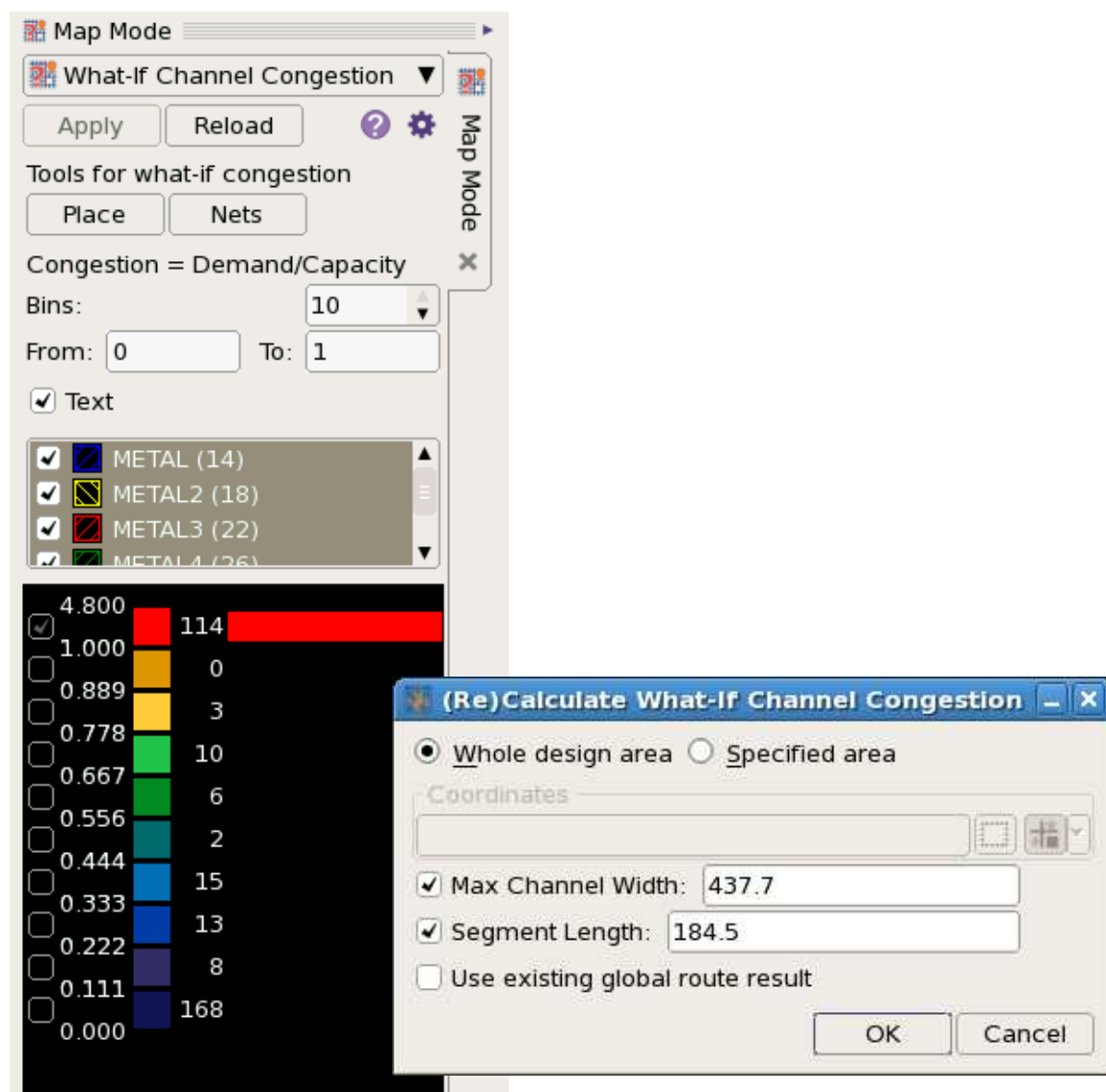
- Select **Expand Selected Hierarchy Trees** and select the number of levels to show the subhierarchies for the selected hierarchy
- Select **Copy Hierarchy Path** to copy the instance path for the current hierarchy to the clipboard

For information on filtering results in the table, see [Filtering Information in the Hierarchy Browser](#).

What-if Channel Congestion

The Fusion Compiler tool supports interactive, what-if analysis to identify congested global routing channels. To open the What-if Channel Congestion map mode in the layout window, select **What-if Channel Congestion** in the **Design Planning** section of the **Task Assistant**, or choose **View > Map > What-if Channel Congestion**. The What-if Channel Congestion map mode appears on the right of the layout window as shown in [Figure 57](#).

Figure 57 What-if Channel Congestion Map Mode and Reload Dialog Box



Performing Global Routing for What-if Channel Congestion

To begin what-if congestion analysis,

1. Click the Reload button to open the Calculate What-If Channel Congestion Map Data dialog box.
2. Select Whole design area or Specified area to choose the area of the design to evaluate.

3. (Optional) Change the value in the Max Channel Width box to set the size of the largest channel to display; see [Filtering by Minimum Channel Dimension](#) for more information about filtering by channel width.
4. (Optional) Change the Segment Length; see [Displaying Channel Segments](#) for more information about segment length.
5. (Optional) Select Use existing global route result to use the existing route information from the `route_global -congestion_map_only true` command.

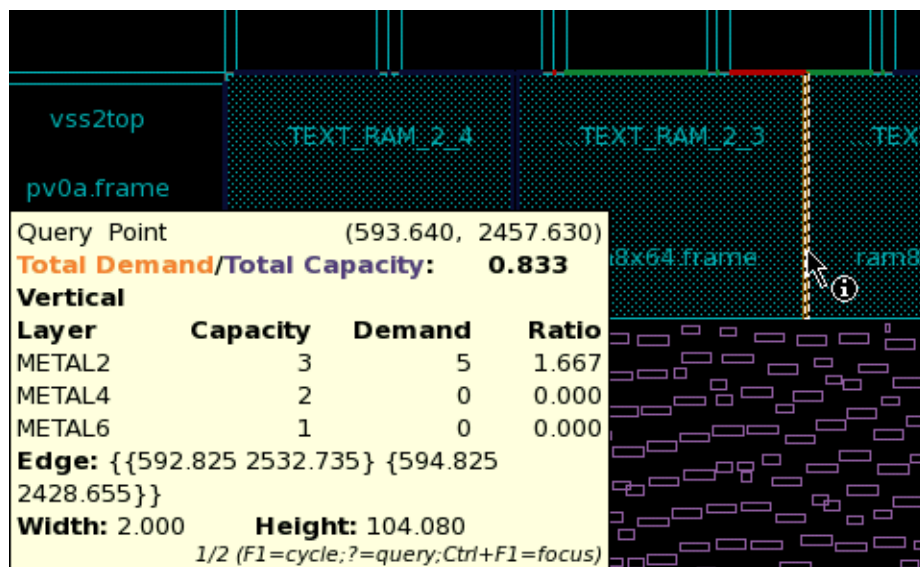
You can use the `remove_routes -global_route` command to remove global routes from your design.

6. Click OK to create the congestion map.

Viewing Congestion Information

After global routing completes, the tool highlights each congested channel in the design. To display additional information about the channel, move the pointer over the channel and view the infotip. The infotip displays the demand and capacity for the channel and for each layer of metal in the preferred routing direction within the channel. [Figure 58](#) shows the infotip displayed for a congested channel.

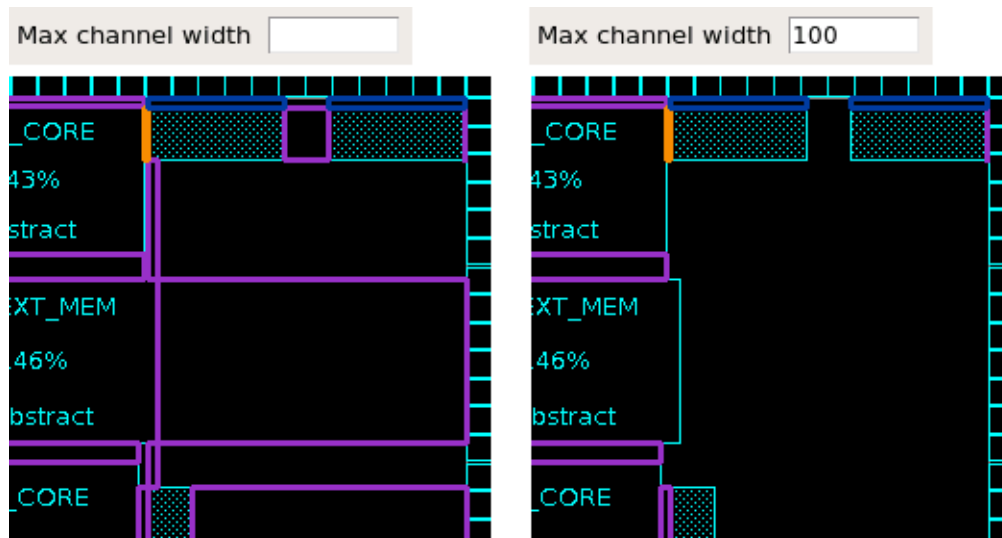
Figure 58 What-if Channel Congestion Infotip



Filtering by Minimum Channel Dimension

To display only narrow channels in your design, click the Reload button, enter a channel width value in the Max channel width box, and click OK. The tool updates the display and hides channels where the smallest dimension (height or width) is greater than the specified maximum channel width value as shown in [Figure 59](#).

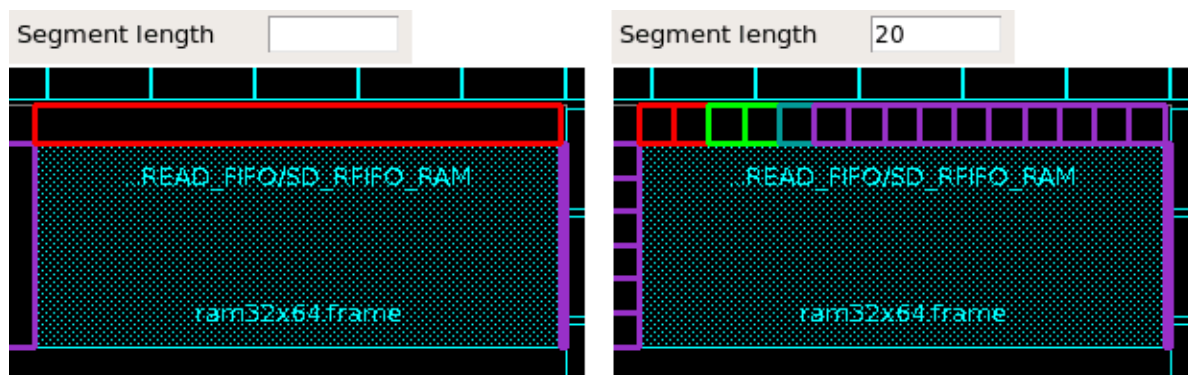
Figure 59 Filtering by Minimum Channel Dimension



Displaying Channel Segments

To perform a more detailed analysis and partition the channel into smaller segments, enter a value in the Segment length box. The tool subdivides each channel and creates segments of the specified length as shown in [Figure 60](#). Move the pointer along the channel to view detailed congestion information for each channel segment in the infotip.

Figure 60 Channel with Segments



Note that the segment dimension might not exactly match the value you specified. In some cases, the tool grows or shrinks the segment to create segments of a uniform size.

Specifying the Region for Analysis

By default, the tool generates a what-if congestion map for the entire design. You can speed up the congestion analysis by specifying a region in which to limit the analysis. Click

the  button to the right of the Region box, draw a rectangle in the layout window, and click Apply to restrict the analysis to the drawn area. Alternatively, specify the coordinates for a bounding box in the Region box and click OK to analyze only the area you specified.

Scaling the Congestion Color Mapping

The tool colors each channel based on the ratio of required routes to the number of available routes in the channel. By default, any channel where the ratio exceeds 1.0 (the number of required routes exceeds the number of available routes) is colored red, and a channel where the ratio is less than 0.2 is colored purple.

To change the way the channels are colored, enter values in the From: and To: boxes and click Apply. The tool redisplay the congestion map and colors the channels based on the values.

Correcting Congestion Problems

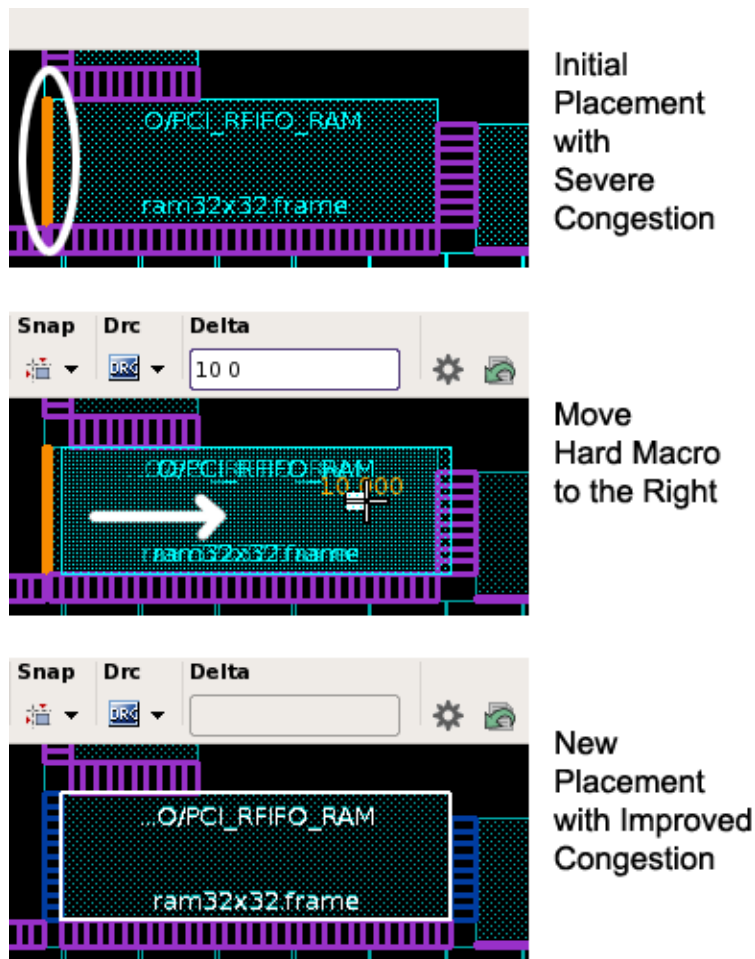
To reduce routing congestion interactively with What-if Channel Congestion analysis, do the following:

1. Select the hard macro which abuts the congested area
2. Press M or select the Move tool from the toolbar
3. Move the hard macro with the mouse to create a wider channel in the congested area

Alternatively, press the Tab key and enter the x- and y-offset in the Delta box in the Move toolbar and press Enter. The tool moves the hard macro by the specified distance. After you move the hard macro to the new location, the tool redisplay the congestion map based on the updated placement.

Repeat these steps until you are satisfied with the congestion map and placement. The layout before and after repair is shown in [Figure 61](#).

Figure 61 What-if Congestion Repair



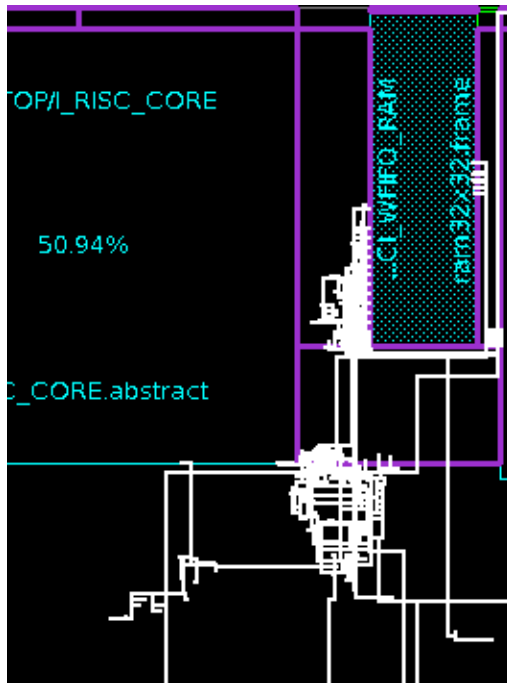
Updating the Placement

After adjusting the channel widths, update the cell placement by clicking the Place button. The tool runs the `create_placement -floorplan -effort medium` command to update the placement while keeping cells near their original locations.

Showing Nets in a Channel

To show the net connections that occupy a specific channel, click the Nets button and click in a channel. The tool displays the nets that pass through the channel and their path outside the channel. [Figure 62](#) shows the layout before and after displaying the nets that pass through the channel. To clear the display, click anywhere outside the channel.

Figure 62 *Displaying Nets in the Channel*



Removing the Congestion Map

To remove the What-if Channel Congestion map from the layout, close the What-If Channel Congestion map mode panel.

Hierarchy Exploration

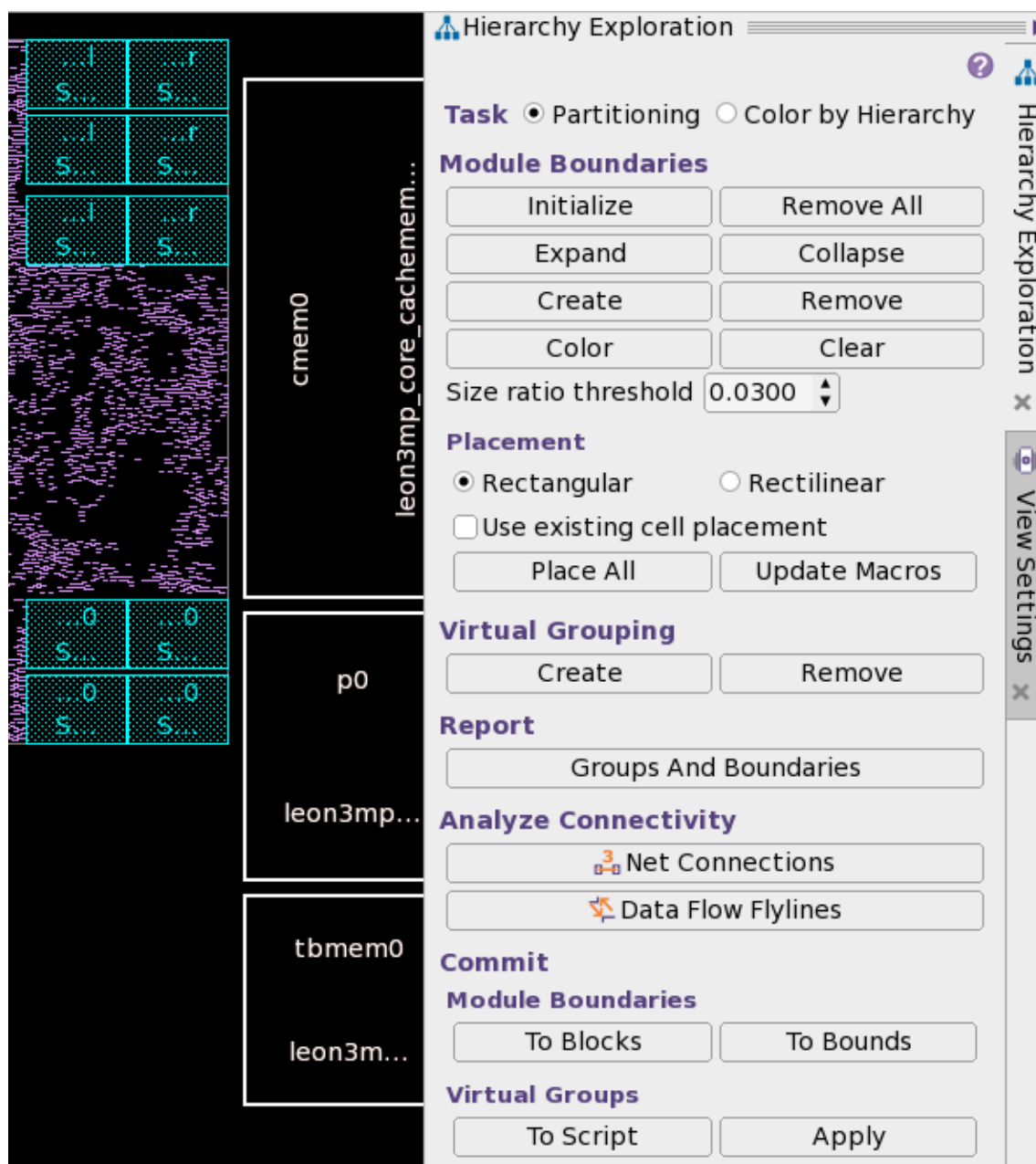
In the Fusion Compiler tool, module boundaries represent uncommitted blocks in the logical hierarchy. Using module boundaries, you can quickly assess the approximate block size that is required for a given utilization for a top-level block and arrange module boundaries within the floorplan for rough estimation of block placement.

To open the Hierarchy Exploration panel in the layout window, select Hierarchy Exploration in the Design Planning section of the Task Assistant or choose View > Assistants > Hierarchy Exploration in the GUI. The Hierarchy Exploration panel appears on the right of the layout window.

Creating Module Boundaries

To create module boundaries, click the Initialize button in the Hierarchy Exploration panel. The tool inserts module boundaries for the top-level modules as shown in [Figure 63](#).

Figure 63 Initialized Module Boundaries



You can perform the same operation with the `explore_logic_hierarchy -create_module_boundary` command.

You can create module boundaries for selected modules using the Hierarchy Browser in the GUI. To create module boundaries with the Hierarchy Browser,

1. Choose View > Hierarchy Browser
2. Select a logical hierarchy block
3. Select Create Module Boundary from the context menu, or click the Create button in the Module Boundaries section of the Hierarchy Exploration panel

Expanding, Collapsing, and Removing Module Boundaries

To expand a module boundary into child modules, select the module boundary and click Expand. To collapse a module boundary into its parent module, select the module boundary and click Collapse. To remove one or more module boundaries, select the module boundaries and click Remove. To remove all module boundaries, click Remove All.

You can also perform these operations by using the context menu.

1. Select one or more module boundaries
2. Right-click in one of the selected module boundaries and choose the operation (Expand, Collapse, or Remove Module Boundary) from the context menu

Coloring Module Boundaries

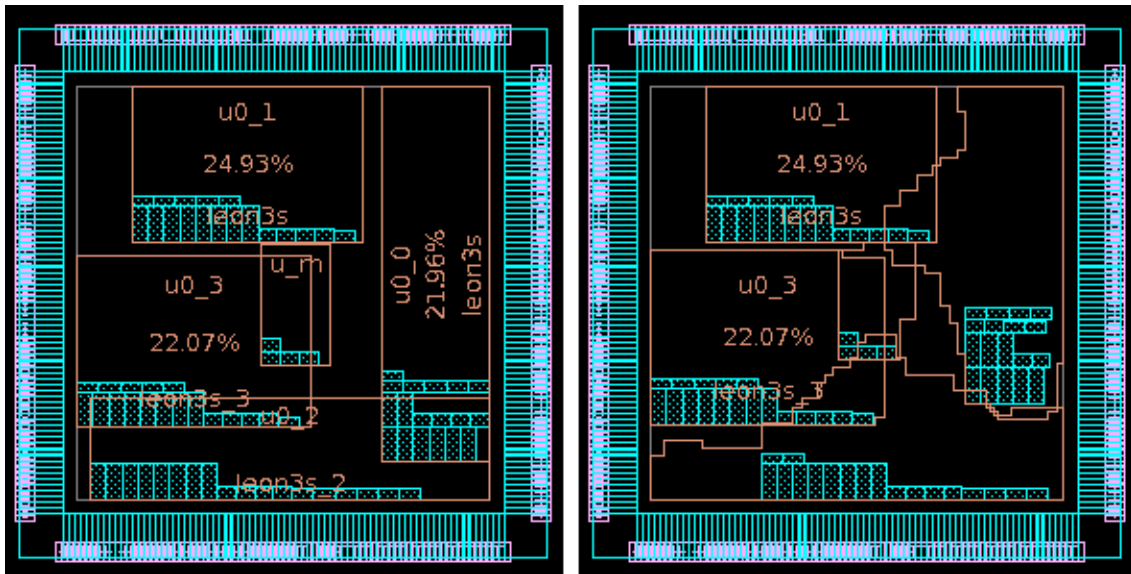
To apply a unique color to each module boundary, click Color. Click Color again to apply a different color scheme. To remove any coloring from the module boundaries, click Clear.

Placing Module Boundaries

To place module boundaries into the core area,

1. Choose the placement type, rectangular or rectilinear. See [Figure 64](#) for an example of both placement types
2. Select Use existing cell placement to place module boundaries based on the existing placement
3. Click Place All to place the module boundaries into the core area

Figure 64 Rectangular and Rectilinear Module Boundary Placements

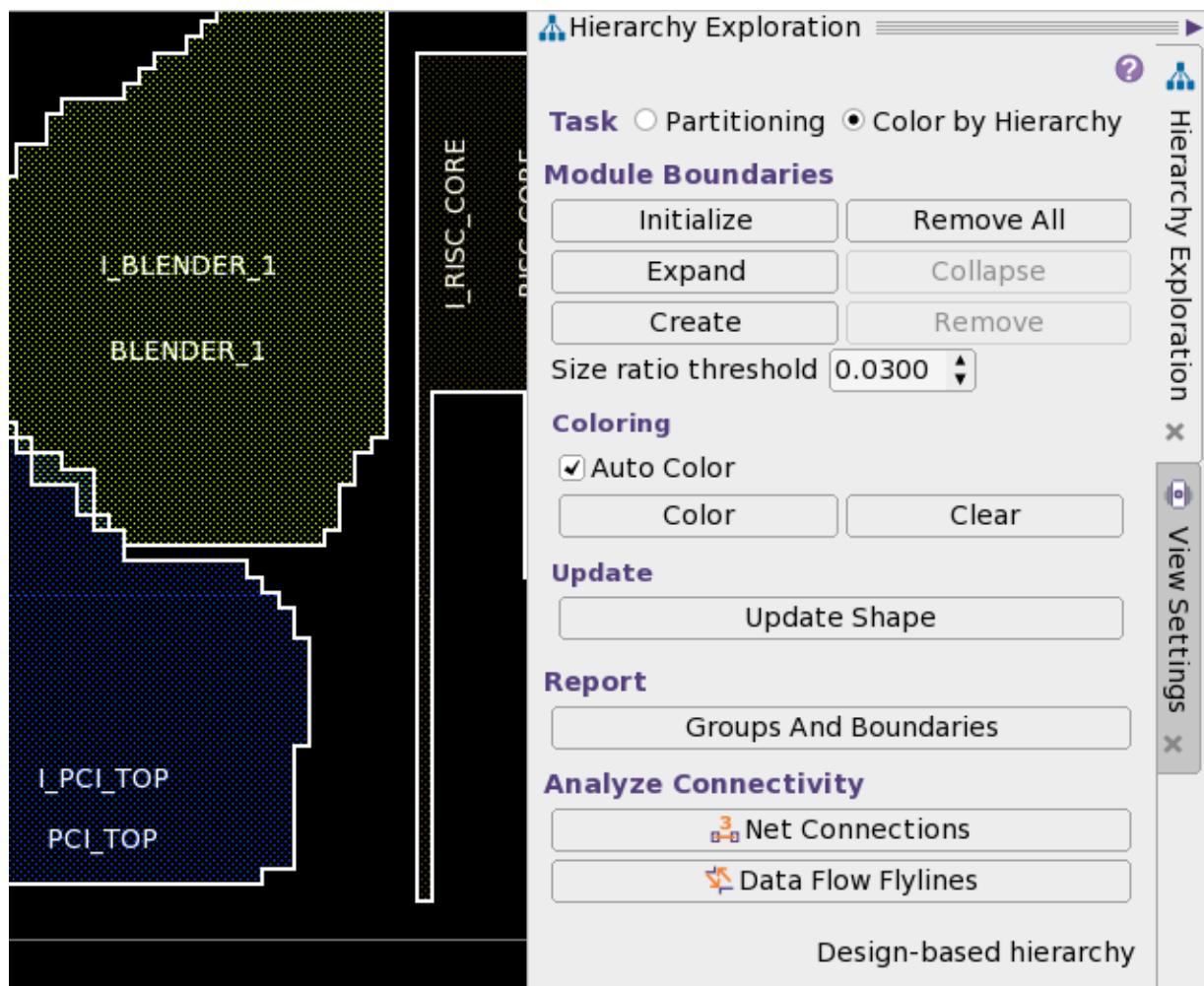


To move inside the module boundary any hard macros that belong to a module bound, click Update Macros.

Hierarchy Exploration and Module Color

The Hierarchy Exploration panel provides tools to add color to module boundaries and their corresponding standard cells and macros within the design boundary. The tool also draws a boundary around different hierarchies within the design. You can access these functions by choosing “Color by Hierarchy” in the Task section at the top of the panel.

Figure 65 Color by Hierarchy in Hierarchy Exploration



Viewing Hierarchies in the Layout

To highlight different hierarchies in the layout,

1. Choose View > Assistants > Hierarchy Exploration from the menu.
2. Select “Color by Hierarchy” in the Task section of the panel.
3. Create the module boundaries by clicking Initialize in the Module Boundaries section of the panel.

The tool highlights and selects all top-level hierarchies.

4. (Optional) Change the module color by clicking Color and remove all module colors by clicking Clear.
5. (Optional) Change the hierarchies that are highlighted by opening the Hierarchy Browser and selecting one or more hierarchies.

What-if Partitioning

What-if partitioning helps you to examine the relation between different module boundaries and organize the hierarchy of module boundaries by creating virtual groups. During what-if partitioning, you

- View the connectivity between different module boundaries by using the Net Connections or Data Flow Flylines tools
- Create virtual groups that contain connected module boundaries
- Investigate different partitioning strategies by placing, combining, or removing virtual groups
- Write out the module boundary groupings for resynthesis

To display the connectivity between module boundaries,

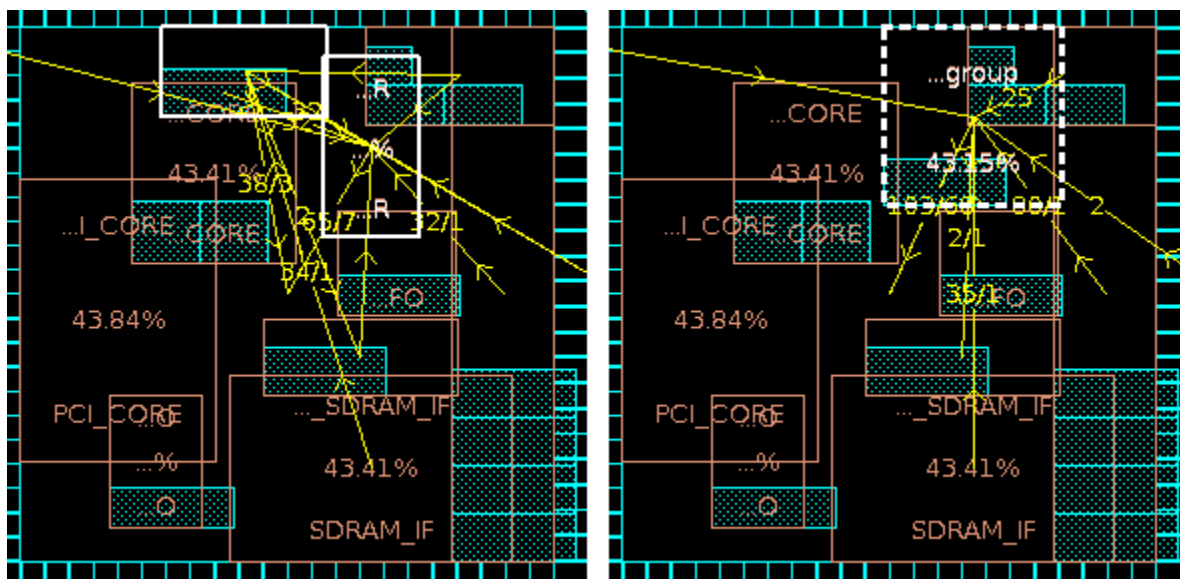
1. Click the Net Connections button in the Hierarchy Exploration panel.
2. Select Selected to Selected in the Connectivity Display section of the Net Connections panel.
3. Click on a module boundary, then click on one or more additional boundaries to view the connectivity between them.

To create a virtual group,

1. Select only the module boundaries or module groups that you want to combine into a new module group.
2. Click the Create button in the Virtual Grouping section of the Hierarchy Exploration panel.
3. (Optional) Ungroup a virtual group by selecting the group and clicking the Remove button.

In the left image in [Figure 66](#), two module boundaries are selected, and net connections for the selected modules are displayed. The two modules are good candidates for grouping because they share common net connections. In the right image, the modules are grouped after clicking the Create button. You can move the pointer over the module group to display information about the group in the infotip.

Figure 66 Creating a Virtual Group of Two Modules



If your selection includes an existing virtual group and other module boundaries, the module boundaries are added to the existing virtual group.

To report the current module boundaries and groups in the layout after creating a virtual group of module boundaries, click the Groups and Boundaries button in the Report section. The tool opens the Groups and boundaries panel and displays a list of virtual groups and module boundaries in the layout. The tool also writes the information to the `module_boundary.results` file as follows:

```
fc_shell> sh cat module_boundary.results
virtual group:  RISC_CORE_group
                I_ORCA_TOP/I_RISC_CORE
                I_ORCA_TOP/I_PCI_TOP
                I_ORCA_TOP/I_BLENDER_4
module boundary: I_CLOCK_GEN
module boundary: I_ORCA_TOP/I_SDRAM_TOP
...
***** End Of Report *****
```

To create a script which contains Tcl commands to create a new level of hierarchy using the virtual group, click the To Script button in the Virtual Groups section. The tool opens the Script Editor and writes `group_cells` and `explore_logic_hierarchy` commands you can use to re-create a new level of hierarchy from the virtual group. An example script is as follows:

```
### Virtual Grouping Commands For Group: RISC_CORE_group
group_cells { I_ORCA_TOP/I_RISC_CORE I_ORCA_TOP/I_PCI_TOP \
              I_ORCA_TOP/I_BLENDER_4 } -cell_name RISC_CORE_group
if { [get_bounds -quiet RISC_CORE_group] ne {} } {
```

```
explore_logic_hierarchy -virtual_ungroup RISC_CORE_group
explore_logic_hierarchy -create_module_boundary \
    -cell [get_cells -quiet -hierarchical RISC_CORE_group]
}
### End Virtual Grouping Commands For Group: RISC_CORE_group
```

To commit selected module boundaries to blocks, click To Blocks in the Commit section of the Hierarchy Exploration panel.

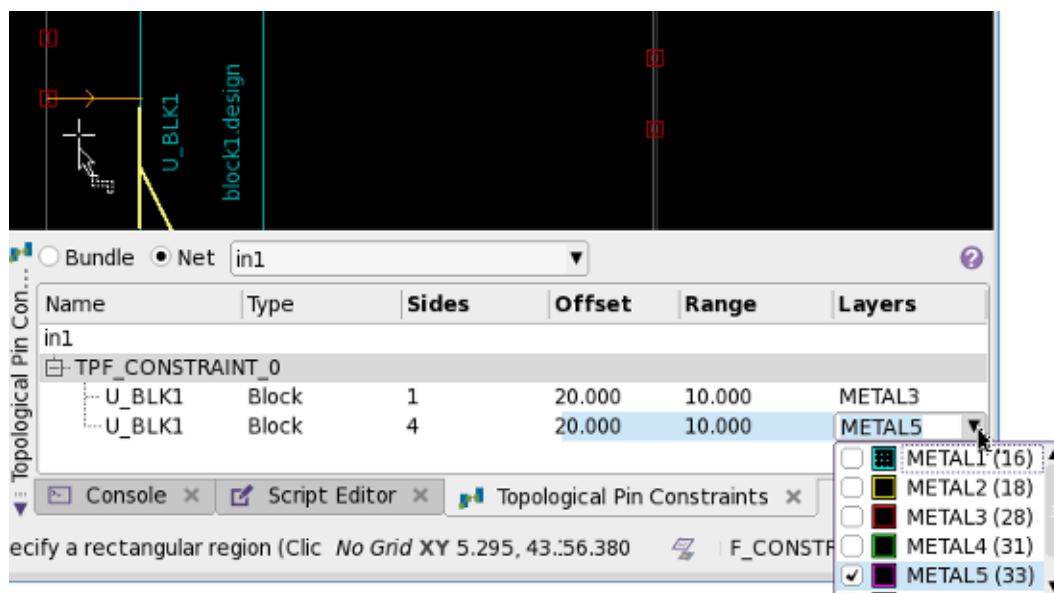
To commit selected module boundaries to bounds, click To Bounds in the Commit section of the Hierarchy Exploration panel.

Topological Pin Constraints

The Topological Pin Constraints tool enables you to create pin constraints in the GUI by specifying a net and drawing a rectangular region in which the feedthrough must be routed. To start Topological Pin Constraints tool, choose Create > Topological Pin

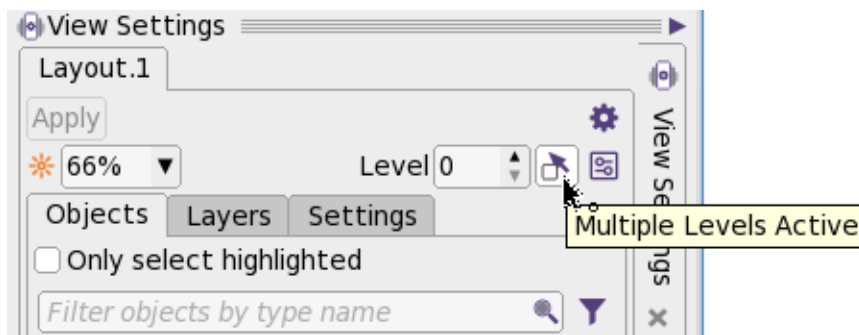
Constraint from the menu or click the  button. To view or edit information about the pin constraint, click the button circled in [Figure 67](#) to open the Topological Pin Constraints dialog box.

Figure 67 Topological Pin Constraints



To create topological pin constraints for a design with multiple levels of hierarchy, click the multiple levels active box in the View Settings panel as shown in the following figure.

Figure 68 Multiple Levels Active



5

Using Flyline Analysis Tools

The Fusion Compiler GUI provides tools to view net connections, feedthrough paths, timing paths, and other path information as flylines in the layout.

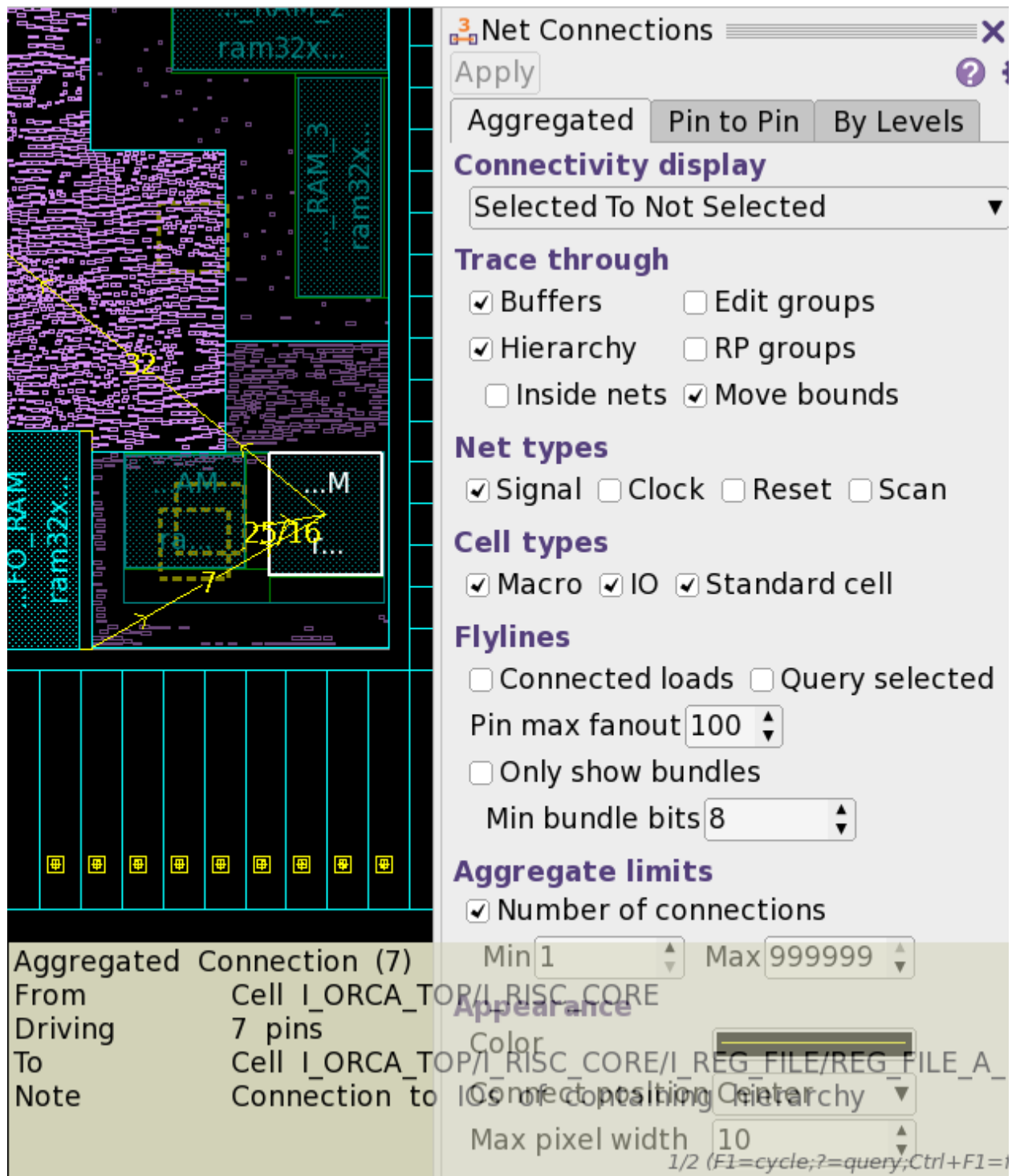
The following topics describe design planning tasks supported by the GUI:

- [Net Connections Flylines](#)
- [Path Analyzer Flylines](#)
- [Data Flow Flylines](#)
- [Register Tracing Flylines](#)
- [Clock Trunk Planning Flylines](#)

Net Connections Flylines

The Fusion Compiler tool provides the Net Connections Flylines tool to view net connections between hierarchies, macros, standard cells, and ports in your design. Use the tool to view the number of connections between these objects and to trace individual nets. To open the Net Connections Flylines panel, choose View > Flylines > Net Connections or click the Net Connections Flylines button. The Net Connections Flylines panel is shown in [Figure 69](#).

Figure 69 Net Connections Flylines Panel



Flyline Display Modes

The Net Connections panel supports the following modes for flyline display:

- Aggregated (default)
- Pin to Pin
- By Levels

These modes are described below. Each mode is accessed by clicking the appropriate tab (Aggregated, Pin to Pin, or By Levels) at the top of the panel, then clicking Apply.

Aggregated Flylines

In Aggregated flylines mode, the tool groups the driver and load pins of the flylines based on their parent hierarchical cell, move bound, edit group, RP group, or voltage area. Aggregated flylines are displayed as a single line together with a count of load and/or driver pins. The count is displayed at the center of each unique connected parent object pair. The types of parent objects to display are determined by the “Trace through” options you select. If the trace-through object is not selected in the panel, then it is included as a possible parent object type. If the original selected object is an edit group, an RP group, a move bound, or a voltage area, then these object types are also candidate parent object types.

Pin to Pin Flylines

In Pin to Pin mode, the tool displays flylines for individual pins. The flylines are drawn from each start pin to all traced load pins (if the start pin is a driver pin). If the start pin is a load pin, the tool draws flylines to all traced driver pins.

By Levels

In By Levels mode, the tool displays flylines based on the tracing depth you specify with the Driver and Load options. Except for buffer cells, the tool counts each standard cell as a level. If the Ignore buffers option is selected, buffers are not counted as levels. This mode also supports different flyline colors for the selected driver and load pins.

You can drill down and view details for individual flylines by clicking an aggregated flyline. The tool opens the Net Connections Explorer dialog box and displays the list of nets represented by the aggregated flyline. For more information about the Net Connections Explorer dialog box, see [Net Connections Explorer](#). To view individual flylines, you can also deselect the Aggregate by block option. When this option is deselected, the tool replaces the aggregated flyline with individual flylines.

Note:

The tool displays flylines based on which pin you select in the path and the type of object you select. The tool displays different flylines depending on whether the pin you select is on a hierarchy boundary.

Setting Flyline Display Options

Use the Connectivity Display options to change the connections that are displayed based on the objects you selected:

- Selected to All: Display flylines between any selected objects and any other object
- Selected to Selected: Display flylines only between the selected objects
- Selected to Not Selected: If one object is selected, display flylines between the selected object and any other object; if two or more objects are selected, display flylines from each selected objects that does not connect to another selected object

Use the Trace through options to control how a net is traced:

- Buffers: Select this option to continue tracing through buffer cells until a nonbuffer cell is reached. When deselected, the tool stops tracing when a buffer cell or any other cell is reached.
- Hierarchy: Select this option to continue tracing through a hierarchical pin. When deselected, the tool stop tracing when a hierarchical pin is reached.
- Edit groups: Select this option to stop tracing at the edit group boundary. When deselected, the tool ignores edit groups when tracing.
- RP groups: Select this option to stop tracing at the RP group boundary. When deselected, the tool ignores the RP group when tracing.
- Inside nets: Select this option to trace nets within a block. By default, nets outside the block are traced. This option can be combined with the "Through points" option in the Pin to Pin tab to trace feedthrough connections when one of the block pins is selected.
- Move bounds: When this option is disabled, tracing stops at nets that connect from pins at the move bound boundary. The boundary occurs when cells inside the move bound connect to cells outside the move bound. If you select a cell inside the move bound, the tool does not trace flylines to cells outside the move bound. If you select a cell outside the move bound, the tool does not trace flylines to cells inside the move bound. By default, this option is selected and the tool traces through move bounds.

Use the Net Types options to specify which type of nets to trace: Signal, Clock, Reset, Scan, Tie High, and Tie Low.

Use the Cell Types options to specify one or more types of cells to trace: macros, I/Os, or standard cells.

Use the Flylines options to further control which flylines are displayed:

- **Connected loads:** Select this option to display flylines between two loads. When deselected, the tool displays flylines between only drivers and loads.
- **Query selected:** Select this option to update the Query panel when you click an object. The panel displays information about the source, destination, and number of connections for the currently displayed flylines.
- **Pin max fanout:** Hide flylines with a fanout greater than the specified number.
- **Only show bundles:** Select this option to display flylines only for bundle signals.
- **Min bundle pins:** Hide flylines for bundles with less than the specified number of pins.

Use the Aggregated Connections options to control the display of aggregated flylines based on connection count:

- **Number of connections:** Select this option to use the Min and Max settings to filter aggregated flylines; aggregated flylines are hidden if the total connection count is outside the specified range
- **Min:** Hide an aggregated flyline if Number of connections is selected and the total number of connections is less than the specified value
- **Max:** Hide an aggregated flyline if Number of connections is selected and the total number of connections is greater than the specified value

Use the Include options to control the type of cells that are traced:

- **Macros:** Select this option to include macro cells in the trace.
- **I/Os:** Select this option to include I/O cells in the trace.
- **Standard cells:** Select this option to include standard cells in the trace.

Use the Appearance options to control the width and position of flylines:

- **Connect Position:** Specifies where to draw the startpoint or endpoint for the flyline (center, inside, lower left, lower right, upper left, or upper right)
- **Max pixel width:** Limits the drawn width of the flyline to the specified number of pixels
- **Nets per pixel:** Specifies the width of the flyline based on the number of net connections; for example, a setting of 100 nets/pixel and a net connection containing 300 connections causes the tool to draw a 3 pixel line ($300 \text{ nets} / 100 \text{ nets per pixel} = 3 \text{ pixels}$)
- **Color:** Specifies the color of the flyline

Displaying Connections to Route Corridors

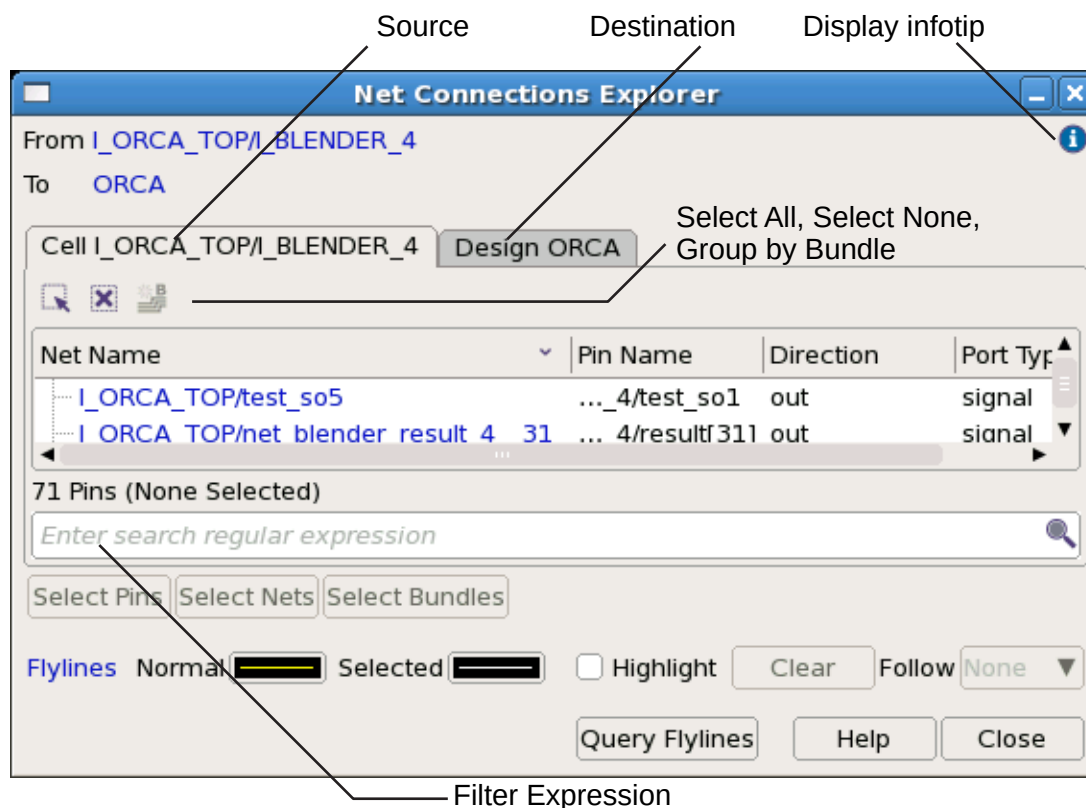
When you select a route corridor, the tool gets the pins of the nets for the route corridor and draws a flyline from the pin center to the nearest edge of the route corridor shape. If the route corridor contains intermediate cells that connect to the route, the tool draws flylines to these cells by using a different color.

Net Connections Explorer

Use the Net Connections Explorer to query individual nets in an aggregated flyline, or display individual net segments in a single flyline.

Open the dialog box by clicking any flyline. The dialog box shows the source block or cell, the destination block or cell, and a list of nets that are represented by the flyline. You can click the name following the From: or To: text to select the block, cell, or port. Move the pointer over the Info icon at the upper right of the dialog box to view summary information about the net connections displayed in the dialog box.

Figure 70 Net Connections Explorer Aggregated Flyline Mode



The dialog box displays two tabs: one tab containing pins and nets for the source object, and one tab for pins and nets of the destination object. Click the appropriate tab to show the nets and pins associated with the object.

To filter the list of nets, enter a filter expression in the box. The expression can be a simple matching expression or a Tcl regular expression. By default, the filter expression is applied to net names. To apply the expression to pin names, use the syntax "Pin Name:string", where "string" is the search expression and "Pin Name" is the header for the column.

Click the Select Pins or Select Nets button to select the pins or nets in the layout for highlighted net. Click Select Bundles to select the net bundles for the highlighted net.

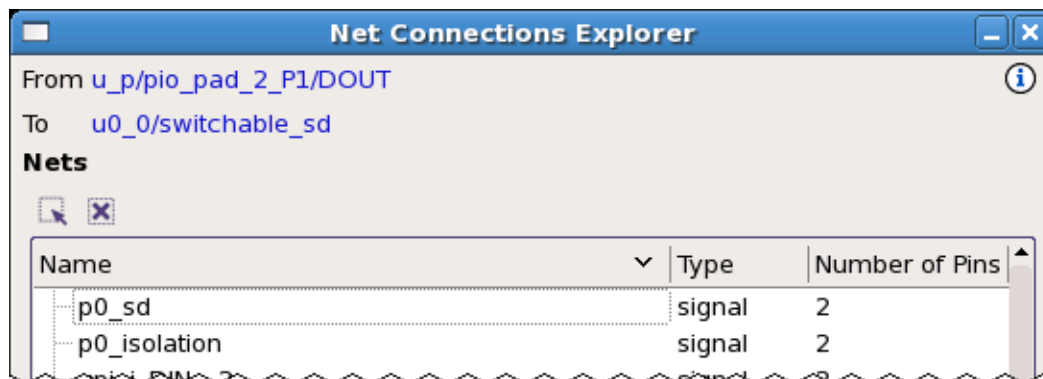
To change the color of either unselected and selected flylines, click the flyline box to the right of Normal or to the right of Selected. To continue displaying a previously selected flyline in the layout, select Highlight. To clear a previously displayed flyline, click Clear.

Use the Follow setting to control how the tool centers the selected flyline in the view. To retain the same zoom factor and pan the layout to center the selected flyline in the view, select Pan in the Follow box. To pan the layout and zoom in to fit the flyline to the view, select Zoom in the Follow box.

When tracing a single flyline, the display changes as shown in [Figure 71](#).

To display the current list of flyline connections in the Query panel, click Query Flylines. To view help information for the Net Connections Explorer dialog box, click Help.

Figure 71 Net Connections Explorer Single Flyline Mode

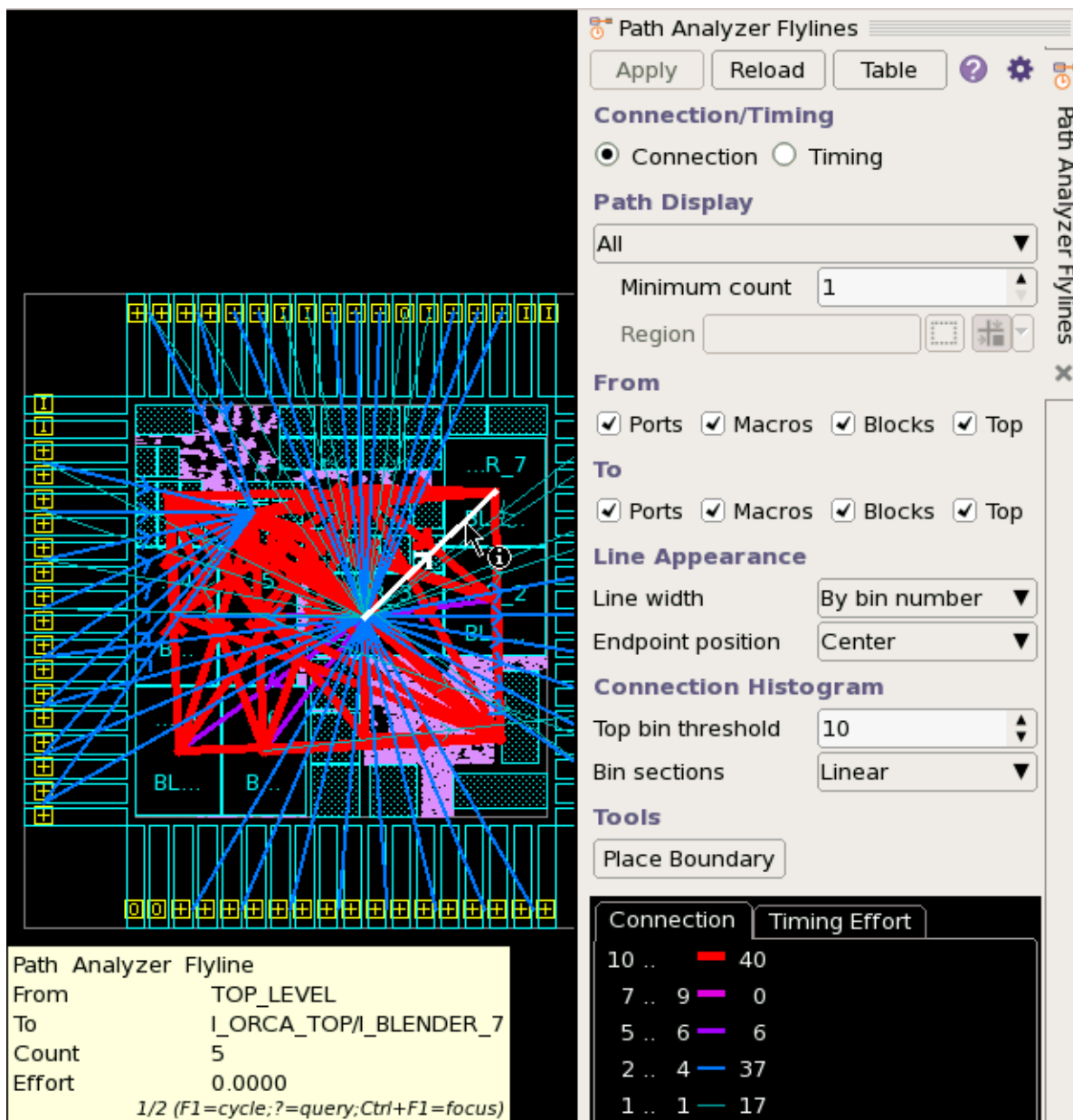


Path Analyzer Flylines

The Fusion Compiler tool supports the Path Analyzer Flylines tool to view timing path connections between top-level signals, macros, ports, and hierarchies in your design. Use the Path Analyzer Flylines tool to identify potential layout-related timing issues and perform an interactive analysis to examine the effect of placement on timing. The

tool provides two analysis modes, connection mode and timing mode, to prioritize the information displayed by the flylines. To open the Path Analyzer Flylines panel, choose View > Flylines > Path Analyzer Flylines or click the Path Analyzer Flylines button. The Path Analyzer Flylines panel is shown in [Figure 72](#).

Figure 72 Path Analyzer Flylines Panel



The tool provides two analysis modes, connection mode and timing mode, to prioritize the information displayed by the flylines. Click the Reload button at the top of the panel to reload the flylines.

If your layout contains module boundaries, click the Place Boundary button in the Tools section to position any currently displayed module boundaries on the block layout display. The tool centers each module boundary over the standard cells for the module. The tool represents connections to standard cells by drawing a flyline to the module boundary, while macros and hierarchy blocks contained inside of the module are not considered. Connections to macros are represented with flylines directly to the macros, not to the module boundary. If a module contains no standard cells, the module boundary is placed at the coordinate (0,0) and no flylines are drawn to the module boundary.

Module boundaries provide information about the physical location of cells in a design hierarchy. You can use the Hierarchy Browser to create module boundaries and use the boundaries with the Path Analyzer Flylines tool. When a flyline connects to a cell within a module boundary, the Path Analyzer Flylines tool draws a flyline to the module boundary. If a module boundary is not present, the Path Analyzer Flylines tool draws a flyline to the center of the block. To examine the hierarchy, choose View > Hierarchy Browser to start the Hierarchy Browser tool.

Viewing Timing Paths by Number of Connections

After clicking the “Reload” button, the Connection mode in the “Connection/Timing” section is selected by default. The Connection mode displays timing path connection flylines between ports, modules and macros in the design. The tool partitions and groups the flylines by number of connections; flylines with the greatest number of connections are shown in red. If you see only light blue flylines, you can adjust the binning by changing the value for Threshold for Top Bin in the Line Appearance section.

If you see red or bright purple flylines that extend over long distances, you might need to change your floorplan to shorten the paths. A large number of long paths can cause routing congestion and contribute to total negative slack problems when timing your block.

Move the mouse pointer over a flyline to view more information. The tool displays an infotip as shown in [Figure 73](#); in this example, all the path for the highlighted flyline have same direction. The infotip shows the path endpoints, the count of the flylines between the two objects, and the relative timing effort required to optimize the timing for those paths. If the effort is greater than 1.0, you might encounter difficulty when closing timing for the path.

Figure 73 Single Direction Flyline with Infotip

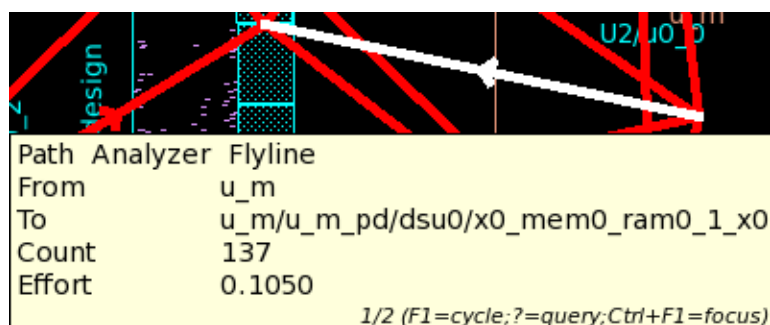
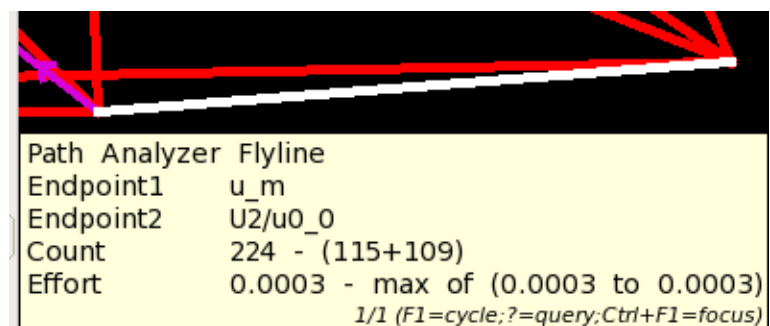


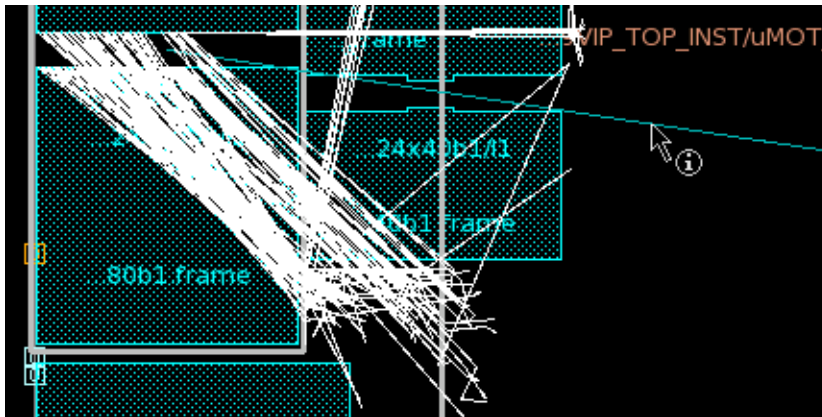
Figure 74 shows a flyline with signals that travel in both directions. The direction arrow on the flyline is not displayed, the infotip displays the signal count as "Total - (endpoint1 to 2 + endpoint2 to 1)" and the Effort as "Total maximum - max of (endpoint1 to 2 max to endpoint2 to 1 max)".

Figure 74 Flyline for Mixed Directions



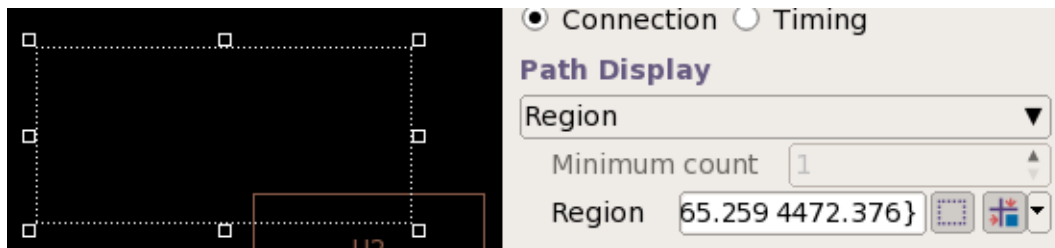
Click the flyline to expand the flyline and view the individual connections between the objects. The tool displays a flyline for each individual timing path as shown in Figure 75. To remove the detail flylines, select another object or click outside the layout.

Figure 75 Detailed Connection Flylines



To view only flylines which pass through a specific region of the design, select Region in the Path Display box and click the dotted square button as shown in [Figure 76](#). Use the mouse to draw a rectangle on the layout to specify the region for analysis. The tool removes flylines which do not pass through the region.

Figure 76 Creating an Area Rectangle



See [Filtering by Design Object and Other Constraints](#) for more information about filtering flylines.

Viewing Timing Paths by Criticality

Select Timing in the "Connection/Timing" section to display timing path connection flylines for the most timing critical paths between objects in the design. In timing mode, the tool bins the flylines by the criticality of the connection and displays the most critical flylines in red. Move your pointer over the flylines and click the flylines as described in [Viewing Timing Paths by Number of Connections](#) to view more information about the flylines.

Use the Distance Sensitivity box to control the weighting applied to flyline length when calculating optimization effort. As you change the Distance Sensitivity, the tool recalculates the optimization effort for the flyline. For larger values of Distance Sensitivity, the tool puts more weight on longer flylines. You can view the updated effort value in the infotip.

In general, technologies with higher wire resistance might require a higher Distance Sensitivity setting. If you see red flylines only if the sensitivity is very high (7 or more), then your design probably does not have a timing problem.

Changing the Flyline Display

Click the Display tab to view and modify the settings for line thickness and binning for the Path Analyzer Flylines display. Select Apply line widths to change the line widths of the flylines based on count of connections between objects. Select Log Scale to change the binning to use a logarithmic scale rather than a linear scale. Enter a new value in the Threshold for Top Bin box to set the maximum value the highest bin.

In connection mode, the Display tab uses color to represent the relative number of connections for each flyline. If the Apply line widths option is selected, the tool displays flylines by using different widths, with the highest bin shown by using the widest line.

In timing mode, color represents the criticality of the timing flyline. If the Apply line widths option is selected, the tool displays the relative number of connections by using different widths, where the widest line represents the highest number of connections.

Filtering by Design Object and Other Constraints

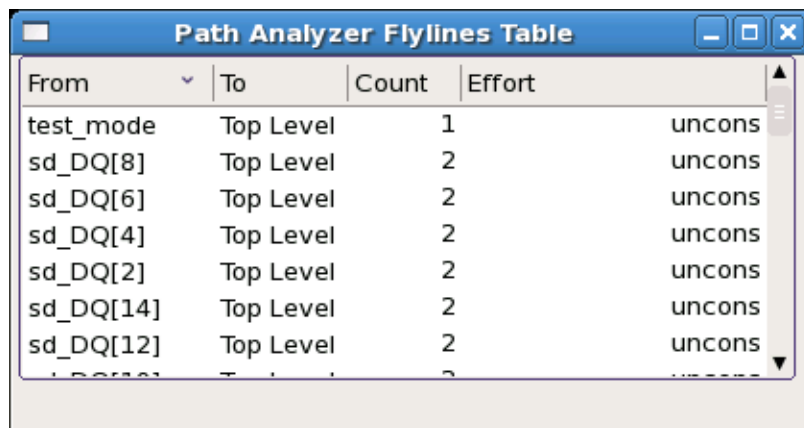
Select object types for analysis by selecting them in the "From" and "To" settings section. Select an entry in the Path Display box to specify the source and destination objects that are used when creating flylines:

- All: Display flylines without any selection criteria. The From and To settings are enabled.
- Region: Display flylines through a user-specified rectangle. The From and To settings are enabled.
- Selected From/To All: Display flylines that either begin or end with the selected objects and any other object. The From and To settings are disabled.
- Selected to Selected: Display flylines that begin or end with the selected objects. The From and To settings are disabled.
- All to Selected: Display flylines that begin at any object and end at the selected objects. The From and To settings are disabled.
- Selected to All: Display flylines that begin at the selected objects and end at any other object. The From and To settings are disabled.

Checking Timing Paths Info in Table

Click the Table button at the top of the palette to display the flyline endpoints, connection count, and timing effort in a table. [Figure 77](#) shows an example table.

Figure 77 Path Analyzer Flyline Table



From	To	Count	Effort
test_mode	Top Level	1	uncons
sd_DQ[8]	Top Level	2	uncons
sd_DQ[6]	Top Level	2	uncons
sd_DQ[4]	Top Level	2	uncons
sd_DQ[2]	Top Level	2	uncons
sd_DQ[14]	Top Level	2	uncons
sd_DQ[12]	Top Level	2	uncons

Path Analyzer Flylines Best Practices

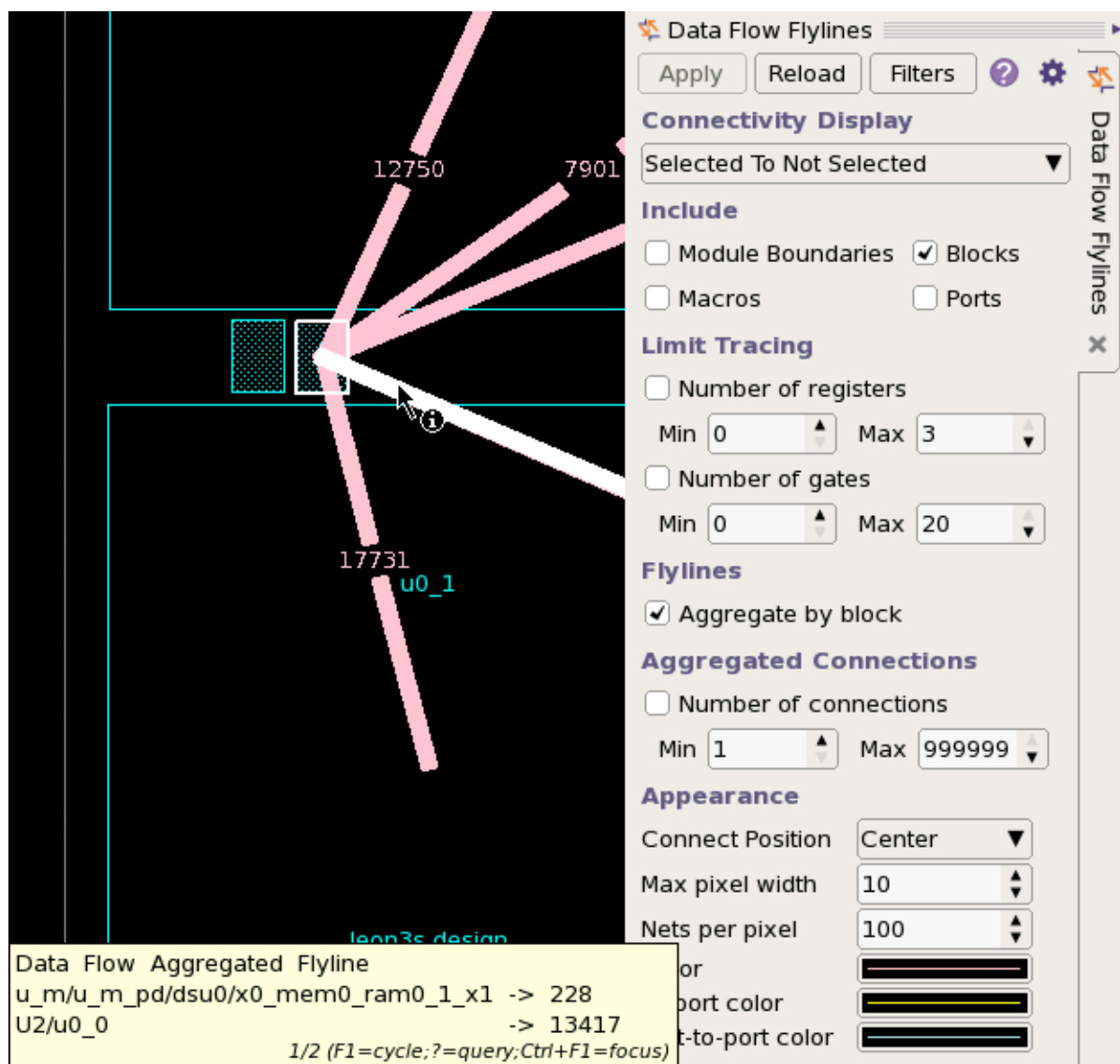
Consider the following suggestions when using the Path Analyzer Flylines tool:

- For most applications, you should deselect the Top option in the From and To sections to remove flylines that do not connect to any particular macro or module boundary. This selection makes the remaining flylines and other information easier to see.
- In connection mode, enter a Minimum count value greater than zero to specify a lower limit on the number of connections required between objects before displaying a flyline.
- In timing mode, toggle the "Unconstrained" option in the Path Display section to include flylines for timing paths without timing constraints, that is, the effort is 0. Toggle "Unachievable" to include flylines for paths which cannot meet the current timing constraints, that is, the effort is 10 or greater. Enter a minimum effort level in the "Minimum Effort" to hide paths with an effort level less than or equal to the specified value.

Data Flow Flylines

Use the Data Flow Flylines tool to view connections between macros, blocks, module boundaries, and I/O ports. The tool helps you understand macro placement quality and view macro-to-macro paths in your design. To open the Data Flow Flylines panel, choose View > Flylines > Data Flow Flylines or click the Data Flow Flylines button.

Figure 78 Data Flow Flylines Panel



Computing Data Flow Flylines

Before tracing flylines, you must compute the flylines to load them into tool as follows:

1. Click the Reload button
2. In the Compute Data Flow Flylines dialog box, select the number of register levels, gate levels, and maximum fanout to trace

3. Click the Filters button to specify nets and pins to ignore during tracing; see [Data Flow Flylines Pin and Net Filters](#) for more information
4. Click OK to compute the data flow flylines

Tracing Flylines

After initializing the data flow flylines, you can trace flylines interactively in the layout window. A typical workflow includes the following steps:

1. Click a macro, block, module boundary, or I/O port to select it; the tool displays flyline connections to the objects selected in the Include section
2. Move the pointer over the flyline and view the number of connections in the infotip
3. Click the flyline to open the Data Flow Connections and Filters dialog; the dialog box shows the list of connections represented by the flyline, see [Viewing Flyline Details and Filtering Flylines](#) for more information
4. Click a path in the list to show the flyline for that path
5. Expand the path by clicking the ([+]) button and click a line that represents a segment of the path; the tool shows the detailed path flyline in red for the segment
6. Continue to trace other segments in the path by clicking on them, or by pressing the Up Arrow or Down Arrows key to scroll through the list of segments

Setting Tracing Options

The Data Flow Flylines tool supports options to control flyline tracing. Use the Connectivity Display setting to control which connections are displayed:

- All: Display all paths
- Selected To All: Display paths from selected objects to all connected objects
- Selected To Selected: Display paths from selected objects to other selected objects that are connected
- Selected To Not Selected: Display paths from selected objects to connected objects that are not selected

Use the Include section to select the type of objects to include when tracing.

Use the Limit Tracing section to adjust the number of registers and number of gates to include in the flyline. You can only reduce the number of levels based on the numbers you specified in the Compute Data Flow Flylines dialog; to increase the available levels, click Reload and compute the flylines again. Note that by default, buffers and inverters

are not included in the gate count for minimum and maximum number of gates between the source and destination pins. As a result, the detailed path connectivity can show more gates than expected.

Aggregated connections are flylines which combine flylines with the same source and destination object into a single flyline. To change the behavior of aggregated flylines,

- Select or deselect the Aggregate by block option to show aggregated flylines or show individual flylines
- Select the Number of connections option and choose values for the minimum and maximum number of connections for an aggregated flyline; the tool hides aggregated flylines with a number of connections outside this range

Use the Appearance section to change the connection point, line width, and color of the flylines:

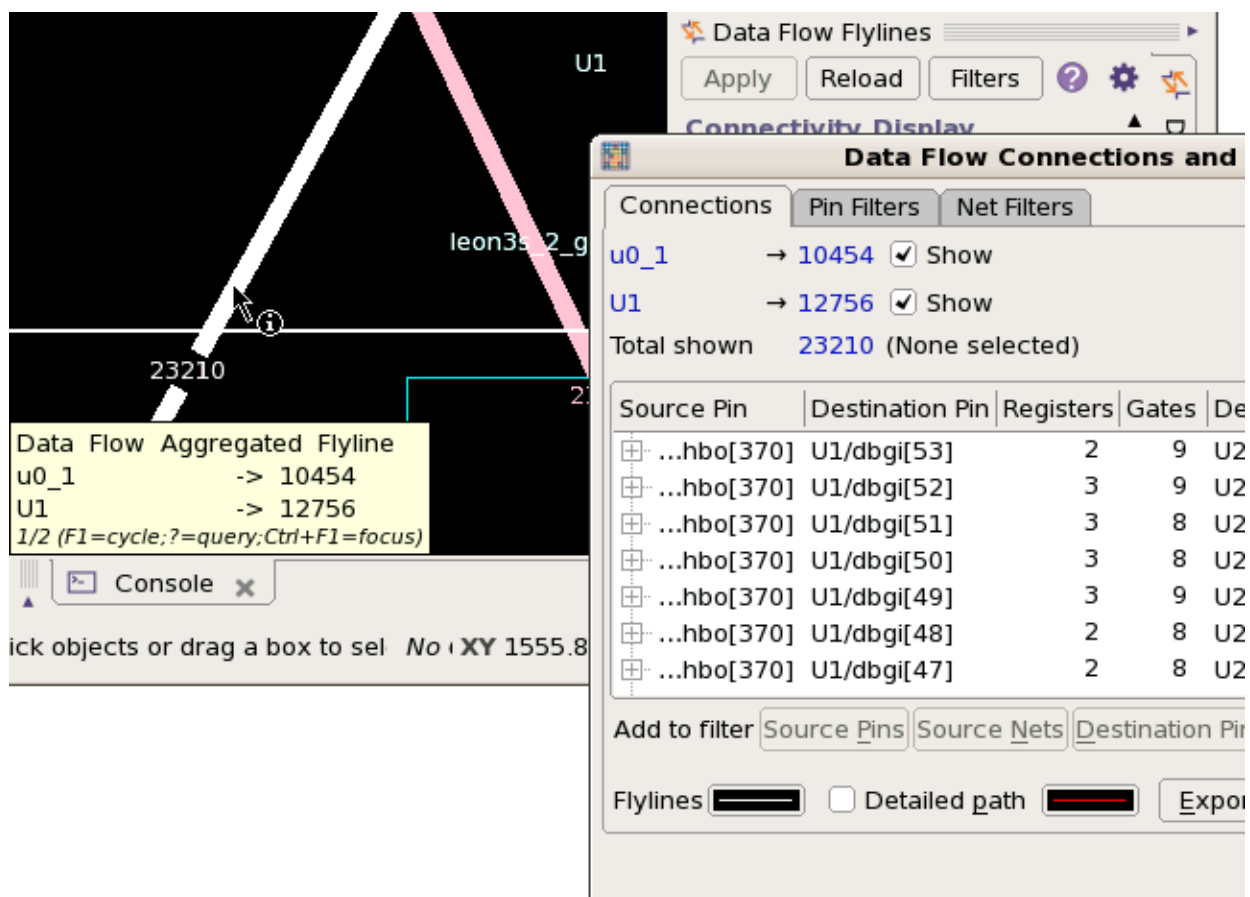
- Select a Connect Position option in the list to change where the flylines are drawn in the layout; the tool draws the flyline to the specified location on the object
- Use the Max pixel width setting to limit the drawn width of the flyline to the specified number of pixels
- Use the Nets per pixel setting to specify the width of the flyline based on the number of net connections; for example, a setting of 100 nets/pixel and a net connection containing 300 connections results in a 3 pixel line ($300 / 100 = 3$)
- Use the Color setting to change the color used when displaying a flyline between macros, blocks, and module boundaries
- Use the To port color setting to change the color of flylines to ports
- Use the Port-to-port setting to change the color of port-to-port flylines

Viewing Flyline Details and Filtering Flylines

Use the Data Flow Connections and Filters dialog box to view individual connections within an aggregated flyline, view single flylines, or set pin and net filters to use when tracing flylines. To open this dialog box, click an aggregated flyline generated by the Data Flow Flylines tool.

Note that after you create a filter, you must click Reload in the Data Flow Flylines panel to recompute the flylines and apply the filter.

Figure 79 Data Flow Connections and Filters



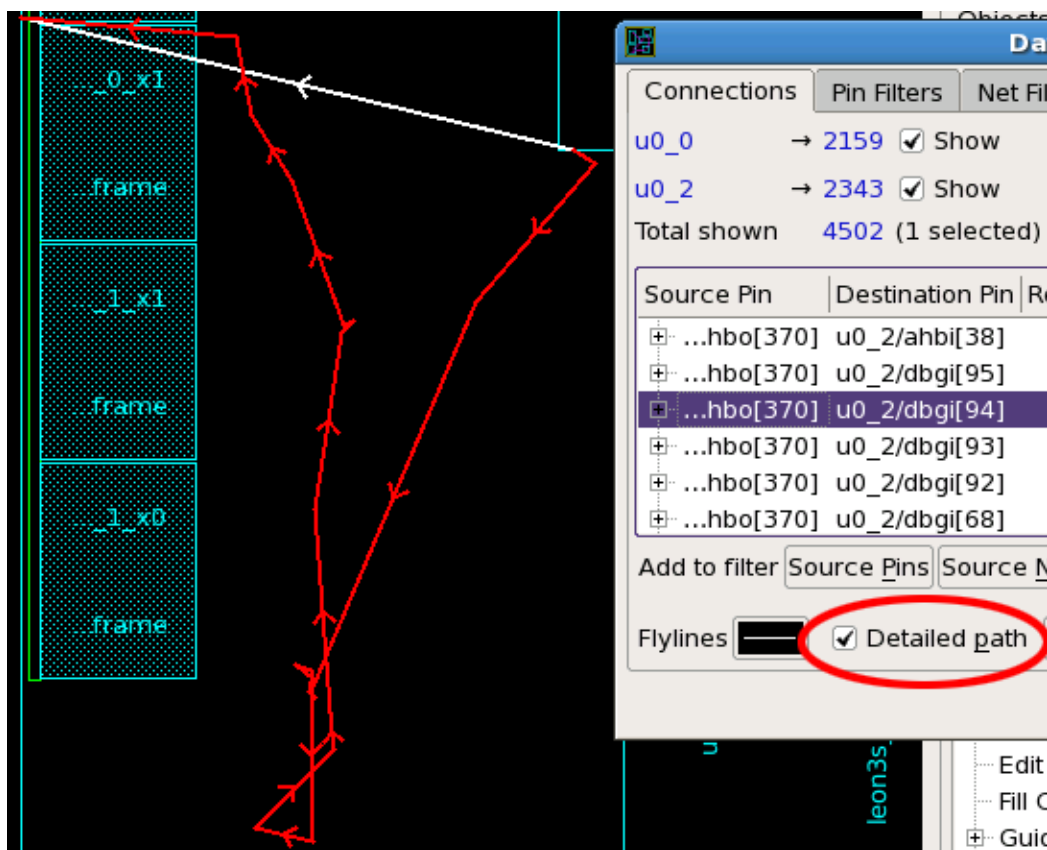
The Connections tab displays a list of connections within the selected flyline. At the top of the tab, the tool shows the names of the two objects connected by the flyline and the number of connections that originate from either object. If there are connections in both directions between the objects, click the Show button to the right of the object name to add to list or remove from the list the connections which originate from that object.

To trace flylines by using the Connections tab,

1. Click a path in the list; the tool shows the flyline for that path
2. Select the Detailed path option to display all intermediate connections in the path; the tool shows a red flyline for each pin-to-pin connection
3. Click the Source Pins, Source Nets, Destination Pins, or Destination Nets buttons to copy the selected path to the respective filter; to apply the filters, you must click Reload to recompute the flylines

4. Expand the path by clicking the ([+]) button to show the source and destination pins for the intermediate connections
5. Click an entry in the list that represents a segment of the path; the tool shows the detailed path flyline in red for the segment
6. Press the Up Arrow or Down Arrow key to scroll through the path and highlight different segments in the layout
7. Press and hold the mouse button and drag the pointer down the list to add additional segments in the path

Figure 80 Flyline and Detailed Path



To improve performance, the tool might not load all path details. In this case, the tool displays the message "Path is not loaded. Double-click to load." in the Detailed Path column. You can load the path by double-clicking it, or load all remaining unloaded paths by clicking the Load All Paths button.

To save a comma-separated list of source pins, destination pins, number of registers, and number of gates for paths in the list, click the Export button.

Data Flow Flylines Pin and Net Filters

To open the Data Flow Connections and Filters dialog box, click the Filters button in the Data Flow Flylines panel. The Pin Filters and Net Filters tabs contain the list of pins and nets to ignore during flyline tracing. To add an entry to the list, enter the name in the Patterns box and click Add. Alternatively, enter an expression compatible with the `get_pins` or `get_nets` command to specify one or more pins or nets.

In the Pin Filters or Net Filters tabs, use the buttons on the right modify the list:

- Use the Add Selected Pins button or Add Selected Nets button to add to the list the pins or nets that are selected in the layout
- Use the Add Pins of Selected button or Add Nets of Selected button to add all pins or nets of the selected object
- Use the Remove button to remove the selected item from the list
- Use the Remove All button to clear the list
- Use the Write to File button to write the list of pins or nets, one item per line, to a file
- Use the Read from File button to load a list of pins or nets from a file

Using Tcl Commands to Analyze Data Flow Flylines

When you click the Reload button in the Data Flow Flylines panel, the tool opens the Compute Data Flow Flylines dialog box and prompts you for information about the number of levels and maximum fanout to trace. You can also use the `compute_dff_connections` Tcl command directly and specify options to the command to generate the register tracing information. For example, the following `compute_dff_connections` command generates path information and specifies tracing for two register levels, 15 levels of gates, and a maximum fanout count of 50.

```
fc_shell> compute_dff_connections -max_reg 2 -max_gate 15 \  
-max_fanout 50
```

After tracing the flylines with the `compute_dff_connections` command, use the `get_dff_connections` command to return all paths or return only paths that meet a specified criteria. By default, the command returns a collection of all paths generated by the `compute_dff_connections` command. Note that the values specified using the

`-max_reg` and `-max_gate` options must be less than or equal to the values specified to the corresponding options in the `compute_dff_connections` command.

The following examples return specific paths extracted by the `compute_dff_connections` command and return attributes for individual paths.

- Return all paths.

```
fc_shell> get_dff_connections
```

- Return paths with no registers and no gates.

```
fc_shell> get_dff_connections -max_reg 0 -max_gate 0
```

- Return all paths to the `pad[0]` port, from the `pad[0]` port, and through the `pad[0]` port.

```
fc_shell> get_dff_connections -to pad[0]
fc_shell> get_dff_connections -from pad[0]
fc_shell> get_dff_connections -through pad[0]
```

- Return all paths through the `pad[0]` and `u1/u2/o1` ports.

```
fc_shell> get_dff_connections -through pad[0] -through u1/u2/o1
```

- Examine the `start.full_name`, `end.full_name`, and other attributes returned for an individual path in the collection.

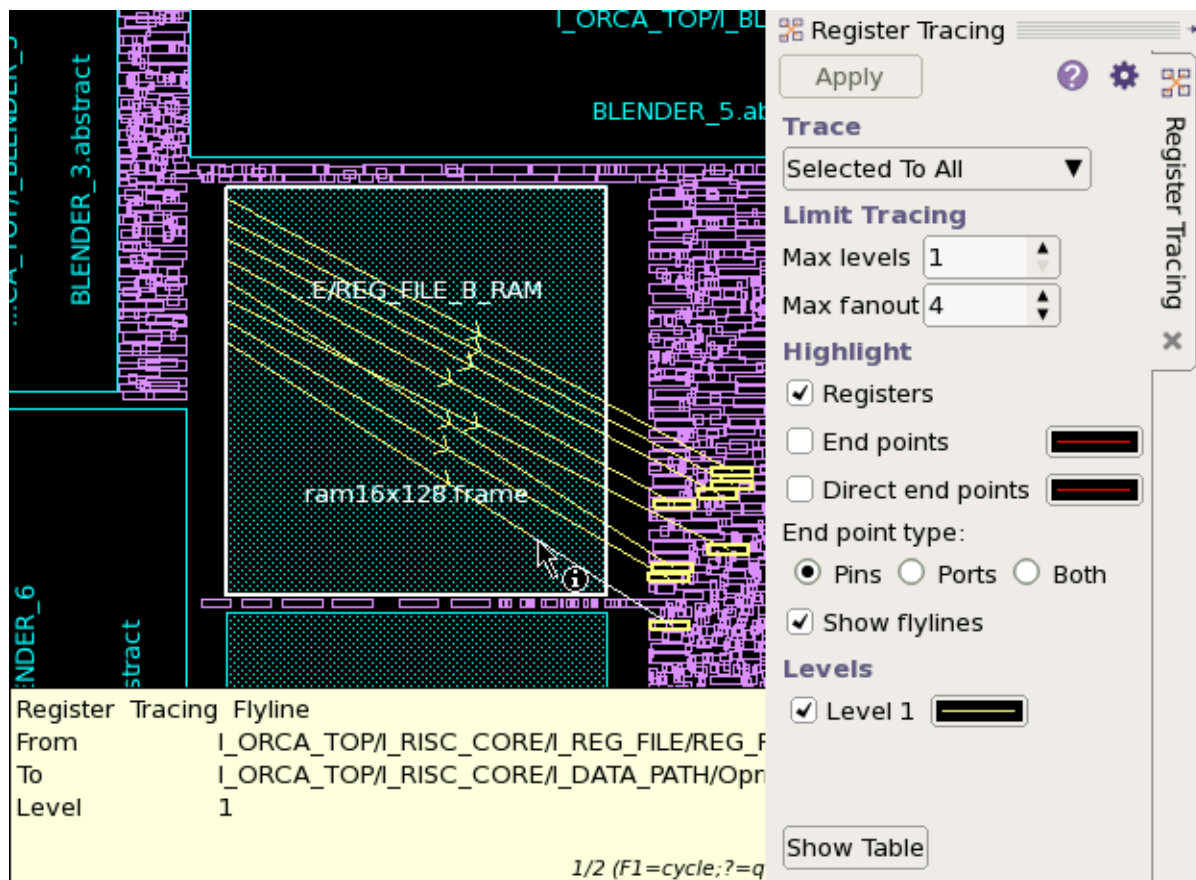
```
fc_shell> set_paths [get_dff_connections -from {pad[6]}]
{{pad[6]::I_ORCA_TOP/I_CONTEXT_MEM ...
fc_shell> set_path [index_collection $paths 0]
{{pad[6]::I_ORCA_TOP/I_CONTEXT_MEM/I_CONTEXT_RAM_0_1/A1[0]}}
fc_shell> get_attribute $path start.full_name
pad[6]
fc_shell> get_attribute $path end.full_name
I_ORCA_TOP/I_CONTEXT_MEM/I_CONTEXT_RAM_0_1/A1[0]
fc_shell> get_attribute $path name
pad[6]::I_ORCA_TOP/I_CONTEXT_MEM/I_CONTEXT_RAM_0_1/A1[0]
fc_shell> get_attribute $path registers
0
fc_shell> get_attribute $path gates
10
```

Register Tracing Flylines

The Fusion Compiler tool provides the Register Tracing Flylines tool to display register connections between hierarchies, macros, and ports in your design. Use the tool to view flylines for register connections between objects, limit tracing based on number of levels of registers or fanout, display connected pins and ports, and list the registers in the path. To open the Register Tracing Flylines panel, choose View > Flylines > Register Tracing

or click the Register Tracing Flylines button (). The Register Tracing Flylines panel is shown in [Figure 81](#).

Figure 81 Register Tracing Flylines Panel



Displaying Flylines

The Register Tracing Flylines tool displays flylines between the driving register pin and receiving register pin for the nets that meet the specified criteria. You can change the flylines that are displayed by changing the settings in the panel as follows and clicking the Apply button to commit the changes.

- Trace Selected to All: Trace the output ports by the specified number of levels and display the flylines
- Trace Selected to Selected: Trace connections between the selected objects and display connections with an endpoint in one of the selected objects

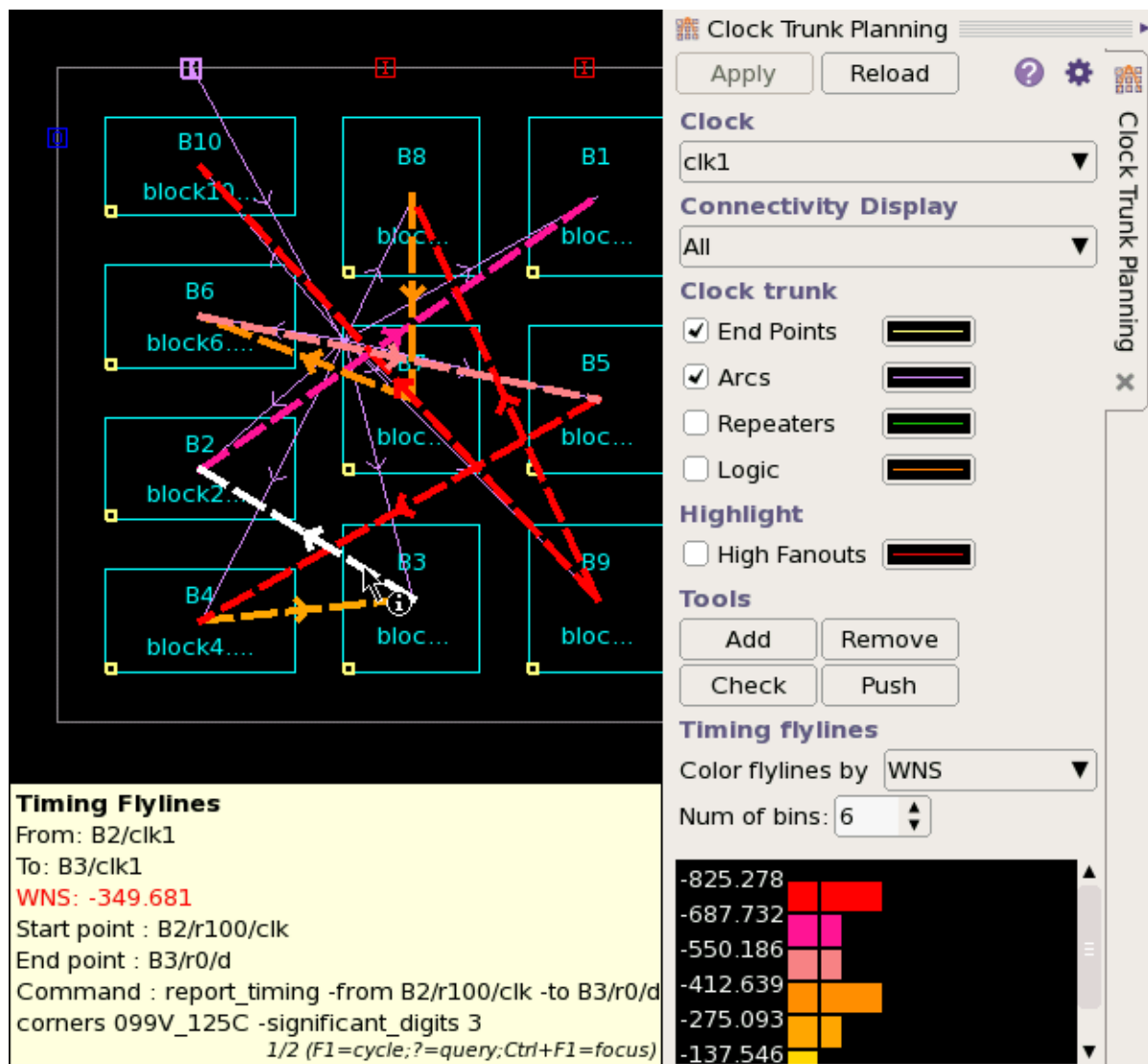
- **Trace All to Selected:** Trace all possible ports and macro pins to the select objects and display only those connections with an endpoint in the selected object
- **Limit Tracing Max levels:** Set the maximum number of register levels to trace
- **Limit Tracing Max fanout:** Specifies the fanout level at which to stop tracing
- **Highlight registers:** Display and highlight cells that are registers
- **Highlight End points:** Display and highlight registers and ports in the specified highlight color
- **Highlight Direct End points:** Display and highlight endpoints that are driven directly by the source pin
- **End point type Pins:** Consider only pins as endpoints
- **End point type Ports:** Consider only ports as endpoints
- **End point type Both:** Consider both pins and ports as endpoints
- **Show flylines:** Toggle flyline visibility
- **All Levels:** Toggle visibility for all tracing levels up to the maximum level set in the Limit Tracing section
- **Level 1, etc:** Toggle visibility for the specified level
- **Show Table:** Display a dialog box that contains a list of startpoints, endpoints, and intermediate registers that meet the tracing criteria. Click on a line in the list to display the flyline in the layout. Click Export to write out the list.

Clock Trunk Planning Flylines

The Fusion Compiler tool provides the Clock Trunk Planning Flylines tool to view and modify the clock trunk for clock signals in the design. You can use the tool to view individual clock trunks, add buffers, remove buffers, check and repair clock trunk violations, and push down cells for the displayed clock trunk. Timing information, such as WNS, TNS, and other parameters, can be viewed by displaying an infotip for different clock trunk planning flylines in the design. To open the Clock Trunk Planning Flylines panel, choose View > Flylines > Clock Trunk Planning from the menu or click the

Clock Trunk Planning button (). The Clock Trunk Planning Flylines panel is shown in [Figure 82](#).

Figure 82 Clock Trunk Planning Palette



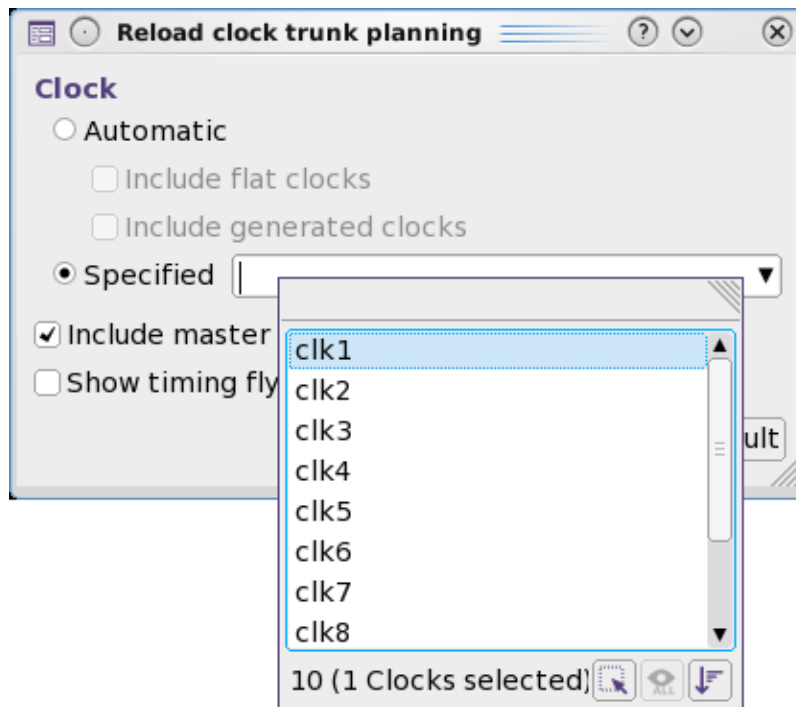
Loading Clock Data

To load clock information into the Clock Trunk Planning tool,

1. Click the reload button
2. In the Reload clock trunk planning dialog shown in [Figure 83](#), choose the clocks to load

- Choose Automatic to load all clock trunks
 - Choose Specified to choose one or more individual clock trunks from the list. Hold the Ctrl key and click to add clocks to the list, or hold the Shift key and click to add multiple clocks.
3. Click OK

Figure 83 Reload Dialog Box



The tool loads the specified clock trunks. If you chose Automatic or selected multiple clock signals, the tool selects one of the signals and displays it. To change the clock trunk that is displayed, choose another clock signal from the Clock drop-down list.

Changing Clock Trunk Highlighting

Use controls in the Clock trunk section to change the color of the clock trunk elements displayed in the layout. Select the box next to the element name to display it in the layout.

Change the color of a clock trunk element by clicking the color bar next to the element name and selecting a color from the list.

- End Points: Defines the color of the clock end points
- Arcs: Defines the color of the flylines between the clock origin and a clock buffer or block
- Repeaters: Defines the color of repeater cells
- Logic: Defines the color of logic cells
- High Fanouts: Defines the color of high-fanout nets

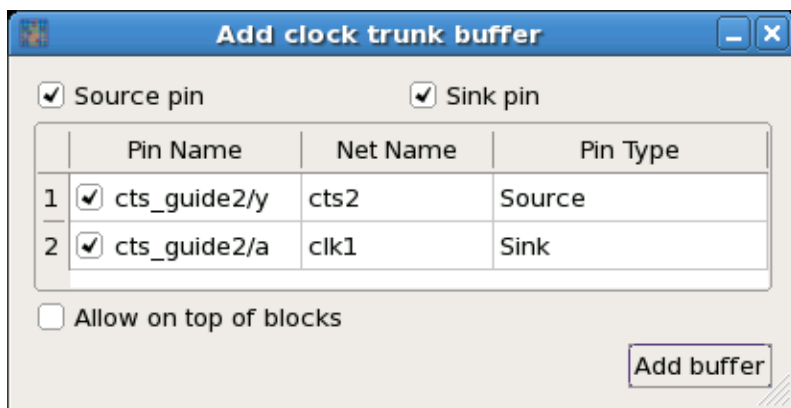
Adding and Removing Clock Trunk Cells

To insert a clock buffer cell on a clock trunk,

1. In the palette, select the clock signal clock trunk for which to insert the buffer in the Clock box.
2. In the layout, select the port, cell, or pins to buffer. The object must be on the currently selected clock trunk.
3. Click the Add button.
4. In the dialog box shown in [Figure 84](#), select the pins to buffer: source pin, sink pin, or both pins.
5. Select the “Allow on top of blocks” button to allow the tool to insert a buffer if the buffer overlaps the block.
6. Click Add buffer

The tool applies the `create_clock_buffer` command to insert a clock buffer cell for each pin selected in step 4.

Figure 84 Add Clock Trunk Buffer



To remove one or more clock buffer cells,

1. Select the cells in the layout
2. Click the Remove button

Displaying Timing Flylines Between Blocks

The Clock Trunk Planning tool can display timing flylines between blocks in the design. Select a timing metric (WNS, WNS(OCV), TNS, Slack, Slack(OCV), Paths, or VioPaths&TNS) from the “Color flylines by” box in the Timing flylines section near the bottom of the palette. The tool bins the paths, adds flylines to the layout according to bin, and displays a histogram of values at the bottom of the palette. To change the number of bins displayed, set a new value in the “Num of bins” box.

To remove timing flylines from the layout, select None on the “Color flylines by” box.

6

Using Map and Visual Modes

The Fusion Compiler GUI provides tools to view data about the design as an overlay on the layout. Using map modes and visual modes, you can quickly identify problem areas in your design, make corrections, and evaluate the fix.

The following topics describe map modes supported by the GUI:

- [Map Mode Overview](#)
- [Global Route Congestion Map](#)
- [Channel Congestion Map](#)
- [Cell, Pin, Block Pin, and Power Density Maps](#)
- [Effective Voltage Areas Map](#)
- [Effective Site Arrays Map](#)
- [Wire Utilization Map](#)
- [Visual Mode Overview](#)
- [Snapshot Visual Mode](#)
- [Highlight Visual Mode](#)
- [Clock Tree Visual Mode](#)
- [Scan Chain Visual Mode](#)
- [Net Connections in Area Visual Mode](#)
- [Relative Placement \(RP\) Groups Visual Mode](#)
- [Relative Placement \(RP\) Net Connection Visual Mode](#)
- [Hierarchy Visual Mode](#)
- [Cell Slack Visual Mode](#)
- [Imported Path Pins Visual Mode](#)
- [Voltage Areas Visual Mode](#)

Map Mode Overview

Map modes are analysis tools you can use to highlight specific floorplan, placement, or routing results in the active layout view. When you enable a map mode, the GUI dims the visible objects in the layout view and opens the Map Mode panel. To view or hide a map mode in the active layout view, you can choose View > Map and select a map mode. Alternatively, click the map mode button on the toolbar. The name of the active map mode appears in the list at the top of the Map Mode panel as shown in [Figure 85](#). The Global Route Congestion Map is shown in [Figure 86](#).

Figure 85 *Map Modes*

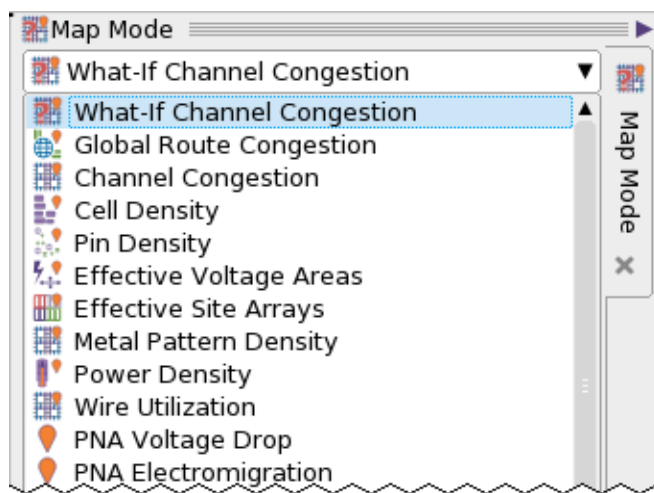
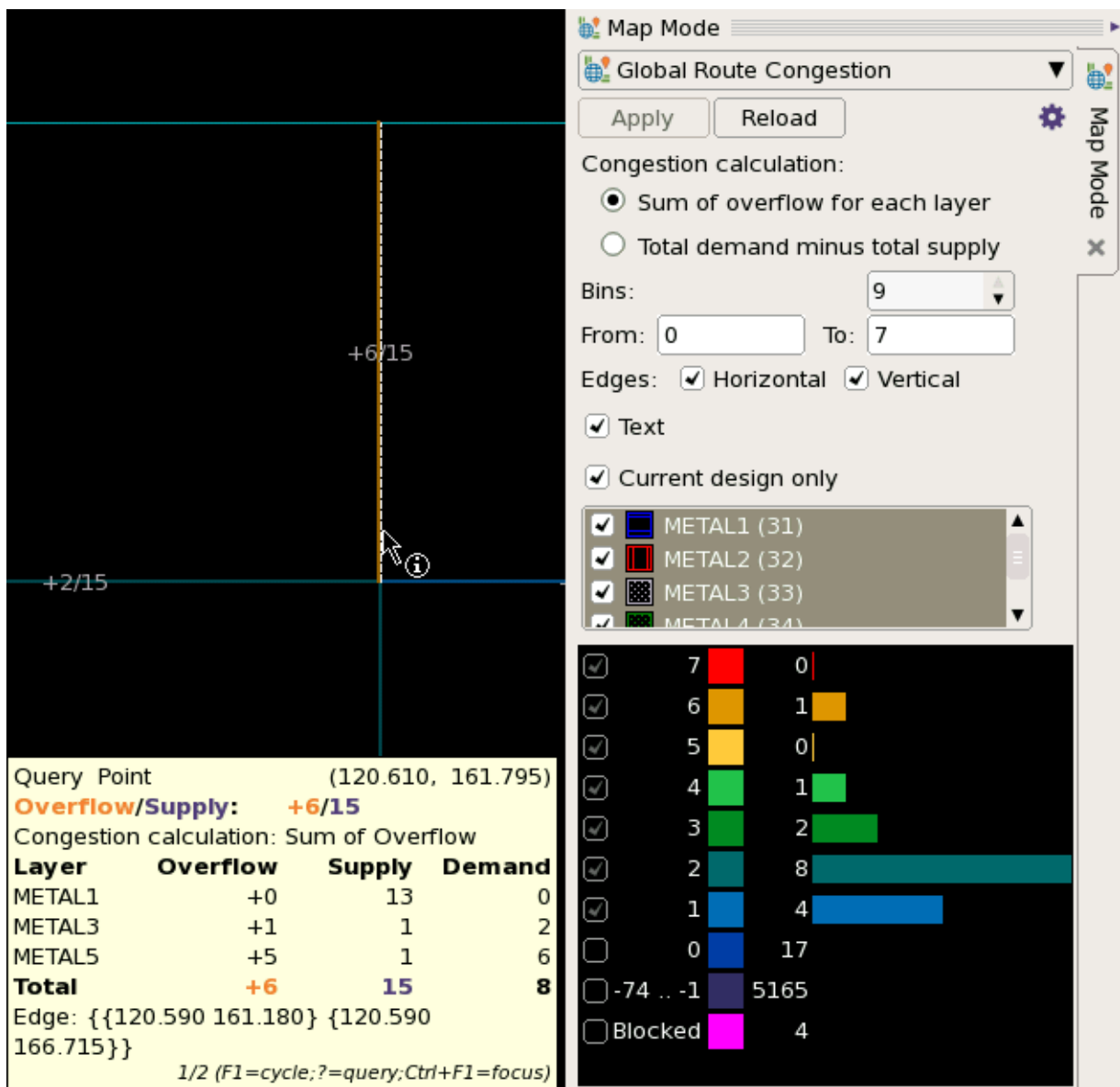


Figure 86 Global Route Congestion Map

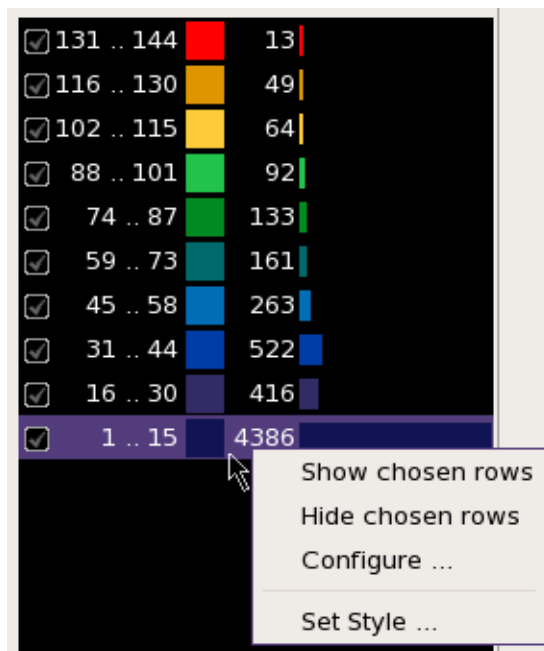


Configuring the Histogram

At the bottom of the map mode panel, the tool displays a histogram of values for the current map mode. You must click Reload to generate the histogram for the first time. To modify the histogram,

- Display or hide histogram bars and their corresponding data in the layout by selecting or deselecting the box at the left of the bar; alternatively, right click anywhere in the row and choose Show chosen rows or Hide chosen rows
- Change the color of the histogram bar and corresponding grid in the layout by right clicking anywhere in the row and choosing Set Style
- Change other display options by right clicking anywhere in the row and selecting Configure; see [Configuring Map Options](#) for more information
- Click Apply to update the map based on the new settings

Figure 87 Map Mode Histogram



Configuring Map Options

You can perform additional configuration of each map mode by clicking the Show Options button () and selecting configure:

- Exaggeration scheme: Set the Top, Middle, and Bottom values to specify the color exaggeration scheme in map modes and visual modes; these values set the minimum, median, and maximum number of pixels to use when displaying congestion bars in the layout
- Show styles: Displays or hides the color key square in the histogram
- Show exaggeration: Displays or hides the calculated exaggeration value next to each histogram bar
- Show counts: Displays or hides the count of edges for each histogram bin
- Show histogram: Displays or hides the histogram bars
- Show only visible bins in histogram: When selected, displays a histogram bar only if the checkbox for the bin is selected

Global Route Congestion Map

The Fusion Compiler tool supports the Global Route Congestion map to highlight areas of the design which contain a high demand for global routes. Congested areas might lead to problems during detail route. To display the Global Route Congestion Map, choose View > Map > Global Route Congestion or click the Global Route Congestion button ()

To generate the global route congestion map, click the Reload button. The tool displays a dialog box with the `route_global` command and options to use to generate the map data. Click OK to run the command. After completion, the tool displays the congestion map.

After the tool displays the congestion map, move the pointer to the layout window and zoom in to view the map in more detail. You can move the pointer over a congestion bar to view more information in the infotip.

Refining the Global Route Congestion Map

You can modify settings in the Global Route Congestion Map panel to change how the congestion map is displayed. Use the following settings to change the congestion map:

- Rule level: Select the rule level to apply to the congestion calculation
- Congestion calculation: Select Sum of overflow for each layer or Total demand minus total supply to select the method used to calculate congestion; see [Congestion Map Calculations](#) for information about how this setting affects the display
- Bins: Change the number of unique bins used to display the map by changing the Bins value; a larger number of bins partitions the map into a greater number of value ranges
- From: and To: Change the range of overflow values displayed in the map and the histogram by changing these values; these values represent the overflow in routing capacity for the route cell, where higher numbers represent more severe congestion
- Edges: Select the edges, Horizontal and Vertical, of the global route cell for which to display congestion values
- Text: Select this option to display in the layout a number that represents the calculated congestion values; if the values are not visible, you must zoom in to view them and make sure that range is selected in the histogram
- Hierarchical: Select this option to display a congestion map within a hierarchical block
- Layers: Select the routing layers to consider when calculating congestion

At the bottom of the panel, the tool displays a histogram of the congestion values. To modify the histogram after generating the map,

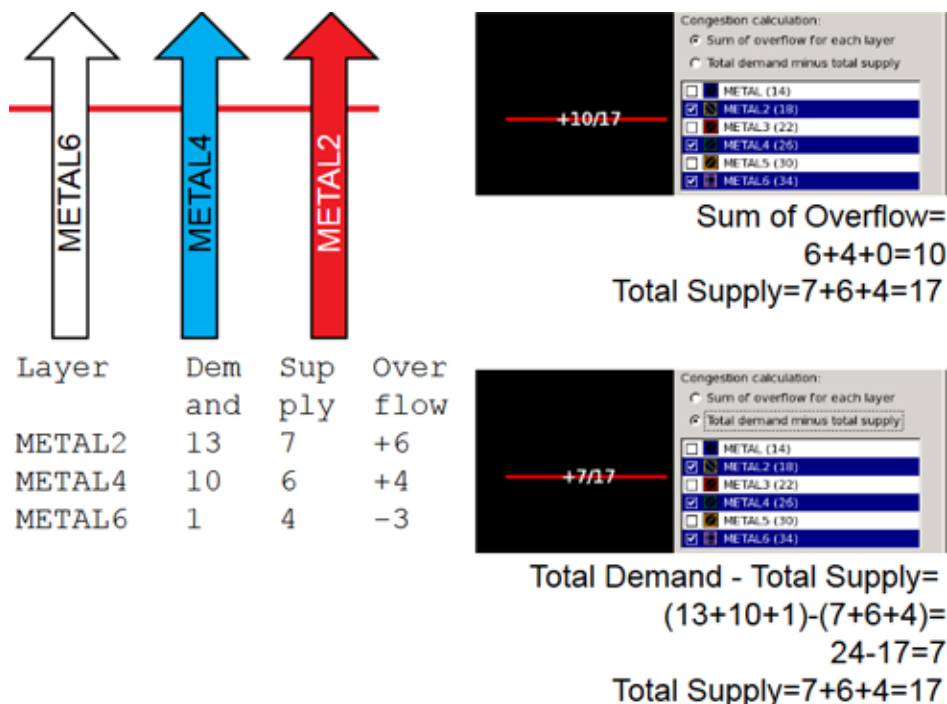
- Select or deselect layers to include or exclude them from the congestion calculation
- Display or hide histogram bars and their corresponding grids in the layout by selecting or deselecting the box at the left of the bar; alternatively, right click anywhere in the row and choose Show chosen rows or Hide chosen rows
- Change the color of the histogram bar and corresponding grid in the layout by right clicking anywhere in the row and choosing Set Style
- Click Apply to update the map based on the new settings

Congestion Map Calculations

The Global Route Congestion map supports two calculation modes: “Sum of overflow for each layer” and “Total demand minus total supply”. The mode determines the congestion value that the tool displays when a global route cell has congestion; the sum-of-overflow mode is the default.

The following shows an example calculation for both modes. The example explains one direction through a global route cell in a design with congestion.

Figure 88 Congestion Example



In this example, Layer METAL2 is heavily congested, with a demand of 13 routing tracks and a supply of only 7 tracks, for an overflow of 6. Layer METAL 4 is moderately congested with an overflow of 4, and layer METAL6 is not congested and contains underflow of 3. Other layers are not used in the congestion calculation is this example.

The sum-of-overflow calculation is the default calculation mode and considers only the overflow for each layer. In this example, the overflow on METAL2 is 6, the overflow on METAL4 is 4, for a total sum of overflow of 6+4=10. No overflow exists on METAL6 and this layer is not used in the calculation, even though it is selected. In a region that contains only underflow, the tool sums the underflow for each selected layer.

The Total Demand Minus Total Supply calculation sums the total demand and subtracts the total supply. In this example, the total demand is 24, the total supply is 17, and the result is 24-17=7. Note that the Total Demand Minus Total Supply calculation produces a more optimistic congestion result in regions which contain both overflow and underflow.

Channel Congestion Map

The Channel Congestion map displays areas of the design in channels between objects which contain a high demand for global routes. You can generate the Channel Congestion map for a small area of the design, modify the design, and regenerate the map to perform quick iterations to repair congested areas of the design. To display the Global Route Congestion Map, choose View > Map > Global Route Congestion or click the Global Route Congestion button ().

To generate the channel congestion map, click the Reload button. The tool displays a dialog box; select Whole design area or Specified area to define the area to analyze for channel congestion. Click OK to run the command. After completion, the tool displays the channel congestion map.

After the tool displays the channel congestion map, move the pointer to the layout window and zoom in to view the map in more detail. You can move the pointer over a congestion bar to view more information in the infotip.

You can modify settings in the Channel Congestion Map panel to change how the congestion map is displayed. Do the following to change the congestion map:

- Congestion calculation: Select Sum of overflow for each layer or Total demand minus total supply to select the method used to calculate congestion; see [Congestion Map Calculations](#) for information about how this setting affects the display
- Bins: Change the number of unique bins used to display the map by changing the Bins value; a larger number of bins partitions the map into a greater number of value ranges
- From: and To: Change the range of overflow values displayed in the map and the histogram by changing these values; these values represent the overflow in routing capacity for the route cell, where higher numbers represent more severe congestion
- Edges: Select the edges, Horizontal and Vertical, of the global route cell for which to display congestion values
- Text: Select this option to display in the layout a number that represents the calculated density values; if the values are not visible, you must zoom in to view them
- Hierarchical: Select this option to display a congestion map within a hierarchical block
- Layers: Select the routing layers to consider when calculating congestion

For more information about modifying the histogram, see [Configuring the Histogram](#).

Cell, Pin, Block Pin, and Power Density Maps

Density maps display areas of high density of an object type or calculated value in the core area or rectangular area within the core. The Fusion Compiler tool provides the Cell Density Map to analyze the density of standard cells, the Pin Density Map to analyze the density of pins in a given region, and the Power Density map to analyze power density in the layout. The Block Pin Density map displays the density of pins on block pin edges. To display the appropriate cell density map,

- Choose View > Map > Cell Density or click the Cell Density button () to display the Cell Density map.
- Choose View > Map > Pin Density or click the Pin Density button () to display the pin density map.
- Choose View > Map > Block Pin Density or click the Block Pin Density button () to display the block pin density map.
- Choose View > Map > Power Density or click the Power Density button () to display the power density map.

To generate the map, click the Reload button. The tool displays a dialog box; select Whole design area or Specified area to define the area to analyze. Click OK to run the command. After completion, the tool displays the cell density map.

After the tool displays the map, move the pointer to the layout window and zoom in to view the map in more detail. You can move the pointer over a grid point to view more information in the infotip. For the cell density, pin density, and power density maps, the Grid size label displays the number of horizontal and vertical grid points in the current map. The Grid size value is determined by the number of standard cell heights per grid point and the size of the rectangle you specified in which to calculate the map.

You can modify the following settings in the map panel to change the display:

- Grid dimension: Select the number of standard cell heights used to define the grid for the cell density, pin density, or power density maps
- Solid filled grids: Fills each grid cell with the histogram color that represents the density for that section of the layout for the cell density, pin density, or power density maps
- Bins: Changes the number of unique bins used to display the map by changing the Bins value; a larger number of bins partitions the map into a greater number of value ranges
- From: and To: Sets the range of density values displayed in the cell density, pin density, or power density map

- Text: Displays in the layout a number that represents the calculated density value at each grid point; if the values are not visible, you must zoom in to view them
- Current design only: Displays a density map only for the current hierarchy level

For more information about modifying the histogram, see [Configuring the Histogram](#).

Effective Voltage Areas Map

The Effective Voltage Areas map displays the default voltage area and other user-created voltage areas for the design. To display the Effective Voltage Areas map, choose View > Map > Effective Voltage Areas or click the Effective Voltage Areas button ().

After the tool displays the map, move the pointer over a voltage area to view more information in the infotip. Click the voltage area to display information about the voltage area in the Query panel.

You can modify settings in the Effective Voltage Areas map panel to change how the map is displayed. Do the following to change the map display:

- Show as fractured rectangles: Select this item to display the voltage areas as fractured rectangles
- Hierarchical: Select this option to display a pin density map within a hierarchical block

For more information about modifying the histogram, see [Configuring the Histogram](#).

Effective Site Arrays Map

The Effective Site Arrays map displays the currently defined site arrays for the design. To display the Effective Site Arrays map, choose View > Map > Effective Site Arrays or click the Effective Site Arrays button ().

After the tool displays the map, move the pointer over the site array to view more information in the infotip. Click the site array to display information about the site array in the Query panel.

Select the Current design only option to display site arrays only for the current level of hierarchy.

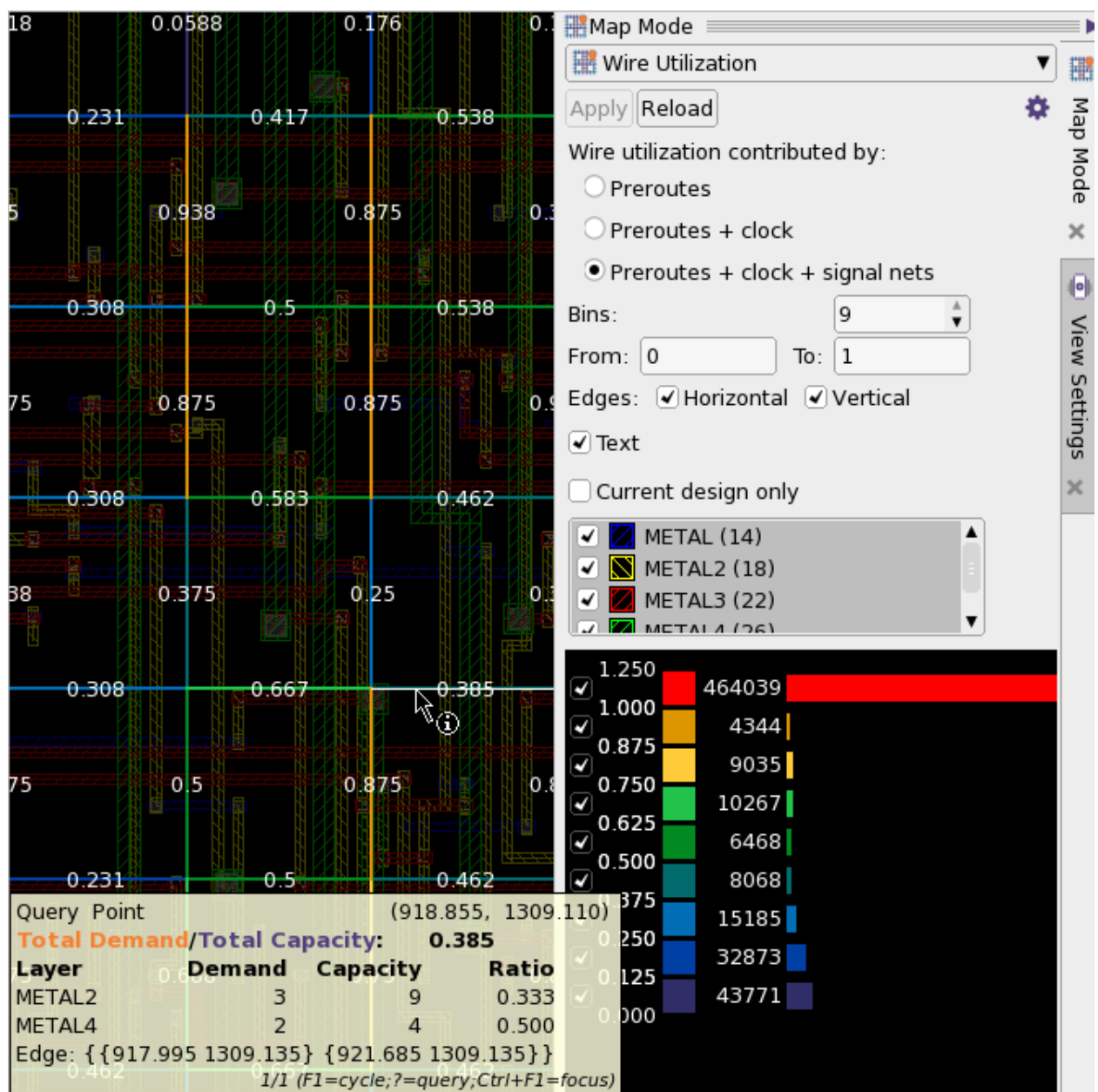
For more information about modifying the histogram, see [Configuring the Histogram](#).

Wire Utilization Map

The Wire Utilization map displays utilization values on the layout. Utilization is the demand for routing tracks divided by the capacity, where demand is the number of tracks occupied by preroutes and detail route shapes.

To display the wire utilization map, choose View > Map > Wire Utilization from the menu. The Wire Utilization map is shown in the following figure.

Figure 89 Wire Utilization Map



Modify the settings in the Wire Utilization map to change how the map is displayed:

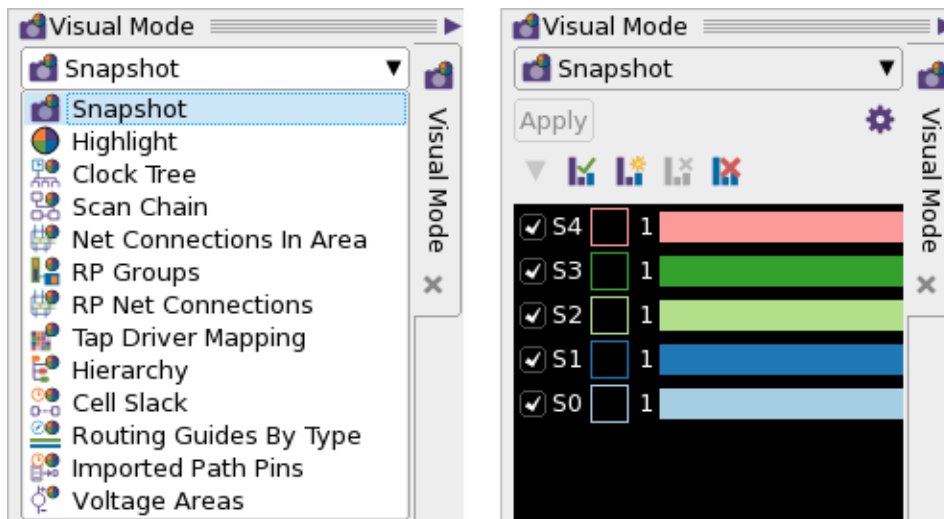
- Wire utilization contributed by: Select Preroutes, Preroutes + clock, or Preroutes + clock + signal nets: Includes only the selected route types in the calculation of the demand number
- Bins: Changes the number of unique bins used to display the map by changing the Bins value; a larger number of bins partitions the map into a greater number of value ranges
- From: and To: Changes the range of overflow values displayed in the map and the histogram by changing these values; these values represent the utilization for the route cell
- Edges: Selects the edges, Horizontal, Vertical, or both, for which to display utilization values
- Text: Displays in the layout a number that represents the calculated utilization values; if the values are not visible, you must zoom in to view them
- Current design only: Limits the utilization calculation to only the current level of hierarchy
- Layers: Selects the routing layers to consider when calculating the utilization

For more information about modifying the histogram, see [Configuring the Histogram](#).

Visual Mode Overview

Visual modes highlight design information in colored overlays in the layout view. Use visual modes to highlight different elements of your design and organize the elements into bins. Visual mode support is provided for clock trees, scan chains, relative placement groups, and so on as shown in [Figure 90](#). To view or hide a map mode in the layout view, choose Highlight > Color By and select a visual mode, or click the Visual Modes button (). Visual modes support style, histogram, and other configuration as described in [Configuring Map Options](#).

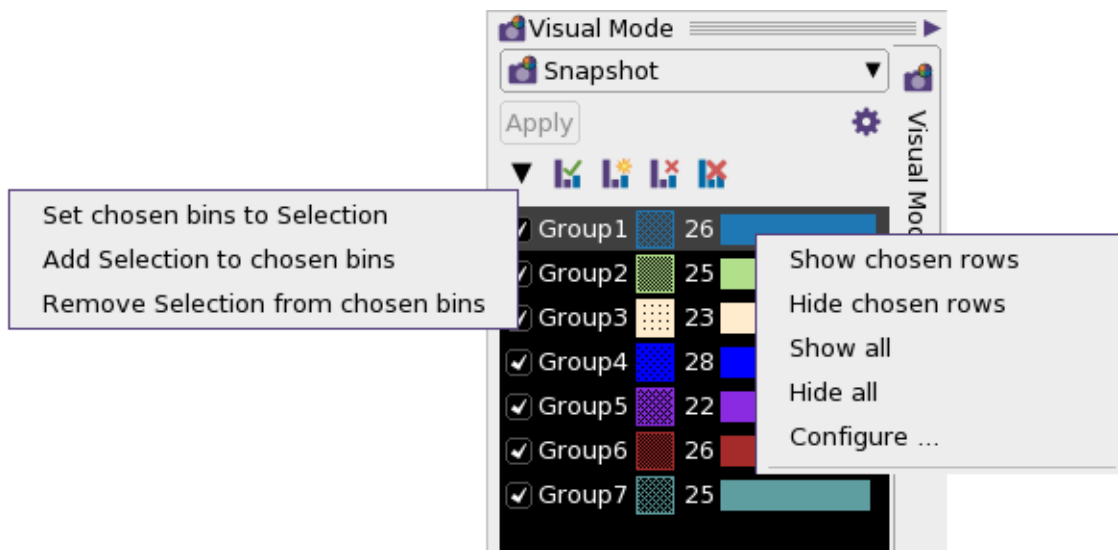
Figure 90 Visual Modes



Snapshot Visual Mode

Use the Snapshot Visual Mode to collect a group of selected objects into a bin and apply a highlight color. Enable the Snapshot Visual Mode by choosing Highlight > Color By > Snapshot from the menu.

Figure 91 Snapshot Visual Mode



You can perform the following operations with the Snapshot Visual Mode:

- Color the currently selected objects, create a new bin, and add the selection to the bin by clicking the Copy selection to new bin button ()
- Create a new empty bin by clicking the Create a new bin button ()
- Add selected objects to a bin by clicking the histogram bar to select the bin, clicking the down arrow at the left of the panel, and choosing Add selection to chosen bins.

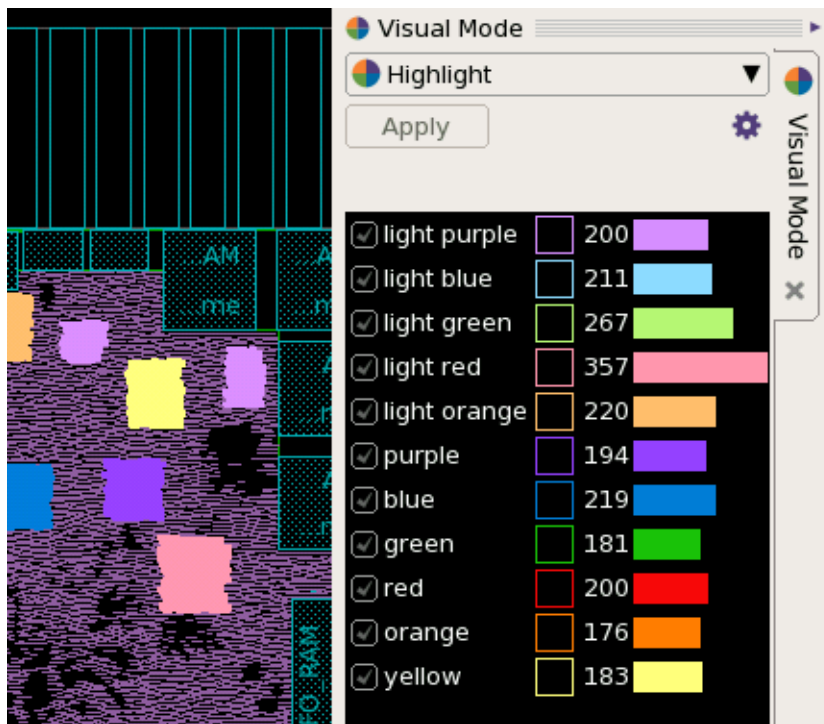
Use the same context menu to remove selected objects from a bin or replace objects in a bin with the current selection.

- Delete a bin by clicking the row to select it and clicking the Delete chosen row button ()
- Remove all rows from the histogram by clicking the Delete all bins button ()
- Change the highlight color by right-clicking the histogram row, choosing Set Style, choosing a new color, and clicking OK.
- Change the ordering of the histogram bars or hide histogram bars by right-clicking a histogram bar and selecting the appropriate item in the context menu (Move to top, Move up, Move down, or Move to bottom).

Highlight Visual Mode

Use the Highlight Visual Mode to create a histogram that represents the count of highlighted objects grouped by highlight color. Before using the Highlight Visual Mode, you can highlight objects in the layout by selecting them and choosing Highlight > Selected. Enable the Highlight Visual Mode by choosing Highlight > Color By > Highlight from the menu.

Figure 92 Highlight Visual Mode



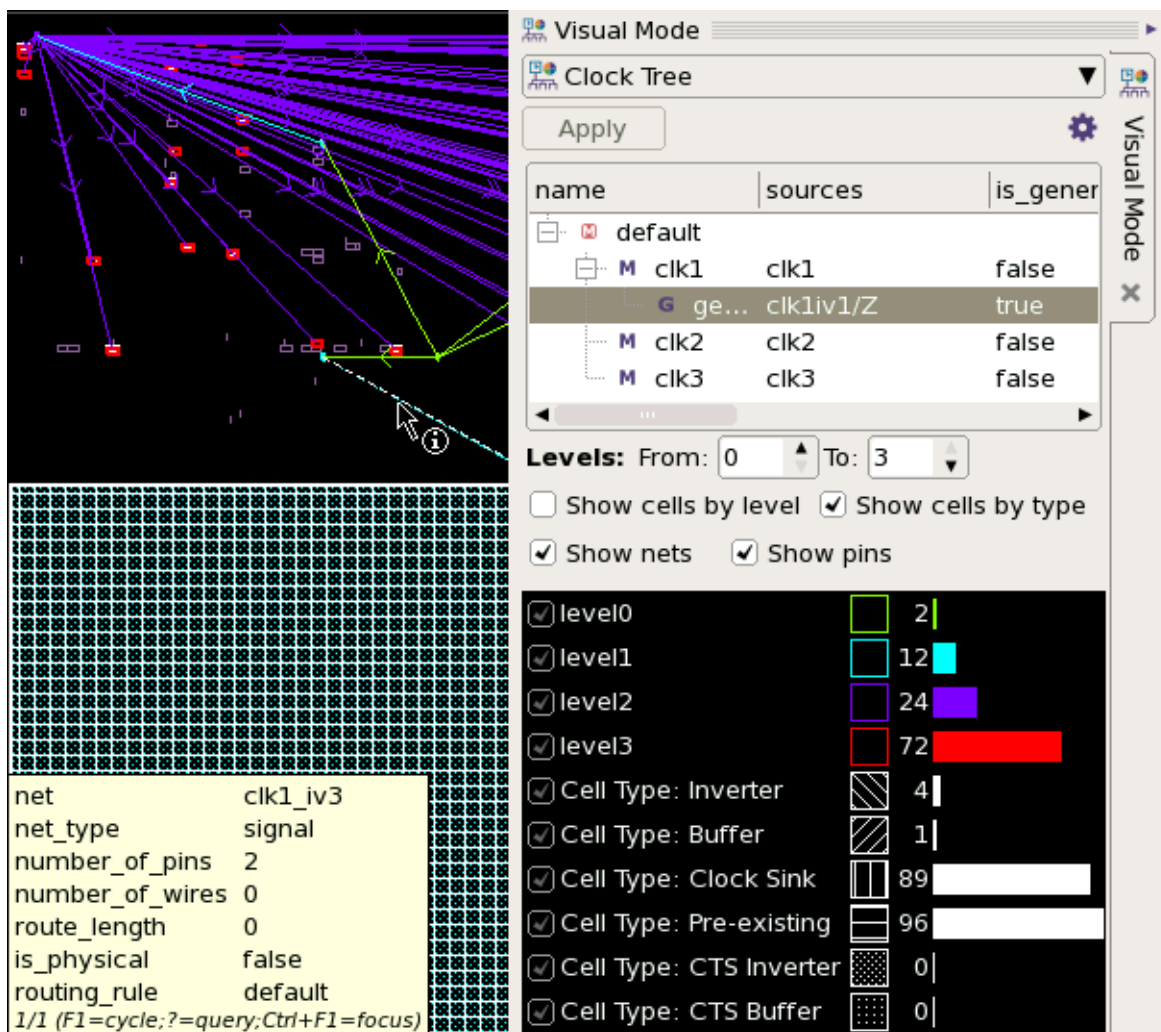
You can perform the following operations by right-clicking a histogram bar in the Highlight Visual Mode panel:

- Show or hide a histogram row and corresponding highlight color in the layout by choosing Show chosen rows or Hide chosen rows from the context menu.
- Configure exaggeration scheme and other display options by choosing Configure from the context menu.
- Change the color of the highlight by choosing Set Style from the context menu.
- Rearrange the order of the histogram bars by choosing Move to top, Move up, Move down, or Move to bottom from the context menu.
- Select or unselect highlighted objects in the layout by choosing Modify Global Selection and choosing an option to set the selection, append the selection, or remove from selection.

Clock Tree Visual Mode

Use the Clock Tree Visual Mode to view and overlay information about the clock trees in your design.

Figure 93 Clock Tree Visual Mode



The panel displays a table containing a list of clock sources in the design. The table displays the type of clock using the following notation:

- M: Master clock
- G: Generated clock
- V: Virtual clock

The panel also displays a histogram that shows the number of clock sources, nets, and pins per level and number of clock elements of each type. You can perform the following operations with the Clock Tree Visual Mode:

- Sort the clock sources by name, port/pin, or `is_generated` attribute by clicking the column heading in the clock table.
- Set the range hierarchy levels that are displayed by setting the values in the Levels: section of the panel.
- Show cells by level by selecting the Show cells by level option.
- Add information regarding the type of cell for clock buffers, inverters, and other cells in the layout by selecting Show cells by type.
- Display clock net flylines in the layout and add the count of clock nets to the histogram by selecting the Show nets option.

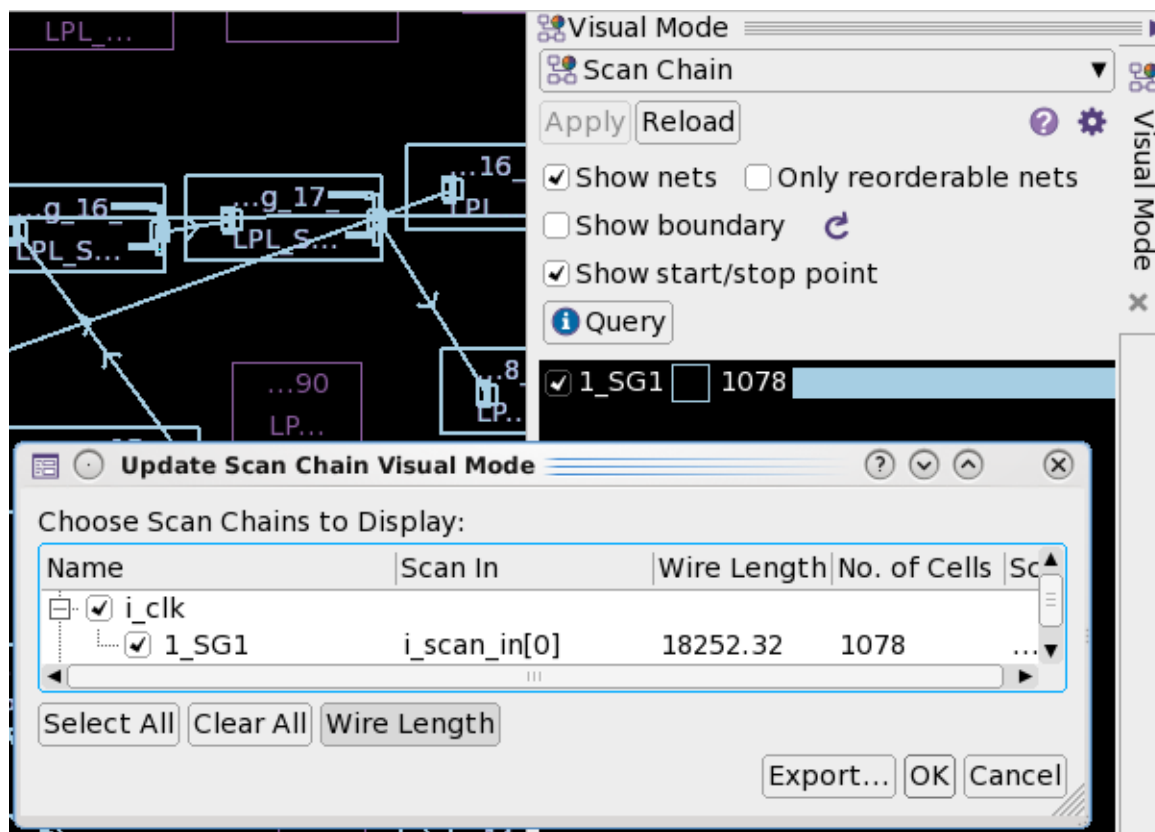
When selected, move the pointer over the clock net flyline to display additional information. If you perform global routing, the tool displays global routes instead of flylines.

- Display pins of cells in the clock tree and add the count of pins to the histogram by selecting the Show pins option.
- Change the highlight color by right-clicking the histogram row, choosing Set Style, choosing a new color, and clicking OK.
- Change the ordering of the histogram bars or hide histogram bars by choosing the appropriate item in the context menu.

Scan Chain Visual Mode

Use the Scan Chain Visual Mode to display connections between scan input, scan registers, and scan output for one or more scan chains. Enable the Scan Chain Visual Mode by choosing Highlight > Color By > Scan Chain from the menu.

Figure 94 Scan Chain Visual Mode



After clicking Reload in the panel, you can perform the following operations in the Update Scan Chain Visual Mode dialog box:

- Select the box next to the scan chain name and click the OK button to display the selected scan chains in the layout window.
- Click Select All button or the Clear All button to perform the action on the scan chains in the list.
- Click the Wire Length button to add a wire length column to the list.
- Click a column and drag it to a new position to reorder the columns.
- Move the mouse over a column header, click the right mouse key, and select an item from the context menu to configure or sort the columns in the list.

You can perform the following operations with the Scan Chain Visual Mode:

- Click the Query button to start the Query tool.

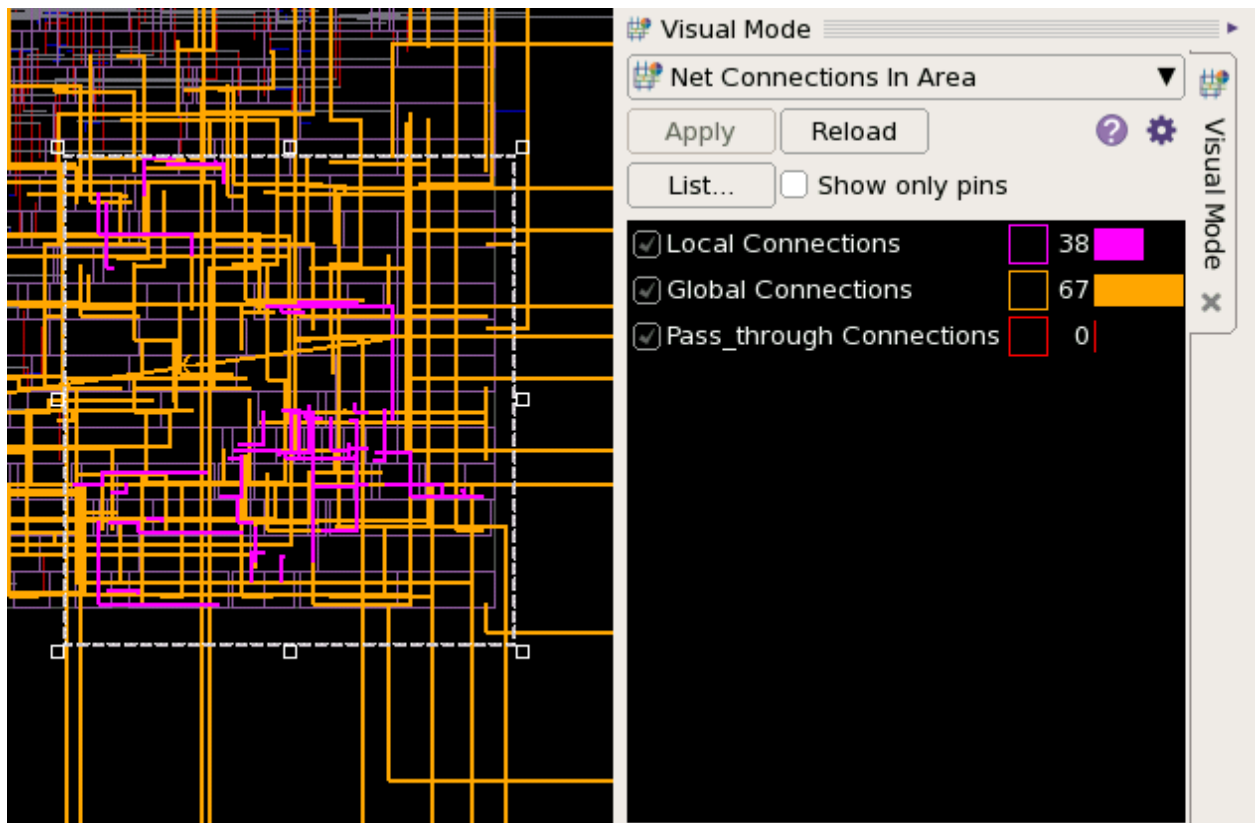
Use the Query tool to display additional information about scan chain cells and pins.

- Select the Only reorderable nets option to display only nets that can be reordered.
- Select the Show boundary option to display an overlay boundary for the scan chain.
- Select the Show start/stop point option to show the port or pin connections to the first and last scan chain registers.
- Change the highlight color by right-clicking the histogram row, choosing Set Style, choosing a new color, and clicking OK.
- Change the ordering of the histogram bars or hide histogram bars by choosing the appropriate item in the context menu.

Net Connections in Area Visual Mode

Use the Net Connections in Area Visual Mode to view nets that intersect a specified area of the layout and help identify the causes of congestion and placement issues during the initial placement stage. Enable the Net Connections Visual Mode by choosing Highlight > Color By > Net Connections in Area, or by clicking the Net Connections in Area button ).

Figure 95 Net Connections in Area



The tool highlights nets in the area you specify and classifies them into three types of nets:

- Local connections: Nets that have all pins and terminals within the specified area
- Global connections: Nets with at least one pin or terminal inside the specified area and at least one pin or terminal outside the specified area
- Pass-through connections: Nets that intersect the specified area but have no pins or terminals inside

To display net connections within a specified area,

1. Click the Reload button and specify the coordinates for the bounding box in which to display the nets.

To define a rectangular bound, click the area selection button () and either type the x- and y-coordinates for the upper-left and lower-right corners of the rectangle in the Coordinates box or drag the pointer in the layout view to form the rectangle.

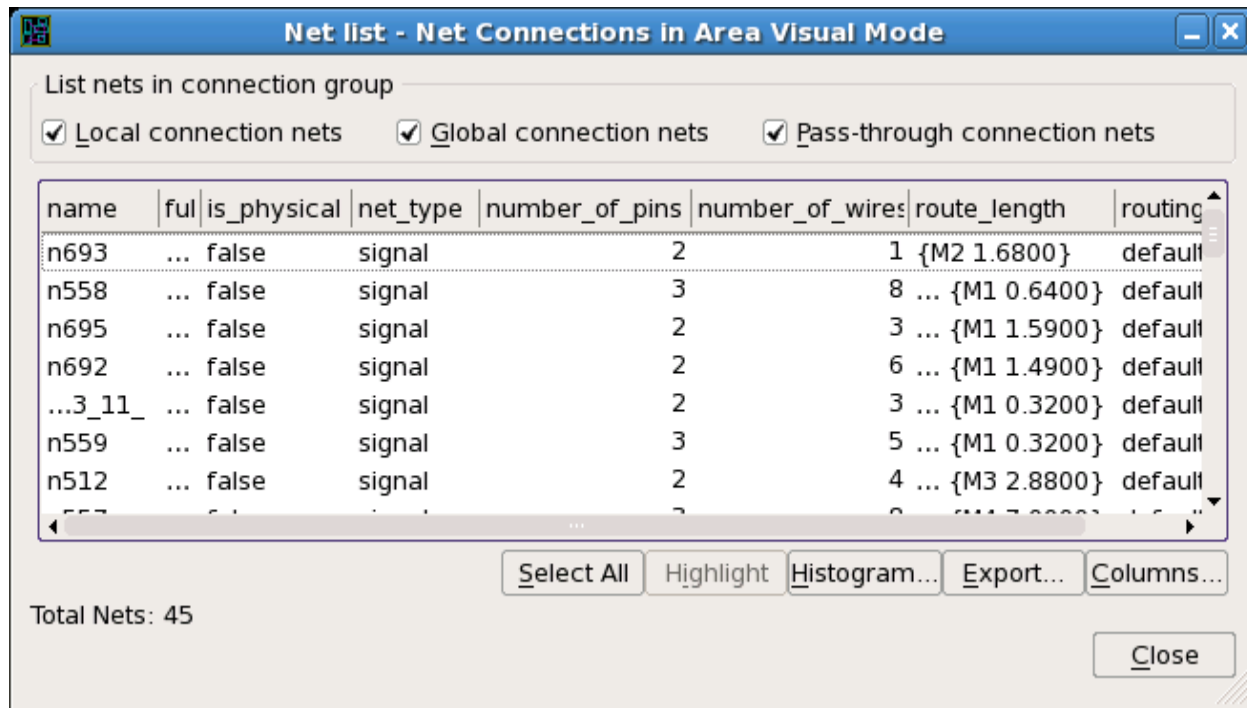
2. Select the types of nets to display (Signal, Clock, Power, Ground, Tie High, or Tie Low).
3. Click OK.

The tool displays the number of nets for each category and colors the nets in the layout window as shown in [Figure 95](#).

Note that [Figure 95](#) shows the layout after performing global routing. If you use the Net Connections in Area Visual Mode before performing global routing, the tool displays flylines instead of global routes.

Select Show only pins to hide the nets highlighted by the tool and highlight only the pins. For more information about modifying the histogram, see [Configuring the Histogram](#). Click the List button to display a dialog box containing a list of nets as shown in [Figure 96](#).

Figure 96 List of Net Connections in Area



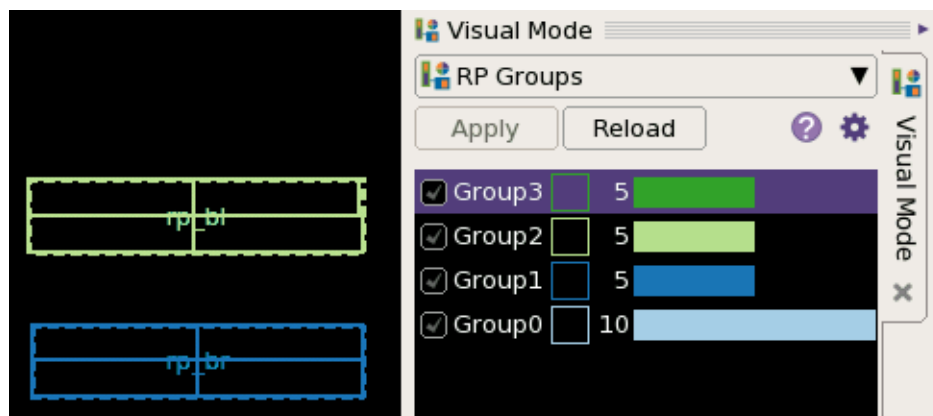
You can perform the following actions by using the Net list dialog box:

- Click the Local connection nets, Global connection nets, or Pass-through connection nets options to toggle the visibility for that net type.
- Select a net name in the list of nets to highlight the net in the layout.
- Click the Select all button to select all nets in the list.
- Click the Histogram button to configure a simple histogram that displays the count of the number of pins or number of wires in the area; click OK in the dialog box to generate the histogram and display it in the histogram panel in the layout window.
- Click the Export button to write out the contents of the table into a file in comma-separated value format.
- Click the Columns button to configure which columns of data are shown.

Relative Placement (RP) Groups Visual Mode

Use the RP Groups Visual Mode to highlight relative placement groups. Enable the RP Groups Visual Mode by choosing Highlight > Color By > RP Groups from the menu.

Figure 97 RP Groups Visual Mode



You can perform the following operations with the RP Groups Visual Mode:

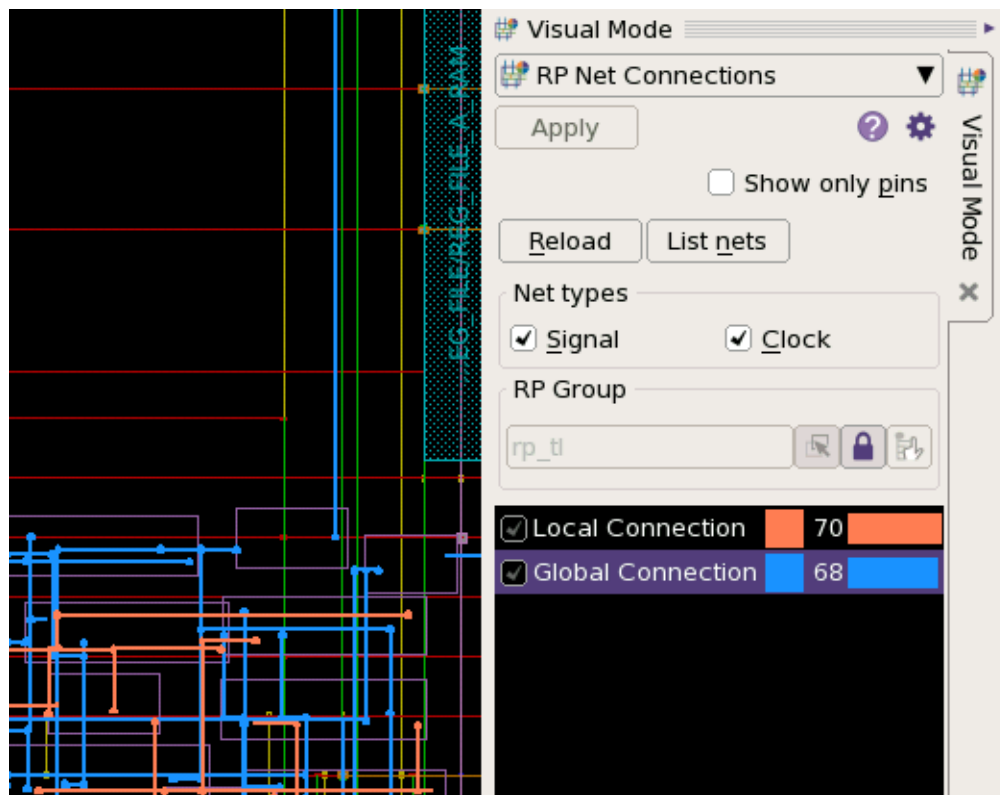
- Highlight the relative placement groups in the design by clicking Reload and clicking OK.
- Highlight the relative placement groups based on group names by selecting Color by RP group.
- Highlight the relative placement groups based on their location in the hierarchy by selecting Color by RP hierarchical level.

- Highlight only the cells in the relative placement groups by selecting the Color cell only option.
- Increase or decrease the number of groups that are highlighted by entering a new value in the Maximum number of colors box; you can enter an integer value between 5 and 20.
- Select or unselect relative placement groups in the layout by right-clicking a histogram bar, choosing Modify Global Selection, and choosing an option to set the selection, append the selection, or remove from selection.
- Change the ordering of the histogram bars or hide histogram bars by choosing the appropriate item in the context menu.

Relative Placement (RP) Net Connection Visual Mode

Use the RP Net Connections Visual Mode to highlight nets that connect relative placement groups. Enable the RP Visual Mode by choosing Highlight > Color By > RP Net Connection from the menu.

Figure 98 *RP Net Connections Visual Mode*

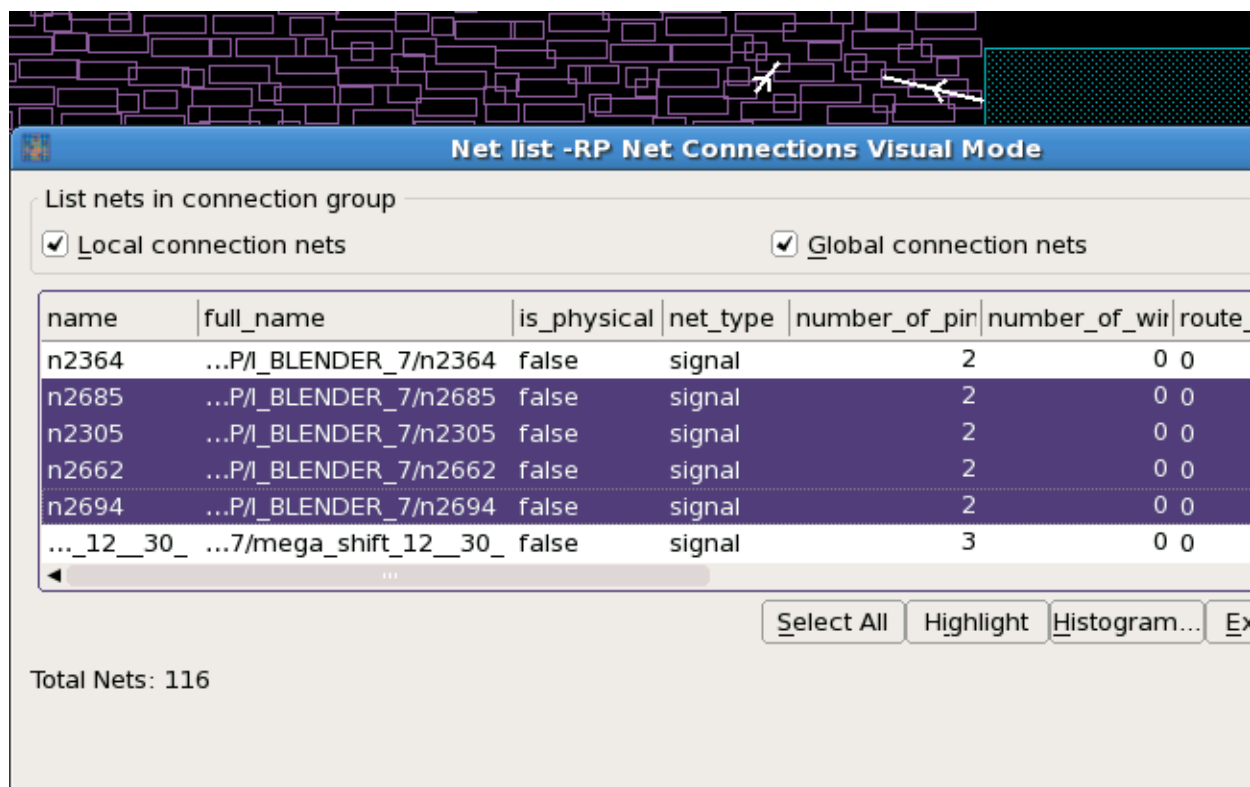


To display RP net connections,

1. Specify one or more RP group names in the RP Group box or click the Browse for RP group button () and select one or more groups.
2. Select the Signal option to display signal nets, the Clock option to display clock nets, or both options to display both types of nets.
3. Click the Reload button to display Net Connections for the specified RP group in the layout.
4. (Optional) Click the List nets button to open a dialog box and display the local and global connections for the specified RP group as shown in [Figure 99](#).

Use the dialog box to select, highlight, or export nets as described in [Net Connections in Area Visual Mode](#).

Figure 99 RP Net Connections List



You can perform the following operations with the RP Net Connections Visual Mode:

- Highlight only the pins connected to the RP net connections by selecting the Show only pins option.
- Show or hide RP net connection flylines by selecting or deselecting the check box next to Local Connection or Global Connection.

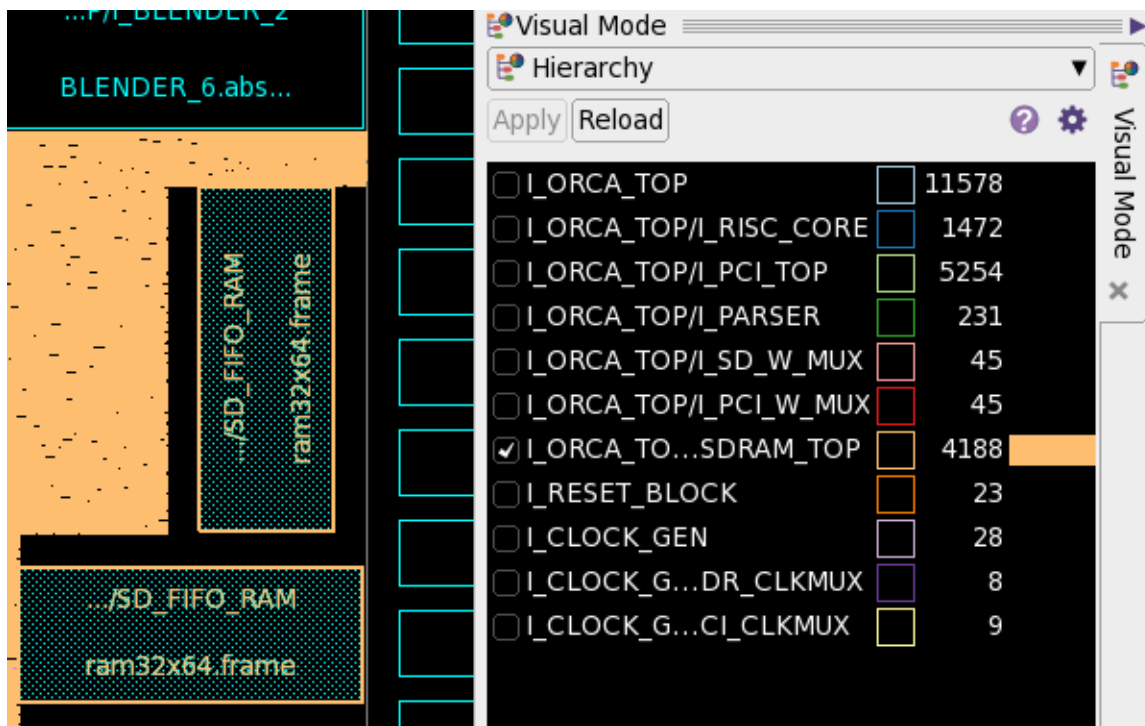
Local connections are nets that connect only to cells in the same relative placement group. Global connections connect at least one cell in the relative placement group to at least one cell outside the relative placement group.

- Select or deselect relative placement net connections in the layout by right-clicking a histogram bar, choosing Modify Global Selection, and choosing an option to set the selection, append the selection, or remove from selection.
- Change the ordering of the histogram bars or hide histogram bars by choosing the appropriate item in the context menu.

Hierarchy Visual Mode

Use the Hierarchy Visual Mode to highlight and select different levels of hierarchy or specific hierarchical cells in your design. Enable the Hierarchy Visual Mode by choosing Highlight > Color By > Hierarchy from the menu.

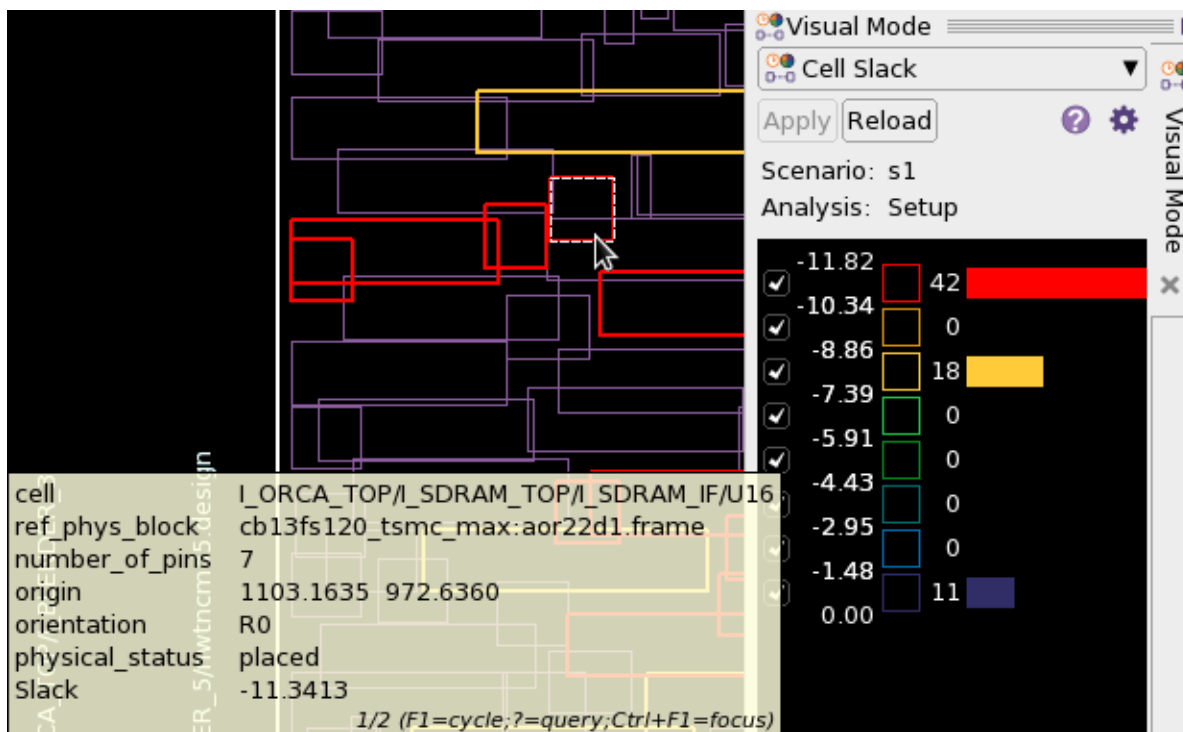
Figure 100 Hierarchy Visual Mode



Cell Slack Visual Mode

Use the Cell Slack Visual Mode to view and overlay information about setup and hold timing slack in your design. Open the panel by choosing Highlight > Color By > Cell Slack from the menu.

Figure 101 Cell Slack Visual Mode



Begin cell slack analysis by clicking the Reload button and selecting options in the dialog box. After setting the options, click OK to generate the data, The Cell Slack dialog box supports the following operations:

- Specify a scenario for analysis by selecting the scenario name from the box.
- Display setup time slack or hold time slack information by selecting the Setup or Hold option.
- Specify the number of bins into which to partition slack results by choosing Number of bins and selecting an integer value greater than zero.

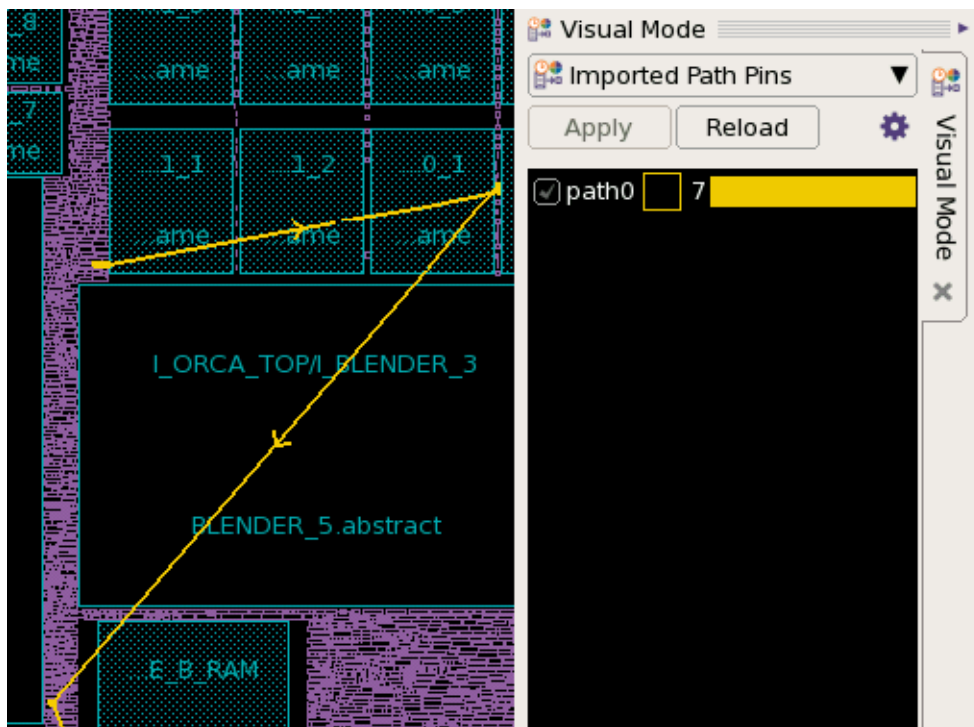
Alternatively, specify a range of bin values by choosing Value range per bin and specifying the minimum and maximum cell slack values. Click the Lower bound strict or Upper bound strict options to remove slack values outside the specified range from the histogram.

- Analyze specific cells by entering a list of cell names in the Cells box, or by selecting the cells and clicking the populate from selection button ().
- Change the color palette used in the histogram by choosing the Temperature or Gradient option.

Imported Path Pins Visual Mode

Use the Imported Path Pins Visual Mode to highlight timing paths in the layout. Enable the Imported Path Pins Visual Mode by choosing Highlight > Color By > Imported Path Pins from the menu.

Figure 102 Imported Path Pins Visual Mode



To import a timing path and highlight the path in the layout,

1. Choose Highlight > Color By > Imported Path Pins to open the Imported Path Pins Visual Mode.

2. Update the timing for your design with the `update_timing` command.

```
fc_shell> update_timing -full
```

3. Report the timing for a path with the `report_timing` command and save the result to a file.

```
fc_shell> report_timing \  
-from I_RESET_BLOCK/sdram_rst_n_buf_reg \  
-to I_ORCA_TOP/I_SDRAM_TOP/I_SDRAM_IF/DQ_out_1_reg_6_ -nosplit \  
-physical > timing.rpt
```

4. Click the Reload button.

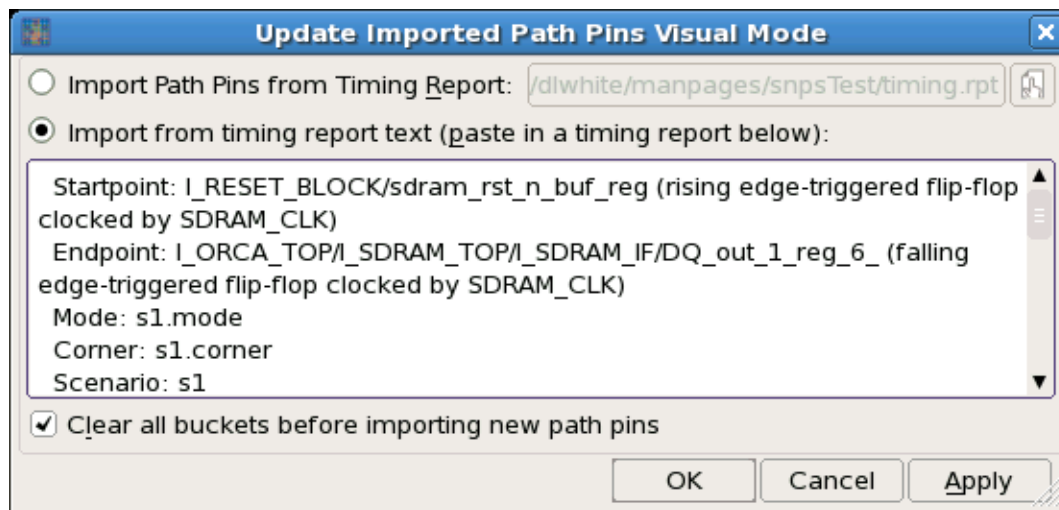
5. Select Import Path Pins from Timing Report and click the file chooser button.
6. Select the timing.rpt file you wrote out in step 3 and click OK.

The Imported Path Pins Visual Mode panel displays a histogram bar for the path with the number of pins in the path. The tool also displays the flylines for the path in the layout view.

7. Repeat steps 3 through 6 to import and display additional paths. Choose the Clear all buckets before importing new path pins option to delete the histogram before importing the new path.

Instead of selecting the Import Path Pins option and importing the file as shown in steps 5 and 6, you can select Import from timing report text and paste the contents of the timing report into the text box as shown in [Figure 103](#).

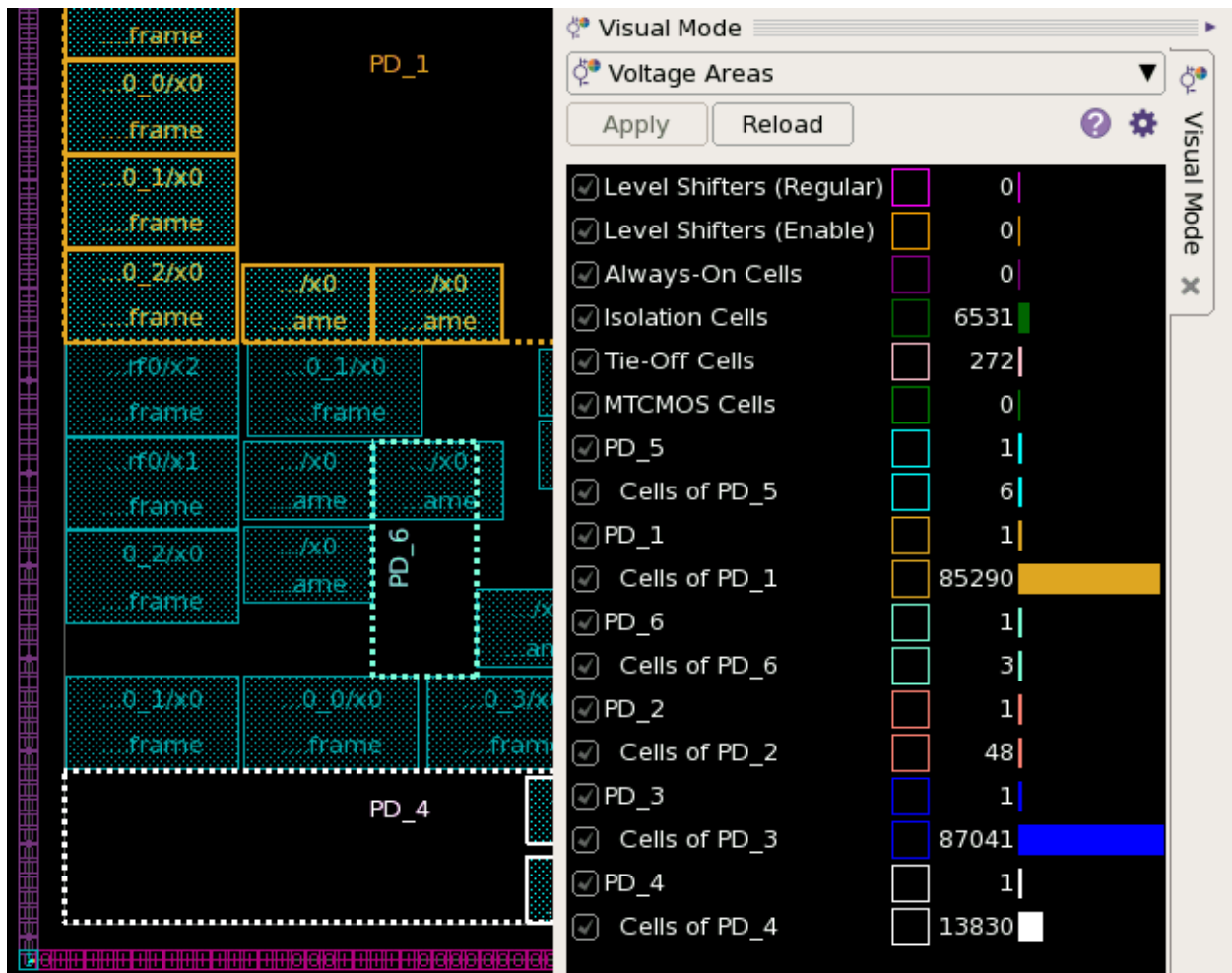
Figure 103 Import From Timing Report Text



Voltage Areas Visual Mode

Use the Voltage Areas Visual Mode to highlight and select power domains, level shifters, isolation cells, and other cells related to the power domains in your design. Enable the Voltage Areas Visual Mode by choosing Highlight > Color By > Voltage Areas from the menu.

Figure 104 Voltage Areas Visual Mode



Click the Reload button to populate the Voltage Areas Visual Mode panel. Right-click a histogram bar and choose an item from the context menu to change the color, move the bar up or down in the list, select related objects in the layout, show or hide the histogram bar, and perform other operations.

7

Viewing Schematics

The Fusion Compiler tool supports design schematics to examine logic elements (blocks and gates) in a design. A design schematic displays a graphic representation of a hierarchical cell in a schematic view. You can move up or down in the design hierarchy to view the schematic for each design in the hierarchy.

The following topics describe the schematic view:

- [Schematic View](#)
- [WordView in the Schematic View](#)
- [Annotating Attributes on the Schematic View](#)
- [Adding Pin Markers to the Schematic View](#)

Schematic View

To view a schematic, use one of the following methods:

- Choose View > Schematic View in the main menu bar or in the Hierarchy Browser
- Click the  button on the toolbar

Use schematic views to visually analyze timing and logic for use in further optimization operations. For example, you can

- View graphic representations of the logic elements (blocks and gates) in a design or subdesign
- Examine timing problems by focusing on critical paths in your logic design

To customize a schematic view, use the View Settings panel located in the tab on the right side of the tool.

See Also

- [Setting and Saving View Properties](#)

WordView in the Schematic View

WordView models wide combination logic gates that can be easily represented in schematic views. In the GUI, WordView allows for a cleaner schematic view of unmapped logic. The tool uses a wide logic gate instead of cascaded gates, and it shows bused instances in a collapsed representation. For purposes of area estimation, WordView more accurately models the original logic represented in the RTL.

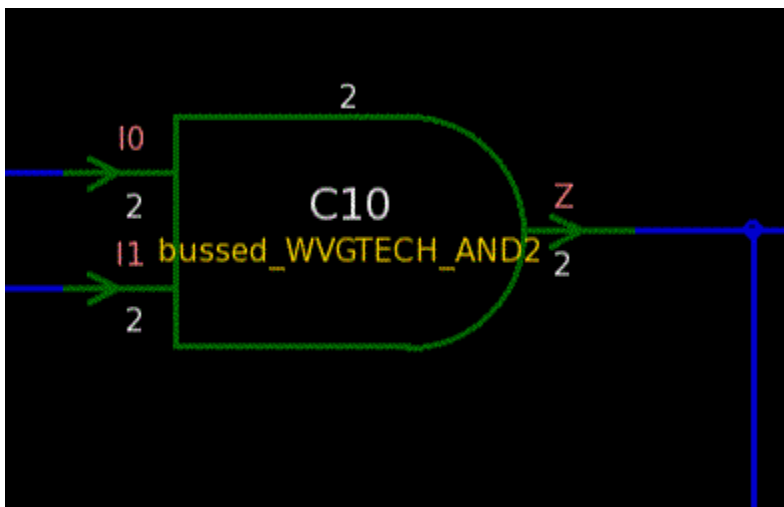
WordView uses a WVGTECH library as opposed to a GTECH library. In order for the tool to infer these gates, you must have a WVGTECH library available.

When you are in the Schematic View, the GUI uses WordView gates to display:

- Bused instances
- Wide gates
- Operators
- Bused registers to multibit cells

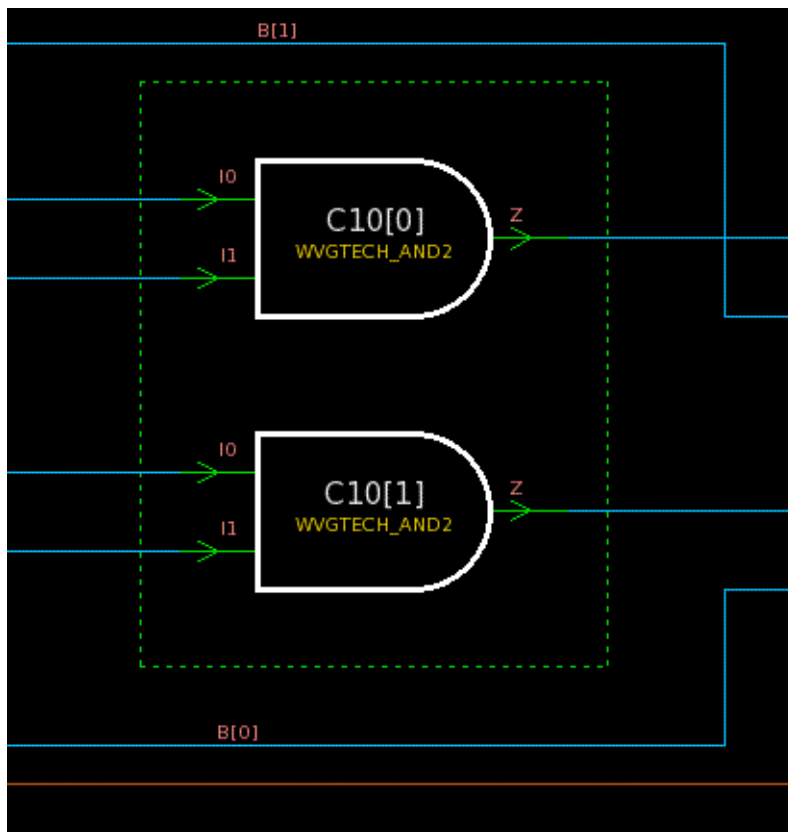
The example in [Figure 105](#) shows a collapsed bused instance. In this case, it is a WVGTECH_AND gate with four inputs pins.

Figure 105 Collapsed View of a Bused Instance



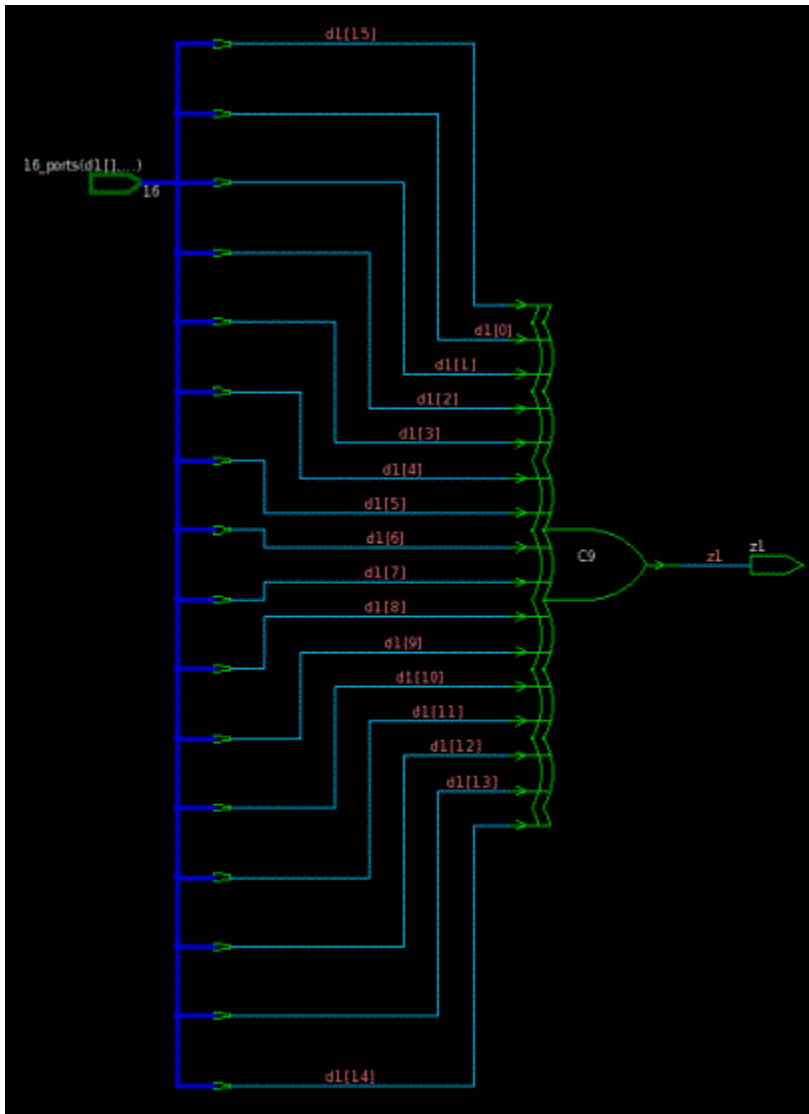
You can expand the WVGTECH gate by selecting it and opening the context menu and selecting the Expand > Selected Objects option. By default, bused instances are displayed in the GUI with a dashed border (see [Figure 106](#)). Bused nets are shown with one thick, blue line and bus inverters have the same logic symbol with a thick, green line. You can change the view from scalar to bus representation for a bused object using the Schematic View's context menu and select the Expand or Collapse menu options.

Figure 106 Expanded View of a Bused Instance



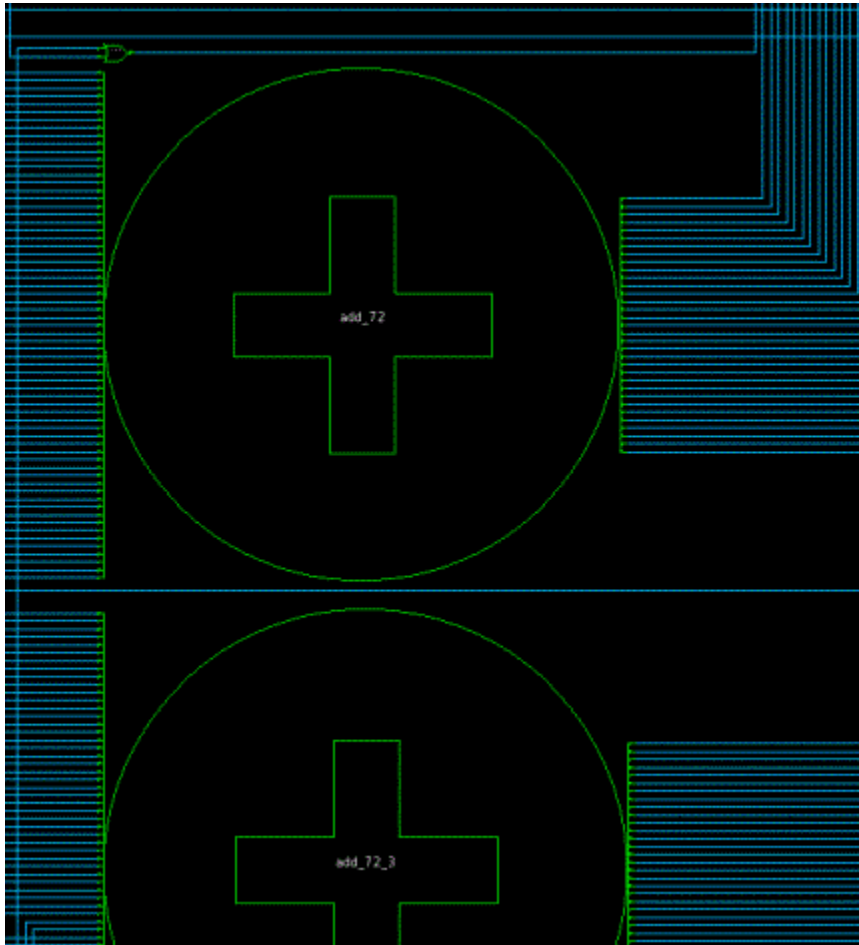
An example of the GUI representation of a WordView wide gate is shown in [Figure 107](#). The XOR gate has 16 input pins and is named WVGTECH_XOR16.

Figure 107 Example of Wide Gates



The GUI represents operators as basic symbols as shown in [Figure 108](#). In this example, the tool is representing the addition operator.

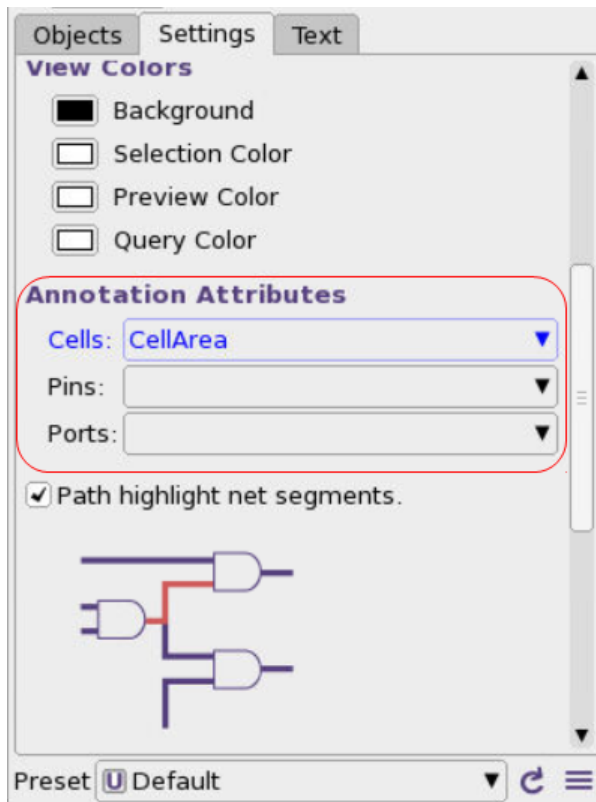
Figure 108 Example of Operators



Annotating Attributes on the Schematic View

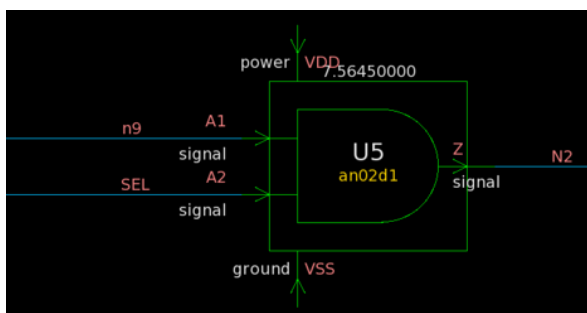
To aid in debugging, you can annotate attributes on cells, pins, and ports in the schematic view. To do this, select the attribute to annotate in the Annotation Attributes section of the Settings tab on the View Settings panel, as shown in [Figure 109](#). You can annotate only a single attribute on an object.

Figure 109 Annotation Attributes Section of the Settings Tab



In the following figure, the attribute annotations are shown in white. The `area` attribute is annotated on the cell and the `port_type` attribute is annotated on the pins.

Figure 110 Attribute Annotations on Cell and Pins



You could use this capability to debug a timing violation by annotating an attribute such as `late_rise_transtion` on the pins on the path or to analyze the power by annotating an attribute such as `switching_activity` or `toggle_rate` on the pins on the path.

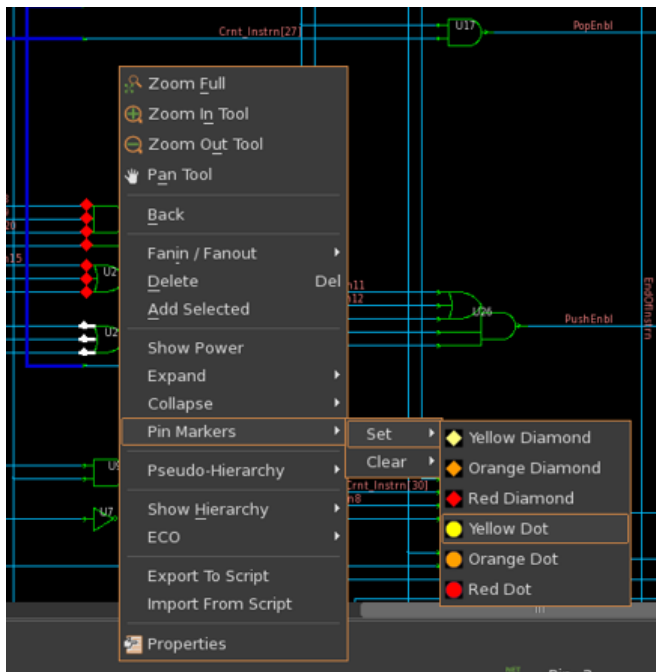
Adding Pin Markers to the Schematic View

To highlight content in the schematic view, you can add markers on pins in the schematic view. When you add a marker, you can select the following characteristics of the marker:

- Shape: diamond or dot
- Color: yellow, orange, or red

To add a marker to a pin, right-click the pin, and choose Pin Markers > Set > *marker_type*, where *marker_type* is one of Yellow Diamond, Orange Diamond, Red Diamond, Yellow Dot, Orange Dot, or Red Dot. [Figure 111](#) shows the interface for adding pin markers.

Figure 111 Adding Pin Markers



8

RTL Cross Probing

The Fusion Compiler tool offers cross-probing capabilities so you can review, verify, and check your RTL design. You can cross-probe cells, pins, or timing paths in a design and examine the corresponding RTL file.

You can cross-probe between a netlist and source file using the following GUI windows:

- Hierarchy browser
- Schematic view
- Layout view
- Source browser window

Note:

To use the cross-probing capabilities, you must have an elaborated or synthesized design.

Cross-Probing From the Hierarchy Browser

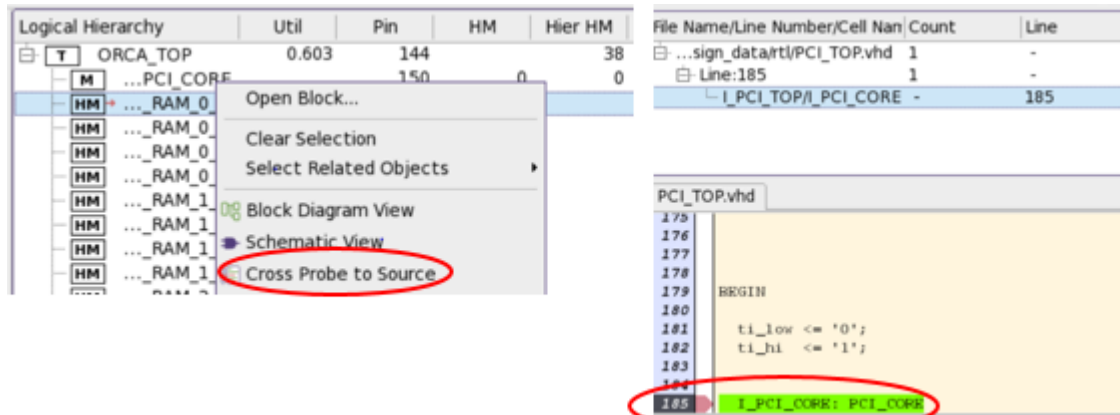
To cross-probe objects in the hierarchy browser,

1. Select an object to cross-probe.
2. Right-click to choose Cross Probe to Source.

The tool locates the RTL file in which the object originates and opens it in the source browser window. A red arrow points to the highlighted RTL code of the object, as shown in the following figure:

Chapter 8: RTL Cross Probing

Cross-Probing From the Schematic View

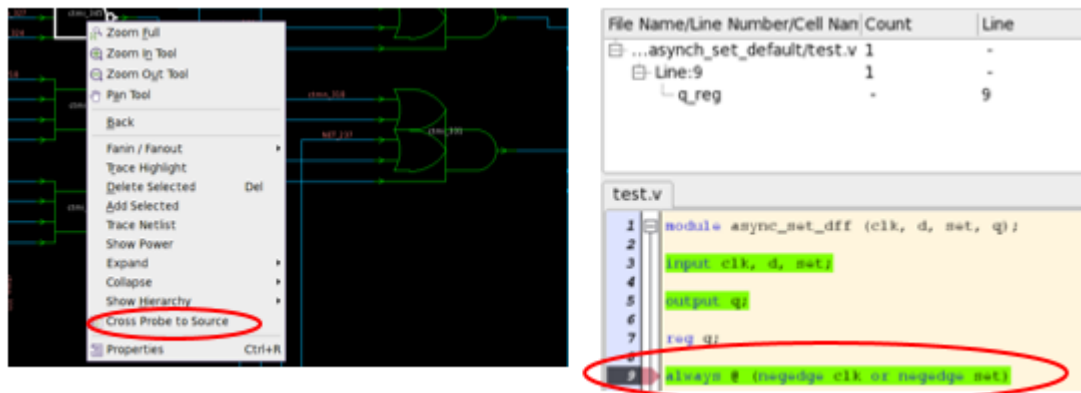


Cross-Probing From the Schematic View

To cross-probe an object in the schematic view,

1. Select an object to cross-probe.
2. Right-click to choose Cross Probe to Source.

The tool locates the RTL file in which the object originates and opens it in the source browser window. A red arrow points to the highlighted RTL code of the object, as shown in the following figure:

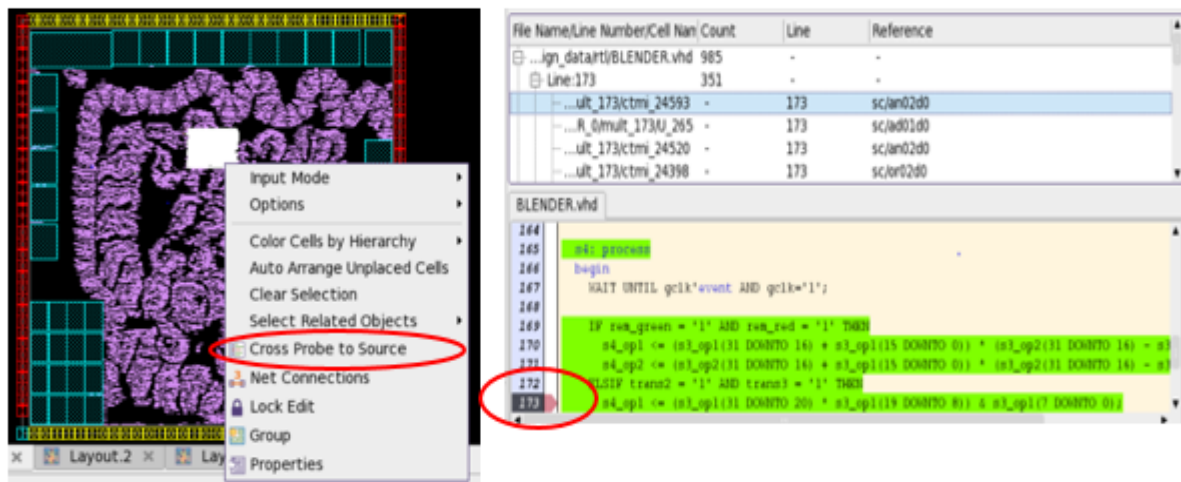


From the Layout View

To cross-probe an object in the layout view,

1. Select an object to cross-probe.
2. Right-click to choose Cross Probe to Source.

The tool locates the RTL file in which the object originates and opens it in the source browser window. A red arrow points to the highlighted RTL code of the object, as shown in the following figure:



From the Source Browser Window

To cross-probe an object in the source view,

1. Choose Open RTL Files from the menu.
2. Select a file to cross-probe from the Open RTL Files dialog box and click Open.

The source browser window opens and the available cross-probing lines are highlighted.

1. Select the line to cross-probe.
2. Right-click and choose Select Objects of Selected Line(s).



The inferred objects are cross-highlighted in the open windows, such as the hierarchical browser, schematic, and layout windows.

Command-Line Cross-Probing

Use the following commands to cross-probe from the command line:

- `report_cross_probing`: Reports cross-probing data for the specified cells and ports.

```
fc_shell-fe> report_cross_probing \
  [get_cells {I_CONTEXT_MEM/ram_read_addr_reg[0]}]

*****
Report : report_cross_probing
...
*****
Type      Reference  FileName          Line  Origin  Object
-----
cell      sdcrq1      CONTEXT_MEM.vhd   64    rtl     I_CONTEXT_MEM/
                                             ram_read_addr_reg[0]
FileName      Full Pathname
-----
CONTEXT_MEM.vhd /slowfs/myfile/ORCA/design_data/rtl/CONTEXT_MEM.vhd
```

Sometimes a cell undergoes a series of operations, such as ungrouping, name changes, and optimizations that collapse the hierarchy and rename the cell. To display the original RTL hierarchical names of specific cells and identify where the cells originated, use the `-original_hierarchy` option.

For example,

```
fc_shell-fe> report_cross_probing [get_cells U1*] -original_hierarchy
*****
Report : report_cross_probing
...
*****
Type      Reference      FileName      Line      Origin      Object/
          Reference      Pathname      Number      Name        Original Hierarchy
-----
cell      HS65_STF_AOI          test1.v      21536      rtl          U129
          HS65_STF_AOI          test1.v      21536      rtl          I_CORE/I_SUB1
cell      HS65_STF_AND2        test2.v      21548      rtl          U104
          HS65_STF_AND2        test2.v      21548      rtl          I_CORE/I_SUB2

Filename      Full Pathname
-----
test1.v        /slowfs/mytest/rtl/test1.v
test2.v        /slowfs/mytest/rtl/test2.v
```

- `report_cross_probing_files`: Reports a list of cross-probing files and their status.

```
fc_shell-fe> report_cross_probing_files
*****
Report : cross_probing_files
...
*****
Status:
    OK - No issues found    Missing - File is missing
    No Permissions - User has no read permissions

-----
Filename      Status
/slowfs/myfile/small_test/design_data/rtl/ORCA_TOP.vhd    OK
```

- `get_cross_probing_info`: Gets cross-probing information about cells and ports and returns the results as a string.

```
fc_shell-fe> get_cross_probing_info [get_cells -hierarchical]
{ {/usr/design/m.v:21} } { {/usr/design/m.v:22} } {
{/usr/joe/design/m.v:21} }
{ {/usr/design/m.v:22} } { {/usr/design/m.v:28} } {
{/usr/design/m.v:12} }
{ {/usr/design/m.v:13} } { {/usr/design/m.v:14} } {
{/usr/design/m.v:4} }
```

- `cross_probing_filter`: Filters a collection of cells based on the file name and line number.

Chapter 8: RTL Cross Probing

Command-Line Cross-Probing

```
fc_shell-fe> cross_probing_filter [get_cells -hierarchical] \  
-source "*m.v:21"  
{mid1/bot1/c1 mid1/bot2/c1}
```

9

Timing Analysis Using the GUI

Use the GUI to view timing analysis results and perform interactive queries on problematic timing paths.

The following topics describe timing analysis in the Fusion Compiler tool.

- [Opening the Timing Analysis Window](#)
- [Displaying the Timing Analysis Endpoint Summary](#)
- [Creating Path Collections](#)
- [Analyzing Logic Levels](#)

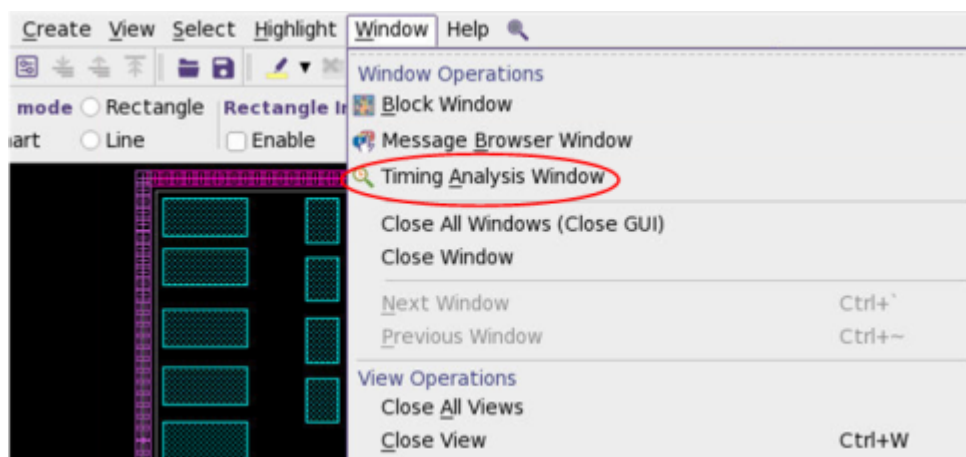
Opening the Timing Analysis Window

After you have compiled your design, you can review timing analysis results by using the GUI.

To gather timing analysis results using the GUI,

1. Use the Timing Analysis window to get the endpoint summary information.

In the main menu bar, select Window > Timing Analysis Window.



2. Create a collection of paths per endpoint summary

3. Analyze the timing slack for the selected collection of the paths

See Also

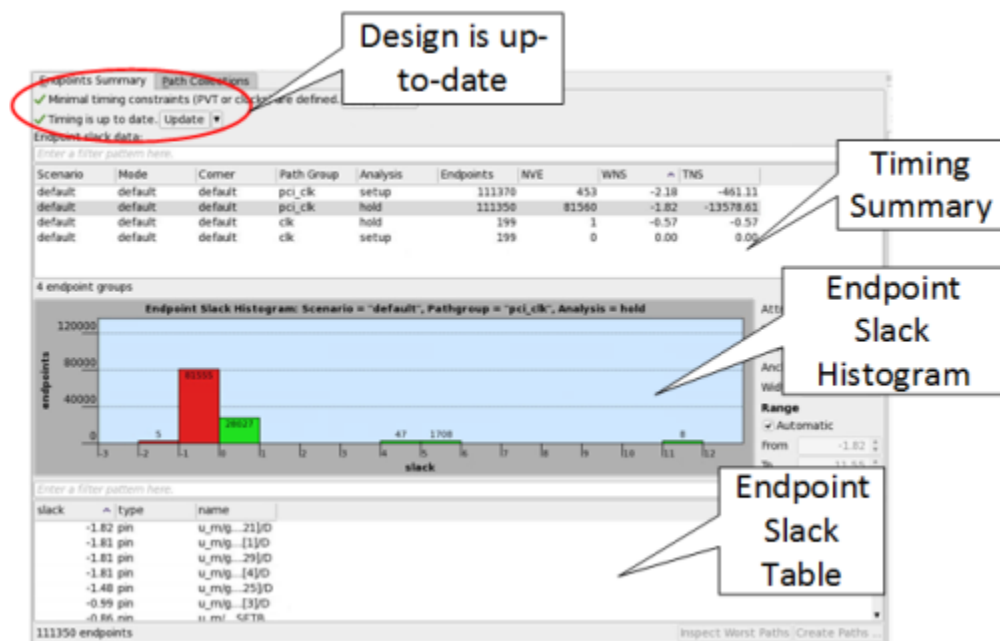
- [Displaying the Timing Analysis Endpoint Summary](#)
- [Creating Path Collections](#)
- [Analyzing Logic Levels](#)

Displaying the Timing Analysis Endpoint Summary

The Timing Analysis Window shows the timing summary of the current design. This should match the results of running the `report_qor -summary` command.

Before using this information for analysis, make sure that the timing is up-to-date. The green check mark indicates that the timing is up-to-date. If it is not, a red mark is displayed. To update the timing, click the Update button.

Figure 112 Timing Analysis Summary Window

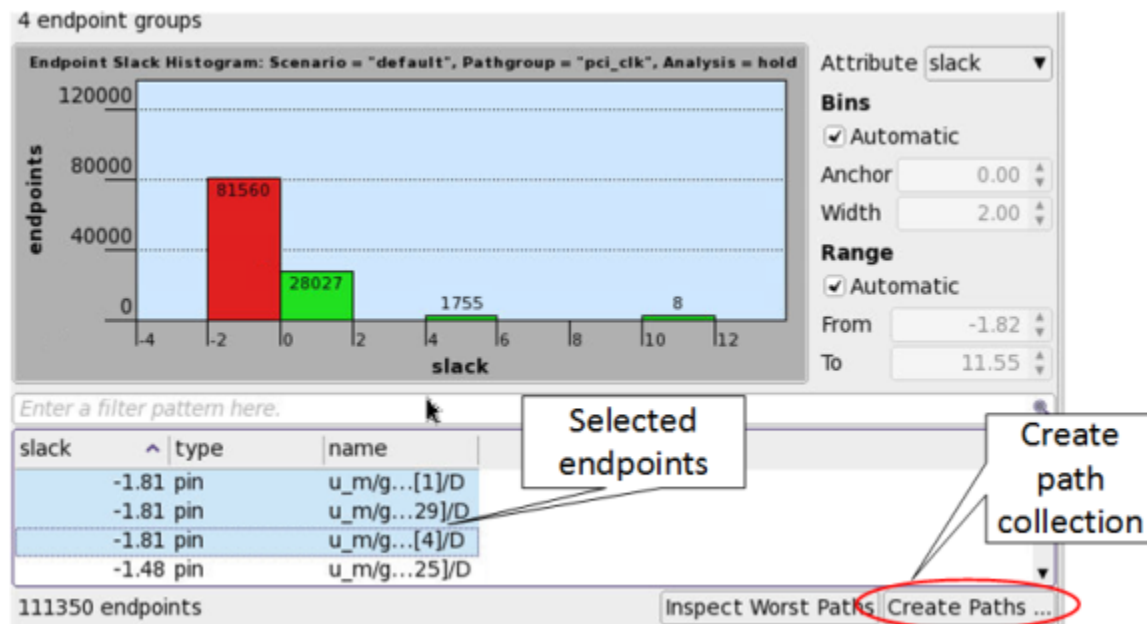


Creating Path Collections

When you are in the Timing Analysis Window and select the Endpoints Summary tab, you can select which paths to analyze by creating a path collection. Do this by selecting

the endpoints in the Endpoint Slack Table at the bottom of the window. Then, select the Create Paths button as shown in [Figure 113](#).

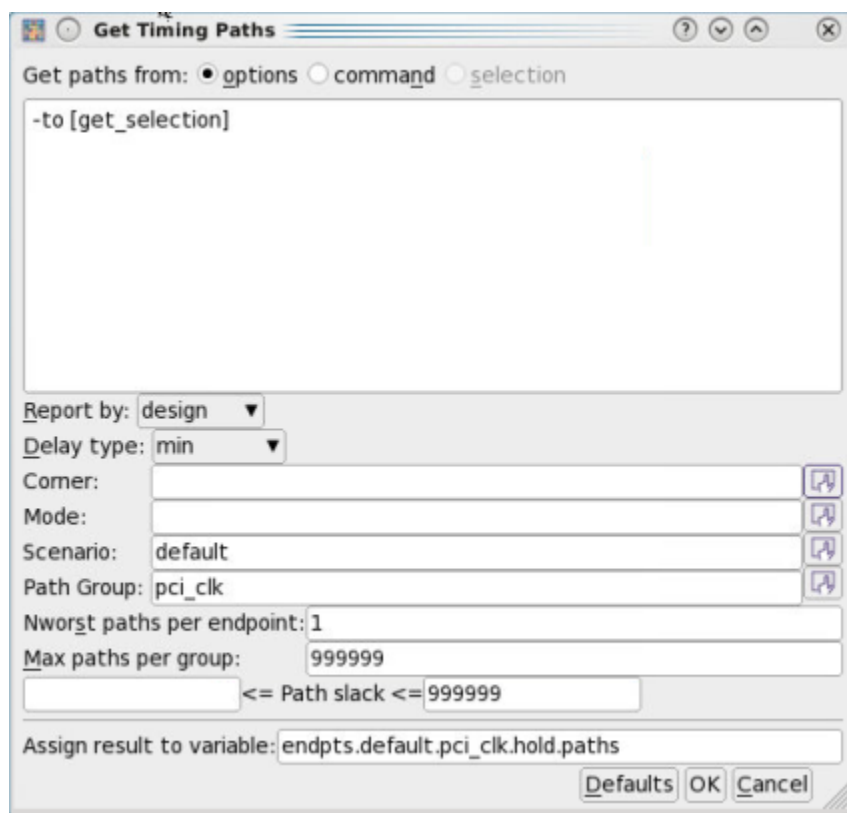
Figure 113 Creating Path Collections



Getting Timing Paths

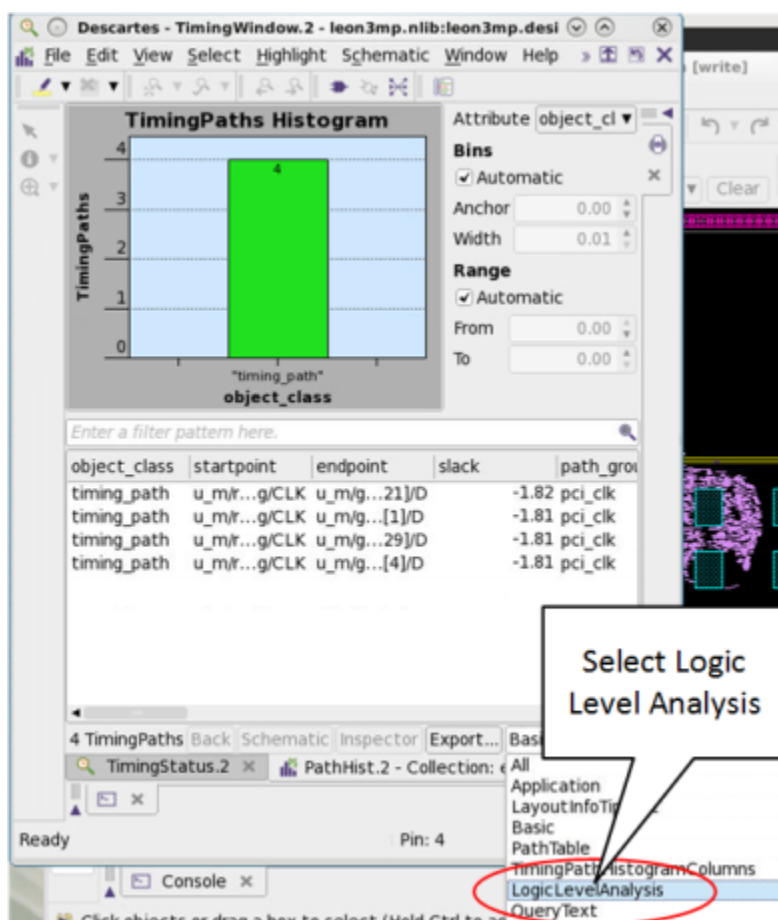
After you select the Create Paths button, the Get Timing Paths dialog box appears as shown.

Figure 114 Get Timing Paths Dialog Box



Use this dialog box to fill out the options to report paths by design, group, scenario, mode, or corner. Clicking the OK button populates the summary of endpoints in histogram format. You can choose to perform logic level analysis on the summary of endpoints displayed by selecting the Logic Level Analysis option from the pulldown menu as shown in [Figure 115](#).

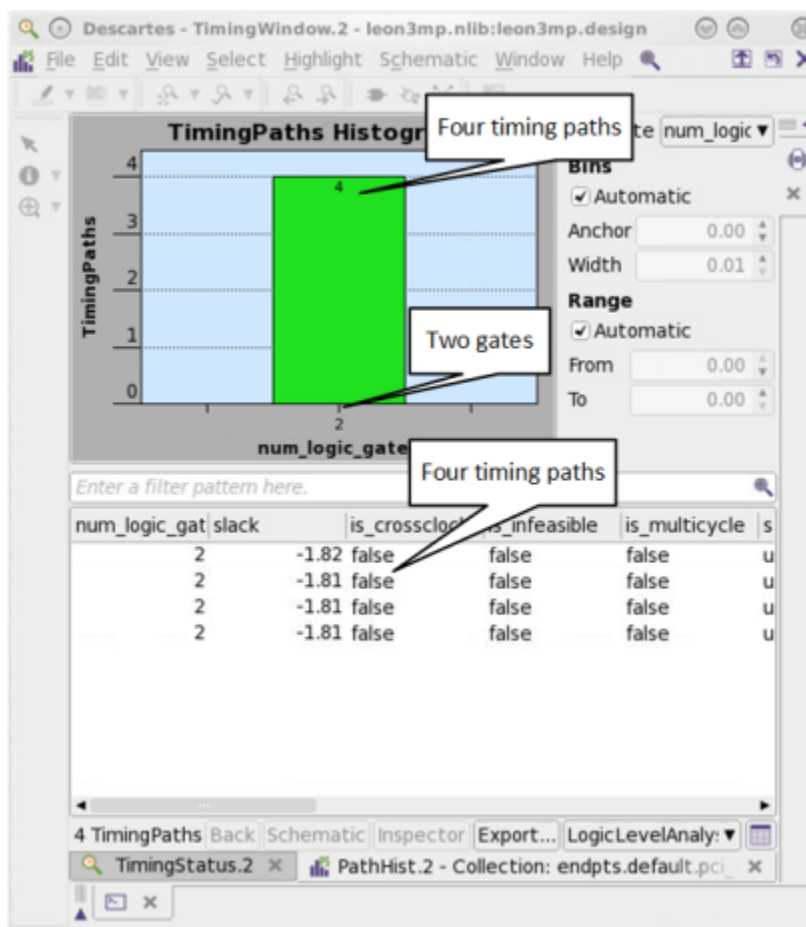
Figure 115 Selecting Logic Level Analysis



Analyzing Logic Levels

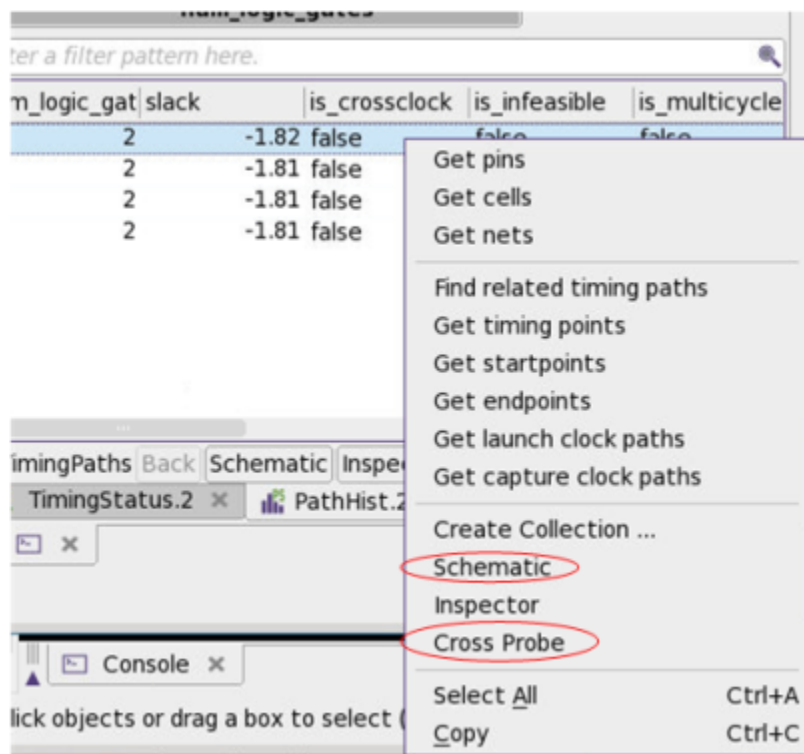
When you perform logic level analysis in the GUI, the path collection window shows an updated histogram of timing paths.

Figure 116 Path Collection Window With Logic Levels



If you select a path in the endpoint slack table and then open the context menu, you can select options for further investigation such as viewing the schematic of the selected path or bringing up the associated RTL using the Cross Probe option.

Figure 117 Topics for Further Analysis



10

Using UPF and Power Analysis Tools

For multivoltage designs, the Fusion Compiler GUI provides a UPF viewer, power state table view, and other tools to view the UPF power intent for your design and support the analysis of your design.

The following topics describe the UPF and power analysis tools:

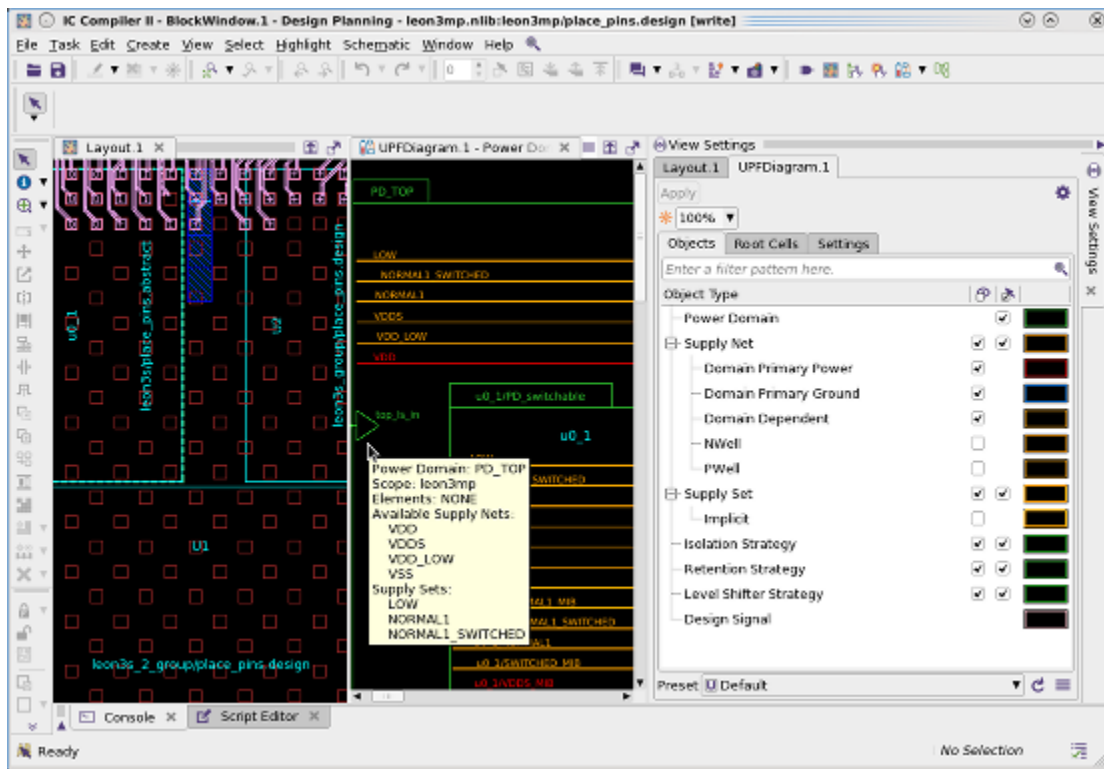
- [Viewing the UPF Power Intent](#)
- [Viewing the Power State Table](#)

Viewing the UPF Power Intent

The UPF diagram presents a graphical representation of the UPF power intent for your design. The UPF diagram displays UPF power domains, supply networks, power switches, isolation and retention strategies, and other elements of your UPF power intent. The UPF diagram supports zooming and panning, object selection, and querying of objects.

To view the UPF diagram, choose View > Multivoltage UPF > Power Domain / Root Cell from the menu, or click the Power Domain button (). [Figure 118](#) shows the GUI with the layout, UPF diagram, and view settings displayed.

Figure 118 UPF Diagram View



Power Domain View

When you first open the UPF diagram, the tool displays the power scope hierarchy in the power domain view. The power domain view displays information organized according to the power domain hierarchy. Power domains within the top-level power domain are represented as child modules of the top-level. Click on a power domain to select it, or right-click the power domain to select related cells, macros, or voltage areas, show the supply network view, or show the power state table.

For each power domain, supply sets are displayed near the top of the rectangle; each thick orange line represents a unique supply set. Click the line to select it. Right-click the line to view the context menu, then choose Show Supply Network View or Power State Table to view additional information about the supply set. Supply nets are displayed below the supply sets; each thin orange line represents a supply net.

Use the zoom and pan tools and commands to magnify and traverse the view. Use the arrow keys to scroll vertically or horizontally through the view. Select individual objects or objects in a rectangular area by using the Selection tool. Preview object information in an infotip by holding the pointer over the object. Display object information in the Query panel by using the Query tool.

Click the UPFDiagram tab in the View Settings menu to view the mapping of object type and color used in the power domain view. Click on a color square to customize the color for that object type.

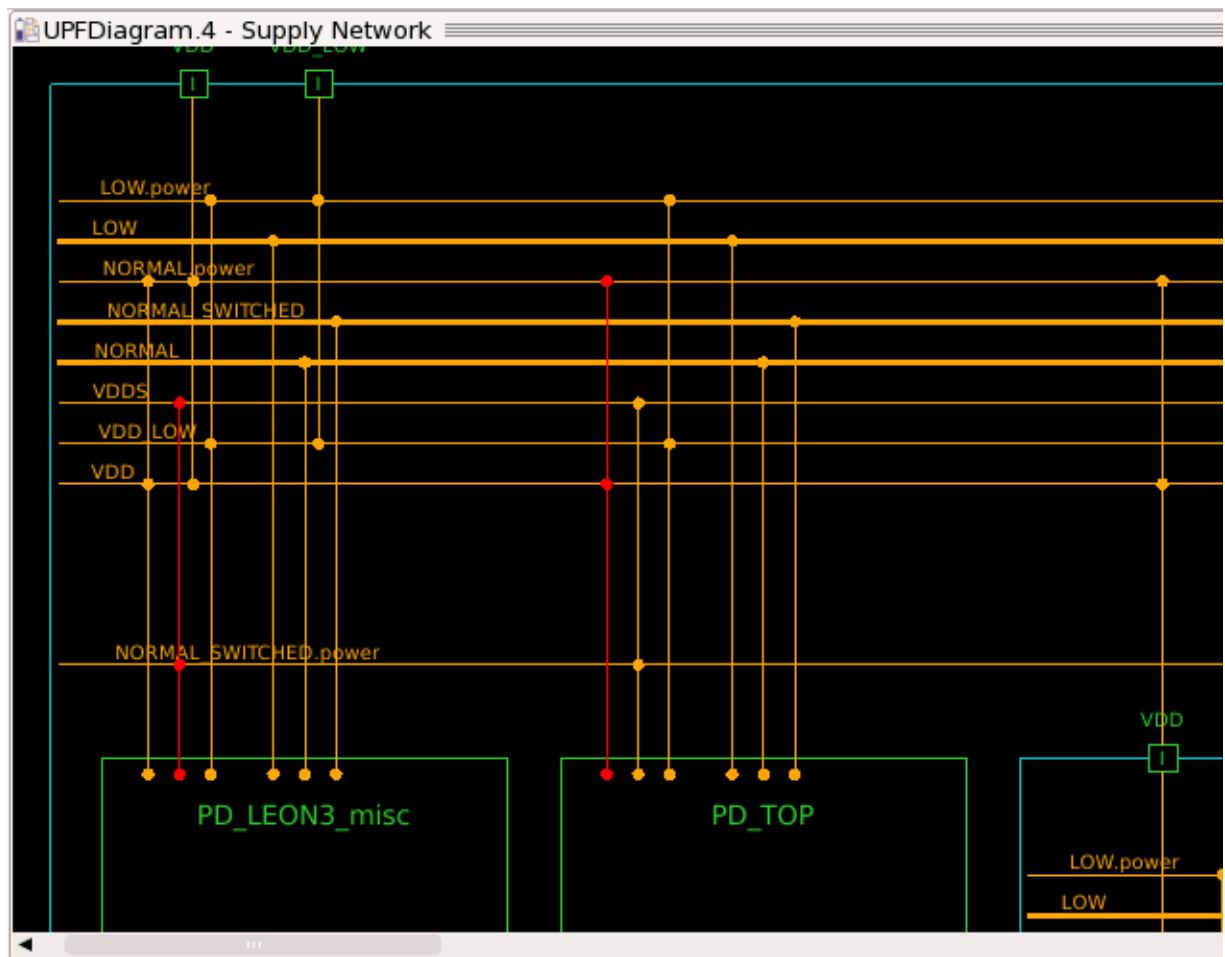
If isolation cells are present, they are displayed below the supply nets. Hold the pointer over the isolation cell to view additional information in the infotip. Click the line to select it, or right-click the line to view the context menu. Use the context menu to select related objects or show the Supply Network View or the Power State Table. Ground nets are displayed at the bottom; each thin blue line represents a ground net.

Ports are displayed along the boundaries of the power domains. Input ports are denoted with the letter "I", and output ports with the letter "O".

Supply Network View

The supply network view provides an alternate view of the power domains in your design. The view displays the connections between different power domains, isolation cells, switch cells, and other elements of the power domain. Similar to the power domain view, the supply network view organizes the view by power scope hierarchy. Right-click anywhere in the view and choose Show Supply Network View to show the supply network view as shown in [Figure 119](#).

Figure 119 Supply Network View



Viewing the Power State Table

You can use the Power State Table panel with the UPF diagram view to analyze and debug your isolation and level-shifter strategies. The tool displays the states for each supply in a power state table and shows their relationship in the UPF diagram view.

[Figure 120](#) shows the Power State Table panel.

Figure 120 Power State Table

Name	Scope
LEON3_MP_...	leon3mp
pst	U2/u0_0
pst	U2
pst	U1/u0_3
pst	U1/u0_2
pst	U1
pst	u0_1

Highlight mode: None

☐ Show state type/voltage

	VDD	VDDS	PD_switchable.primary.power	VDD_LOW	VSS
state_1	NORMAL	misc_on	pds_on	LOW	GND
state_2	NORMAL	misc_on	pds_off	LOW	GND
state_3	NORMAL	misc_off	pds_off	LOW	GND

The Power State Table panel supports the following analysis types:

- Always-on analysis compares the on-off states between power and ground supplies

Always-on analysis compares the on-off states between supplies, including both power and ground supplies. This analysis produces one of the following states: More AO, Less AO, Equally AO, and Unrelated AO.

- Multivoltage level-shifter analysis compares the voltage relationships between power supplies

Multivoltage level-shifter analysis compares the voltage relationships between supplies. This analysis produces one of the following states: LH, HL, HL_LH, or None.

11

Using the Error Browser

The Fusion Compiler GUI provides the error browser tool to view error and informational messages, highlight related objects in the layout view, and disposition errors. Using the error browser tool, you can quickly select a group of errors to review, locate a particular error, and investigate the problem in the layout.

The following topics describe the error browser tool:

- [Error Browser Overview](#)
- [Loading and Saving Error Data](#)
- [Managing the Error Hierarchy](#)
- [Managing the Error List and Viewing Error Details](#)
- [Using Error Filters](#)
- [Highlighting Errors](#)
- [Marking Errors as Fixed and Changing Error Status](#)
- [Writing Out DRC Error Data](#)

Error Browser Overview

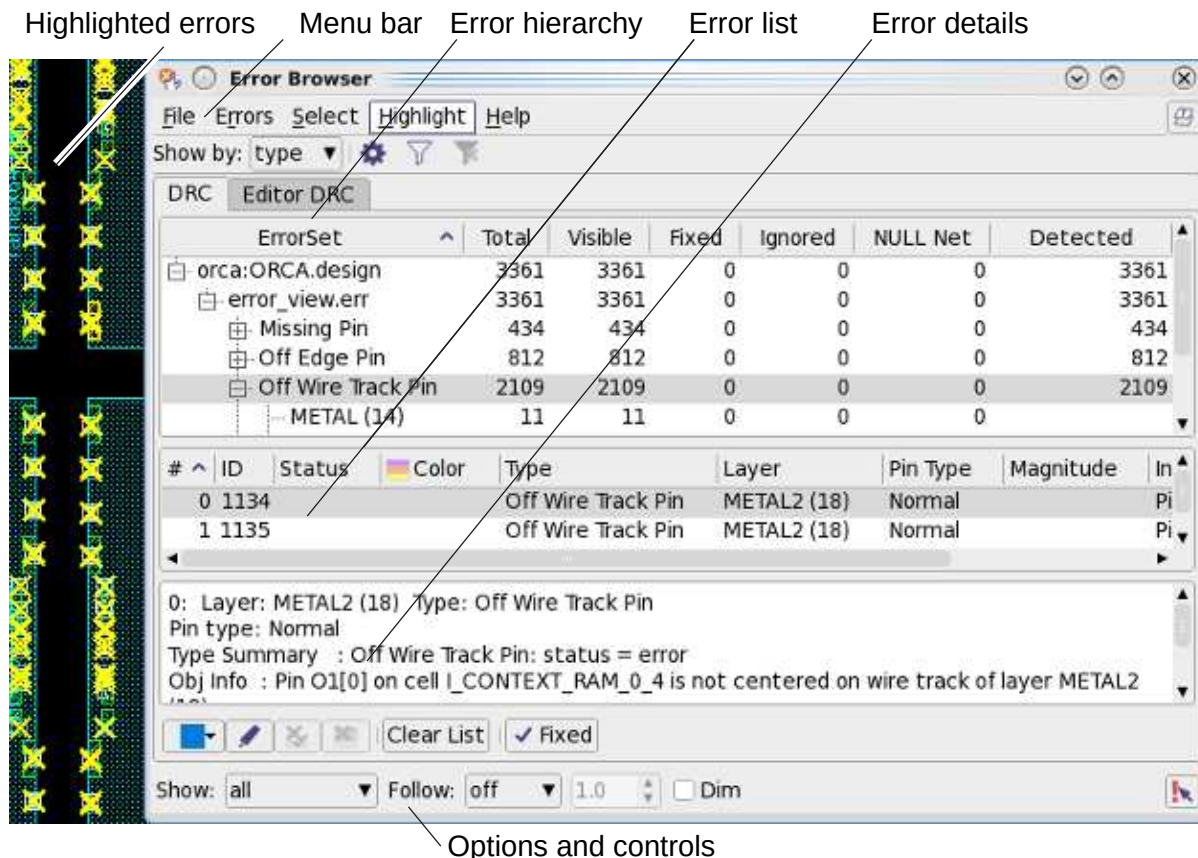
The error browser provides functions to help you review errors, disposition errors, and save the updated error file to disk. As you select errors in the error browser, the layout viewer pans to the error location and zooms in to isolate the error in the view. Note that only the active layout view changes when the error browser zooms or pans to display the selected errors.

The error browser groups together errors based on error types, where error types are assigned by the GUI task or Tcl command that generated the error. Each error contains a brief description, bounding box coordinates for the error, and other information about the error.

To start the error browser, chose View > Error Browser. When you open the error browser for the first time, the tool displays a dialog box containing the list of available error message files, the checking task that generated the file, and the time the file was last

modified. Select an entry in the list and click Open Selected to continue. The error browser is shown in [Figure 121](#).

Figure 121 Error Browser



A typical flow for debugging errors with the error browser includes the following steps:

1. Verify the design and save the results.
2. Open the saved error file in the error browser.
3. Select a group of errors to review.

You can group errors by either error type or the layer associated with the errors. Alternatively, choose a group of errors by their physical location in the design.

4. Analyze and debug each error by selecting the error in the list and viewing the error context in the layout view.

5. Correct the error in the layout view and mark the error as fixed or ignored in the error browser.
6. (Optional) Hide the fixed errors.

Loading and Saving Error Data

By default, the error browser reads and writes error data files associated with the current block. The files are stored with a .err extension and are saved in the library directory for the block. You can also save and load an XML-format error data file not associated with the block.

To open an existing error data file that is associated with the current block,

1. Choose File > Open to open the Open Error Data Dialog Box.
2. Select the error data file to open.
3. Click the Open Selected button.

To open an error data file with a .err extension or an unmanaged XML-format error data file,

1. Choose File > Read Error File to open the file browser.
2. Select the file that you want to open.

If your file has a .xml extension, choose Any files in the Files of type box in the Open Error Data File dialog box.

3. Click OK.

To save changes to the error data after updating the fixed status or making other changes,

1. Choose File > Save.
2. Click the Save button.

The tool writes the current state of the Error Browser to the output file. This menu operation is disabled if there are no open error data files or if there are no changes to the error data.

To save the error data in other formats,

- Choose File > Export as CSV to write out the error data as a comma-separated values.
- Choose File > Export to Error File to write out the information as error data.

Managing the Error Hierarchy

The error browser displays two tabs below the menu bar: DRC and Editor DRC. The list in the DRC tab shows the hierarchy of errors in the current error data file. You can perform the following operations with the error hierarchy list:

- Click the [+] button to the left of the ErrorSet column to expand the list, or click the [-] button to collapse the list.
- Click a column header to sort the column or change the sort order for the column.
- Move the pointer over a row in the list to display information in the infotip.
- Click a row in the list to pan and zoom to the error in the layout view, and display additional information in the error list section of the browser.
- Click a row, hold down the Shift key, and click another row further down the list to select the range of rows.
- Click a row, hold down the Ctrl key, and click a second row to select multiple rows. Continue holding the Ctrl key and click additional rows to add them to the selection.
- Right click the mouse to display a context menu; select an item from the menu to open new error data, save the current error data, mark the current error as fixed, and so on.
- Choose Errors > Next Error Group or Errors > Previous Error Group from the Error Browser menu to navigate to another error group in the error hierarchy.

Managing the Error List and Viewing Error Details

Below the error hierarchy list, the tool displays the error list. The error list contains a list of individual errors for the error type you selected in the error hierarchy section of the tool. You can perform the following operations with the error list:

- Click a column header to sort the column or change the sort order for the column.
- Click and drag a column header to reorder the columns.
- Move the pointer over a row in the list to display information in the infotip.
- Click a row in the list to pan and zoom to the error in the layout view, and display additional information in the error list section of the browser.
- Right click the mouse to display a context menu; select an item from the menu to toggle the fixed state of the error, highlight errors, and so on.
- Hold the Ctrl or Shift key and click additional rows in the list, or click and drag the mouse down the list to select more than one error.

- Clear the list by selecting the context menu item or by choosing Errors > Clear Error List
- Choose Errors > Next Error Group or Errors > Previous Error Group from the Error Browser menu to navigate to another group in the error list.

The error browser displays details about the selected errors in the error detail section below the error list. Hold the pointer over the error detail pane to display an infotip that contains the error detail information.

Using Error Filters

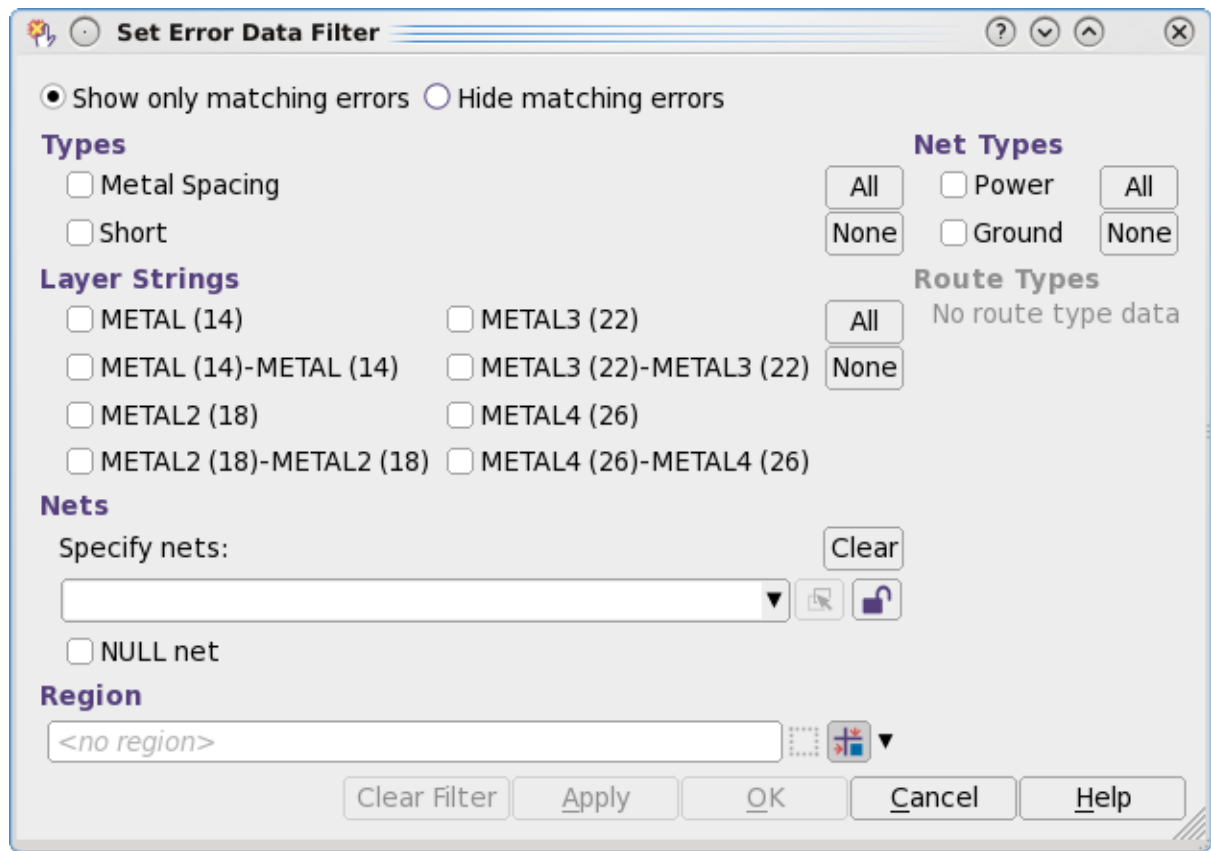
Use error filtering to temporarily remove the errors from the error browser and the layout view. Apply filters by using the Set Error Data Filter dialog box.

To create an error filter,

1. Choose Errors > Set Filter.
2. Choose Show only matching errors or Hide matching errors to set the filter type.
3. Create the filter criteria by selecting the error type, layer setting, nets, and other settings.
4. Click Apply or OK to apply the filter, or Clear Filter to remove the filter.

When you create a filter, the filter is applied to the currently open error data and any other error data opened in the current session. Filters do not accumulate; applying a new filter overrides any previously set filter. To clear a filter, choose Errors > Clear Filter or click Clear Filter in the Set Error Data Filter dialog box.

Figure 122 Set Error Data Filter Dialog



Highlighting Errors

You can annotate errors in the layout view by applying different highlight colors. Use the Highlight menu to highlight errors and to remove highlighting. Alternatively, use the highlight buttons at the bottom of the GUI to perform these tasks. The Highlight menu provides the following menu commands:

- By Type, Layer, Net: Opens the Highlight Errors dialog box to highlight errors based on error type, layer string, or associated nets.
- Selected: Highlights selected errors with the current highlight color.
- All Errors in List: Highlights all visible errors in the error list with the current highlight color.
- Clear Selected: Clears highlight coloring from selected errors.
- Clear Current Color: Clears all highlighting of the current highlight color.

- Clear All: Clears all error highlighting.
- Set Current Color: Provides a menu of colors to select as the current highlight color.
- Next Color: Advances the current highlight color to the next color in the Set Current Color menu.
- Auto Cycle Colors: Toggles the option for changing the current highlight color to the next color after each Highlight > Selected operation.
- Error Highlight Counts: Opens a dialog box that displays the number of errors in each highlight set and a simple histogram chart.

Marking Errors as Fixed and Changing Error Status

To change the status of an error, select one or more errors in the error list and do one of the following:

- Click the Fixed button to toggle the fixed status value in the Status column.
- Click the right mouse key and choose Set Status, then select Error, Fixed, or Ignored as needed. Alternatively, choose Errors > Set Status from the Error Browser menu and make the same selection.

These actions change the status of the selected errors. The value in the “Status” column changes to a check mark to indicate that the error is fixed, clears to indicate an unfixed error, or changes to an (X) to indicate an ignored status.

Writing Out DRC Error Data

The error browser provides a graphical tool for reviewing errors reported by various checking commands in the tool. To write a text report of the errors in a specific error data file, use the `report_drc_errors` command and use options to report specific error types or errors on specific layers.

The command writes out three types of reports: `error_type`, `error_layer`, and `detailed`. Use the `-report_type` option to specify the report type. The `error_type` report, generated by default or by specifying the `-report_type error_type` option, is the default report type and lists the types and counts for each error type as shown in the following example. The `error_layer` report type, generated by the `-report_type error_layer` option, is similar to the `error_type` report, but groups errors first by layer, then by error type.

ErrorSet	Total	Visible	Fixed	Ignored
-----	-----	-----	-----	-----
Design : test	54	3	0	0
Data : zroute.err	54	3	0	0
Net	3	1	0	0

METAL3 (28)	1	1	0	0
Route Spacing	6	2	0	0
METAL2 (18)-METAL3 (28)	1	1	0	0
METAL3 (28)-METAL4 (31)	1	1	0	0

The detailed report type, generated by the `-report_type detailed` option, reports error id, error type, and other information for each error similar to the comprehensive error information displayed in the error browser. The detailed report is shown in the following example.

```
Error ID      : 1
Error Type    : Route Spacing
Error Layer   : METAL3 (28)-METAL4 (31)
Number of errors : 6
Description   : Spacing violation between
                hmg_out44_10_ and RES_I109_X9
                on layers {METAL3 (28) METAL4 (31)} --
                direction: vertical, required: 1.2 um, actual: 0.6 um

Status       : error
Bbox         : {293.8000 364.4000} {294.2000 366.8000}

Error ID      : 3
Error Type    : Route Spacing
Error Layer   : METAL2 (18)-METAL3 (28)
Number of errors : 6
Description   : Spacing violation between A0_9_ and
                B0_9_ on layers {METAL2 (18) METAL3 (28)} --
                direction: upper left to lower right, required:
                1.2000 um, actual: 0.6000 um

Status       : error
Bbox         : {261.0000 301.8000} {261.4000 303.0000}
```

The following examples write out different error reports based on the command options used:

- Write out the default `error_data` error report for all errors in the error data file.
- Write out a detailed report only for the METAL3 layer for errors of type Route Spacing and Net.

```
fc_shell> report_drc_errors -error_data zroute.err
```

```
fc_shell> set type [get_drc_error_types \
    -error_data zroute.err {{Route Spacing} {Net}}]
fc_shell> report_drc_errors -error_data zroute.err \
    -report_type detailed -error_type $type -layers {METAL3}
```

12

Using the Message Browser

The Fusion Compiler GUI provides the enhanced message system (EMS) database and message browser tool to manage and view error and informational messages related to rule violations in the layout. The EMS database is generated by the `check_design` command and other checking commands. Using the message browser, you can select individual messages, select messages by message type, highlight the affected objects in the layout view, and create waivers for violations. The message browser also displays an HTML-format man page for the currently selected message. If you have multiple EMS databases, the message browser can merge the databases into a single consolidated database.

The following topics describe the message browser:

- [Preparing the EMS Database](#)
- [Message Browser Layout](#)
- [Reclassifying a Message](#)
- [Creating a Waiver](#)
- [Merging EMS Databases](#)

Preparing the EMS Database

Before using the message browser, you must create an EMS database. The database is created by design checking tools such as `check_design`, `check_hier_design`, `check_netlist`, `check_routability`, and `check_timing`. Run the `create_ems_database` command before the checking command and the `save_ems_database` command afterward. For example,

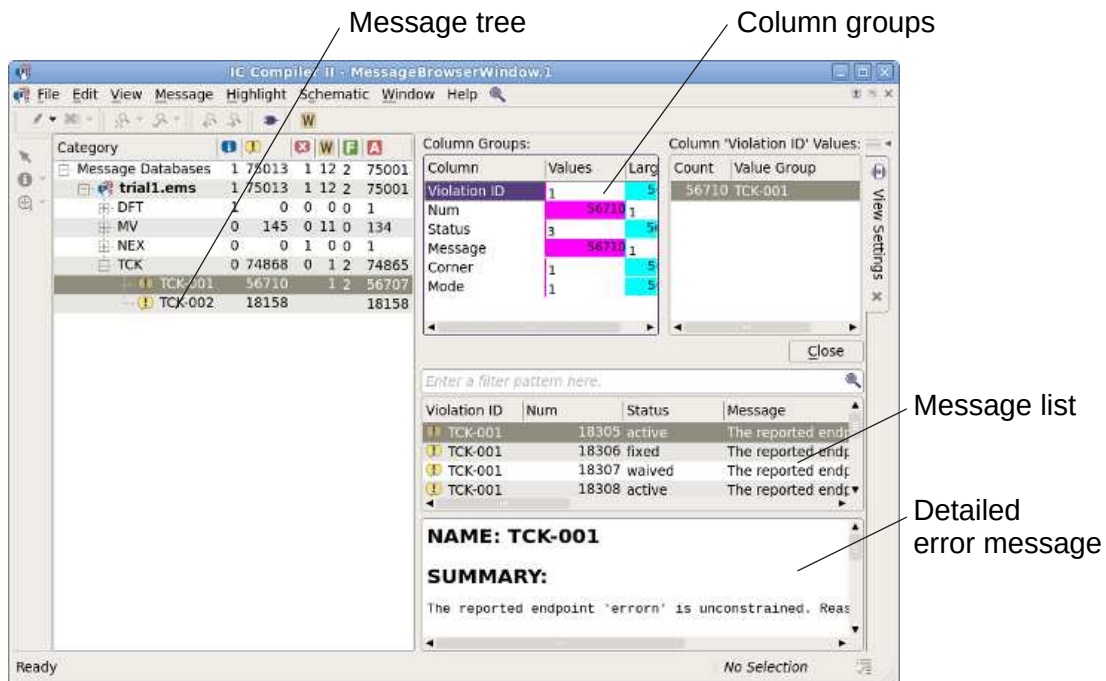
```
fc_shell> create_ems_database check_hier.ems
fc_shell> check_hier_design -stage timing
fc_shell> save_ems_database
```

Choose Window > Message Browser Window from the layout window to open the message browser. Alternatively, use the `check_design` command with the `-open_message_browser` option to automatically open the message browser after the analysis.

Message Browser Layout

Figure 123 shows the different areas displayed in the message browser.

Figure 123 Message Browser

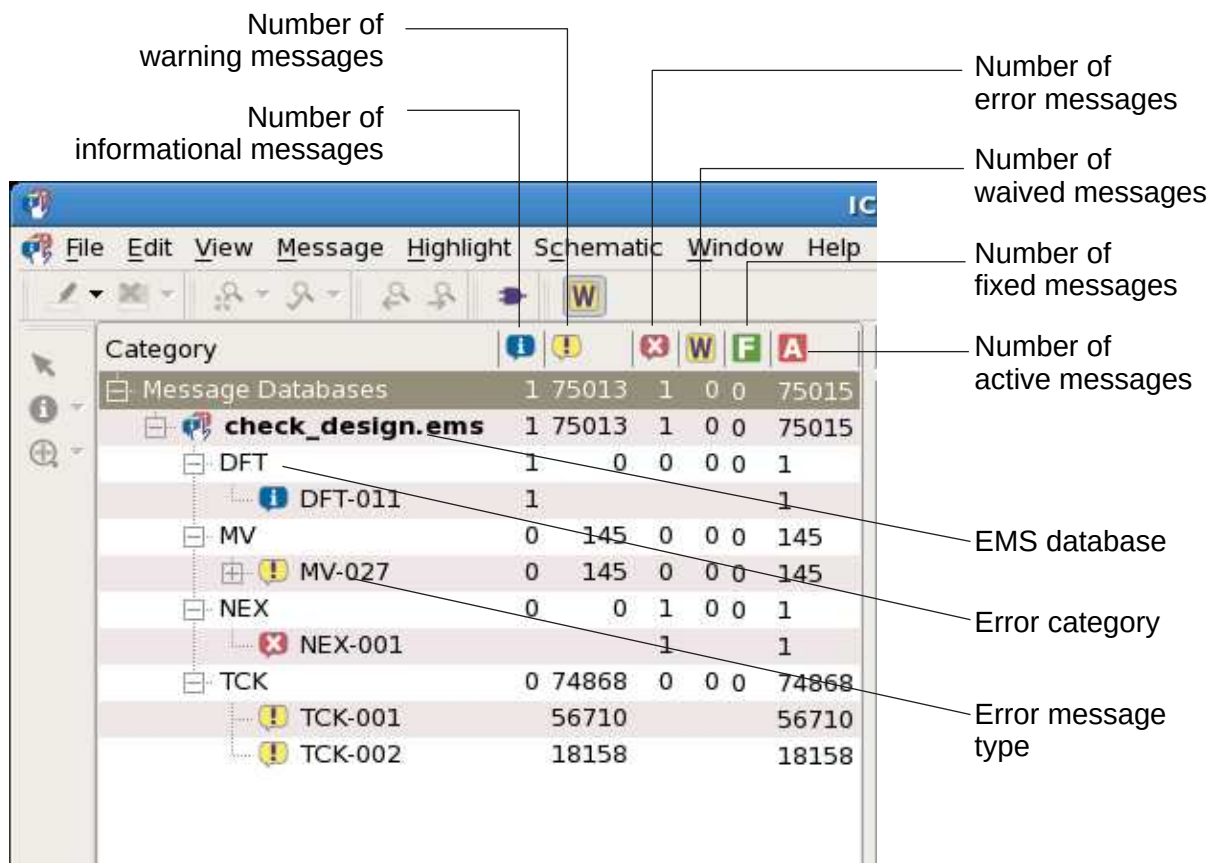


The following sections describe the panes in the message browser.

Message Tree

The message tree lists errors in a tree view at the left of the browser as shown in Figure 124. The tree view displays the name of the current EMS database and the hierarchy of errors and warnings in the database.

Figure 124 Message Browser Tree



Message List

The message list displays a complete list of messages for the currently selected error message type as shown in Figure 125. Each entry includes the Violation ID, internal error message index, current status (active, waived, or fixed), message text, and additional information if available. Move your mouse pointer over a message row to view additional information about the message. Click a row in the list to select it. After selecting the message row, the tool updates the detailed error message display. Click and drag the pointer to select more than one row, or press the Shift or Control key while clicking the row to add the row to the selection. Press Control-A to select all rows.

Figure 125 Message List

Violation ID	Num	Status	Message
TCK-001	18306	active	The reported endpoint 'errorn' is u
TCK-001	18307	active	The reported endpoint 'sa[14]' is u
TCK-001	18308	active	The reported endpoint 'sa[13]' is u
TCK-001			[12]' is u
TCK-001			[11]' is u
TCK-001			[10]' is u
TCK-001			[9]' is un
TCK-001			[8]' is un
TCK-001			[7]' is un
TCK-001			[6]' is un
TCK-001			[5]' is un
TCK-001			[4]' is un

Violation ID: TCK-001
Num: 18308
Status: active
Message: The reported endpoint 'sa[13]' is unconstrained.
Reason: 'no check'. Mode:'default'.
Corner: default
Mode: default
Message Database:
/remote/us01home21/dlwhite/manpages/check_design.ems

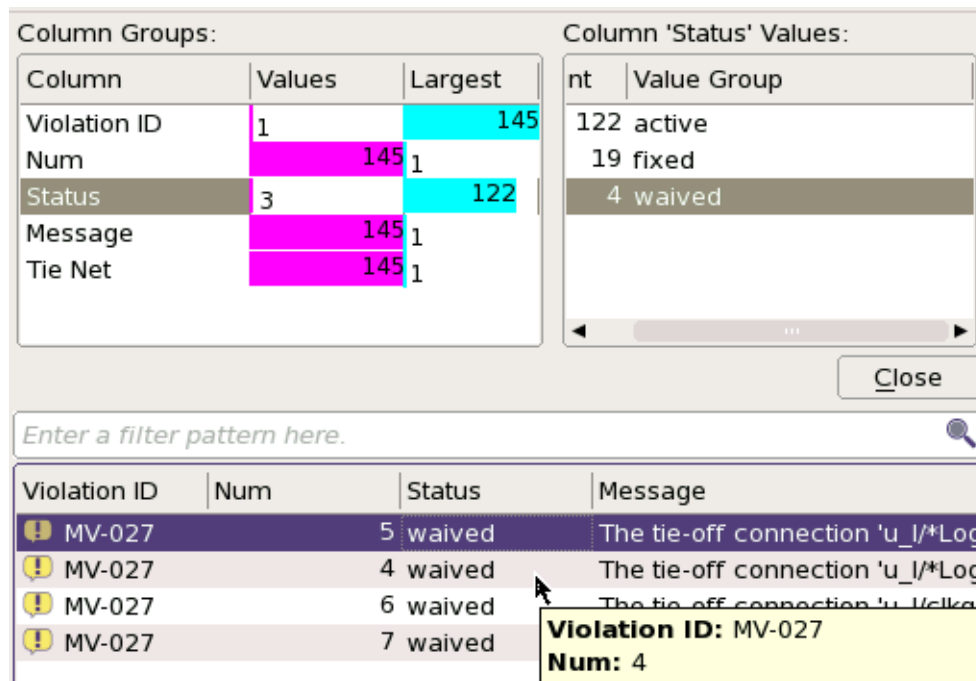
In the messages area, right click the mouse to display the context menu; select an entry from the menu to perform an action as follows:

- Waive: Reclassify the selected messages as waived; this action persists only for the current session
- Mark Fixed: Reclassify the selected messages as fixed; this action persists only for the current session
- Mark Active: Reclassify the selected messages as active; this action persists only for the current session
- Select: Select the related object in the layout view
- Highlight: Highlight the related object in the layout view
- Properties: Display the properties for the object
- Show Column Groups: Opens the Column Groups and Column Status Values panes
- Create Waiver: Opens the Create Waiver dialog; use this dialog to create a waiver as described in [Creating a Waiver](#)
- List Waivers: Opens the Message Waivers dialog; use this dialog to display the current list of waivers
- Copy: Copy the text for the currently selected rows

You can filter the list of messages by entering a filter pattern or by using column groups. As you enter characters in the search box, the tool updates the display to show only the messages containing the search string.

Column groups combine information counts from the columns in the message list area into a histogram view. The values column displays the number of unique results for each column. Click in a row in the Column Groups box to display additional information in the column values area on the right. As you click a row in the column values area, the list of messages is filtered to display only messages that match the currently selected value group. Click the Close button to close the column groups panel.

Figure 126 Column Groups



Detailed Error Message

The message browser displays a detailed message summary and a description of the currently selected error. As you click different error messages in the message list, the detailed error message shows the message summary and detailed description for the error.

To change the position of the detailed error message, choose View > Html Pane from the menu and select On Right or On Bottom.

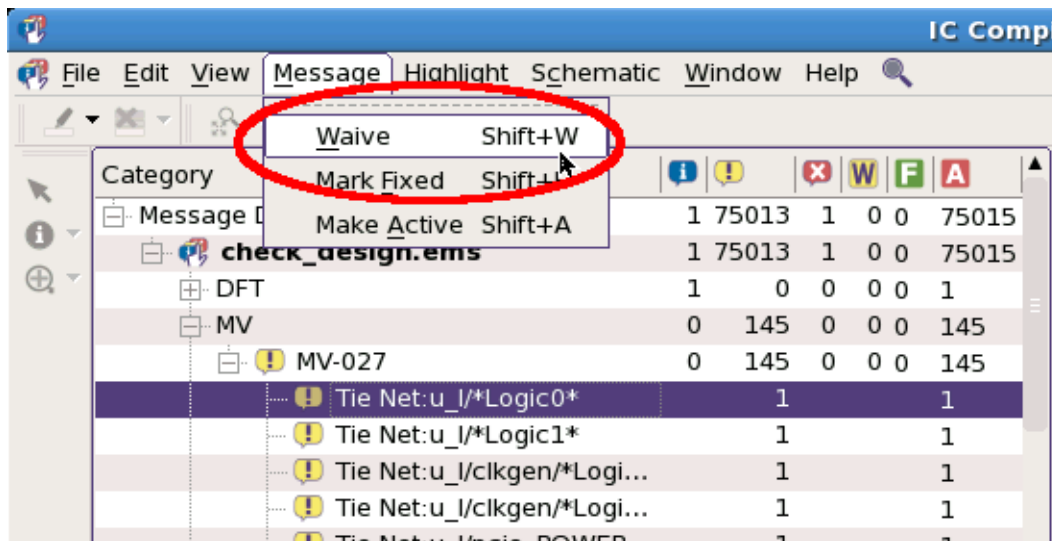
Reclassifying a Message

Use the message browser to reclassify individual messages as active, fixed, or waived. To reclassify a message only for the current session,

1. Select the message in the Tree View.
2. Choose Waive, Mark Fixed, or Mark Active from the Message Menu as shown in [Figure 127](#). Alternatively, click the message and select the action from the context menu.

The tool updates the count in the list view.

Figure 127 *Reclassify a Message*



Note that these changes are active only for the current session and are not saved when you exit the message browser.

Creating a Waiver

Create a waiver for a message to attach additional information to the message. This information is not saved to the EMS database. However, you can write out the waivers as Tcl commands and apply them in a later session. To create a waiver,

1. Select a message row to be waived.
2. Right-click and select Create Waiver from the context menu.

The tool displays the error waiver form above the message list as shown in [Figure 128](#).

3. Complete the error waiver form by entering information in the Name and other boxes. Box labels preceded by an asterisk (*) are required entries.

4. Modify the criteria as needed.

Click the plus sign button (+) to add criteria, click the minus sign button (-) to remove a criterion, or click Remove All to remove all criteria.

Note that the criteria are applied to all messages in the list.

5. Click Apply to create the waiver.

The tool applies the criteria to all messages in the list and updates the status to Waived for the matching messages.

To display a list of waivers for the current session, choose View > List Message Waivers from the menu. The tool displays a list of current waivers as shown in [Figure 129](#).

Figure 128 Create Waiver

*Name: Domain: MV Rule: MV-027

Comment:

*Set up criteria:

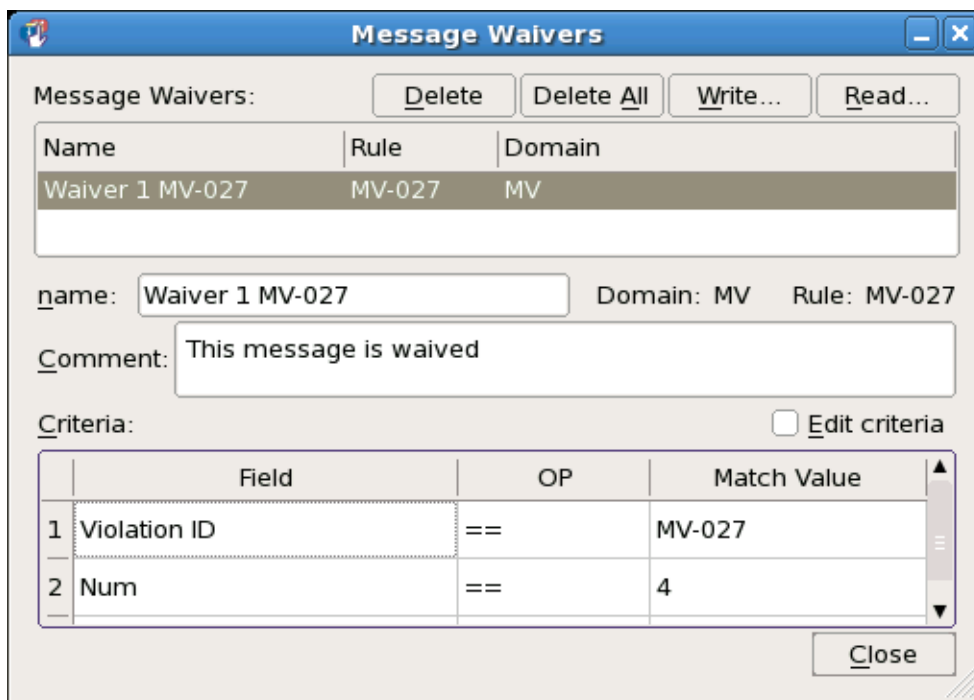
	Field	OP	Match Value
1	Violation ID	==	MV-027
2	Num	==	5
3	Status	==	active
4	Message	==	The tie-off connection 'u_l/*Lo...

Enter a filter pattern here.

Violation ID	Num	Status	Message
MV-027	5	active	The tie-off connection 'u_l/*Logic0*'
MV-027			The tie-off connection 'u_l/*Logic1*'
MV-027			The tie-off connection 'u_l/*Logic2*'

- Waive
- Mark Fixed
- Create Waiver
- List Waivers
- Copy

Figure 129 List Message Waivers



Click on the waiver to view the criteria used to create the waiver. To edit the waiver, select Edit criteria, modify the criteria settings, and click Commit. You can also use this dialog to write out the current waivers as a Tcl file, read in waivers stored in a Tcl file, or delete current waivers.

To use the waivers in another session, you must save them to disk. To save the waivers to disk, click Write from the Message Waivers dialog or choose File > Write Waiver to Script from the message browser menu. The tool writes out `gui_create_message_waiver` commands to the file. To read in waivers saved in a Tcl file, click Read from the Message Waivers dialog or choose File > Read Waiver from Script from the message browser menu, then select the Tcl file that contains the waivers.

Merging EMS Databases

The message browser can load multiple EMS databases in the same session. Only one database can be active at a time. To load more than one database,

1. Choose File > Open Message Database from the menu.

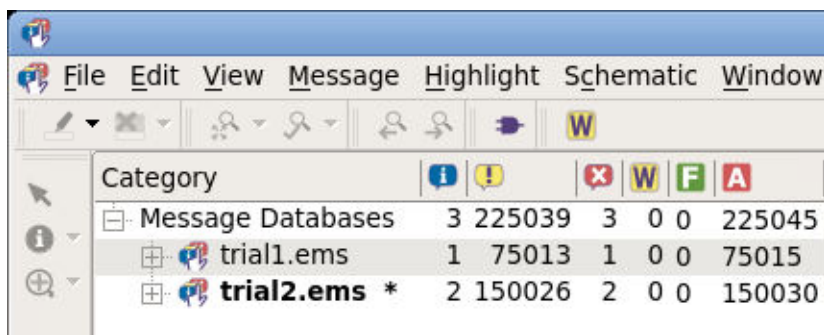
Alternatively, open the database with the `open_ems_database` Tcl command as in the following example:

```
fc_shell> open_ems_database MV.ems
```

2. (Optional) If you have opened multiple EMS databases, click the database to be merged to select it and choose File > Set Current Message Database from the menu.
3. Choose File > Merge Message Database from the menu, select a file, and click OK.

The tool loads the database file you selected and sets it as the current database as shown in [Figure 130](#). The newly loaded database file contains its original messages and the messages from the current database before the merge.

Figure 130 Merging Databases



For more information about managing EMS databases with Tcl commands, see the man pages for the following commands:

- `close_ems_databases`
- `create_ems_database`
- `get_current_ems_database`
- `get_ems_databases`
- `open_ems_database`
- `save_ems_database`
- `set_current_ems_database`

13

Creating Charts

The Fusion Compiler tool contains a comprehensive plot generation utility you can use to generate interactive plots based on design, layout, and other data. See the following topics for more information:

- [Creating a Model](#)
- [Creating a Plot](#)

Creating a Model

To create a chart, begin by importing the chart data from a comma-separated values (CSV) file to create a model. Use the `gui_create_charts_model` command to read the CSV and save the information as a Tcl variable.

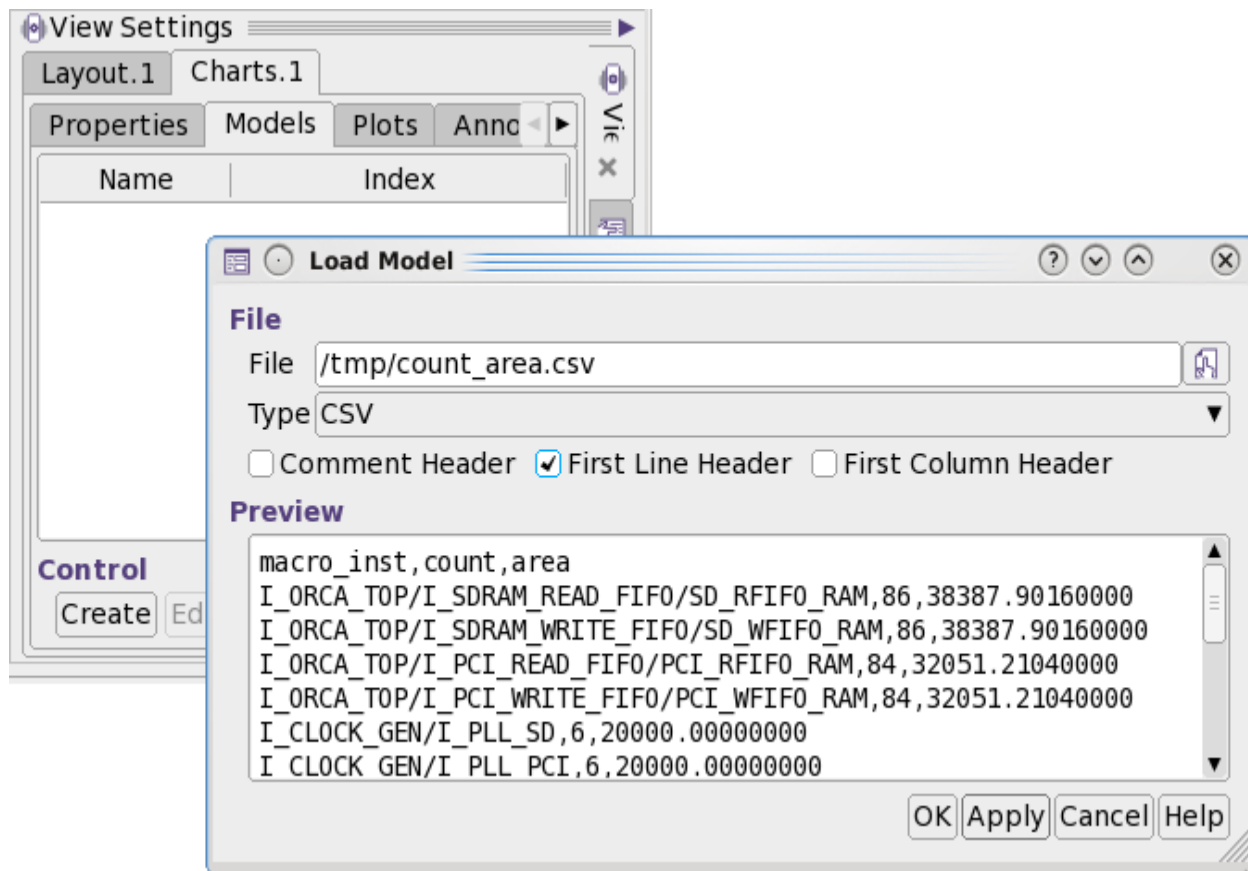
The following example writes out a simple CSV file containing the instance name, number of pins, and area for each macro in the design. The `gui_create_charts_model` command loads the CSV file and creates a chart model.

```
set macros [get_cells -hier -filter "design_type==macro"]
set csv_file /tmp/count_area.csv
echo "macro_inst,count,area" > $csv_file
foreach_in_collection c $macros {
    echo [get_object_name $c],[get_attribute $c number_of_pins],
    [get_attribute $c area] >> $csv_file
}
set model [gui_create_charts_model -csv $csv_file -first_line_header ]
```

Note that the `gui_create_charts_model` command automatically derives the data type for each column of data.

Alternatively, you can import the CSV file by using the GUI. Click the Charts tab in the View Settings panel, then click the Models tab. Click the Create button to open the Load Model dialog box. Enter the file name, select options to specify the header. The tool displays a preview of the data as shown in the following figure. Click OK to create the model.

Figure 131 Load Model Dialog Box

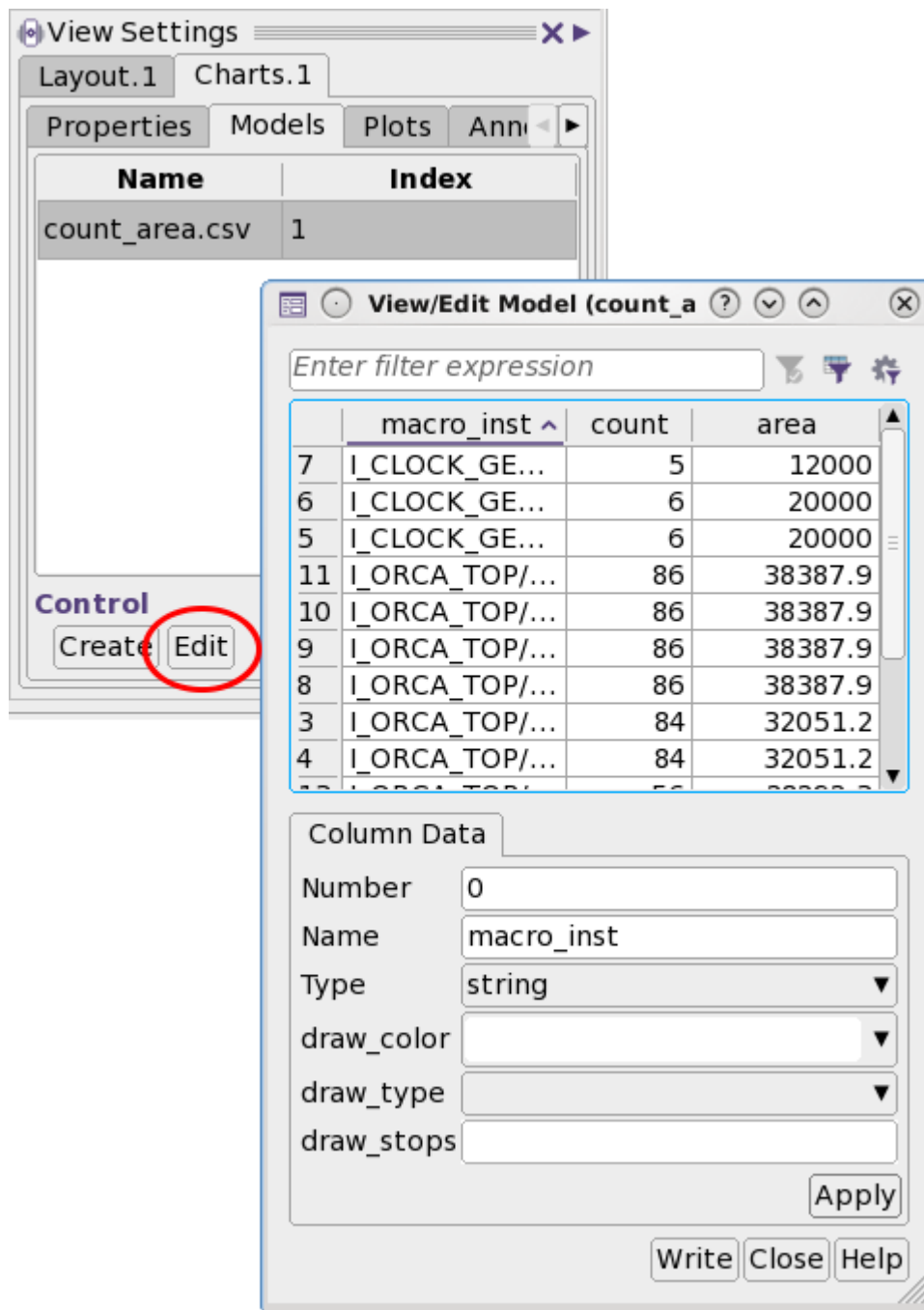


Checking the Model

After creating the model, check the model in the GUI by clicking the Charts tab in the View Settings panel. Click the Models tab to show the list of models created with the `gui_create_charts_model` command.

To view details for a model, including column names, data types, and other properties, click the model in the table and click the Edit button to show the table of data. Click each column to show the column name, data type, and other attributes for the data column. To change the properties for the column, select a entry in the Column Data section and click Apply.

Figure 132 View/Edit Model Dialog Box



Adding or Deleting Data From the Model

After importing the CSV file, you can modify the data type by adding columns, deleting columns, and performing other operations with the `gui_change_charts_model` command. The following example adds a new column called “double_count” which is twice the value of the data in column(1).

```
gui_change_charts_model -model $model -add \  
    -expr {column(1) + column(1)} -header double_count
```

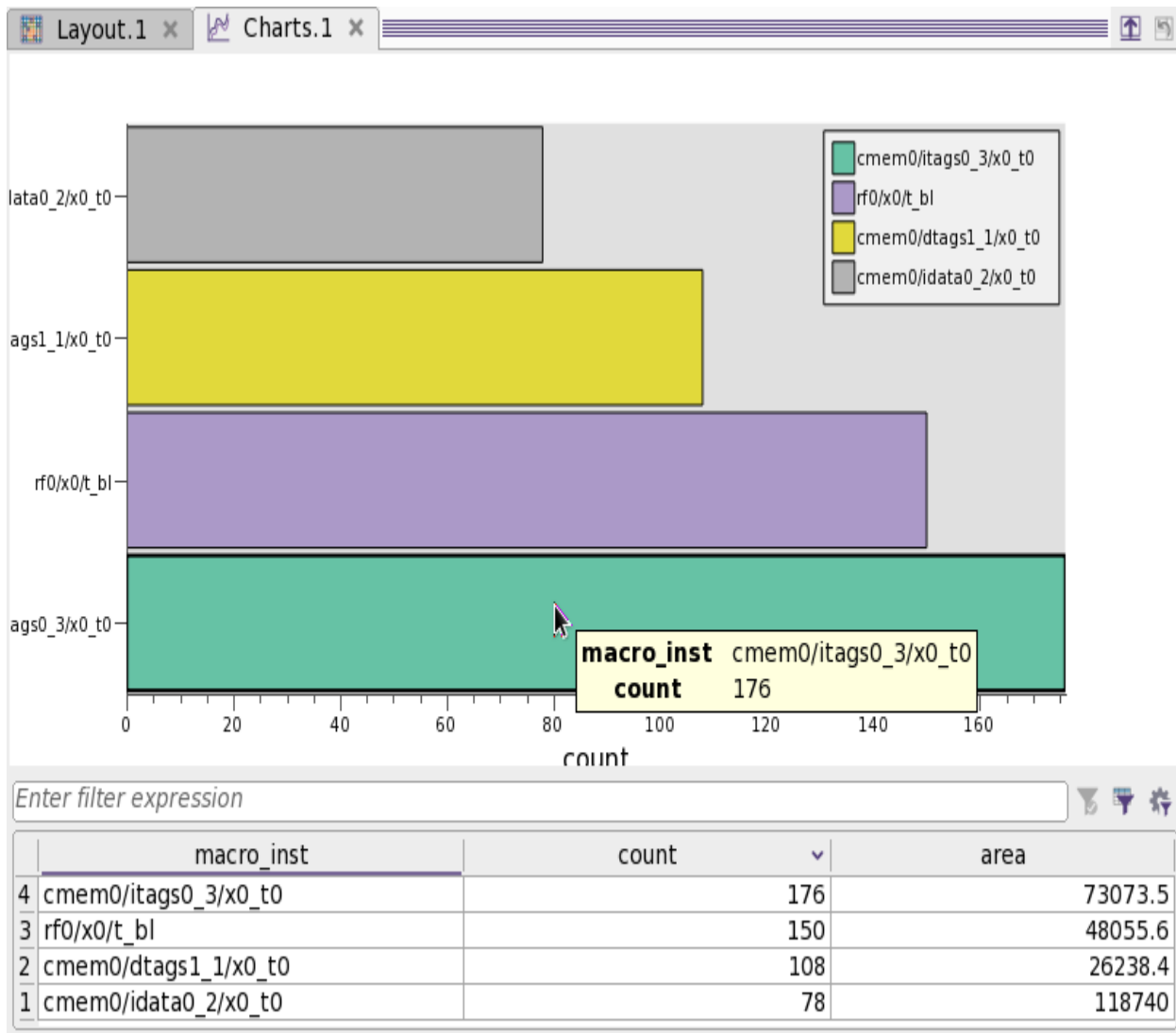
Creating a Plot

After importing the CSV file to create a model, you can create a plot to view the data. Use the `gui_create_charts_plot` command to create a bar chart, distribution plot, XY chart, and so on.

The following example writes out a CSV file that contains the pin count and area for four macros in the design. The `gui_create_charts_plot` command creates the initial plot based on the CSV file contents. Next, the plot is converted to a horizontal bar chart by choosing Horizontal from the context menu. The data table is displayed by choosing Show Table from the context menu.

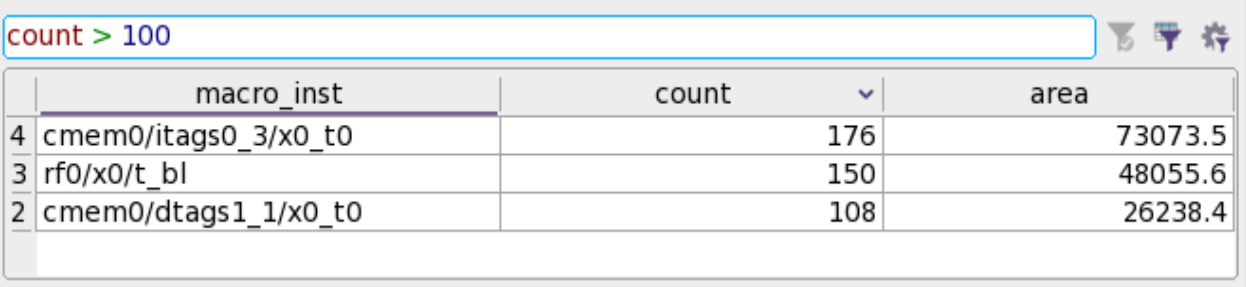
```
# Create the Charts window  
set view [gui_create_window -type Charts]  
  
# Specify the macros and output CSV file  
set macros [get_cells { cmem0/idata0_2/x0_t0 cmem0/dtags1_1/x0_t0  
    rf0/x0/t_b1 cmem0/itags0_3/x0_t0 }]  
set csv_file count_area.csv  
  
# Write out the CSV file  
echo "macro_inst,count,area" > $csv_file  
foreach_in_collection c $macros {  
    echo [get_object_name $c],\  
    [get_attribute $c number_of_pins],\  
    [get_attribute $c area] >> $csv_file  
}  
  
# Create the model  
set model [gui_create_charts_model -csv $csv_file -first_line_header]  
  
# Create the bar chart from the model  
set plot [gui_create_charts_plot -view $view -type barchart \  
    -model $model -columns {{name 0} {value {1}}}]
```

Figure 133 Bar Plot



After creating the chart, click the column headers to sort the data. To filter the table data, enter a filter expression in the filter expression box. For example, to show only macros with a count greater than 100, enter “count > 100” in the filter expression box as shown in the following figure.

Figure 134 Filtering Plot Data



The screenshot shows a data filtering interface. At the top, there is a text input field containing the filter expression `count > 100`. To the right of the input field are three icons: a funnel, a list, and a gear. Below the input field is a table with four columns: an index column, a `macro_inst` column, a `count` column, and an `area` column. The table contains three rows of data, all of which have a `count` value greater than 100.

	macro_inst	count	area
4	cmem0/itags0_3/x0_t0	176	73073.5
3	rf0/x0/t_bl	150	48055.6
2	cmem0/dtags1_1/x0_t0	108	26238.4

14

Using the PG Pattern Viewer

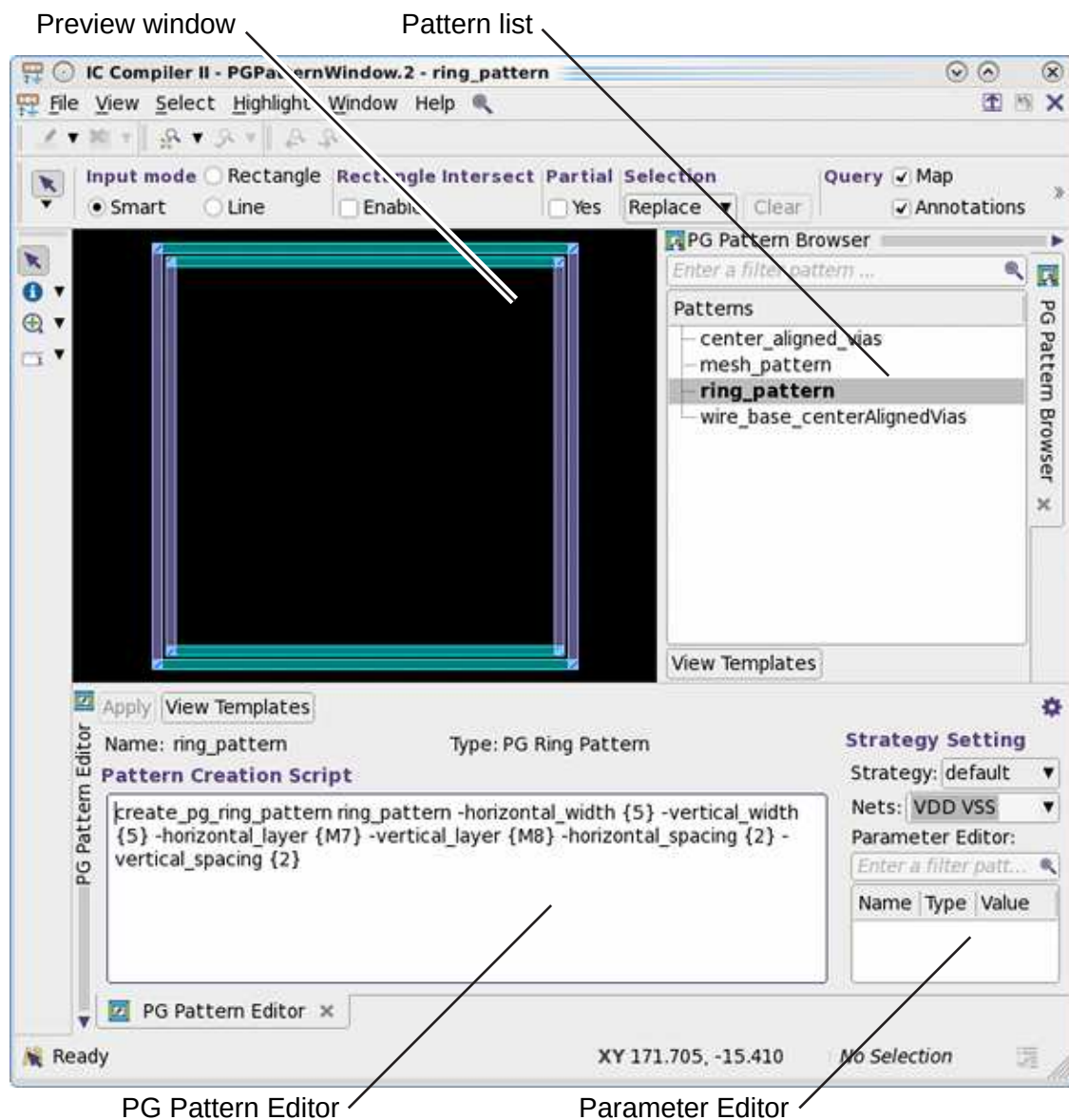
The Fusion Compiler GUI provides the PG Pattern window for creating and previewing PG patterns. Use the PG Pattern window to view a collection of prepackaged PG pattern templates. After selecting a template for a PG pattern, copy the pattern template to the PG Pattern Editor and edit the pattern to conform to your design requirements. If the pattern uses parameters, you can set parameters in the PG Pattern window. Preview the pattern and make any necessary changes to the Tcl command. Only after the pattern is finalized, return to the layout window and instantiate the pattern in your design.

See the following topics for more information about the PG Pattern window:

- [Selecting a PG Pattern Template](#)
- [Previewing a Template PG Pattern](#)
- [Previewing Existing Pattern and Strategy Definitions](#)
- [Highlighting Individual Patterns](#)

The PG Pattern window is shown in the following figure.

Figure 135 PG Pattern Window



Selecting a PG Pattern Template

The PG Pattern window contains a set of prepackaged templates that you can use as a starting point for creating your own pattern. Open the PG Pattern window by choosing

Window > PG Pattern Window from the menu. To view and select a template in the PG Pattern Window,

1. Open the PG templates dialog box by clicking the View Templates button in the PG Pattern window.
2. (Optional) In the PG Pattern Templates dialog box, enter a string in the filter pattern box to filter the list of templates. The matching is case-insensitive.

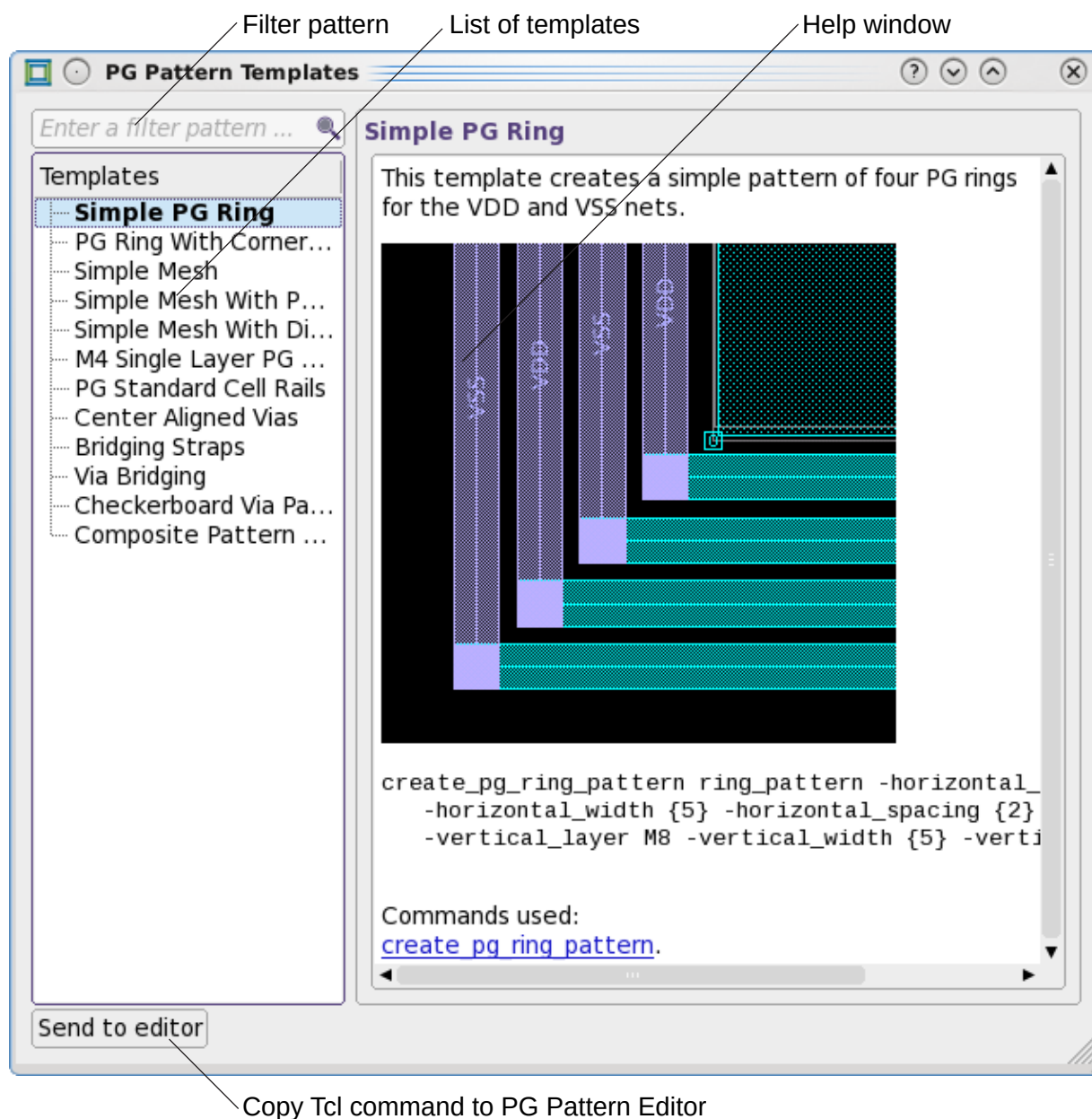
For example, enter “Mesh” or “mesh” to display only templates that contain the word Mesh.

3. Select a template by clicking a template name in the list.

The tool displays the template in the PG Pattern Template window. A typical template includes a brief description, an image of the pattern, and the Tcl command used to create the pattern.

4. Click the Send to Editor button to copy the template to the PG Pattern Editor in the PG Pattern window.

Figure 136 PG Pattern Templates



Previewing a Template PG Pattern

The pattern preview function displays a template pattern in the preview window. To use the pattern preview function,

1. Copy a pattern template to the PG Pattern Editor as described in [Selecting a PG Pattern Template](#).
2. (Optional) Modify the layer, spacing, width, and other parameters of the pattern to match the current design.
3. Click the Apply button to display the pattern in the preview window.

The tool runs the `create_*_pattern` command specified in the PG Pattern Editor panel and adds the pattern name to the list of patterns in the PG Pattern Browser. Use the `report_pg_patterns` command in the layout window to write out a detailed report for the current patterns defined for the design. If any errors occur while creating the pattern, the tool writes error messages to the console.

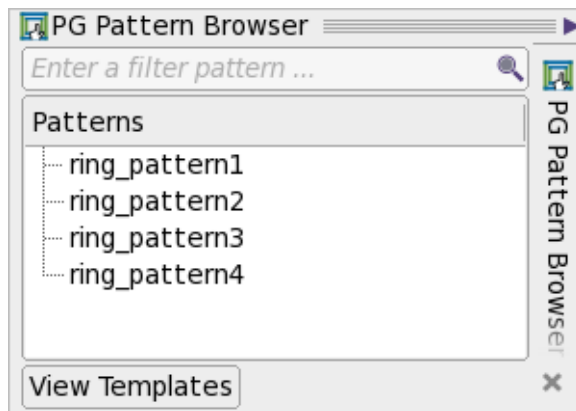
Previewing Existing Pattern and Strategy Definitions

You can use the PG Pattern window to preview previously defined patterns and strategies as follows:

1. Create the PG pattern by running the `create_pg_ring_pattern`, `create_pg_ring_pattern`, or other pattern creation command in the layout window.

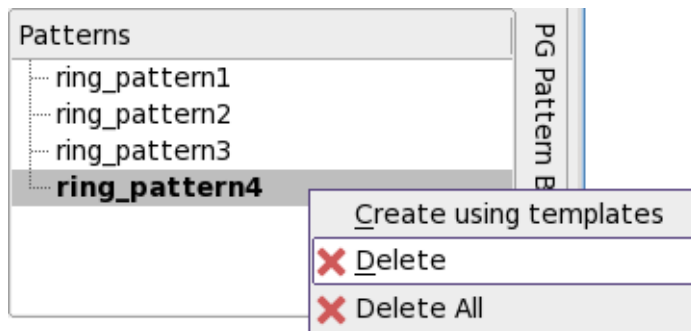
You can create the patterns before or after opening the PG Pattern window. Any PG patterns that you define in the layout window will be listed in the Patterns list under PG Pattern Browser. In this example, four ring patterns are defined.

Figure 137 *Predefined Patterns*



2. (Optional) Click the right mouse button and choose Delete from the context menu to remove a pattern from the list.

Figure 138 Pattern List Context Menu



3. Click a pattern in the list to select it and display the corresponding `create_*_pattern` command in the Pattern Creation Script editor.
4. (Optional) Modify the pattern in the editor.
5. (Optional) Create one or more PG strategies with the `set_pg_strategy` command.
6. (Optional) Select the strategy to create the pattern in the Strategy box under Strategy Setting.

Only the strategies available for the selected pattern are displayed in the Strategy box. The tool also creates a default strategy for the pattern you selected.

7. Click Apply to create the pattern in the preview window based on the selected strategy and pattern.

Note that the pattern is created only in the preview window, not the layout window for the current design.

8. (Optional) Select a new pattern from the pattern list or a new strategy from the strategy box and repeat the previous step to generate a new pattern.

Highlighting Individual Patterns

The PG Pattern viewer supports the highlighting of individual child patterns within a pattern. Use this feature to highlight the power straps and vias for a child pattern within a

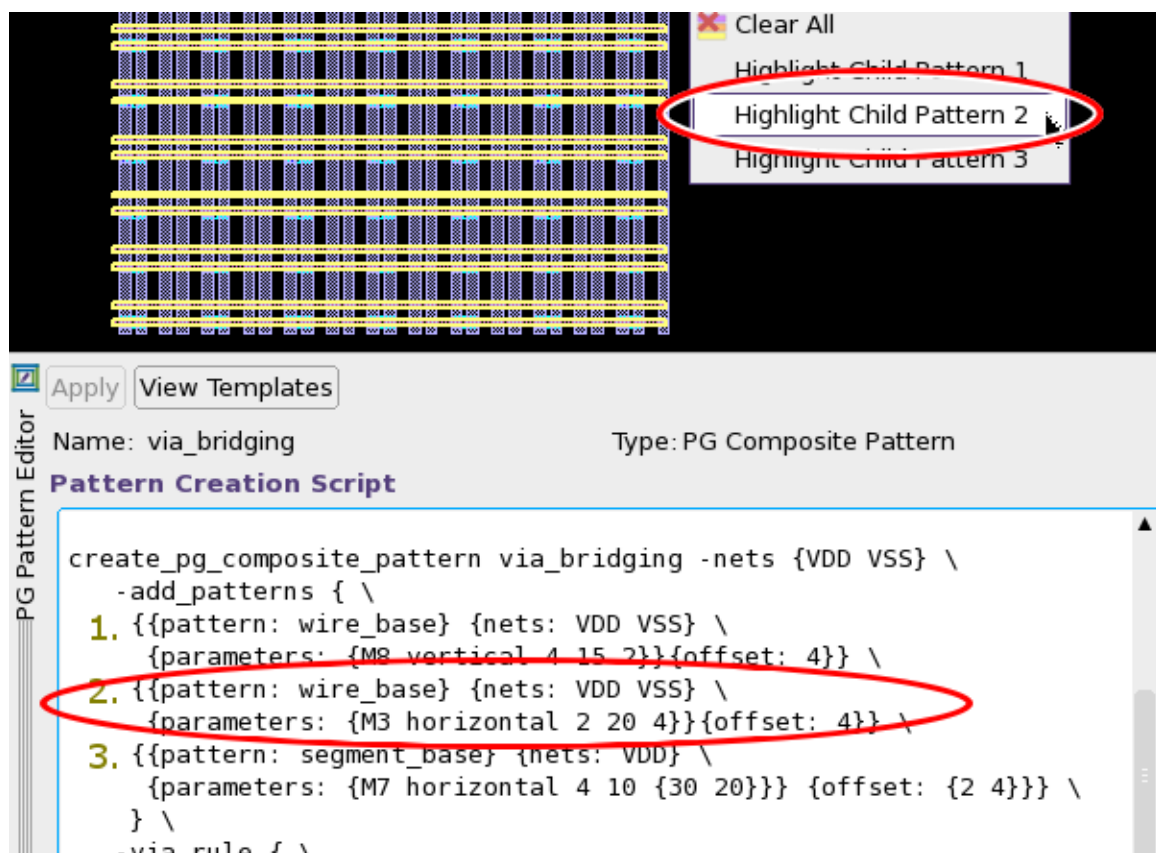
pattern definition and make adjustments to the metal layer, layer width, spacing, and so on. To view an individual child pattern within a pattern,

1. Select the pattern for the list of pattern templates, such as the via bridging pattern.
2. Click the right mouse button and select the child pattern number from the list.

The tool highlights the metal shapes that compose the pattern number.

The following figure shows the sample via bridging composite pattern with second pattern highlighted.

Figure 139 Highlight Child Patterns



15

Creating Routes

The Fusion Compiler GUI provides interactive and programmatic interfaces to quickly and easily create routes. Use the Create Route tool to route one or more nets or bundles between pins, ports, and other route shapes in the design. Similar to the Create Route tool, use the Create RDL tool to route single nets to create orthogonal and 45-degree routes on the redistribution layer or other metal layers. The `create_trunk`, `create_trunk_pin_to_pin`, and `create_trunk_pin_to_trunk` commands create route segments on specified nets based on command line options.

The following topics describe how to create routes by using the GUI:

- [Creating Routes Interactively](#)
- [Performing Interactive Design Rule Checking on Routes](#)
- [Creating Routes With Tcl Commands](#)

Creating Routes Interactively

The Create Route tool enables you to interactively route one or more nets or bundles. Options to the tool specify how the routes are created. The tool supports blockage avoidance and DRC feedback to help create high quality, violation-free routes and connection. Snapping, auto-alignment, and auto-welding are provided to quickly and easily connect new routes to pins or preexisting routes. Bundle routing is supported for the simultaneous routing of several signals at one time.

Starting the Create Route Tool

To start the Create Route tool, do one of the following:

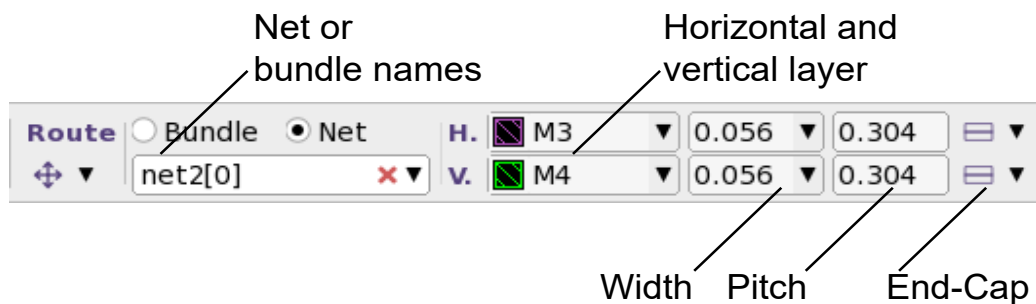
- Click the Create Route button ()
- Choose Create > Route from the menu
- Press Shift+R

The tool starts and displays the Create Route toolbar.

Specifying Layer, Width, Pitch, and End-Cap Style

The Create Route tool provides settings and options on the toolbar to specify the layer, width, pitch, and end-cap style for routes as shown in the following figure. Other settings are available by clicking the More, DRC, or Snap buttons on the toolbar.

Figure 140 Layer Name, Width, Pitch, and End-Cap Settings



Use any of the following methods to specify the layer name, width, and pitch for a net or bundle:

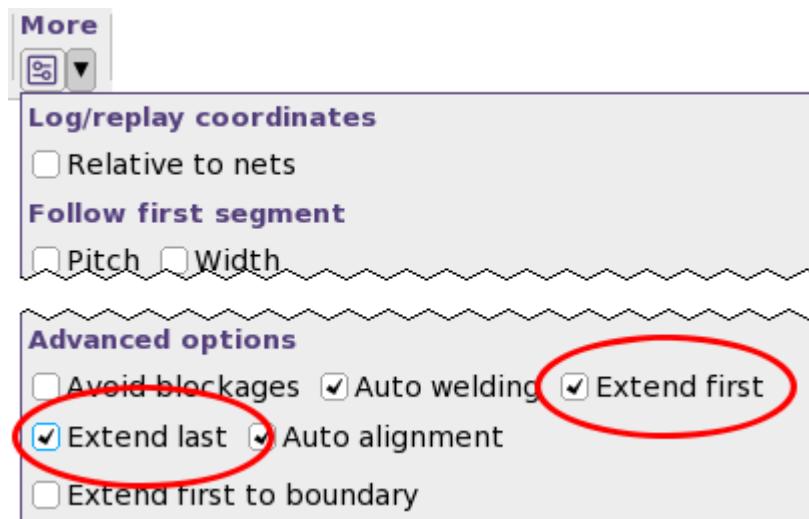
- Enter net names or a bundle names in the box; the tool sets the horizontal and vertical layers to match the lowest metal layers for routing.
- Select a pin before starting the Create Route tool to infer the net name. The tool sets the layer, width, and so on based on the selected pin.
- Select the Net option in the Auto section of the toolbar, then click an existing route to infer the net name from that route. To infer the layer, width, or end-cap style from the existing route, select the corresponding option in the Auto section.
- In the More settings panel in the toolbar, enable the Connect to pin by layer option and specify a layer name. The tool uses the specified layer when automatically connecting to pins, unless the pins are the starting or ending point for the route.

Creating Routes Between Pins

To create routes for a collection of pins,

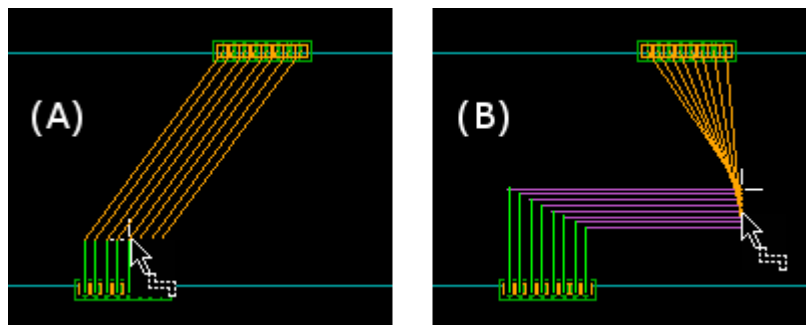
1. Select the source or destination pins for the nets to be routed.
2. Start the Create Route tool by pressing Shift+R or by clicking the Create Route tool button ().
3. Click the More button in the toolbar and select the Extend first option and the Extend last options in the Advanced options section.

Figure 141 Settings for Connecting Routes Pin-to-Pin



4. Begin drawing the route by placing the pointer in the route path near the selected pins and clicking the mouse as shown in image (A) in the following figure.

Figure 142 Routing From Selected Pins

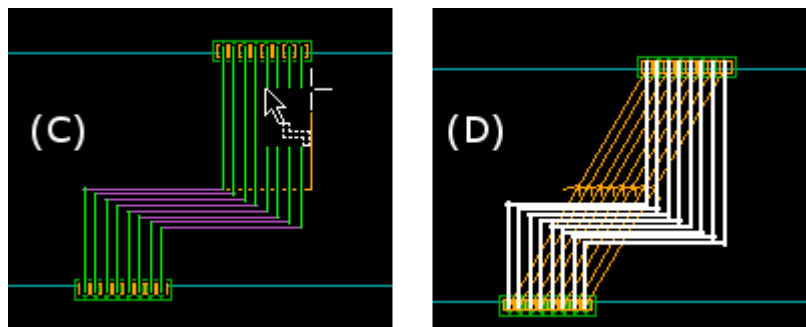


5. Click to create a corner, then move the mouse pointer in the orthogonal direction as shown in image (B).

Repeat this step as needed to create additional route segments.

6. Move the mouse pointer near the destination pins until the tool shows the completed route as shown in image (C) in the following figure.

Figure 143 Completing the Route



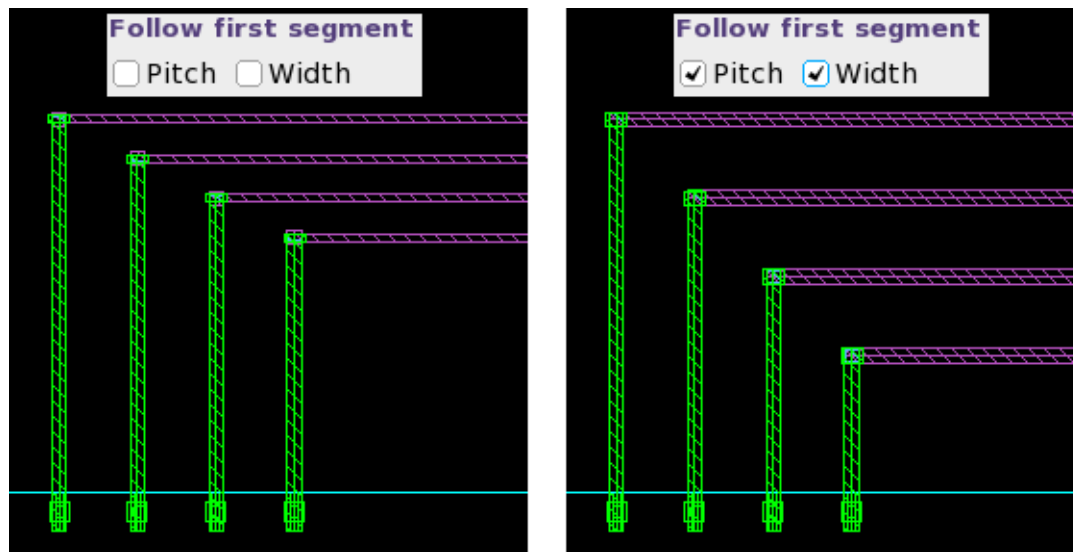
7. Press the Enter key or click the right mouse key and choose Apply from the context menu to complete the route.

The tool completes the route as shown in image (D).

Applying Route Segment Width and Spacing Based on the First Segment

To create all route segments with the same width and spacing as the first route segment, click the More button on the toolbar and enable the Pitch and Width options in the Follow first segment section. When enabled, the tool creates all subsequent route segments to match the first segment as shown in the following figure.

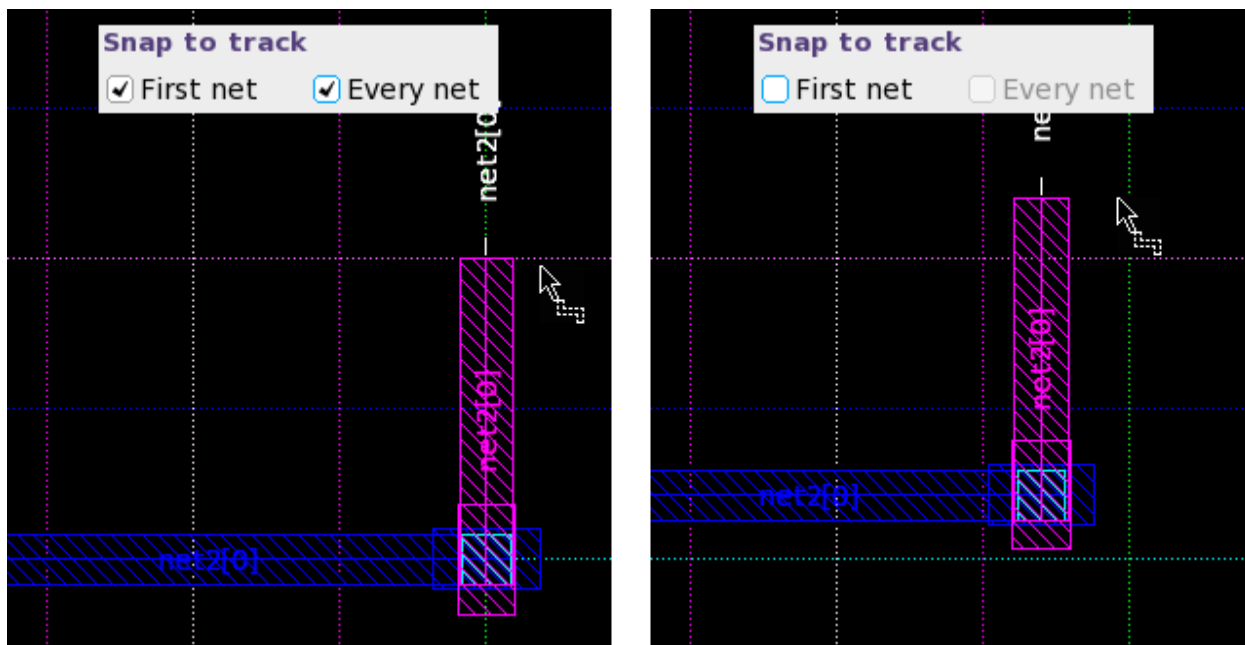
Figure 144 Follow First Segment



Creating Off-Track Routes

To create off-track routing segments, click the More button on the toolbar and disable the First net and Every net options in the Snap to track section. By default, the First net option is enabled and the first routing segment is created on the routing track. The following figure shows these options.

Figure 145 Snap to Track



Creating Successive Routes Above or Below the Drawn Path

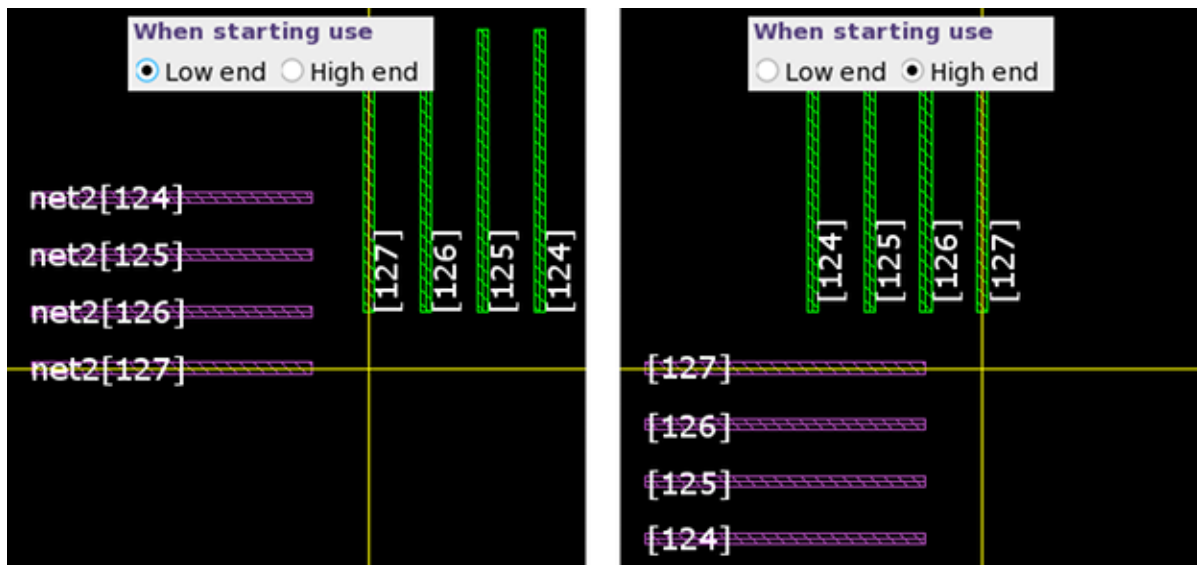
The Create Route tool starts with the first net in the bundle or the net name box and creates the route along the path you draw with the mouse. Subsequent nets are created in parallel to the first net; the direction is determined by the Low end and High end options in the When starting use section of the More settings panel.

In the following example, the bundle named bundle4 is created with the `create_bundle` command and is used to route the four nets as follows:

```
fc_shell> create_bundle -name {bundle4} \  
           {net2[127] net2[126] net2[125] net2[124]}
```

The first net in the list is net2[127] and is created along the path you draw with the mouse. By default, subsequent nets are created above or to the right of net2[127] as shown in the left image in the following figure. The yellow highlight represents the path of the mouse used to draw the route.

Figure 146 Low End and High End Routing Options

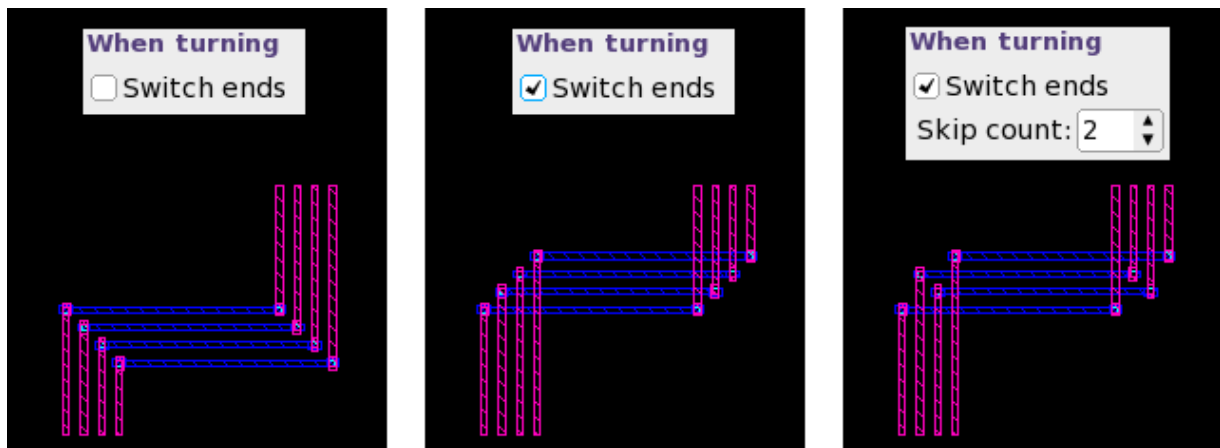


To create subsequent routes to the bottom or to the left of the mouse path, click the More button in the toolbar and choose the High end option. The tool creates routes as shown in the right image in the previous figure.

Changing the Route Ordering for Orthogonal Routes

The Create Route tool supports different methods for ordering nets when creating an orthogonal route. By default, the tool maintains the ordering of the nets and does not overlap the routes when creating new route segments. To reverse the ordering of the routes when creating an orthogonal route, click the More button and enable the Switch ends option in the When turning section. To stagger the routes at the turn, enable the Skip count option and set the Skip count to a value greater than 1. The following figure shows the routing result after setting these options.

Figure 147 Route Turn Styles

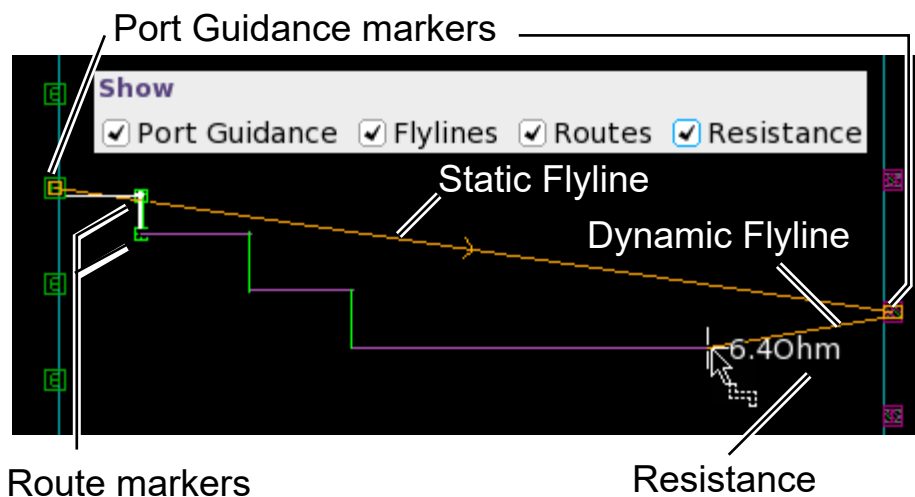


Displaying Route Guidance Flylines and Annotations

The Create Route tool displays flylines and other guidance to help you create the route. When you start the Create Route tool and enter a net name, the tool displays a static flyline between the pins and ports on the route. After you begin the route, the tool adds a dynamic flyline to show the shortest path to the closest pin connection; the dynamic flyline updates as you move the mouse to create the route segments. After you complete the route, the dynamic flyline is removed. By default, the tool displays flylines between the pins and ports on the route, and port guidance markers on the ports in the route.

To hide flylines and port guidance markers, click the More button and deselect the Port Guidance and Flylines options in the Show section of the panel. To display existing routes and a resistance value for the current route that dynamically updates as you create route segments, select the Routes and Resistance options in the panel. The following figure shows the options in the panel and the layout annotations associated with the options.

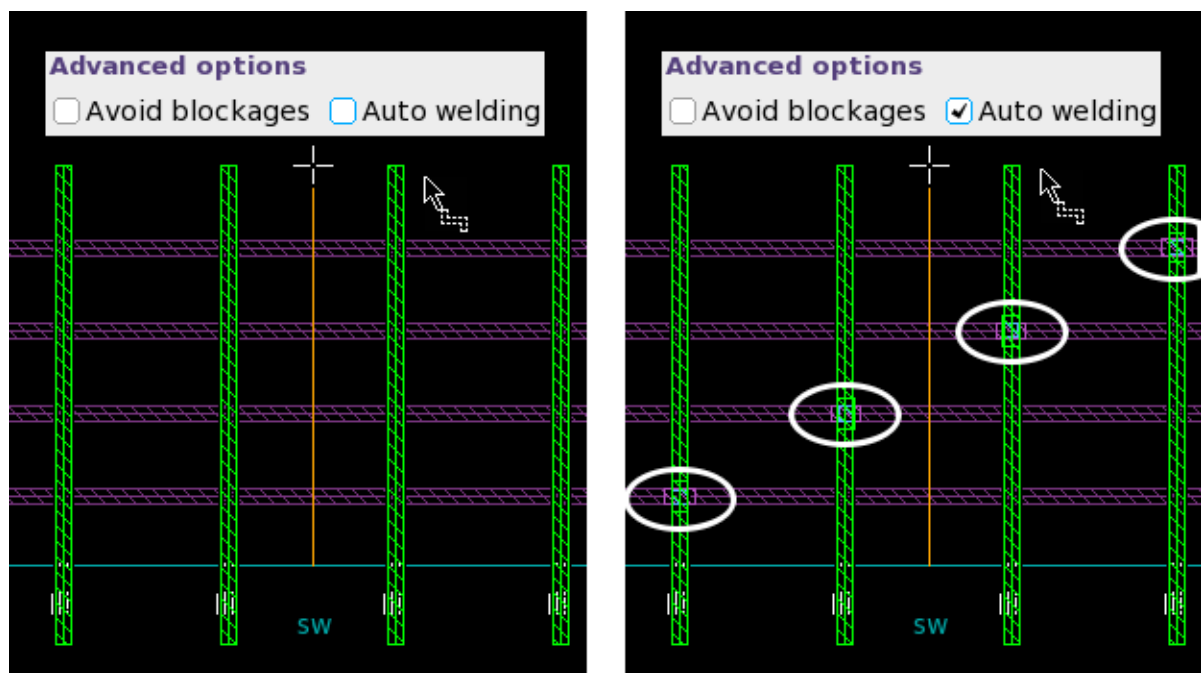
Figure 148 Show Flylines and Other Annotations



Preventing Connections to Existing Routes

To prevent the tool from automatically creating vias when crossing a route of the same net, click the More button and deselect the Auto welding option in the Advanced options section of the panel. The following figure shows the routing result created by changing this option.

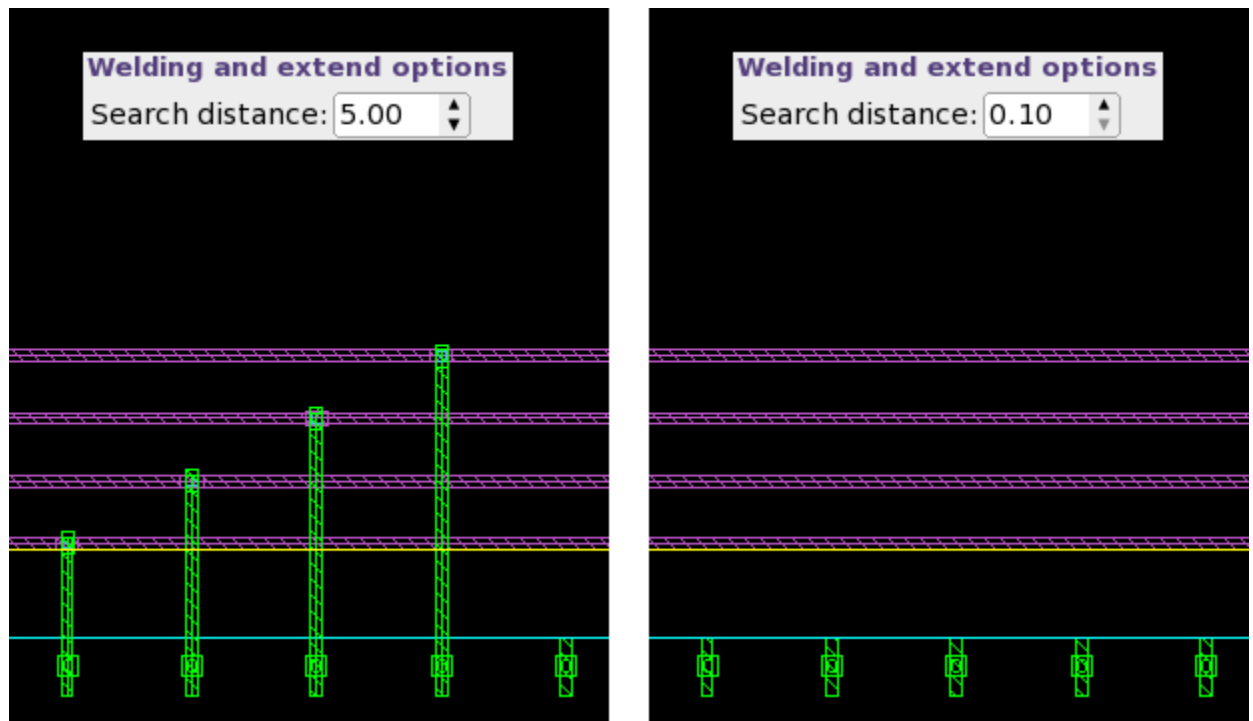
Figure 149 Auto Welding



Setting a Search Distance and Specifying Layers to Search

The Create Route tool automatically connects routes to the pins that are within the search distance of the current mouse position. To prevent the tool from connecting to pins when the Auto Weld option is enabled, click the More button and set the value in the Search distance box in the Welding and extend options section to 0. To allow the tool to connect to pins at a greater distance, increase the distance to allow the tool to search for pins farther from the route. The following figure shows the layout result for the same pins and drawn route (along the yellow highlight) at different search distances.

Figure 150 Search Distance



The tool searches all routing layers for connections. To limit the search to only adjacent layers, click the More button and select the Search Layers: Adjacent option in the Welding and extend options section of the panel. When selected, the tool only connects to pins that are on a layer adjacent to the current routing layer. To specify a range of layers to search, select the Specified option in the same section and select one or more layers in the layer box.

Specifying Other Welding and Connection Options

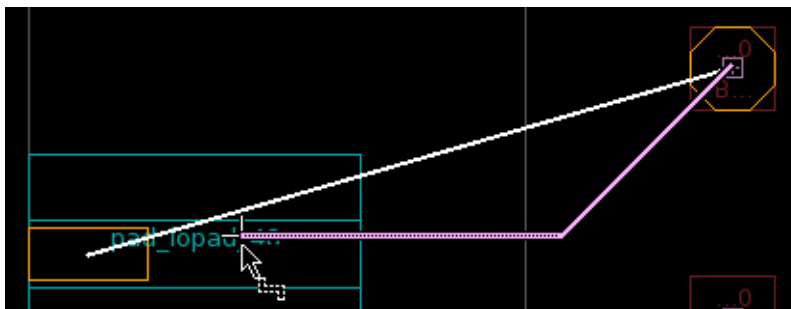
The Create Route tool supports additional options for specifying the connection to existing routes or pins.

- Ignore pin access direction: When selected, the tool automatically connects routes to pins regardless of pin access direction. By default, pin access direction is considered.
- Skip PG nets: When selected, avoid creating a connection to a PG net. This option is used to skip connections to ground nets when the nets are used for net shielding.
- Honor via to via spacing: When selected, the tool honors same-net via-to-via spacing rules.
- No via on pin: Avoids inserting a via when making a connection to a pin.
- Adjust to pin width: When selected, the tool sets the width of the route to the width of the pin, if the wire width is less than the pin width. By default, this option is deselected and the width of the wire is specified in the toolbar. Note that if the wire width is greater than the pin width, the pin width is used for the wire width.
- Align with pin center: When deselected, the tool aligns the connecting wire with the nearest parallel pin edge or extends the wire within the parallel pin edge boundaries. Track snapping and spacing requirements might affect the final wire position. By default, this option is selected and the tool aligns the route with the center of the bounding box for the pin if possible. When the pin is wider than the connecting wire, the tool might shift the wire off-center to honor track snapping and spacing settings.

Creating 45-Degree and RDL Routes

The Create RDL tool supports many of the same features of the Create Route tool. In addition, the Create RDL tool can be used to create redistribution layer (RDL) routes and general 45-degree routes on one metal layer.

Figure 151 Create RDL Tool



To create a 45-degree route with the Create RDL tool,

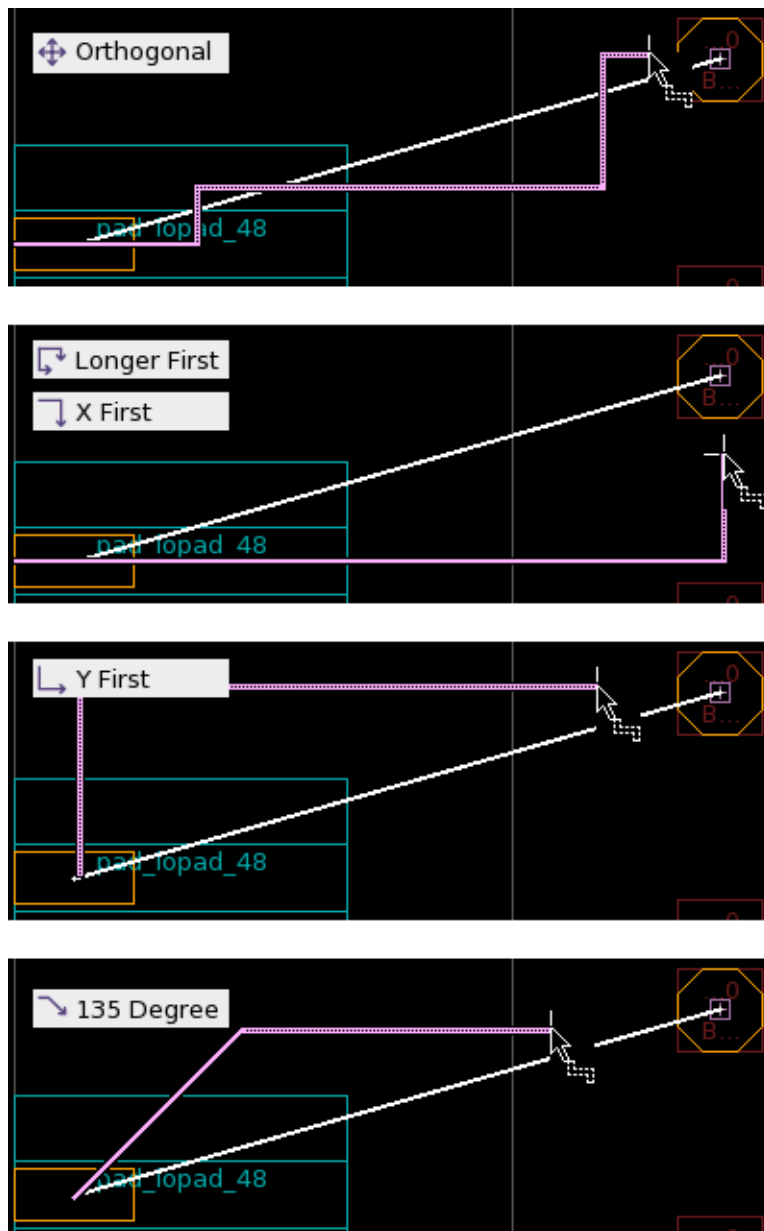
1. Start the Create RDL tool by choosing Create > RDL from the menu or by clicking the Create RDL button ().
2. Enter the net or bundle name to route in the net name box.
3. (Optional) Click the More button and modify settings in the More options panel.

The Create RDL tool supports many of the same settings as the Create Route tool, such as extend first, extend last, high end, low end, and so on as described in the previous sections.

4. Click the Route button at the left of the toolbar and choose the angle to use to route the net.

This setting determines how the route is created as shown in the following figure.

Figure 152 Create RDL Drawing Modes



5. Click the pin, port, or net connection to begin the route.
6. Click to complete a route segment and begin a new segment.
7. Press Enter to complete the route.

Recording and Replaying Create Route Actions

The Create Route and Create RDL tools support a record feature that captures the mouse actions used to create a route and saves them to the Script Editor. The script can be saved to disk and run at a later time. Alternatively, the script can be run directly from the Script Editor. The record feature can be configured with options to tolerate small changes in the position of the pins; this allows you to re-create a route after minor design changes.

To create a route and enable recording,

1. Start the Create Route tool by choosing Create > Route, clicking the Create Route button () , or pressing Shift+R.

Start the Create RDL tool by choosing Create > RDL or by clicking the Create RDL button ().

2. (Optional) Set any necessary routing options in the toolbar.
3. (Optional) If the pin or segment coordinates in the route are expected to change when the route is re-created, click the More button and enable the Relative to nets option in the Log/replay coordinates section.

When this option is enabled, the tool captures coordinates relative to the bounding box for the entire route, rather than capturing absolute coordinates. This option is useful if the script will be applied to create routes in a different area of the design.

4. Click the Record button () on the Create Route tool toolbar.
5. Create the route by clicking the startpoint, creating net segments, and terminating the route.
6. When the route is complete, open the Script Editor and click the Save As button () to save the script to a file.

As you create the net segments, the Create Route tool captures the actions in the Script Editor. A sample script captured by the record function is as follows:

```
...
# Info : recording session started
...
gui_mouse_tool -window [gui_get_current_window -types Layout -mru] \
  -start CreateRouteTool
set_edit_setting -default
set_snap_setting -default
...
gui_set_mouse_tool_option -tool CreateRouteTool -all -default
gui_set_mouse_tool_option -tool CreateRouteTool \
  -option {BundleOrNetNames} \
  -value {net2[127] net2[126] net2[125] net2[124]}
```

```
...
gui_mouse_tool -window [gui_get_current_window -types Layout -mru] \
-add_point {455.756 907.825} -scale 0.02
gui_mouse_tool -window [gui_get_current_window -types Layout -mru] \
-add_point {568.946 1194.223} -scale 5.6331
...
gui_mouse_tool -window [gui_get_current_window -types Layout -mru] -apply
gui_mouse_tool -window [gui_get_current_window -types Layout -mru] -abort
...
# Info : recording session ended
```

To re-create the route, run the script by clicking the Run Script button in the Script Editor. Alternatively, save the script to a file and execute the script with the `source` command.

Performing Interactive Design Rule Checking on Routes

The layout view provides the Editor DRC tool to identify and diagnose DRC violations caused by interactive editing operations. The Editor DRC tool detects design rule violations, displays error shapes in the active layout view, and records the violations in an Editor DRC error data file. Note that the Editor DRC tool runs only a quick check and displays errors using a set of the DRC rules defined in the technology file; the Editor DRC tool does not replace a comprehensive DRC tool such as IC Validator.

In addition to the Create Route tool, the following layout editing tools work with the Editor DRC tool:

- Copy Objects tool
- Create Fill tool
- Create RDL tool
- Create Shape tool
- Create Terminal tool
- Create Via tool
- Quick Connect tool
- Stretch Connected tool
- Stretch tool

Verifying Route Edits Interactively

To enable or disable the Editor DRC tool for interactive editing operations, click the DRC button () on the toolbar and select the DRC enabled option.

When you enable the Editor DRC tool for an interactive editing operation, the tool provides immediate visual feedback if it detects a violation. The layout view displays a different visual cue for each type of violation.

The Editor DRC tool supports rule checking on the following types of objects:

- Horizontal and vertical net shapes and user shapes, including wires, vias, via arrays, paths, polygons, and rectangles
- Port shapes and terminals
- Standard cells, hard and soft macros, and black boxes
- Routing blockages

Verifying Edited Nets

To check selected nets for Editor DRC tool violations,

1. Select one or more nets you need to check.
2. Choose Edit > DRC > Run DRC on Selected Shapes.

Alternatively, you can use the `gui_check_drc_errors` command.

If an error is found, the tool creates the error data and opens the Error Browser with the error data displayed.

Setting Editor DRC Options

The Editor DRC tool provides rule checking for a set of DRC rules defined in the technology file. The interactive editing tools use the Editor DRC tool to detect routing design rule violations that occur during interactive editing operations. You can set the maximum number of violations that the tool can detect and control whether the tool opens the error browser and loads the errors automatically when violations are detected.

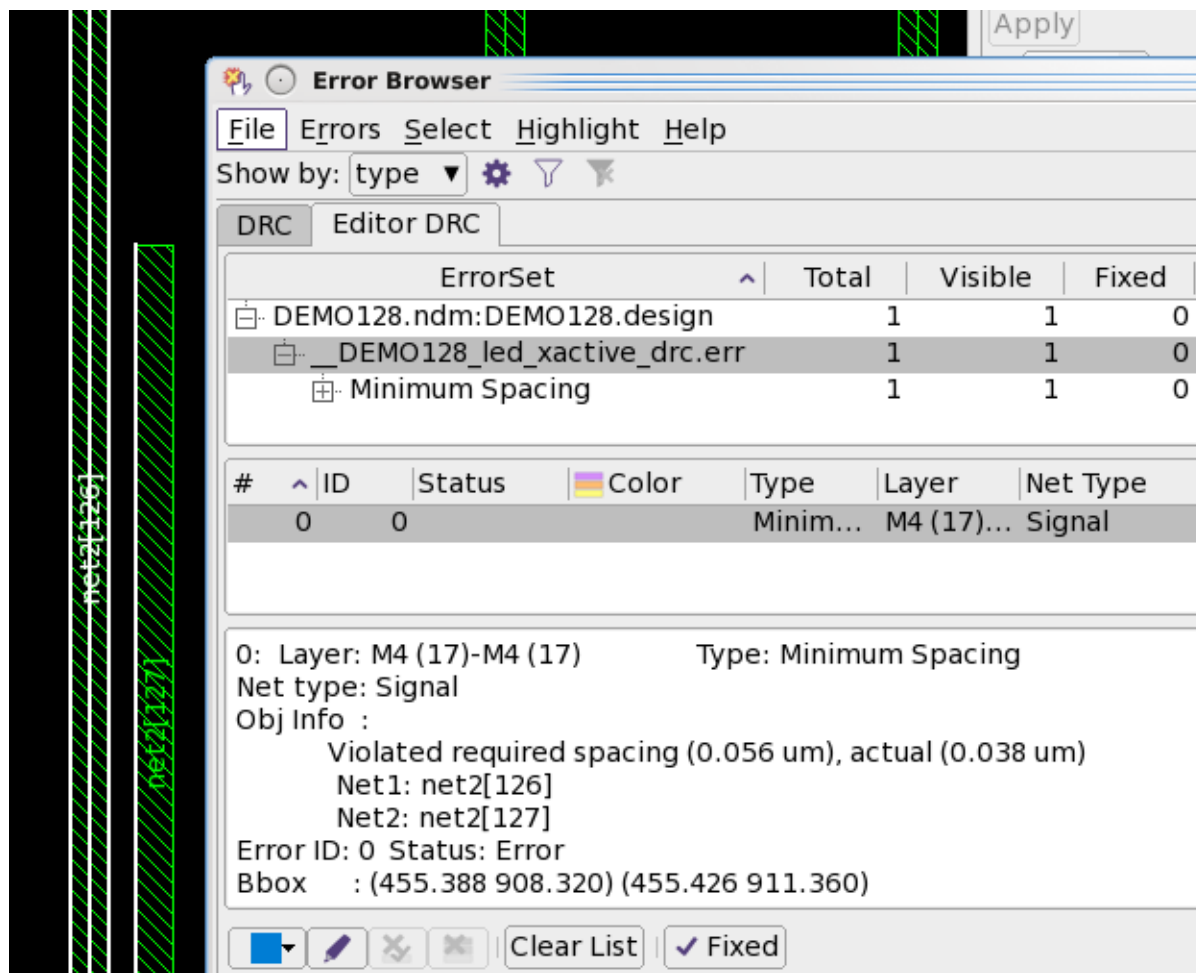
To set Editor DRC tool options, click the DRC button () on the toolbar when a tool that supports the Editor DRC is active.

When you change an option setting in the Editor DRC options dialog box, the change occurs immediately for all applicable tools in all open layout views. If GUI command output logging is enabled in the Application Preferences dialog box, the tool also records the change by displaying a `set_edrc_setting` command in the console log view and history view.

Examining Errors Reported by Editor DRC

When the Editor DRC tool detects design rule violations, it displays error shapes in the active layout view, records the errors in an Editor DRC error view, and automatically loads information about the error into the error browser as shown in the following figure.

Figure 153 Editor DRC



The error view contains error records for violations detected during interactive editing operations and violations detected when you run checks on selected shapes. The Editor DRC tool updates the error data when you fix a violation, reverse an editing operation that caused a violation, disable the design rule check used to detect a violation, or remove errors in the error browser.

The Editor DRC error view is associated with the current design. It remains open and loaded in the error browser until you close the design or change the current design. However, the tool does not save the error view when you save or close the design. If you

close the current design, the Editor DRC tool closes the error view and removes the errors from the error browser.

Creating Routes With Tcl Commands

The Fusion Compiler GUI supports Tcl commands to create route trunks for specified nets and bundles. Use the `create_trunk`, `create_trunk_pin_to_trunk`, and `create_trunk_pin_to_pin` commands to create route trunk segments and connections as described in this section.

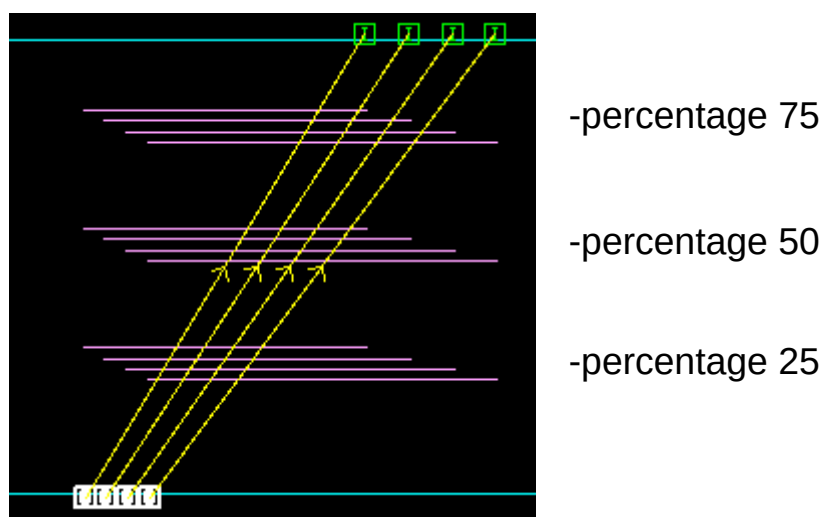
Creating Route Segments in One Direction

To create route trunks in one direction on a single metal layer, use the `create_trunk` command as follows:

```
fc_shell> set nets [get_nets {net1 net2 net3 net4}]
fc_shell> create_trunk -nets $nets -layer M3 -gravity percentage \
    -percentage 25
```

The command creates a route segment for each net or bundle specified by the `-nets` argument. In this example, the `-gravity percentage` option specifies how the route segments are aligned with the bounding box for the nets. The `-percentage 25` option specifies that the first route segment is placed at the bottom edge of the bounding box plus 25 percent of the height of the bounding box. The following figure shows different routing results for percentage values of 25, 50, and 75.

Figure 154 Create Trunk



Specify the following options to the `create_trunk` command control how the routes are created.

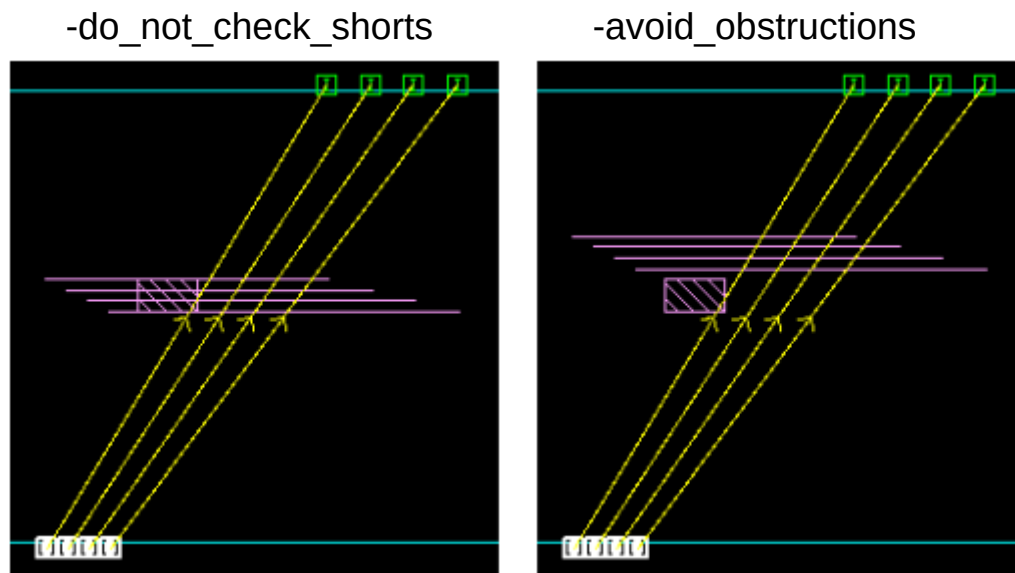
- Specify the routing direction for the layer with the `-direction horizontal | vertical` option. By default, the command creates routes in the preferred direction.
- Align routes with the driver for the net by specifying `-gravity driver`. Align routes with the load for the net by specifying `-gravity load`.
- Create the routes on specific tracks within the bounding box for the net by using the `-track_numbers`, `-track_coords`, `-track_step`, and `-track_ref_bbox` options. For example, to create routes within the bounding box from (50,50) to (100,100), starting from the second track within the box, specify the following command:

```
fc_shell> create_trunk -nets $nets -layer M3 \  
    -track_ref_bbox {{50 50} {100 100}} -track_numbers 2
```

- Place routes on the next available unobstructed track by using the `-avoid_obstructions` option. By default, the command issues a warning message and skips the net if an obstruction is found.
- Disable checks for shorts and obstructions and preview the route with the `-do_not_check_shorts` option.

The following figure shows the layout result created by using the `-do_not_check_shorts` and `-avoid_obstructions` options.

Figure 155 Ignoring Obstructions and Shorts, and Avoiding Obstructions

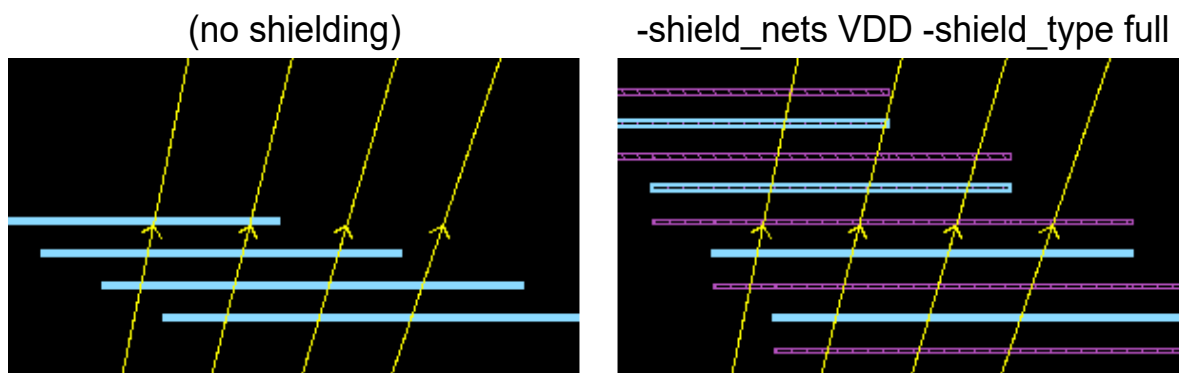


- Create shielding nets by using the `-shield_nets` and `-shield_type` options. The `-shield_type` option supports full, partial, or cap shields on the routes.

The following figure shows the default layout result without shield nets and the layout result after running the following command to create shielding nets. The nonshield nets specified by `$nets` are highlighted for reference.

```
fc_shell> create_trunk -nets $nets -layer M3 -shield_nets VDD \
    -shield_type full
```

Figure 156 Shield Nets



- Avoid snapping routes to tracks by using the `-do_not_snap_to_track` option.
- Create more than one route per net by using the `-num_wires` option.
- Restrict the routes only certain track widths by using the `-width` option. By default, routes are created on the tracks with the default width.

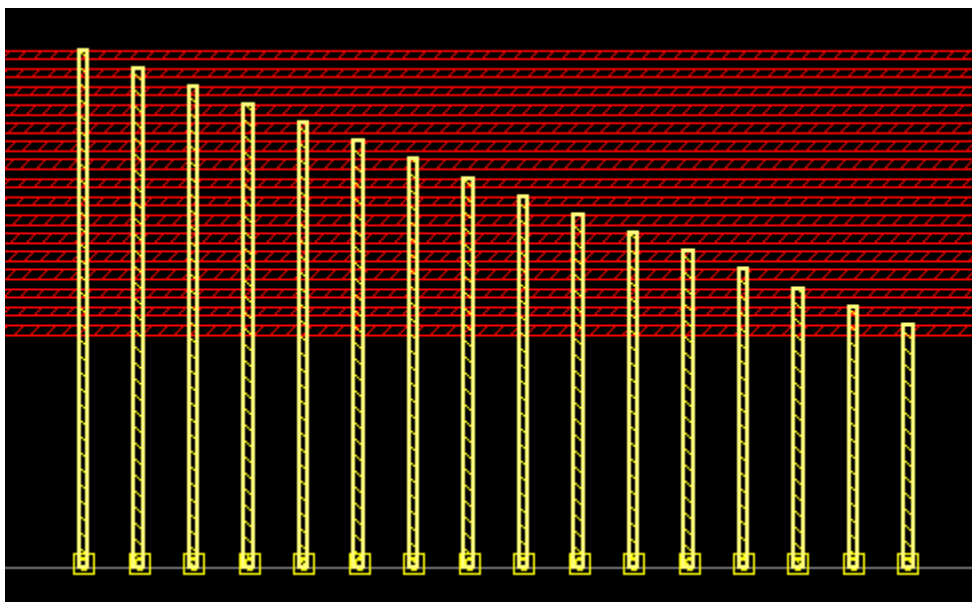
Creating Routes From Pins to Existing Route Segments

To create routes between pins and existing route segments, use the `create_trunk_pin_to_trunk` command as follows:

```
fc_shell> create_bundle b1 [get_nets {net1 net2 net3 net4}]
fc_shell> create_trunk_pin_to_trunk -nets b1
```

The command creates a route segment between an existing route trunk and the closest pin associated with the nets in the route trunk as shown in the following figure. In this example, the route is created on the same layer as the pin.

Figure 157 Connect Pins to Existing Routes

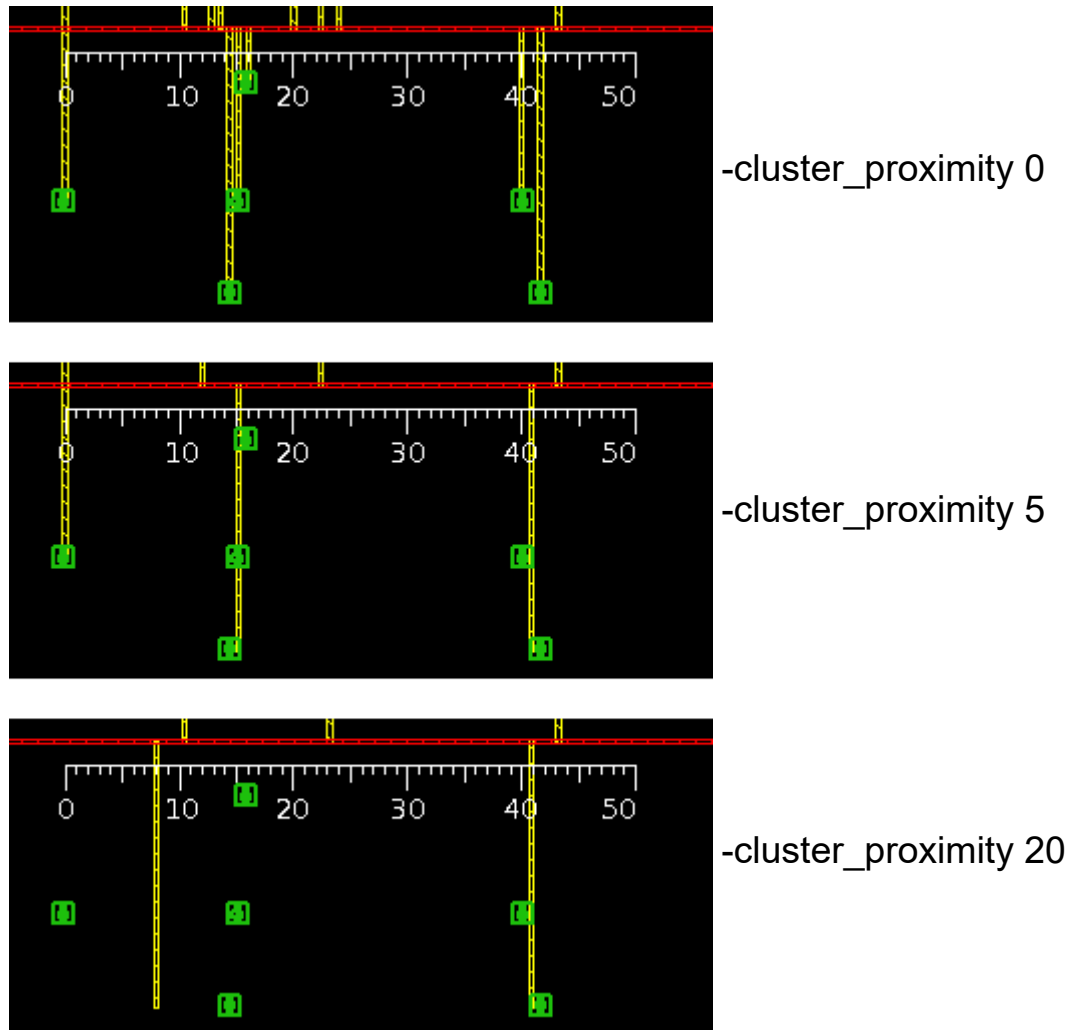


Specify the following options to the `create_trunk_pin_to_trunk` command to control how the routes are created.

- Specify the layer to use to connect the pins to the routes with the `-layer` option. The command creates routes in the preferred direction for the specified layer and skips connections in the nonpreferred direction.
- Create route segments in a specific direction with the `-direction horizontal | vertical` option. Connections in the other direction are skipped.
- Avoid snapping routes to tracks by using the `-do_not_snap_to_track` option.
- Place routes on the next available, unobstructed track by using the `-avoid_obstructions` option. By default, the command issues a warning message and skips the net if an obstruction is found.
- Disable checks for shorts and obstructions and preview the route with the `-do_not_check_shorts` option.
- Avoid snapping routes to tracks by using the `-do_not_snap_to_track` option.
- Create shielding nets by using the `-shield_nets` and `-shield_type` options. The `-shield_type` option supports full, partial, or cap shields on the routes.
- Reduce the number of route shapes created from the trunk by specifying the `-cluster_proximity` option. Use this option to specify the range in microns within which to group individual pin-to-trunk shapes to a single shape. The command creates the route segment from the trunk, but might not complete the connection to the pin.

The following figure shows the layout result after specifying three different values for the cluster proximity.

Figure 158 Cluster Proximity



To complete the missing connections, run the `create_trunk_pin_to_trunk` command again and specify the layer for the orthogonal direction. The following example first creates the vertical routes, then calls the `create_trunk_pin_to_trunk` command again to complete the connection to the pins.

```
fc_shell> create_trunk_pin_to_trunk -nets $nets -layer METAL2 \
    -cluster_proximity 40
fc_shell> create_trunk_pin_to_trunk -nets $nets -layer METAL3
```

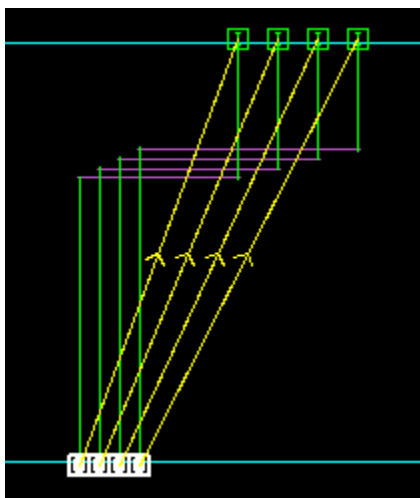
Creating Routes From Pin to Pin

To create route segments between pins, use the `create_trunk_pin_to_pin` command as follows:

```
fc_shell> create_bundle b1 [get_nets {net1 net2 net3 net4}]
fc_shell> create_trunk_pin_to_pin -nets b1 \
    -extensions {0.67 1.0 0.33} -layer_legs {M4 M3 M4}
```

This command produces the following layout result:

Figure 159 Route Trunk Pin to Pin



In this example, the `-extensions` option specifies the relative length of the route segments, starting at the driving pins and ending at the load pins. To reverse the direction of a route segment, specify a negative extension value. To route from the destination pins, see the description of the `-start_from` option in the following list. The corresponding layers for the segments are specified with the `-layer_legs` option.

Specify the following options to the `create_trunk_pin_to_pin` command to control how the routes are created.

- Specify the specific track widths to use for a given layer with the `-layer_legs` option. For example, the following command places routes on layers M2 and M3 only on tracks with a width of 0.4 or 0.8. To use any available track width, replace `{0.4 0.8}` with the `all` argument. By default, the command places routes only on tracks with the default width.

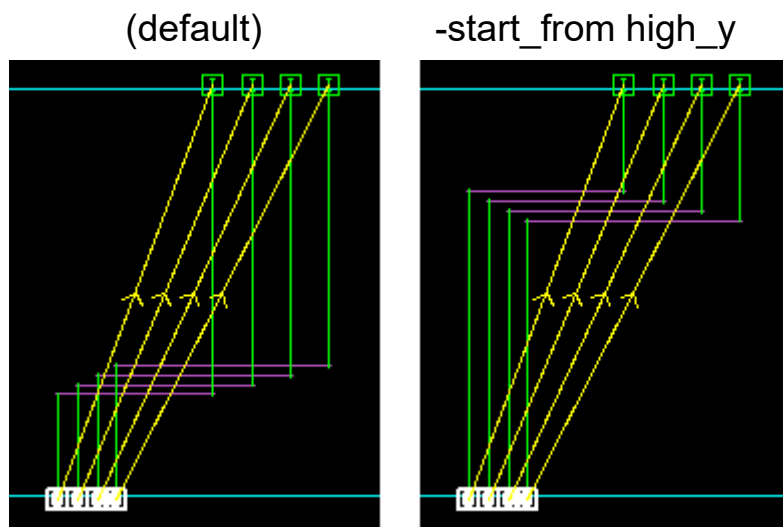
```
fc_shell> create_trunk_pin_to_pin -nets $nets \
    -layer_legs {{METAL2 {0.4 0.8}} {METAL3 {0.4 0.8}}} \
    -extensions {1 1}
```

- Specify the length of each segment with the `-extensions` option. The option takes a list of extensions, each element in the list specifies the length of one segment in the route. Extensions can be specified as a fraction of the net bounding box, a percentage of the net bounding box, a distance in microns, or as user-defined units.
- Specify the layer to use for each segment with the `-layer_legs` option together with a list of layers. Each layer corresponds to the segment specified by the `-extensions` option.
- Specify a bounding box in which to create the route segments with the `-net_bbox_override` option. The actual bounding box for the specified nets is ignored and the command does not try to connect the pins.
- Specify the relative location of the pins which to begin the route with the `-start_from` option.

In the following example, the first command begins the routes from the source pins at the bottom of the figure and produces the layout result shown on the left; the second command begins the routes from the destination pins at the top of the figure and produces the layout result shown on the right.

```
fc_shell> create_trunk_pin_to_pin -nets b4 \
    -extensions {0.25 1.0 0.75} -layer_legs {M4 M3 M4}
fc_shell> create_trunk_pin_to_pin -nets b4 \
    -extensions {0.25 1.0 0.75} -layer_legs {M4 M3 M4} \
    -start_from high_y
```

Figure 160 Start From Location



- Avoid snapping routes to tracks by using the `-do_not_snap_to_track` option.
- Create shielding nets by using the `-shield_nets`, `-shield_type`, and `-shield_pullback` options. The `-shield_type` option supports full, partial, or cap shields on the routes.

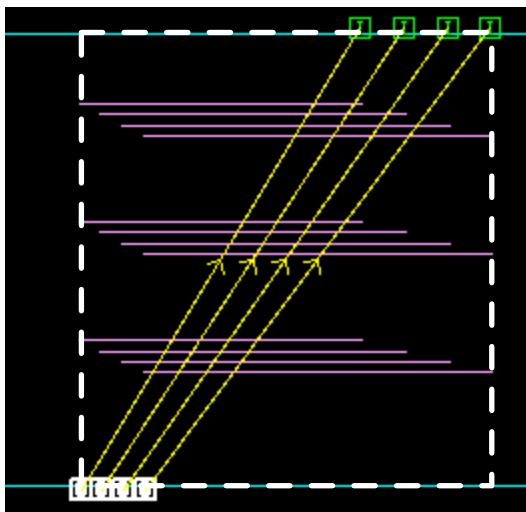
```
fc_shell> create_trunk_pin_to_pin -nets $nets \  
    -extensions {0.5 1.0 0.5} -layer_legs {M2 M3 M2} \  
    -shield_type full -shield_nets VDD
```

- Disable checks for shorts and obstructions and preview the route with the `-do_not_check_shorts` option.
- Shorten the route segment from the starting pin with the `-start_pullback` option, or shorten the route segment at the ending pin with the `-stop_pullback` option. Specify a negative value for the distance to extend the route segment rather than shorten it.

Routing Area Bounding Box

By default, the `create_trunk` and `create_trunk_pin_to_pin` commands create routes within a bounding box formed by the source and destination pins as shown in the following figure.

Figure 161 Route Trunk Default Bounding Box



To create the route trunks within a specific bounding box, specify the `-net_bbox_override` option with the `create_trunk` and `create_trunk_pin_to_pin` commands. The following example creates route trunks for the `net1` and `net2` nets within the box bounded by (100,100) and (200,200).

```
fc_shell> create_trunk -nets {net1 net2} -layer METAL3 \  
    -net_bbox_override {{100 100} {200 200}}
```

16

Extending the Graphical User Interface

The Fusion Compiler GUI provides an extensive application programming interface (API) to support customization of menus, buttons, and other elements of the GUI. Use the API to create custom buttons, menu items, map modes, and link them to custom or built-in functions in the tool. Customizations can be loaded automatically when the tool starts, or added later by sourcing the commands from a script file.

In addition, the tool supports annotations. Use annotations to draw lines, rectangles, polygons, symbols, and text on the layout view.

The following topics describe different types of GUI customization:

- [Creating Menu Items](#)
- [Creating Toolbars and Toolbar Items](#)
- [Adding Buttons to the Favorites Palette](#)
- [Highlighting Objects with Tcl Commands](#)
- [Creating Annotations](#)
- [Creating New Visual Modes](#)

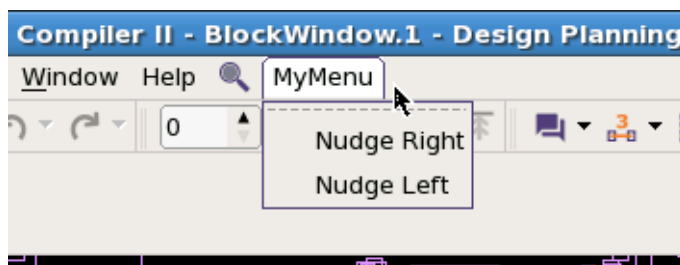
Creating Menu Items

The Fusion Compiler tool supports menu customization with the `gui_create_menu` and other commands. Use the `gui_create_menu` commands to modify the menu bar and add new commands; use the `gui_set_hotkey` command to define or redefine hotkeys.

The following independent examples create increasingly complex menu customizations for new menu entries called “Nudge Left” and “Nudge Right”, which pan the view slightly to the left and right.

- Define two basic menu items.

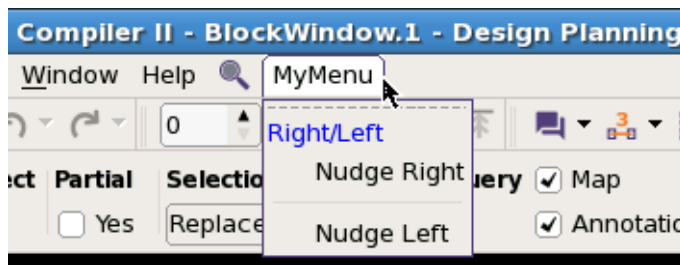
Figure 162 Simple Menu Item



```
gui_create_menu -menu "MyMenu->Nudge Right" \  
  -tcl_cmd {gui_scroll \  
    -window [gui_get_current_window -view] -hrel .05}  
  
gui_create_menu -menu "MyMenu->Nudge Left" \  
  -tcl_cmd {gui_scroll \  
    -window [gui_get_current_window -view] -hrel -.05}
```

- Extend the previous customization to create a header for the menu items and insert a separator.

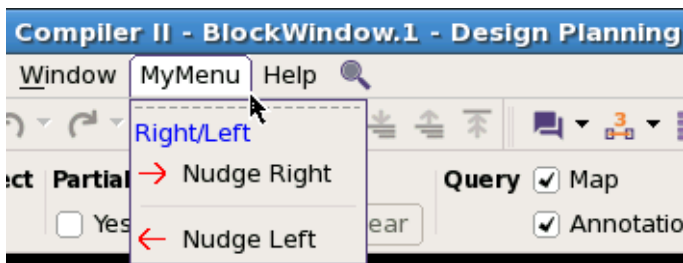
Figure 163 Menu With Header and Separator



```
gui_create_menu -menu "MyMenu->Right/Left" -heading "Right/Left"  
gui_create_menu -menu "MyMenu->Nudge Right" \  
  -tcl_cmd {gui_scroll \  
    -window [gui_get_current_window -view] -hrel .05}  
  
gui_create_menu -menu "MyMenu->____" -separator  
gui_create_menu -menu "MyMenu->Nudge Left" \  
  -tcl_cmd {gui_scroll \  
    -window [gui_get_current_window -view] -hrel -.05}
```

- Extend the previous customization to place the new menu items one position to the left of the Help menu and define a custom icon for each menu item.

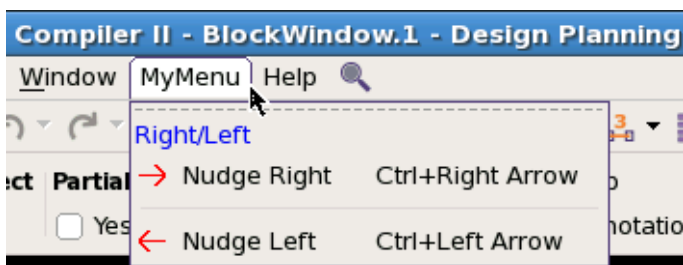
Figure 164 Menu With Specific Placement



```
gui_create_menu -menu "MyMenu->Right/Left" \
    -heading "Right/Left" \
    -anchor_item Help \
    -anchor_offset -1
gui_create_menu -menu "MyMenu->Nudge Right" \
    -tcl_cmd {gui_scroll \
        -window [gui_get_current_window -view] -hrel .05} \
    -icon_file /u/user/icons/right_16_xpm.xpm
gui_create_menu -menu "MyMenu->___" -separator
gui_create_menu -menu "MyMenu->Nudge Left" \
    -tcl_cmd {gui_scroll \
        -window [gui_get_current_window -view] -hrel -.05} \
    -icon_file /u/user/icons/left_16_xpm.xpm
```

- Extend the previous customization to redefine a hotkey definition with the `gui_set_hotkey` command after defining the menu items with the `gui_create_menu` command. The hotkey definition appears in the menu.

Figure 165 Menu with Hotkey Definitions



```
gui_set_hotkey -replace -hot_key Ctrl+Right \
    -menu "MyMenu->Nudge Right"
gui_set_hotkey -replace -hot_key Ctrl+Left \
    -menu "MyMenu->Nudge Left"
```

If the hotkey is not previously assigned, you can assign it with the `-hot_key` option to the `gui_create_menu` command. This example uses `gui_set_hotkey -replace` to overwrite an existing hotkey setting.

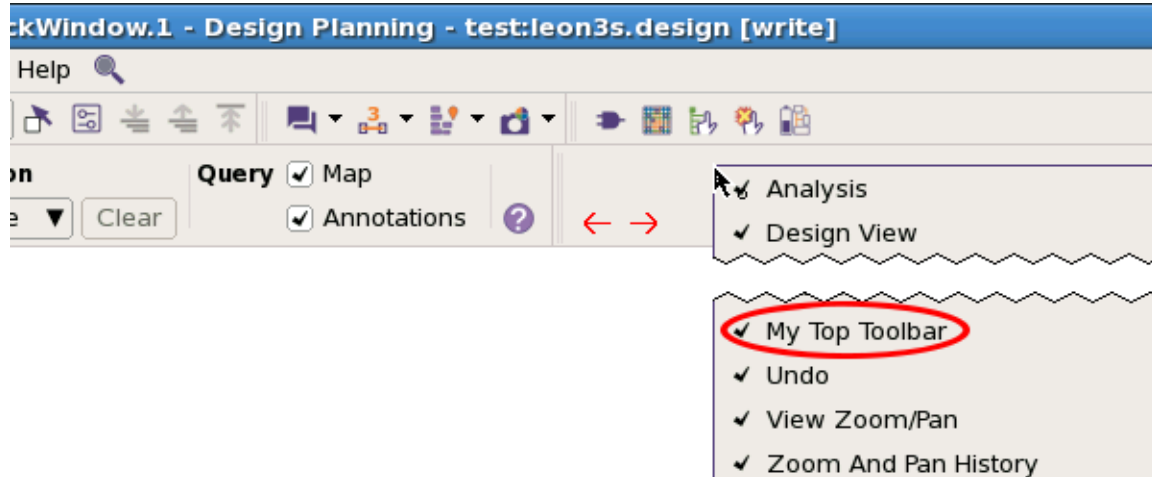
Creating Toolbars and Toolbar Items

Use the `gui_create_toolbar` and `gui_create_toolbar_item` commands to create new toolbars and add menu items to the toolbars. The following example creates a toolbar at the top and inserts toolbar items for the “Nudge Left” and “Nudge Right” menu items defined in the previous section.

```
gui_create_toolbar -name "MyTopToolbar" \  
-title "My Top Toolbar" -dock_side Top  
gui_create_toolbar_item -toolbar MyTopToolbar \  
-menu "MyMenu->Nudge Left"  
gui_create_toolbar_item -toolbar MyTopToolbar \  
-menu "MyMenu->Nudge Right"
```

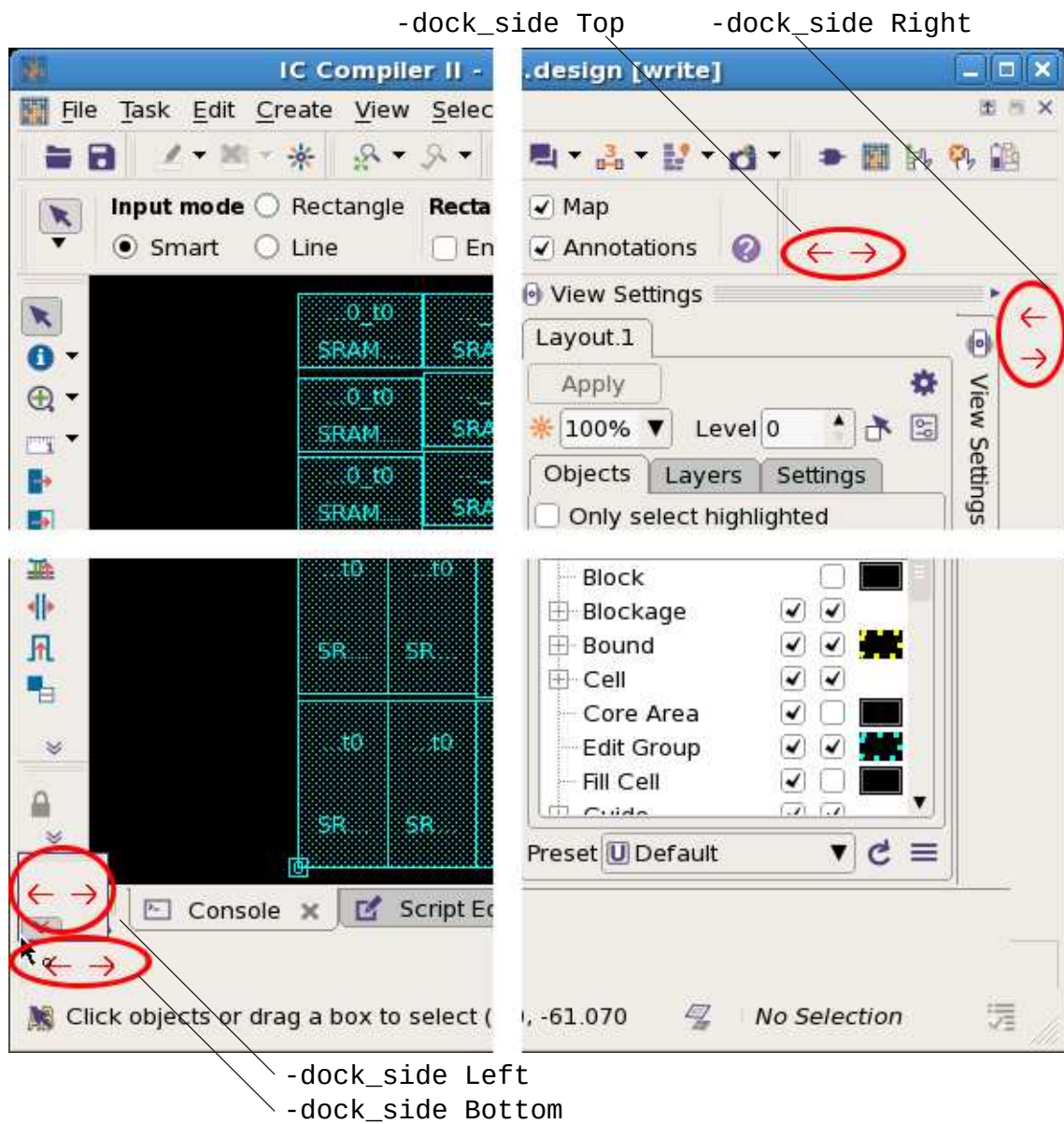
To view the toolbar, make sure the new toolbar is selected by right-clicking in the window and enabling the menu selection. Alternatively, choose View > Toolbars > My Top Toolbar from the menu bar.

Figure 166 Toolbar Menu



You can dock a toolbar at the top, bottom, left, or right of the window. The following figure shows the GUI after creating a new toolbar at each dock location and adding toolbar items for the “Nudge Left” and “Nudge Right” menu items.

Figure 167 Toolbar Dock Locations for `gui_create_toolbar`



Adding Buttons to the Favorites Palette

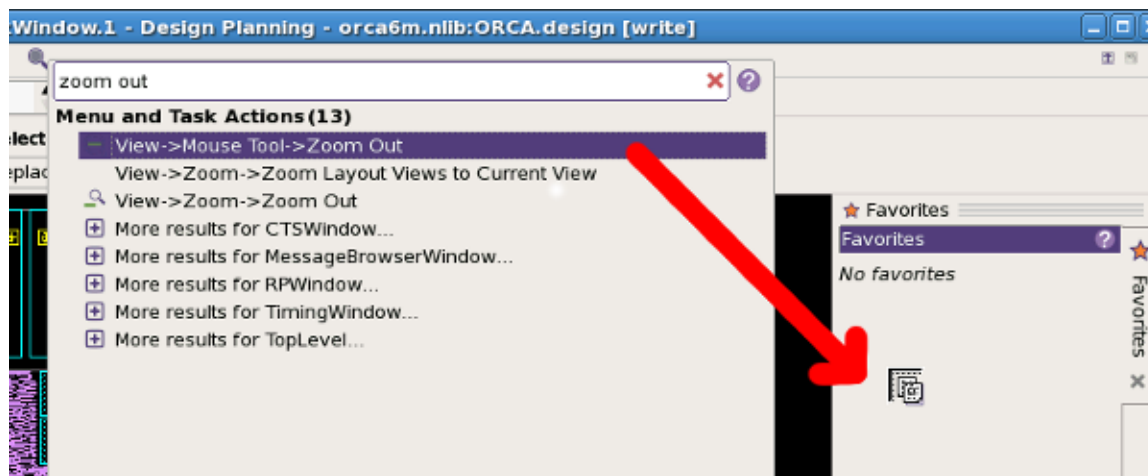
Instead of creating a new toolbar, you can add buttons to the Favorites palette. To add a button item to the Favorites palette,

1. Press F10 to open the Favorites palette.
2. Click a button and drag it onto the Favorites palette.

To find a button item using command search and add it to the Favorites palette,

1. Press F10 to open the Favorites palette.
2. Click the magnifying glass to open command search.
3. Enter the command or task name.
4. Click the item in the result list and drag it onto the Favorites palette.

Figure 168 Dragging a Button to the Favorites Palette



Highlighting Objects with Tcl Commands

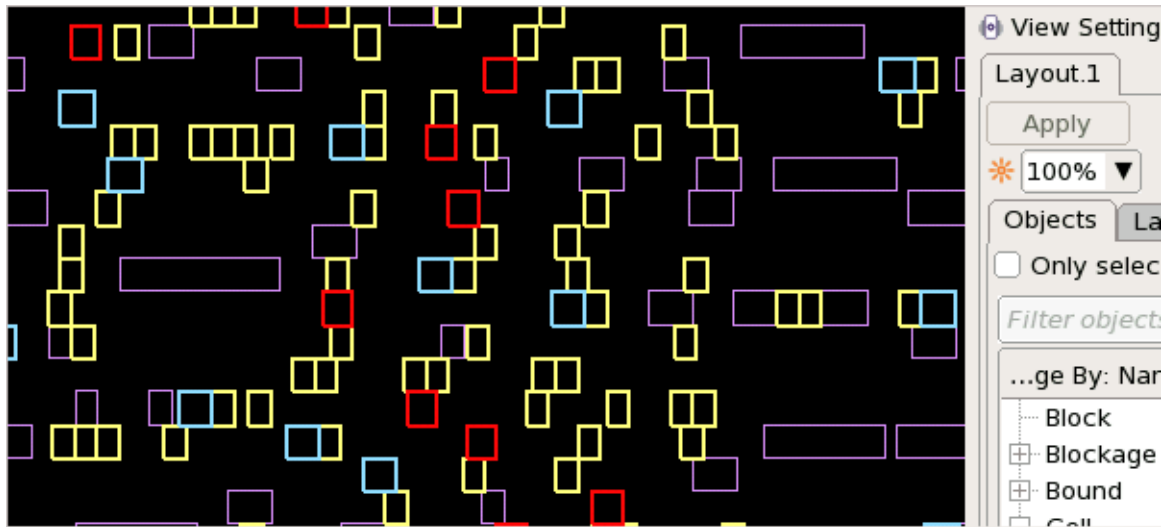
Using Tcl commands, you can highlight objects in the GUI to display them in different colors for easy identification. You can also define a collection of objects to color.

- Highlight NAND2*, NAND3*, and NAND4* cells using three different colors.

```
gui_change_highlight -color yellow
    -collection [get_cells -hier -filter "ref_name=~NAND2*"]
gui_change_highlight -color red
    -collection [get_cells -hier -filter "ref_name=~NAND3*"]
```

```
gui_change_highlight -color light_blue  
-collection [get_cells -hier -filter "ref_name=~NAND4*"]
```

Figure 169 Highlighted Cells



- Remove red highlights.

```
gui_change_highlight -color red -remove
```

- Remove all highlights.

```
gui_change_highlight -all_colors -remove
```

Creating Annotations

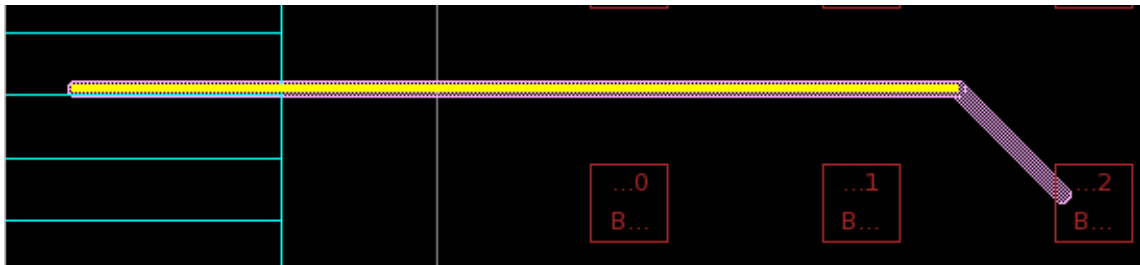
To add information to the layout view, use annotations to create lines, rectangles, polygons, and text. Annotations are not saved to the design database and therefore are not persistent. Use the `gui_add_annotation` command to add annotations; use the `gui_remove_annotations` command to remove annotations.

The following examples create different types of annotations. The final example removes all annotations.

- Add a line annotation

```
gui_add_annotation -window Layout.1 -type line -color yellow \  
-width 4 {{67 2748} {636 2748}}
```

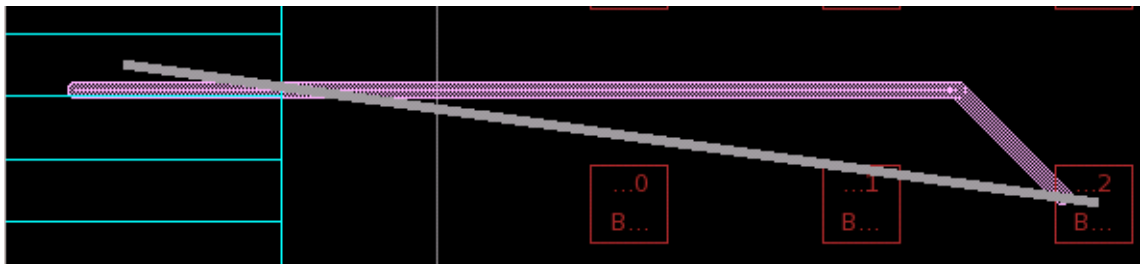
Figure 170 Line Annotation



- Add a gray line from the center of cell pad_iopad_40 to the center of cell left_15_2 and specify the color as hexadecimal value.

```
set cell11 [get_cells pad_iopad_40]
set cell12 [get_cells left_15_2]
gui_add_annotation -window Layout.1 -width 5 -color #999999 \
    -type line [list [list $cell11 center] [list $cell12 center]]
```

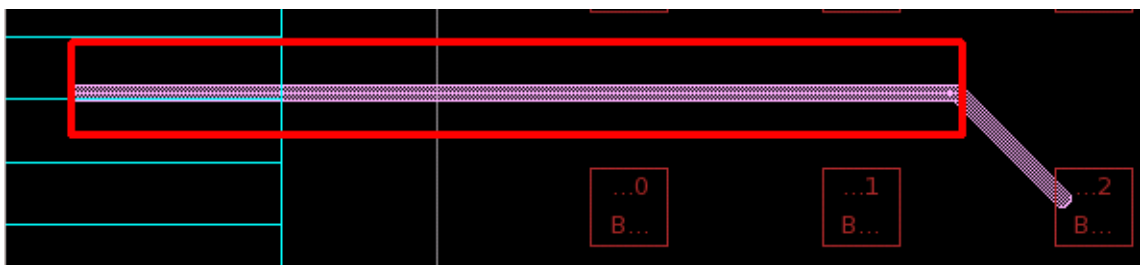
Figure 171 Line Annotation Between Cells



- Add a rectangle annotation.

```
gui_add_annotation -window Layout.1 -type rect -color red \
    -width 4 {{65 2700} {640 2800}}
```

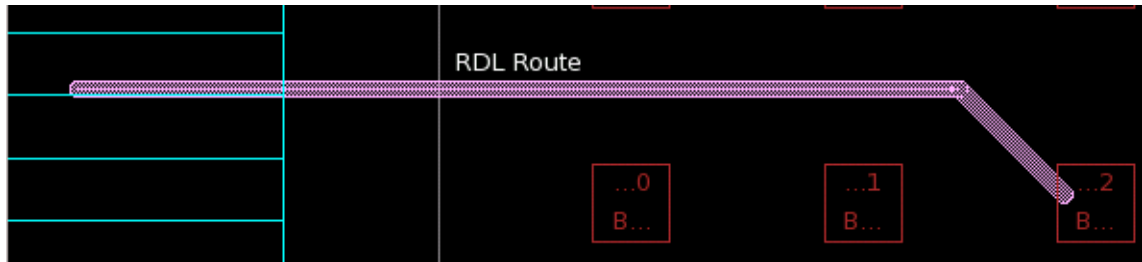
Figure 172 Rectangle Annotation



- Add a text annotation. You must make sure that the bounding box is large enough to fit the text.

```
gui_add_annotation -window Layout.1 -type text -color white \
    -text "RDL Route" {{65 2715} {640 2815}}
```

Figure 173 Text Annotation



- Add multiple annotations and include an infotip for each annotation.

```
set net pad[40]
set shapes [get_shapes -q -of_objects $net]
foreach_in_collection shape $shapes {
    set len [get_attribute $shape length]
    gui_add_annotation -window Layout.1 -type line \
        -color yellow -width 4 \
        -info_tip "RDL Route $net\nLength = $len" \
        [get_attribute $shape points]
}
```

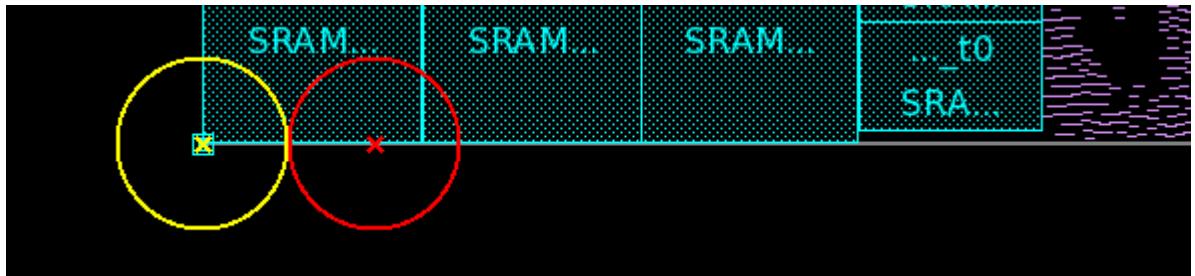
Figure 174 Annotations with Infotip



- Add two circular annotations, each with a radius of 100.

```
gui_add_annotation -type circle -color yellow -width 2 \
    -radius 100 {{0 0}}
gui_add_annotation -type circle -color red \
    -width 2 {{200 0} {300 0}}
```

Figure 175 Circular Annotation

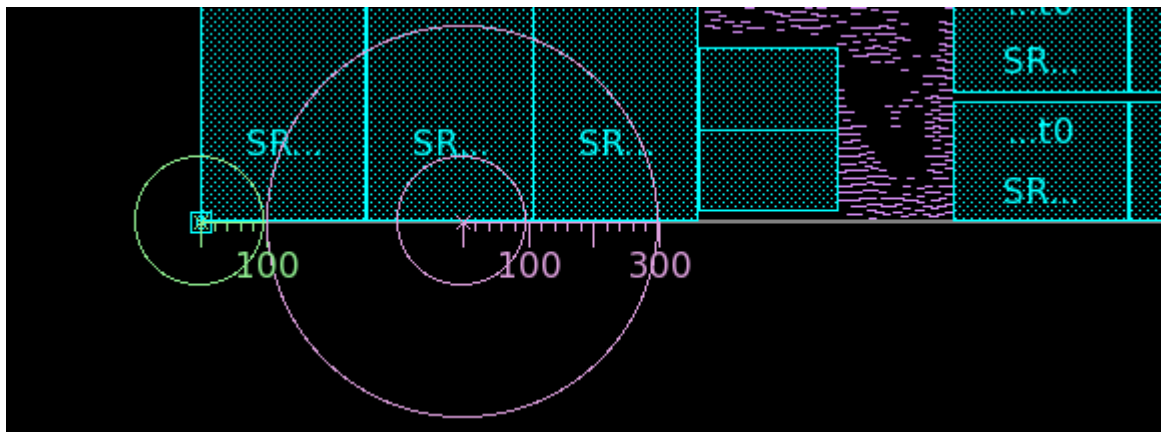


- Create two circular rulers centered at (0, 0) and (400, 0) respectively.

```
gui_add_annotation -type circular_ruler \
    -color light_green {{0 0} {100 0}}
gui_add_annotation -type circular_ruler \
    -color light_purple {{400 0} {500 0} {700 0}}
```

You can specify additional points to create additional circular rulers with the same center. The `-text`, `-pattern`, `-width`, and `-line_style` options are ignored when creating circular rulers with the `gui_add_annotation` command.

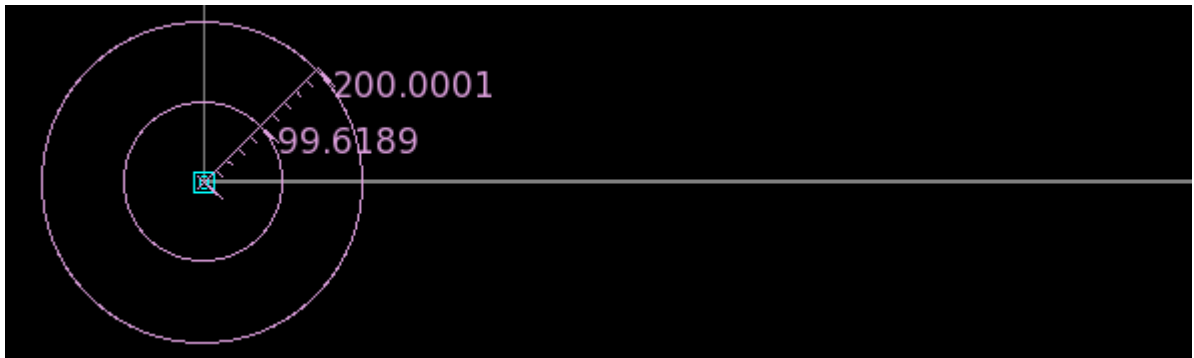
Figure 176 Circular Rulers



- Create a circular ruler at a 45-degree angle.

```
gui_add_annotation -type circular_ruler -color light_purple \
    {{0 0} {70.171 70.711} {141.421 141.421}}
```

Figure 177 Circular 45-Degree Ruler



- Remove all annotations from the current window.

```
gui_remove_annotations \  
-window [gui_get_current_window -view -mru]
```

- Remove all annotations from all windows.

```
gui_remove_annotations [gui_get_annotations]
```

Saving Annotations

To write out annotations created by the `gui_add_annotation` command, use the `gui_write_annotations` command. By default, the `gui_write_annotations` command writes out global annotations. To write out all current layout annotations, specify the command with the `[gui_get_annotations]` argument.

Use the following options with the `gui_write_annotations` command to control the output.

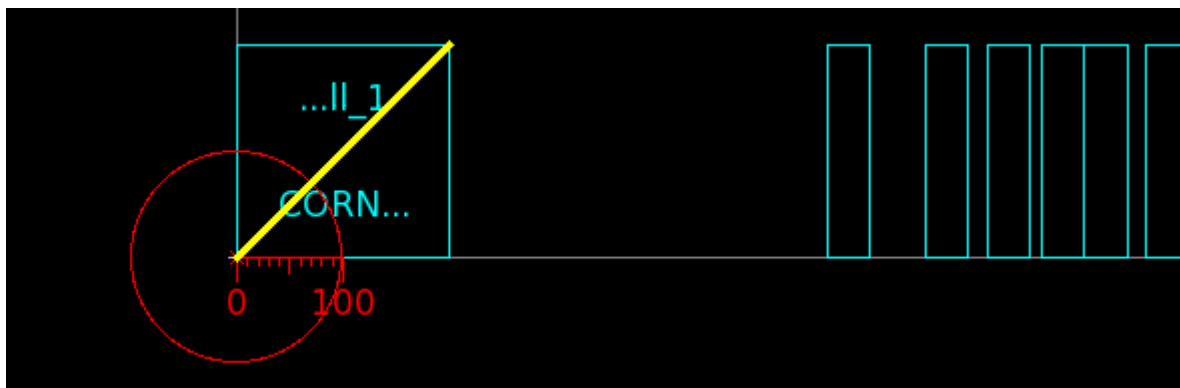
- `-file file_name`: Specify the output file name.
- `-append`: Append the output to the specified file. By default, the command issues an error message if you specify an existing file with the `-file` option.
- `-force`: Overwrite the specified output file.
- `-compress gzip`: Compress the output file using gzip.
- `annotations`: Specify annotations to write to the output file.

The following example creates two annotations: a line annotation from (0, 0) to (200, 200) and a circle annotation at (0, 0). Note that the first annotation uses the `-window` option to specify the window while the second annotation does not.

```
gui_add_annotation -window Layout.1 -type line -color yellow \  
-width 4 {{0 0} {200 200}}  
gui_add_annotation -type circular_ruler \  
-color red {{0 0} {100 0}}
```

These commands create the following annotations in the layout:

Figure 178 Line and Circle Annotations



The `gui_write_annotations -file file_name [gui_get_annotations]` command writes out all global and window-specific annotations.

```
fc_shell> gui_write_annotations \  
-file all_gui_annotations.tcl [gui_get_annotations]  
fc_shell> sh cat all_gui_annotations.tcl  
...  
set Layout_1 Layout.1  
gui_add_annotation -window $Layout_1 -group global -type polyline \  
-color yellow -pattern NoBrush -line_style SolidLine \  
-width 4 {{0.000 0.000} {200.000 200.000}}  
gui_add_annotation -window global -group global \  
-type circular_ruler -color red -pattern NoBrush \  
-line_style SolidLine -width 1 {{0.000 0.000} {100.000 0.000}}
```

The `gui_write_annotations -file file_name` command writes out only the global annotations (specified without the `-window` option).

```
fc_shell> gui_write_annotations -file global_gui_annotations.tcl  
fc_shell> sh cat global_gui_annotations.tcl  
...  
gui_add_annotation -window global -group global \  
-type circular_ruler -color red -pattern NoBrush \  
-line_style SolidLine \  
-width 1 {{0.000 0.000} {100.000 0.000}}
```

Creating New Visual Modes

Visual modes highlight design information in colored overlays. Use visual modes to highlight different elements of your design and organize the elements into bins. The `gui_create_vm` and `gui_create_vmbucket` commands define new visual modes, create individual bins, and associate a collection with each bin.

The following example creates a new visual mode to color macros in different sections of the design hierarchy. The `gui_create_vm` command creates a new visual mode called **Macros**. The `reload_macros_vm` procedure is called when you click the Reload button after starting the Macros visual mode. The `gui_create_vmbucket` command assigns individual bins in the visual mode to specific collections. The `gui_show_map` command opens the Visual Mode palette and displays the new Macros visual mode. After the tool displays the Visual Mode palette, click the Reload button to color the macros. For more information about the Visual Mode palette, see [Visual Mode Overview](#).

```
gui_create_vm -name Macros -update_cmd reload_macros_vm
proc reload_macros_vm {args} {
    set vm Macros
    gui_remove_vmbucket -vmname $vm -all
    set collection_1 [get_cells -physical_context \
        -filter "is_hard_macro && full_name=~I_CLOCK_GEN/*"]
    set collection_2 [get_cells -physical_context \
        -filter "is_hard_macro && full_name=~I_ORCA_TOP/I_PCI_TOP/*"]
    set collection_3 [get_cells -physical_context \
        -filter "is_hard_macro && full_name=~I_ORCA_TOP/I_CONTEXT_MEM/*"]
    set collection_4 [get_cells -physical_context \
        -filter "is_hard_macro && full_name=~I_ORCA_TOP/I_RISC_CORE/*"]

    gui_create_vmbucket -vmname $vm -name clock_gen \
        -title {clock_gen} -color red -collection $collection_1
    gui_create_vmbucket -vmname $vm -name pci_top \
        -title {pci_top} -color green -collection $collection_2
    gui_create_vmbucket -vmname $vm -name context_mem \
        -title {context_mem} -color blue -collection $collection_3
    gui_create_vmbucket -vmname $vm -name risc_core \
        -title {risc_core} -color yellow -collection $collection_4
}
gui_show_map -map Macros -show true
```

Figure 179 Custom Visual Mode

